

Service Manual

LG-LS990

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1. INTRODUCTION

1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

Toll fraud, the unauthorized use of telecommunications system by an unauthorized part (for example, persons other than your company's employees, agents, subcontractors, or person working on your company's behalf) can result in substantial additional charges for your telecommunications services.

System users are responsible for the security of own system. There are may be risks of toll fraud associated with your telecommunications system. System users are responsible for programming and configuring the equipment to prevent unauthorized use. The manufacturer does not warrant that this product is immune from the above case but will prevent unauthorized use of common carrier telecommunication service of facilities accessed through or connected to it. The manufacturer will not be responsible for any charges that result from such unauthorized use.

B. Incidence of Harm

If a telephone company determines that the equipment provided to customer is faulty and possibly causing harm or interruption in service to the telephone network, it should disconnect telephone service until repair can be done. A telephone company may temporarily disconnect service as long as repair is not done.

C. Changes in Service

A local telephone company may make changes in its communications facilities or procedure. If these changes could reasonably be expected to affect the use of the phones or compatibility with the net work, the telephone company is required to give advanced written notice to the user, allowing the user to take appropriate steps to maintain telephone service.

D. Maintenance Limitations

Maintenance limitations on the phones must be performed only by the manufacturer or its authorized agent. The user may not make any changes and/or repairs expect as specifically noted in this manual. Therefore, note that unauthorized alternations or repair may affect the regulatory status of the system and may void any remaining warranty.

E. Notice of Radiated Emissions

This model complies with rules regarding radiation and radio frequency emission as defined by local regulatory agencies. In accordance with these agencies, you may be required to provide information such as the following to the end user.

F. Pictures

The pictures in this manual are for illustrative purposes only; your actual hardware may look slightly different.

G. Interference and Attenuation

A phone may interfere with sensitive laboratory equipment, medical equipment, etc. Interference from unsuppressed engines or electric motors may cause problems.

H. Electrostatic Sensitive Devices

ATTENTION

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign.

Following information is ESD handling:

- Service personnel should ground themselves by using a wrist strap when exchange system boards.
- When repairs are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system boards or parts like EEPROM to the factory, use the protective package as described.

1.3 Specifications

KEY FEATURES

- Android 4.4.2 - Powered Smartphone
- Up to 7 Customizable Home Screens
- 5.46" Touch Screen
- 13MP AF with dual flash, 2.1M VT Image sensor
- 4G Mobile Hotspot
- DivX® H.264, H.263 Capable

ENHANCED USER EXPERIENCE

- Android 4.4.2
- Home Screens – customize 5 or 7 screens with shortcuts to apps, favorites, and widgets
- QWERTY Keyboard with Skype® – use one continuous finger motion to input text when composing a message
- Exchange Support – improved security, remote wipe capability for Exchange administrators, support for Exchange calendars in the Calendar application, auto-discovery for easy setup and syncing of an Exchange account,* and Global Address Lists lookup in the Email application
- Proximity Sensor – locks the touch screen and buttons while talking on the phone
- Motion Sensor – Digital compass , Accelerometer
- USB Charging via Computer
- Airplane Mode (RF Off)

*Available for Exchange 2007 and higher.

DESIGN

- Sleek and Stylish Design
- 5.46" Display
- Capacitive Touch Screen – Glass with fast and accurate response
- Touch Vibration for Tactile Feedback*
- Micro USB Charging Port

*Only available on certain interactions with the touch screen.

MOBILE MEDIA

- Android Market™ – browse and download thousands of free and paid apps from around the world†
- Live Wallpaper – dynamic, animated backgrounds

- Full Web Browsing
- Connect to Social Networks – send quick updates to Twitter™, Facebook®, MySpace™,* or added social networks†
- Mobile Instant Messaging – Google Talk™, AIM®,* Yahoo!® Messenger,* Windows Live™*†
- Gmail™ and Email – mobile personal and corporate email†
- Visual Voice Mail†
- Text, Picture, Video, and Voice Messaging†
- Flash 10.1 for Mobile Devices
- Java Support

*Applications downloaded from Android Market.

CAMERA/VIDEO

- 13MP AF/OIS with dual flash, 2.1M VT Image sensor and Camcorder*
- Camera Resolutions: 13M(4160x3120(4:3)), W10M(4160x2340(16:9)), 9M(3120x3120(1:1)), 3M(2048 x 1536(4:3))
- Camera and Video Zoom: up to 8x
- Image Editor: Zoom, Exposure, Artistic Effect, Color, Adjustment
- Brightness, ISO, White Balance,* Color Effects, Scene Mode, Timer
- Burst Shot – take 20 shots in quick succession
- Video Resolutions: UHD(3840 x 2160), FHD(1920 x 1080), HD(1280 x 720), 120-HD(Slow motion)
- Front-facing camera Resolutions: W2M(1920 x 1080), 1M(1280 x 960)
- *Available in both video and camera modes.

MULTIMEDIA

- Video Player for WMV, MP4, MKV, 3GP, and 3G2 Formats
- Music Player for MP3, WMA, and Unprotected AAC
- Stereo Sound via Headset or *Bluetooth*
- Multitask While Playing Music in the Background
- Music Library – Organized by Artists, Albums, Songs, Genres, Folders and Playlists
- Repeat and Shuffle Music Playback Modes

- microSD Memory Slot with Support for up to 32GB Storage
- USB Mass Storage – transfer files between microSD™ card and PC*

BLUETOOTH®

- Version: 4.0
- Save up to 100 *Bluetooth* Pairings*
- Supported Profiles: headset, hands-free, object push, advanced audio distribution (stereo), audio/video remote control, file transfer, phone book access, message access, serial port
- Listen to Music with Optional Stereo *Bluetooth* Headset**
- Send User-Generated Pictures (JPEG) and Videos via *Bluetooth* Wireless Technology

*Depends on available memory.

**Accessories sold separately.

VOICE/AUDIO

- One-Touch Speakerphone*
- Voice Clarity*
- Voice Commands
- Voice-Activated Dialing
- Text to Speech
- MP3 Music Ringer Support (clips from hit songs)†
- Voice Memo Recording
- More 20 Unique Ringtones + Vibrate and Silent Modes
- TTY Support

TOOLS & DATA

- Android Apps: Browser, Calculator, Calendar (Corporate or Google), Camera, Clock, Contacts, Email, Gallery, Gmail, Market, Messaging, Music, Phone, Settings, Voice Dialer, Voice Recorder, Voice Search
- Metro Navigator® – spoken turn-by-turn directions showing real-time traffic
- Wi-Fi Connectivity – IEEE 802.11 a/b/g/n, ac

1. INTRODUCTION

- Phone Book – unlimited fields* for numbers, emails, physical addresses, IM screen names, notes, nickname, web addresses, special dates, and a picture ID**
- FOTA Capable – upgrade firmware over the air†
- S-GPS and A-GPS Support for Enhanced Location Accuracy

*Depends on available memory.

**Dependent on photos stored in your gallery.

SPECIFICATIONS

- Technology: GSM/ WCDMA /LTE/GPS/ WiFi/ BT /NFC
- Frequencies:

Support Band	TX Freq (MHz)	RX Freq (MHz)
GSM850	824-849	869-894
GSM900	880-915	925-960
DCS1800	1710-1785	1805-1880
PCS1900	1850-1910	1930-1990
W2100(B1)	1920-1980	2110-2170
W1900(B2)	1850-1910	1930-1990
W850(B5)	824-849	869-894
W900(B8)	880-915	925-960
LTE B1	1920-1980	2110-2170
LTE B3	1710-1785	1805-1880
LTE B7	2500-2570	2620-2690
LTE B8	880-915	925-960
LTE B20	832-862	791-821
LTE B28A	718-748	773-803
LTE B28B	703-733	758-788
LTE B40	2300-2400	
BT/WiFi	2400 MHz ~ 2483.5 MHz & 5150 MHz ~ 5825 MHz	
GPS	1574.42 MHz ~ 1576.42 MHz, 1597.55MHz ~ 1605.86MHz	
NFC	13.56MHz	

- Data Transmission: EDGE, GPRS (Class 12) , HSPA+(Cat. 24) HSUPA(Cat. 6), LTE(Cat 3)
- Dimensions: 146.3(H) x 74.6(W) x 8.95(D) (mm)
- Weight: 149g (with Battery)
- Display: 5.46", TFT
- Standard Battery: min. 2940[mAh], typ. 3000[mAh]
- Internal Memory Size: 16Gb(SDRAM DDR3) + 16GB eMMC

ACCESSORIES

- Standard Battery*
- Travel Adapter and USB Data Cable*
- Bluetooth Headset (Optional)
- Bluetooth Hands-free kit (Optional)
- Stereo Headset*
- Vehicle Power Charger (VPC) (Optional)

2. PERFORMANCE

2.1 Product Name

LG-LS990 : CDMA BC0/1/10 + LTE B25/26/41 + WCDMA850/1900/2100 + GSM850/EGSM/DCS/PCS(HSPA+ 21Mbps(cat12)/ HSUPA 5.8Mbps(cat6) / GPRS Class10/EDGE Class10)

2.2 Supporting Standard

Item	Feature	Comment
Supporting Standard	CDMA BC0/BC1/BC10 LTE FDD B25/26, TDD B41 WCDMA B1/B2/B5, EGSM/GSM850/DCS1800/PCS1900 with seamless handover	
Frequency Range	CDMA BC10 : TX 817~821MHz, CDMA BC10 : RX 862~866MHz CDMA BC0 : TX 824~849MHz, CDMA BC0 : RX 869~894MHz CDMA BC1 : TX 1950~1910MHz, CDMA BC1 : RX 1930~1990MHz LTE B25 : TX 1850~1915MHz, LTE B25 : RX 1930~1995MHz LTE B26 : TX 814~849MHz, LTE B26 : RX 859~894MHz WCDMA B5 : TX 824~849MHz, WCDMA B5 : RX 869~894MHz WCDMA B2 : TX 1850~1910MHz, WCDMA B2 : RX 1930~1990MHz EGSM :TX 880~915 MHz EGSM : RX 925~960 MHz GSM850 : TX 824~849 MHz GSM850 : RX 869~894 MHz GSM900 : TX 880~915 MHz	

2. PERFORMANCE

	GSM900 : RX 925~960 MHz DCS1800 : TX 1710~1785 MHz DCS1800 : RX 1805~1880 MHz PCS1900 : TX 1850~1910 MHz PCS1900 : RX 1930~1990 MHz	
Application Standard	WAP 2.0	

2.3 Main Parts & Solution

Item	Part Name	Comment
Digital Baseband	MSM8974AC : Qualcomm	
Analog Baseband	MSM8974AC : Qualcomm	
RF Chip	WTR1625L : Qualcomm	

2.4 HW Features

Item	Feature	Comment
Form Factor	DOP type	
Battery	1) Capacity Standard : Li-Ion, 3000mAh	
	2) Packing Type : <u>Soft Pack</u>	
Size	Standard : 146.3 x 74.6 x 8.95	
Weight	About 149g	With Battery
RX sensitivity	CDMA(BC0) : -109 dBm CDMA(BC1) : -108 dBm CDMA(BC10) : -109 dBm LTE(FDD B25) : -98 dBm LTE(FDD B26) : -98 dBm LTE(TDD B41) : -98 dBm WCDMA(B1) : -109 dBm WCDMA(B2) : -109 dBm WCDMA(B5) : -109 dBm EGSM : -109 dBm GSM850 : -109 dBm DCS 1800 : -108 dBm PCS 1900 : -107 dBm	
TX output power	CDMA/ WCDMA/ GSM/ LTE/ GPRS	CDMA BC0 : 24.8dBm CDMA BC1 : 24.2dBm CDMA BC10 : 24.5dBm WCDMA B1 : 23dBm WCDMA B2 : 23dBm WCDMA B5 : 23dBm EGSM : 32.5dBm GSM850 : 32.5dBm DCS1800 : 29.5dBm PCS 1900: 29.5dBm LTE B25 : 23dBm LTE B26 : 23dBm LTE B41 : 23.5dB
EDGE compatibility	GPRS Class 10	
Display	Main LCD	

2. PERFORMANCE

	TFT Main LCD (5.46", 2560 X 1440 Pixels)	
Built-in Camera	13M AF + 2.1M VT	
Status Indicator	Yes	
Keypad	Full Touch Screen Back Key : 3	Back Key : Power On Volume up/down
ANT	Main : Internal Fixed Type	
System connector	5 Pin u-USB	
Ear Phone Jack	3.5 pi type	
PC synchronization	Yes	
Memory	3GB(SDRAM LPDDR3) + 32GB eMMC	
Speech coding	FR, EFR, HR, AMR	
Vibrator	Built in Vibrator	
Blue Tooth	Bluetooth 4.0+BLE	
MIDI(for Buzzer Function)	SW Decoded 72Poly	
Music Player	MP3/WMA/AAC/MIDI/EAAC+/HE AAC/OGG	
Video Player	MPEG4, H.264	
Camcorder	MPEG4 or H.263	
Voice Recording	Yes	
Speaker Phone mode Support	Yes	
Travel Adapter	Yes	

1) RSSI Display

	Specification			
	LTE	WCDMA & GSM	CDMA	
BAR 6	-84 dBm ~ RF Energy	-75dBm ~ RF Energy	-75dBm ~ RF Energy	Antenna BAR
BAR 6 → 5	-84dBm ~ -89 dBm	-85dBm ~ -75dBm	-85dBm ~ -75 dBm	
BAR5 → 4	-89dBm ~ -94 dBm	-91dBm ~ -85dBm	-90dBm ~ -85 dBm	
BAR4 → 3	-94dBm ~ -99 dBm	-97dBm ~ -91dBm	-95dBm ~ -90 dBm	
BAR 3 → 2	-99dBm ~ -104 dBm	-102dBm ~ -97dBm	-100dBm ~ -95 dBm	
BAR2 → 1	-104dBm~-109dBm	-107dBm ~ -102dBm	-105dBm ~ -100 dBm	
BAR 1 → 0	-109dBm ~ -114dBm	-107dBm~ RF Energy	-109dBm ~ -114 dBm	
BAR 0 → No SVC	-114 dBm~ RF Energy	No pilot channel detection	-114dBm ~ RF Energy	

Measure RSSI after 10 seconds if you change power.

2) Charging Time

3.5 hour under (3000mAh battery, 1.8A T/A)

3) Travel Charger

Input : 100 ~ 240 V, 50/60 Hz

Output : 5.0V, 1800 mA

4) Battery Type

Li-ion Battery Pack, 3000 mAh

Standard Voltage : 3.8 V, Battery Full Charge Voltage : 4.35V

5) Current Consumption

Item	WCDMA Only	GSM Only	LTE	Comment
Standby	8.5mA ↓ (DRX 7)	8.5mA(PP=5) ↓	8.5mA ↓ (DRX 2.56)	GPS & Wifi Off Neighborhood Cell Off, LCD Off
Talk Mode	- Max Pwr : 600mA ↓ @ Avg - Tx 10dBm : 300mA ↓ @ Avg	350mA(Tx Lvl: 5) ↓	300mA ↓ Cat2 @Tx(10dBm)	GPS & Wifi Off Neighborhood Cell Off, LCD Off
Power Off	500uA ↓	500uA ↓	500uA ↓	

6) Battery Bar

Battery Bar	Specification	Battery Bar	Specification
Bar 20(Full)	98%	Bar 9 -> Bar 8	43% -> 42%
Bar 20 -> Bar 19	98% -> 97%	Bar 8 -> Bar 7	38% -> 37%
Bar 19 -> Bar 18	93% -> 92%	Bar 7 -> Bar 6	33% -> 32%
Bar 18 -> Bar 17	88% -> 87%	Bar 6 -> Bar 5	28% -> 27%
Bar 17 -> Bar 16	83% -> 82%	Bar 5 -> Bar 4	23% -> 22%
Bar 16 -> Bar 15	78% -> 77%	Bar 4 -> Bar 3	16% -> 15%
Bar 15 -> Bar 14	73% -> 72%	Bar 3 -> Bar 2	13% -> 12%
Bar 14 -> Bar 13	68% -> 67%	Bar 2 -> Bar 1	8% -> 7%
Bar 13 -> Bar 12	63% -> 62%	Bar 1 -> Bar 0	3% -> 2%
Bar 12 -> Bar 11	58% -> 57%	Power off	1%
Bar 11 -> Bar 10	53% -> 52%	Low battery pop-up	15% , 5%, 0%
Bar 10 -> Bar 9	48% -> 47%		

2.5 SW Features

Item	Feature	Comment
RSSI	0 ~ 6 Levels	
Battery Charging	0 ~ 20 Levels	Using Fuel Gauging (%)
Key Volume	0 ~ 7 Level	
Audio Volume	0 ~ 15 Level	
Time / Date Display	Yes	
Multi-Language	Yes	English, Espanol, Franch, Korea, Chinese, etc
Quick Access Mode	Phone / Messaging / Apps / Web / Contact	
PC Sync	Yes	
Speed Dial	Yes	Voice mail center -> 1 key
Profile	Yes	not same with feature phone setting
CLIP / CLIR	Yes	
Phone Book	Name / Number / Email / Website / Postal addresses / Organizations / Groups / Birthday Notes / Ringtone	There is no limitation on the number of items. It depends on available memory amount.
Last Received Number	Yes	There is no limitation on the number of items. It depends on available memory amount.
Last Missed Number	Yes	There is no limitation on the number of items. It depends on available memory amount.
Search by Number / Name	Name / Number	
Group	Yes	There is no limitation on the number of items. It depends on available memory amount.

2. PERFORMANCE

Fixed Dial Number	Yes	
Service Dial Number	No	
Own Number	Yes	Read / Write
Voice Memo	Yes	Support voice recorder
Call Reminder	No	Missed call popup
Network Selection	Automatic	
Mute	Yes	
Call Divert	Yes	Support call forwarding
Call Barring	Yes	
Call Charge (AoC)	No	
Call Duration	Yes	
SMS (EMS)	There is no limitation on the number of items. It depends on available memory amount.	EMS does not support.
SMS Over GPRS	No	No
EMS Melody / Picture	No	
Send / Receive / Save	No	
MMS MPEG4	Yes	
Send / Receive / Save	Yes	
Long Message	depends on SIM	
Cell Broadcast	Yes	
Download	Yes	
Game	TBD	
Calendar	Yes	
Memo	No	integrated to QMemo+
World Clock	Yes	
Unit Convert	No	
Stop Watch	Yes	
Wall Paper	Yes	
WAP Browser	No	WAP stack and wml are not supported.
Download Melody / Wallpaper	Yes	Over web browser

2. PERFORMANCE

SIM Lock	Yes	Operator Dependent
SIM Toolkit	Class 1, 2, 3, C, E	
MMS	Yes	
EONS	Yes	
CPHS	Yes	V4.2
ENS	No	
Camera	Yes	13M AF VT : 2.1M
JAVA	No	Android do not support JAVA
Voice Dial	Yes	
IrDa	No	
Bluetooth	Yes	Ver. 3.0
FM radio	Yes	
GPRS	Yes	Class 12
EDGE	Yes	Class 12
Hold / Retrieve	Yes	
Conference Call	Yes	Max. 7
DTMF	Yes	
Memo pad	No	integrated to QMemo+
TTY	No	
AMR	Yes	
SyncML	No	
IM	Yes	Gtalk
Email	Yes	

2.6 H/W SPEC.

1) GSM transceiver specification

Item	Specification
Phase Error	Rms : 5° Peak : 20°
Frequency Error	GSM : 0.1 ppm DCS / PCS : 0.1 ppm
EMC (Radiated Spurious Emission Disturbance)	GSM/DCS : < -28dBm
Transmitter Output power	GSM850, EGSM : 5dBm ~ 33dBm ± 3dB DCS/PCS : 0dBm ~ 30dBm ± 3dB
Burst Timing	< 3.69us
Spectrum due to modulation out to less than 1800kHz offset	200kHz : -36dBm 600kHz : -51dBm/-56dBm
Spectrum due to modulation out to larger than 1800kHz offset to the edge of the transmit band	GSM850, EGSM : 1800-3000kHz : < -63dBc(-46dBm) 3000kHz-6000kHz : < -65dBc(-46dBm) 6000kHz < : < -71dBc(-46dBm) DCS, PCS : 1800-3000kHz : < -65dBc(-51dBm) 6000kHz < : < -73dBc(-51dBm)
Spectrum due to switching transient	400kHz : -19dBm/-22dBm(5/0), -23dBm 600kHz : -21dBm/-24dBm(5/0), -26dBm
Reference Sensitivity – TCH/FS	Class II(RBER) : -105dBm(2.439%)
Usable receiver input level range	0.012(-15 - -40dBm)
Intermodulation rejection – Speech channels	± 800kHz, ± 1600kHz : -98dBm / -96dBm (2.439%)
AM Suppression – GSM : -31dBm – DCS : -29dBm	-98dBm/-96dBm (2.439%)

2) WCDMA transmitter specification

Item	Specification
Transmit Frequency	BD1: 1920MHz ~ 1980 MHz BD2: 1850~1910 MHz BD5: 824 MHz ~849 MHz BD8: 880 MHz ~960 MHz
Maximum Output Power	+24 dBm / 3.84 MHz, +1 / -3 dB
Frequency Error	within ± 0.1 PPM
Open Loop Power Control	Normal Conditions : within ± 9 dB, Extreme Conditions : within ± 12 dB
Minimum Transmit Power	< -50 dBm / 3.84 MHz
Occupied Bandwidth	< 5 MHz at 3.84 Mcps (99% of power)
Adjacent Channel Leakage Power Ratio (ACLR)	> 33 dB @ ± 5 MHz, > 43 dB @ ± 10 MHz
Spurious Emissions $ f-f_c > 12.5$ MHz	< -36 dBm / 1 kHz RW @ $9 \text{ kHz} \leq f < 150 \text{ kHz}$ < -36 dBm / 10 kHz RW @ $150 \text{ kHz} \leq f < 30 \text{ MHz}$ < -36 dBm / 100 kHz RW @ $30 \text{ MHz} \leq f < 1 \text{ GHz}$ < -30 dBm / 1 MHz RW @ $1 \text{ GHz} \leq f < 12.75 \text{ GHz}$ < -60 dBm / 3.84 MHz RW @ $869 \text{ MHz} \leq f \leq 894 \text{ MHz}$ < -60 dBm / 3.84 MHz RW @ $1930 \text{ MHz} \leq f \leq 1900 \text{ MHz}$ < -60 dBm / 3.84 MHz RW @ $2110 \text{ MHz} \leq f \leq 2155 \text{ MHz}$ < -67 dBm / 100 kHz RW @ $925 \text{ MHz} \leq f \leq 935 \text{ MHz}$ < -79 dBm / 100 kHz RW @ $935 \text{ MHz} < f \leq 960 \text{ GHz}$ < -71 dBm / 100 kHz RW @ $1805 \text{ MHz} \leq f \leq 1880 \text{ MHz}$ < -41 dBm / 300 kHz RW @ $1884.5 \text{ MHz} < f < 1919.6 \text{ MHz}$
Transmit Intermodulation	< -31 dBc @ 5 MHz & < -41 dBc @ 10 MHz when Interference CW Signal Level = -40 dBc
Error Vector Magnitude	< 17.5 %, when Pout $\geq$ -20 dBm
Peak Code Domain Error	< -15 dB at Pout $\geq$ -20 dBm

3) WCDMA receiver specification

Item	Specification																			
Receive Frequency	BD1: 2110 MHz ~2170 MHz BD2: 1850~1910 MHz BD5: 869 MHz ~ 894 MHz																			
Reference Sensitivity Level	Band1 : BER < 0.001 when Îor = -106.7 dBm / 3.84 MHz Band2 : BER < 0.001 when Îor = -106.7 dBm / 3.84 MHz Band5 : BER < 0.001 when Îor = -106.7 dBm / 3.84 MHz																			
Maximum Input Level	BER < 0.001 when Îor = -25 dBm / 3.84 MHz																			
Adjacent Channel Selectivity (ACS)	ACS > 33 dB where BER < 0.001 when Îor = -92.7 dBm / 3.84 MHz & loac = -52 dBm / 3.84 MHz @ ±5 MHz																			
Blocking Characteristic	BER < 0.001 when Îor = -103.7 dBm / 3.84 MHz & lblocking = -56 dBm / 3.84 MHz @ Fuw(offset) = ±10 MHz or lblocking = -44 dBm / 3.84 MHz @ Fuw(offset) = ±15 MHz																			
Spurious Response	BER < 0.001 when Îor = -103.7 dBm / 3.84 MHz & lblocking = -44 dBm																			
Intermodulation	BER < 0.001 when Îor= -103.7 dBm / 3.84 MHz & louw1 = -46 dBm @ Fuw1(offset) = ±10 MHz & louw2 = -46 dBm / 3.84 MHz @ Fuw2(offset) = ±20 MHz																			
Spurious Emissions	< -57 dBm / 100 kHz BW @ 9 kHz ≤ f < 1 GHz < -47 dBm / 1 MHz BW @ 1 GHz ≤ f ≤ 12.75 GHz																			
Inner Loop Power Control In Uplink	Adjust output(TPC command) <table><tr><td>cmd</td><td>1dB</td><td>2dB</td><td>3dB</td></tr><tr><td>+1</td><td>+0.5 / 1.5</td><td>+1 / 3</td><td>+1.5 / 4</td></tr><tr><td>0</td><td>-0.5 / +0.5</td><td>-0.5 / +0.5</td><td>-0.5 / +0.5</td></tr><tr><td>-1</td><td>-0.5 / -1.5</td><td>-1 / -3</td><td>-1.5 / -4</td></tr></table> group(10equal command group) <table><tr><td>+1</td><td>+8 / +12</td><td>+16 / +24</td></tr></table>	cmd	1dB	2dB	3dB	+1	+0.5 / 1.5	+1 / 3	+1.5 / 4	0	-0.5 / +0.5	-0.5 / +0.5	-0.5 / +0.5	-1	-0.5 / -1.5	-1 / -3	-1.5 / -4	+1	+8 / +12	+16 / +24
cmd	1dB	2dB	3dB																	
+1	+0.5 / 1.5	+1 / 3	+1.5 / 4																	
0	-0.5 / +0.5	-0.5 / +0.5	-0.5 / +0.5																	
-1	-0.5 / -1.5	-1 / -3	-1.5 / -4																	
+1	+8 / +12	+16 / +24																		

4) HSDPA transmitter specification

Item	Specification															
Transmit Frequency	BD1: 1920MHz ~ 1980 MHz BD2: 1850~1910 MHz BD5: 880 MHz ~ 915 MHz BD8: 824 MHz ~ 849 MHz															
Maximum Output Power	Sub-Test 1=2/15 25.7~20.3dBm / 3.84 MHz 2=12/15 25.7~20.3dBm / 3.84 MHz 3=15/8 25.7~19.8dBm / 3.84 MHz 4=15/4 25.7~19.8dBm / 3.84 MHz															
Spectrum Emission Mask	Sub-Test : 1=2/15, 2=12/15, 3=15/8, 4=15/4 <table><tr><td>Frequency offset from carrier Δf</td><td>Minimum requirement</td><td>Measurement Bandwidth</td></tr><tr><td>2.5 ~ 3.5 MHz</td><td>-35-15$\times(\Delta f-2.5)$dBc</td><td>30 kHz</td></tr><tr><td>3.5 ~ 7.5 MHz</td><td>-35-1$\times(\Delta f-3.5)$dBc</td><td>1 MHz</td></tr><tr><td>7.5 ~ 8.5 MHz</td><td>-35-10$\times(\Delta f-7.5)$dBc</td><td>1 MHz</td></tr><tr><td>8.5 ~ 12.5 MHz</td><td>-49dBc</td><td>1 MHz</td></tr></table>	Frequency offset from carrier Δf	Minimum requirement	Measurement Bandwidth	2.5 ~ 3.5 MHz	-35-15 $\times(\Delta f-2.5)$ dBc	30 kHz	3.5 ~ 7.5 MHz	-35-1 $\times(\Delta f-3.5)$ dBc	1 MHz	7.5 ~ 8.5 MHz	-35-10 $\times(\Delta f-7.5)$ dBc	1 MHz	8.5 ~ 12.5 MHz	-49dBc	1 MHz
Frequency offset from carrier Δf	Minimum requirement	Measurement Bandwidth														
2.5 ~ 3.5 MHz	-35-15 $\times(\Delta f-2.5)$ dBc	30 kHz														
3.5 ~ 7.5 MHz	-35-1 $\times(\Delta f-3.5)$ dBc	1 MHz														
7.5 ~ 8.5 MHz	-35-10 $\times(\Delta f-7.5)$ dBc	1 MHz														
8.5 ~ 12.5 MHz	-49dBc	1 MHz														
Adjacent Channel Leakage Power Ratio (ACLR)	Sub-Test : 1=2/15, 2=12/15, 3=15/8, 4=15/4 > 33 dB @ ± 5 MHz > 43 dB @ ± 10 MHz															
Error Vector Magnitude	< 17.5 %, when Pout $\geq$ -20 dBm															

5) HSDPA receiver specification

Item	Specification
Receive Frequency	BD1: 2110 MHz ~2170 MHz BD2: 1850~1910 MHz BD5: 869MHz ~ 894 MHz BD8: 925MHz ~ 960 MHz
Maximum Input Level (BLER or R), 16QAM Only	Sub-Test : 1=2/15, 2=12/15, 3=15/8, 4=15/4 BLER < 10% or R >= 700kbps

6) LTE transmitter specification

Item	Spec.	Min	Typ.	Max	
Maximum Output Power(Class III)	Power class 3 (Environmental Test)	20.3 (20.3)	23.0 (23.0)	25.7 (25.7)	dBm 10MHz
Frequency Error	QPSK	-0.1PPM -15hz	-	+0.1PPM -15hz	
EVM	QPSK 16 QAM			17.5 12.5	%
IQ-component	QPSK, 12RB			-24.2	dBc
In-band emissions for non allocated RB	QPSK, 12RB			-24.2	dB
Spectrum Flatness	QPSK, 50RB	-2		+3.4	dBpp
Occupied Bandwidth	QPSK, 50RB			10	MHz
ACLR (Adjacent Channel Leakage Power Ratio)	E-UTRA ACLR1			-29.2	dB
	UTRA ACLR1			-32.2	dB
	UTRA ACLR2			-35.2	dB
Transmitter Spurious emissions	9 kHz < f < 150 kHz			-36	dBm 1kHz
	150 kHz < f < 30 kHz			-36	dBm 10kHz
	30 kHz < f < 1 GHz			-36	dBm 100kHz
	1GHz < f < 12.75 GHz			-30	dBm 1MHz

7) LTE receiver specification

Item	Spec.	Min	Typ.	Max	
Sensitivity Level (DL : QPSK, 50RB UL : QPSK, 25RB)	Dual Receiver			B2: -95 B4 : -97 B17: -94	dBm /10MHz
	Single Receiver (exception in 3GPP)			B2 : -92 B4 : -94.0 B17: -91.0	dBm /10MHz
Maximum input Level	Dual Receiver (DL : 64QAM, 50RB UL : QPSK, 12RB)			-25.7	dBm /10MHz
ACS (Adjacent Channel Selectivity)	Dual Receiver (DL : QPSK, 50RB UL : QPSK, 12RB) $\Delta f \pm 7.5\text{MHz}$, LTE BW 5MHz			33	dB
In-band Blocking (DL : QPSK, 50RB UL : QPSK, 20RB)	Dual Receiver $\Delta f \pm 7.5\text{MHz}$, LTE BW 5MHz $\Delta f \pm 12.5\text{MHz}$, LTE BW 5MHz			-88.3	dBm /10MHz
Out-of band Blocking (DL : QPSK, 50RB UL : QPSK, 20RB)	Dual Receiver $\Delta f \pm 15\text{MHz}$, ~85MHz, CW	-		-88.3	dBm /10MHz
Narrow Band Blocking (DL : QPSK, 50RB UL : QPSK, 20RB)	Dual Receiver $\Delta f \pm 5.2125\text{MHz}$, CW			-81.3	dBm /10MHz
Intermodulation (DL: QPSK, 50RB UL : QPSK, 20RB)	Dual Receiver $\Delta f \pm 12.5\text{MHz}$, CW $\Delta f \pm 25\text{MHz}$, LTE BW 5MHz			-88.3	dBm /10MHz
Spurious Emission (DL: QPSK, 0RB UL : QPSK, 0RB)	Dual Receiver $30\text{MHz} < f < 1\text{GHz}$			-57	dBm /100MHz
	$1\text{GHz} < f < 12.75\text{GHz}$			-47	dBm /1MHz

8) CDMA (DCN) transmitter specification

Item	Spec.	Min	Typ.	Max	Unit
Frequency Accuracy	-	-300	-	+300	Hz
Time Reference	-	-1	-	+1	uS
Waveform Quality	-	0.944	0.98	-	-
Open Loop Output Power Class III	Ior = 93.5dBm/1.23MHz	10.5	20	25.5	dBm/1.23MHz
	Ior = 65dBm/1.23MHz	-17.5	-8	1.5	dBm/1.23MHz
	Ior = 25dBm/1.23MHz	-57.5	-48	-38.5	dBm/1.23MHz
Maximum Output power(Class III)	Power class 3	23.0	24.8	26.0	dBm/1.23MHz
Maximum Spurious Emission	$\Delta f = \pm 900\text{KHz}$ Maximum Suprious Emission	-42	-48	-	dBc/30KHz
	$\Delta f = \pm 1.98\text{MHz}$ Maximum Suprious Emission	-54	-60	-	dBc/30KHz
Standby Output Power	Ior = 75dBm/1.23MHz (when TX disabled)	-	-	-61	dBm/1MHz
Maximum Controlled Output Power	-	-	-53	-50	dBm/1.3MHz
Conducted Spurious Emissions (Pout = 13dBm)	$F_c \pm \Delta f 885\text{KHz}$	-	-	-42	dBc/30KHz
	$F_c \pm \Delta f 1.98\text{KHz}$	-	-	-54	dBc/30KHz

9) CDMA (DCN) receiver specification

Item	Spec.	Min	Typ.	Max	Unit
Sensitivity	-	-	-106	-104	dBm/ 1.23MHz
Single Tone Desensitization	Ior = 94dBm Jammer = GMSK -45dBm $\Delta f = \pm 1100\text{KHz}$ FER < 1% -> Jammer PWR	-45	-43	-	dBm
Intermodulation Spurious Response Attenuation	Ior = 101dBm Jammer = -43dBm $\Delta f = \pm 900\text{KHz}, \pm 1.7\text{MHz}$ FER < 1% -> Jammer PWR	-43	-41	-	dBm
	Ior = 90dBm Jammer = -32dBm $\Delta f = \pm 900\text{KHz}, \pm 1.7\text{MHz}$ FER < 1% -> Jammer PWR	-32	-30	-	dBm
	Ior = 79dBm Jammer = -21dBm $\Delta f = \pm 900\text{KHz}, \pm 1.7\text{MHz}$ FER < 1% -> Sector PWR	-	-84	-79	dBm/ 1.23MHz
Conducted Spurious Emissions	Tx =off, RX Band (869~894MHz)	-	-	-81	dBm/ 1.23MHz
	Tx =off, TX Band (824~849MHz)	-	-	-61	dBm/ 1.23MHz
	Tx =off, RX/TX Band	-	-	-47	dBm/ 30MHz

10) CDMA (UCPSC) transmitter specification

Item	Spec.	Min	Typ.	Max	Unit
Frequency Accuracy	-	-150	-	+150	Hz
Time Reference	-	-1	-	+1	uS
Waveform Quality	-	0.944	0.98	1	-
Open Loop Output Power Class II	lor = 97dBm/1.23MHz	10.5	20	25.5	dBm/1.23MHz
	lor = 65dBm/1.23MHz	-20.5	-11	-1.5	dBm/1.23MHz
	lor = 25dBm/1.23MHz	-60.5	-51	-41.5	dBm/1.23MHz
Maximum Output power(Class II)	Power class 2	23.0	24.5	26.0	dBm/1.23MHz
Maximum Spurious Emission	$\Delta f = \pm 1.25\text{MHz}$ Output power-> Max	-42	-46	-	dBc/30KHz
	$\Delta f = \pm 1.98\text{MHz}$ Output power-> Max	-50	-55	-	dBc/30KHz
Standby Output Power	lor = 75dBm/1.23MHz (when TX disabled)	-	-	-61	dBm/1MHz
Maximum Controlled Output Power	-	-	-53	-50	dBm/1.23MHz
Conducted Spurious Emissions (Pout = 13dBm)	$F_c \pm \Delta f 1.25\text{KHz}$	-	-	-42	dBc/30KHz
	$F_c \pm \Delta f 1.98\text{KHz}$	-	-	-54	dBc/30KHz

11) CDMA (USPCS) receiver specification

Item	Spec.	Min	Typ.	Max	Unit
Sensitivity	-	-	-106	-104	dBm/ 1.23MHz
Single Tone Desensitization	Ior = 94dBm $\Delta f = \pm 1.25\text{MHz}$ Jammer = GMSK -45dBm FER < 1% -> Jammer PWR	-45	-23	-	dBm
Intermodulation Spurious Response Attenuation	Ior = 94dBm Jammer = GMSK -44dBm $\Delta f = \pm 1.27\text{KHz}, \pm 2.64\text{MHz}$ FER < 1% -> Jammer PWR	-44	-42	-	dBm
Conducted Spurious Emissions	Tx =off, RX Band (1930~1990MHz)	-	-	-81	dBm/ 1.23MHz
	Tx =off, TX Band (1850~1910MHz)	-	-	-61	dBm/ 1.23MHz
	Tx =off, RX/TX Band	-	-	-47	dBm/ 30MHz

12) CDMA (BC10) transmitter specification

Item	Spec.	Min	Typ.	Max	Unit
Frequency Accuracy	-	-300	-	+300	Hz
Time Reference	-	-1	-	+1	uS
Waveform Quality	-	0.944	0.98	1	-
Open Loop Output Power Class III	lor = 93.5dBm/1.23MHz	10.5	20	25.5	dBm/ 1.23MHz
	lor = 65dBm/1.23MHz	-17.5	-8	1.5	dBm/ 1.23MHz
	lor = 25dBm/1.23MHz	-57.5	-48	-38.5	dBm/ 1.23MHz
Maximum Output power(Class III)	Power class 3	23.0	24.5	26.0	dBm/ 1.23MHz
Maximum Spurious Emission	$\Delta f = \pm 900\text{KHz}$ Maximum Suprious Emission	-42	-48	-	dBc/ 30KHz
	$\Delta f = \pm 1.98\text{MHz}$ Maximum Suprious Emission	-50	-60	-	dBc/ 30KHz
Standby Output Power	lor = 75dBm/1.23MHz (when TX disabled)	-	-	-61	dBm/ 1MHz
Maximum Controlled Output Power	-	-	-53	-50	dBm/ 1.3MHz
Conducted Spurious Emissions (Pout = 13dBm)	$F_c \pm \Delta f 885\text{KHz}$	-	-	-42	dBc/ 30KHz
	$F_c \pm \Delta f 1.98\text{KHz}$	-	-	-54	dBc/ 30KHz

13) CDMA (BC10) receiver specification

Item	Spec.	Min	Typ.	Max	Unit
Sensitivity	-	-	-106	-104	dBm/ 1.23MHz
Single Tone Desensitization	Ior = 94dBm Jammer = GMSK -45dBm $\Delta f = \pm 1100\text{KHz}$ FER < 1% -> Jammer PWR	-45	-43	-	dBm
Intermodulation Spurious Response Attenuation	Ior = 101dBm Jammer = -43dBm $\Delta f = \pm 900\text{KHz}, \pm 1.7\text{MHz}$ FER < 1% -> Jammer PWR	-43	-41	-	dBm
	Ior = 90dBm Jammer = -32dBm $\Delta f = \pm 900\text{KHz}, \pm 1.7\text{MHz}$ FER < 1% -> Jammer PWR	-32	-30	-	dBm
	Ior = 79dBm Jammer = -21dBm $\Delta f = \pm 900\text{KHz}, \pm 1.7\text{MHz}$ FER < 1% -> Sector PWR	-	-84	-79	dBm/ 1.23MHz
Conducted Spurious Emissions	Tx =off, RX Band (869~894MHz)	-	-	-81	dBm/ 1.23MHz
	Tx =off, TX Band (824~849MHz)	-	-	-61	dBm/ 1.23MHz
	Tx =off, RX/TX Band	-	-	-47	dBm/ 30MHz

14) GPS receiver specification

Item	Specification
Receive Frequency	1574.42 MHz ~ 1576.42 MHz
Minimum Sensitivity	1 satellite $\geq -142\text{dBm}$, 7 satellites $\geq -147\text{dBm}$ at coarse time aiding

15) WLAN 802.11a transceiver specification

Item	Specification
Transmit Frequency	5150 MHz ~ 5825 MHz (CH36~CH165)
Tx Power Level	$\leq 23\text{dBm}(5150 \sim 5350 \text{ MHz}), \leq 30\text{dBm}(5470 \sim 5825 \text{ MHz})$
Frequency Tolerance	within ± 20 PPM
Chip clock Frequency Tolerance	within ± 20 PPM
Spectrum Mask	≤ -20 @ $\pm 11\text{MHz}$ offset (9MHz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20MHz) ≤ -40 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30MHz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6\text{Mbps}, \leq -8\text{dB}@9\text{Mbps}, \leq -10\text{dB}@12\text{Mbps},$ $\leq -13\text{dB}@18\text{Mbps}, \leq -16\text{dB}@24\text{Mbps}, \leq -19\text{dB}@36\text{Mbps},$ $\leq -22\text{dB}@48\text{Mbps}, \leq -25\text{dB}@54\text{Mbps}$
Spurious Emissions	$< -36 \text{ dBm @ } 30 \text{ MHz to } 47 \text{ MHz}$ $< -54 \text{ dBm @ } 47 \text{ MHz to } 74 \text{ MHz}$ $< -36 \text{ dBm @ } 74 \text{ MHz to } 87,5 \text{ MHz}$ $< -54 \text{ dBm @ } 87,5 \text{ MHz to } 118 \text{ MHz}$ $< -36 \text{ dBm @ } 118 \text{ MHz to } 174 \text{ MHz}$ $< -54 \text{ dBm @ } 174 \text{ MHz to } 230 \text{ MHz}$ $< -36 \text{ dBm @ } 230 \text{ MHz to } 470 \text{ MHz}$ $< -54 \text{ dBm @ } 470 \text{ MHz to } 862 \text{ MHz}$ $< -36 \text{ dBm @ } 862 \text{ MHz to } 1 \text{ GHz}$ $< -30 \text{ dBm @ } 1 \text{ GHz to } 5.15 \text{ GHz}$ $< -30 \text{ dBm @ } 5.35 \text{ GHz to } 5.47 \text{ GHz}$ $< -30 \text{ dBm @ } 5.725 \text{ GHz to } 26.5 \text{ GHz}$
Rx Min input Sensitivity	PER $\leq 10\%$ $-82\text{dBm}@6\text{Mbps}, -81\text{dBm}@9\text{Mbps}, -79\text{dBm}@12\text{Mbps}$ $-77\text{dBm}@18\text{Mbps}, -74\text{dBm}@24\text{Mbps}, -70\text{dBm}@36\text{Mbps}$ $-66\text{dBm}@48\text{Mbps}, -65\text{dBm}@54\text{Mbps}$
Rx Max input Sensitivity	$\geq -30\text{dBm}(6,9,12,18,24,36,48,54\text{Mbps})$ @ PER $\leq 10\%$

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Rx Adjacent Channel Rejection	PER $\leq$ 10%, ACR $\geq$ 16dB@6Mbps, ACR $\geq$ 15dB@9Mbps, ACR $\geq$ 13dB@12Mbps, ACR $\geq$ 11dB@18Mbps, ACR $\geq$ 8dB@24Mbps, ACR $\geq$ 4dB@36Mbps ACR $\geq$ 0dB@48Mbps, ACR $\geq$ -1dB@54Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity
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16) WLAN 802.11b transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2473.5 MHz (CH1~CH11)
Tx Power Level	$\leq$ 20dBm
Frequency Tolerance	within ± 25 PPM
Chip clock Frequency Tolerance	within ± 25 PPM
Spectrum Mask	≤ -30 @ $f_c - 22\text{MHz} < f < f_c - 11\text{MHz}$ and $f_c + 11\text{MHz} < f < f_c + 22\text{MHz}$ ≤ -50 @ $f < f_c - 22\text{MHz}$ and $f > f_c + 22\text{MHz}$
Power ramp on/off time	$\leq 2\mu\text{s}$
Carrier Suppression	$\leq -15\text{dB}$
Modulation Accuracy(Peak EVM)	$\leq 35\%$
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	$\leq -76\text{dBm}$ (1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER $\leq 8\%$
Rx Max input Sensitivity	$\geq -10\text{dBm}$ (1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER $\leq 8\%$
Rx Adjacent Channel Rejection	$\geq 35\text{dB}$ @FER $\leq 8\%$, interference input signal -70dBm@ $f_c \pm 25\text{MHz}$ (11Mbps)

17) WLAN 802.11g transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2473.5 MHz (CH1~CH11)
Tx Power Level	≤ 20dBm
Frequency Tolerance	within ±25 PPM
Chip clock Frequency Tolerance	within ±25 PPM
Spectrum Mask	≤ -20 @ ±11MHz offset (9Mhz ~ 11MHz) ≤ -28 @ ±20MHz offset (11MHz ~ 20Mhz) ≤ -40 @ ±30MHz offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	≤ -5dB@6Mbps, ≤ -8dB@9Mbps, ≤ -10dB@12Mbps, ≤ -13dB@18Mbps, ≤ -16dB@24Mbps, ≤ -19dB@36Mbps, ≤ -22dB@48Mbps, ≤ -25dB@54Mbps
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6Mbps, -81dBm@9Mbps, -79dBm@12Mbps -77dBm@18Mbps, -74dBm@24Mbps, -70dBm@36Mbps -66dBm@48Mbps, -65dBm@54Mbps
Rx Max input Sensitivity	≥ -20dBm(6,9,12,18,24,36,48,54Mbps) @ PER ≤ 10%
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6Mbps, ACR ≥ 15dB@9Mbps, ACR ≥ 13dB@12Mbps, ACR ≥ 11dB@18Mbps, ACR ≥ 8dB@24Mbps, ACR ≥ 4dB@36Mbps ACR ≥ 0dB@48Mbps, ACR ≥ -1dB@54Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

18) WLAN 802.11n transceiver specification

Item	Specification
Transmit Frequency	2400.0 MHz ~ 2473.5 MHz (CH1~CH11) 5150.0 MHz ~ 5825.0 MHz (CH36~CH165)
Tx Power Level	$\leq 20\text{dBm}$ (2400 ~ 2473.5 MHz) $\leq 23\text{dBm}$ (5150 ~ 5350.0 MHz), $\leq 30\text{dBm}$ (5470 ~ 5825.0 MHz)
Frequency Tolerance	within ± 25 PPM(2400 ~ 2473.5 MHz) within ± 20 PPM(5150 ~ 5825.0 MHz)
Chip clock Frequency Tolerance	within ± 25 PPM(2400 ~ 2473.5 MHz) within ± 20 PPM(5150 ~ 5825 MHz)
Spectrum Mask	≤ -20 @ $\pm 11\text{MHz}$ offset (09MHz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20MHz) ≤ -45 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30MHz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6.5\text{Mbps}$, $\leq -10\text{dB}@13\text{Mbps}$, $\leq -13\text{dB}@19.5\text{Mbps}$, $\leq -16\text{dB}@26\text{Mbps}$, $\leq -19\text{dB}@39\text{Mbps}$, $\leq -22\text{dB}@52\text{Mbps}$, $\leq -25\text{dB}@58.5\text{Mbps}$, $\leq -28\text{dB}@65\text{Mbps}$
Spurious Emissions	[2400 MHz ~ 2473.5 MHz (CH1~CH11)] < -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz [5150 MHz ~ 5825 MHz (CH36~CH165)] < -36 dBm @ 30 MHz to 47 MHz < -54 dBm @ 47 MHz to 74 MHz < -36 dBm @ 74 MHz to 87,5 MHz < -54 dBm @ 87,5 MHz to 118 MHz < -36 dBm @ 118 MHz to 174 MHz < -54 dBm @ 174 MHz to 230 MHz < -36 dBm @ 230 MHz to 470 MHz

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	< -54 dBm @ 470 MHz to 862 MHz < -36 dBm @ 862 MHz to 1 GHz < -30 dBm @ 1 GHz to 5.15 GHz < -30 dBm @ 5.35 GHz to 5.47 GHz < -30 dBm @ 5.725 GHz to 26.5 GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6.5Mbps, -79dBm@13Mbps, -77dBm@19.5Mbps -74dBm@26Mbps, -70dBm@39Mbps, -66dBm@52Mbps -65dBm@58.5Mbps, -64dBm@65Mbps
Rx Max input Sensitivity	≥ -20dBm @ PER ≤ 10%(2400 ~ 2473.5 MHz) ≥ -30dBm @ PER ≤ 10%(5150 ~ 5825.0 MHz)
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6.5Mbps, ACR ≥ 13dB@13Mbps, ACR ≥ 11dB@19.5Mbps, ACR ≥ 8dB@26Mbps, ACR ≥ 4dB@39Mbps, ACR ≥ 0dB@52Mbps ACR ≥ -1dB@58.5Mbps, ACR ≥ -2dB@65Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent Sensitivity specified in min input sensitivity

19) WLAN 802.11n(HT20) transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2473.5 MHz (CH1~CH11) (USA) 5150~5850MHz(CH36,CH40,CH44,CH48~116,CH132~165) (USA) 2400 MHz ~ 2483.5 MHz (CH1~CH13) (Europe) 5150~5725MHz(CH36,CH40,CH44,CH48~140) (Europe)
Tx Power Level	$\leq 20\text{dBm}$ (CH1~13) (Europe), $\leq 30\text{dBm}$ (CH1~11) (USA) $< 50\text{mW}$ (CH36~48), $< 250\text{mW}$ (CH52~140), $< 1\text{W}$ (CH149~165) (USA) $\leq 20\text{dBm}$ (CH36~64), $\leq 27\text{dBm}$ (CH100~140) (Europe)
Frequency Tolerance	within ± 25 PPM(2400 ~ 2473.5 MHz) within ± 20 PPM(5150 ~ 5850 MHz)
Chip clock Frequency Tolerance	within ± 25 PPM(2400 ~ 2473.5 MHz) within ± 20 PPM(5100 ~ 5850 MHz)
Spectrum Mask	≤ -20 @ $\pm 11\text{MHz}$ offset (9MHz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20MHz) ≤ -45 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30MHz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}$ @6.5Mbps, $\leq -10\text{dB}$ @13Mbps, $\leq -13\text{dB}$ @19.5Mbps, $\leq -16\text{dB}$ @26Mbps, $\leq -19\text{dB}$ @39Mbps, $\leq -22\text{dB}$ @52Mbps, $\leq -25\text{dB}$ @58.5Mbps, $\leq -28\text{dB}$ @65Mbps
Spurious Emissions	[2400 MHz ~ 2473.5 MHz (CH1 ~ 11)] $< -36\text{ dBm}$ @ 30MHz ~ 1GHz $< -30\text{ dBm}$ above @ 1GHz ~ 12.75GHz $< -47\text{ dBm}$ @ 1.8GHz ~ 1.9GHz $< -47\text{ dBm}$ @ 5.15GHz ~ 5.3GHz [5150 MHz ~ 5850 MHz (CH36 ~ 165)] $< -36\text{ dBm}$ @ 30 MHz to 47 MHz $< -54\text{ dBm}$ @ 47 MHz to 74 MHz $< -36\text{ dBm}$ @ 74 MHz to 87,5 MHz

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	< -54 dBm @ 87,5 MHz to 118 MHz < -36 dBm @ 118 MHz to 174 MHz < -54 dBm @ 174 MHz to 230 MHz < -36 dBm @ 230 MHz to 470 MHz < -54 dBm @ 470 MHz to 862 MHz < -36 dBm @ 862 MHz to 1 GHz < -30 dBm @ 1 GHz to 5.15 GHz < -30 dBm @ 5.35 GHz to 5.47 GHz < -30 dBm @ 5.725 GHz to 26.5 GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6.5Mbps, -79dBm@13Mbps, -77dBm@19.5Mbps -74dBm@26Mbps, -70dBm@39Mbps, -66dBm@52Mbps -65dBm@58.5Mbps, -64dBm@65Mbps
Rx Max input Sensitivity	≥ -20dBm @ PER ≤ 10%(2400 ~ 2473.5 MHz) ≥ -30dBm @ PER ≤ 10%(5150 ~ 5850 MHz)
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6.5Mbps, ACR ≥ 13dB@13Mbps, ACR ≥ 11dB@19.5Mbps, ACR ≥ 8dB@26Mbps, ACR ≥ 4dB@39Mbps, ACR ≥ 0dB@52Mbps ACR ≥ -1dB@58.5Mbps, ACR ≥ -2dB@65Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

20) WLAN 802.11ac(VHT20)

Item	Spec.
Transmit Frequency	2400 ~ 2473.5 MHz (CH1 ~ 11) 5170 ~ 5825 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 165)
Tx Power Level	$\leq 10\text{mW/MHz}$ (CH1 ~ 11) $\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 165)
Frequency Tolerance	within ± 25 PPM(2400 ~ 2473.5 MHz) within ± 20 PPM(5170 ~ 5825 MHz)
Chip clock Frequency Tolerance	within ± 25 PPM(2400 ~ 2473.5 MHz) within ± 20 PPM(5170 ~ 5825 MHz)
Spectrum Mask	2.4G (2400 ~ 2473.5 MHz) ≤ -20 @ $\pm 11\text{MHz}$ offset (9Mhz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20Mhz) ≤ -45 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30Mhz) 5G (5170 ~ 5710 MHz) ≤ -20 @ $\pm 11\text{MHz}$ offset (9Mhz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20Mhz) ≤ -40 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6.5\text{Mbps}$, $\leq -10\text{dB}@13\text{Mbps}$, $\leq -13\text{dB}@19.5\text{Mbps}$, $\leq -16\text{dB}@26\text{Mbps}$, $\leq -19\text{dB}@39\text{Mbps}$, $\leq -22\text{dB}@52\text{Mbps}$, $\leq -25\text{dB}@58.5\text{Mbps}$, $\leq -27\text{dB}@65\text{Mbps}$, $\leq -30\text{dB}@78\text{Mbps}$
Spurious Emissions	[2400 MHz ~ 2473.5 MHz (CH1 ~ 11)] < -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Spurious Emissions	[5170 MHz ~ 5825 MHz (CH36 ~ 165)] < -36 dBm @ 30 MHz to 47 MHz < -54 dBm @ 47 MHz to 74 MHz < -36 dBm @ 74 MHz to 87,5 MHz < -54 dBm @ 87,5 MHz to 118 MHz < -36 dBm @ 118 MHz to 174 MHz < -54 dBm @ 174 MHz to 230 MHz < -36 dBm @ 230 MHz to 470 MHz

2. PERFORMANCE

	< -54 dBm @ 470 MHz to 862 MHz < -36 dBm @ 862 MHz to 1 GHz < -30 dBm @ 1 GHz to 5.15 GHz < -30 dBm @ 5.35 GHz to 5.47 GHz < -30 dBm @ 5.725 GHz to 26.5 GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6.5Mbps, -79dBm@13Mbps, -77dBm@19.5Mbps -74dBm@26Mbps, -70dBm@39Mbps, -66dBm@52Mbps -65dBm@58.5Mbps, -64dBm@65Mbps, -59dBm@78Mbps
Rx Max input Sensitivity	≥ -30dBm @ PER ≤ 10%
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6.5Mbps, ACR ≥ 13dB@13Mbps, ACR ≥ 11dB@19.5Mbps, ACR ≥ 8dB@26Mbps, ACR ≥ 4dB@39Mbps, ACR ≥ 0dB@52Mbps ACR ≥ -1dB@58.5Mbps, ACR ≥ -2dB@65Mbps, ACR ≥ -7dB@78Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

21) WLAN 802.11ac(VHT40)

Item	Spec.
Transmit Frequency	5170 ~ 5825 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 165)
Tx Power Level	$\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 165)
Frequency Tolerance	within ± 20 PPM(5170 ~ 5825 MHz)
Chip clock Frequency Tolerance	within ± 20 PPM(5170 ~ 5825 MHz)
Spectrum Mask	≤ -20 @ $\pm 21\text{MHz}$ offset (19Mhz ~ 21MHz) ≤ -28 @ $\pm 40\text{MHz}$ offset (21MHz ~ 40Mhz) ≤ -40 @ $\pm 60\text{MHz}$ offset (40MHz ~ 60Mhz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB@}13.5\text{Mbps}$, $\leq -10\text{dB@}27\text{Mbps}$, $\leq -13\text{dB@}40.5\text{Mbps}$, $\leq -16\text{dB@}54\text{Mbps}$, $\leq -19\text{dB@}81\text{Mbps}$, $\leq -22\text{dB@}108\text{Mbps}$, $\leq -25\text{dB@}121.5\text{Mbps}$, $\leq -27\text{dB@}135\text{Mbps}$, $\leq -30\text{dB@}162\text{Mbps}$, $\leq -32\text{dB@}180\text{Mbps}$
Spurious Emissions	[5170 MHz ~ 5825 MHz (CH36 ~ 165)] < -36 dBm @ 30 MHz to 47 MHz < -54 dBm @ 47 MHz to 74 MHz < -36 dBm @ 74 MHz to 87,5 MHz < -54 dBm @ 87,5 MHz to 118 MHz < -36 dBm @ 118 MHz to 174 MHz < -54 dBm @ 174 MHz to 230 MHz < -36 dBm @ 230 MHz to 470 MHz < -54 dBm @ 470 MHz to 862 MHz < -36 dBm @ 862 MHz to 1 GHz < -30 dBm @ 1 GHz to 5.15 GHz < -30 dBm @ 5.35 GHz to 5.47 GHz < -30 dBm @ 5.725 GHz to 26.5 GHz
Rx Min input Sensitivity	PER $\leq 10\%$ $-79\text{dBm@}13.5\text{Mbps}$, $-76\text{dBm@}27\text{Mbps}$, $-74\text{dBm@}40.5\text{Mbps}$ $-71\text{dBm@}54\text{Mbps}$, $-67\text{dBm@}81\text{Mbps}$, $-63\text{dBm@}108\text{Mbps}$ $-62\text{dBm@}121.5\text{Mbps}$, $-61\text{dBm@}135\text{Mbps}$, $-56\text{dBm@}162\text{Mbps}$, $-54\text{dBm@}180\text{Mbps}$

2. PERFORMANCE

Rx Max input Sensitivity	$\geq -30\text{dBm}$ @ PER $\leq 10\%$ (5170 ~ 5825 MHz)
Rx Adjacent Channel Rejection	PER $\leq 10\%$, ACR $\geq 16\text{dB}@13.5\text{Mbps}$, ACR $\geq 13\text{dB}@27\text{Mbps}$, ACR $\geq 11\text{dB}@40.5\text{Mbps}$, ACR $\geq 8\text{dB}@54\text{Mbps}$, ACR $\geq 4\text{dB}@81\text{Mbps}$, ACR $\geq 0\text{dB}@108\text{Mbps}$ ACR $\geq -1\text{dB}@121.5\text{Mbps}$, ACR $\geq -2\text{dB}@135\text{Mbps}$ ACR $\geq -7\text{dB}@162\text{Mbps}$, ACR $\geq -9\text{dB}@180\text{Mbps}$ ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

22) WLAN 802.11ac(VHT80)

Item	Spec.
Transmit Frequency	5170 ~ 5825 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 165)
Tx Power Level	$\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 165)
Frequency Tolerance	within ± 20 PPM(5170 ~ 5825 MHz)
Chip clock Frequency Tolerance	within ± 20 PPM(5170 ~ 5825 MHz)
Spectrum Mask	≤ -20 @ $\pm 41\text{MHz}$ offset (39MHz ~ 41MHz) ≤ -28 @ $\pm 80\text{MHz}$ offset (41MHz ~ 80MHz) ≤ -40 @ $\pm 120\text{MHz}$ offset (80MHz ~ 120MHz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB@}29.3\text{Mbps}$, $\leq -10\text{dB@}58.5\text{Mbps}$, $\leq -13\text{dB@}87.8\text{Mbps}$, $\leq -16\text{dB@}117\text{Mbps}$, $\leq -19\text{dB@}175.5\text{Mbps}$, $\leq -22\text{dB@}234\text{Mbps}$, $\leq -25\text{dB@}263.3\text{Mbps}$, $\leq -27\text{dB@}292.5\text{Mbps}$, $\leq -30\text{dB@}351\text{Mbps}$, $\leq -32\text{dB@}390\text{Mbps}$
Spurious Emissions	[5170 MHz ~ 5825 MHz (CH36 ~ 165)] < -36 dBm @ 30 MHz to 47 MHz < -54 dBm @ 47 MHz to 74 MHz < -36 dBm @ 74 MHz to 87,5 MHz < -54 dBm @ 87,5 MHz to 118 MHz < -36 dBm @ 118 MHz to 174 MHz < -54 dBm @ 174 MHz to 230 MHz < -36 dBm @ 230 MHz to 470 MHz < -54 dBm @ 470 MHz to 862 MHz < -36 dBm @ 862 MHz to 1 GHz < -30 dBm @ 1 GHz to 5.15 GHz < -30 dBm @ 5.35 GHz to 5.47 GHz < -30 dBm @ 5.725 GHz to 26.5 GHz
Rx Min input Sensitivity	PER $\leq 10\%$ $-76\text{dBm@}29.3\text{Mbps}$, $-73\text{dBm@}58.5\text{Mbps}$, $-71\text{dBm@}87.8\text{Mbps}$ $-68\text{dBm@}117\text{Mbps}$, $-64\text{dBm@}175.5\text{Mbps}$, $-60\text{dBm@}234\text{Mbps}$

2. PERFORMANCE

	-59dBm@263.3Mbps, -58dBm@292.5Mbps, -53dBm@351Mbps, -51dBm@390Mbps
Rx Max input Sensitivity	$\geq -30\text{dBm}$ @ PER $\leq 10\%$ (5170 ~ 5825 MHz)
Rx Adjacent Channel Rejection	PER $\leq 10\%$, ACR $\geq 16\text{dB}@29.3\text{Mbps}$, ACR $\geq 13\text{dB}@58.5\text{Mbps}$, ACR $\geq 11\text{dB}@87.8\text{Mbps}$, ACR $\geq 8\text{dB}@117\text{Mbps}$, ACR $\geq 4\text{dB}@175.5\text{Mbps}$, ACR $\geq 0\text{dB}@234\text{Mbps}$ ACR $\geq -1\text{dB}@263.3\text{Mbps}$, ACR $\geq -2\text{dB}@292.5\text{Mbps}$ ACR $\geq -7\text{dB}@351\text{Mbps}$, ACR $\geq -9\text{dB}@390\text{Mbps}$ ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input s ensitivity

23) Bluetooth transmitter specification

Item	Spec.		Min	Typ.	Max	Unit
Output Power	DH5 PRBS9	Nominal Pwr	-6	-	+20	dBm
		Peak Pwr	-	-	+23	dBm
Power Density	DH5 PRBS9	@100kHz BW		-	+20	dBm
Frequency Range	DH5 PRBS9	-30dBm @100kHz BW	2400	-	2483	MHz
Tx Output Spectrum (20dB BW)	DH5 PRBS9	$\Delta f = f_H - f_L$		-	1	MHz
Tx Output Spectrum (Adjacent Output Power)	DH5 PRBS9	@M-N = 2		-	-20	dBm
		@M-N >= 3		-	-40	dBm
Modulation Index	DH5 11110000	$f1_{AVG}$	140	-	175	kHz
	DH5 10101010	$f2_{MAX}$	115	-		kHz
		$f2_{AVG}/f1_{AVG}$	0.8	-		
Initial Carrier Freq. Tolerance	DH5 PRBS9		-75	-	+75	kHz
Carrier Frequency Drift	DH1 10101010	Freq. Drift	-25	-	+25	kHz
		Drift Rate	-20	-	+20	kHz/50us
	DH3 10101010	Freq. Drift	-40	-	+40	kHz
		Drift Rate	-20	-	+20	kHz/50us
	DH5 10101010	Freq. Drift	-40	-	+40	kHz
		Drift Rate	-20	-	+20	kHz/50us
Out of Band Spurious Emission	DH1 PRBS9	@ 30MHz ~ 12.75GHz	20			dBc

24) Bluetooth receiver specification

Item	Spec.		Min	Typ.	Max	Unit
Sensitivity (Single Slot Packet)	DH1 PRBS9	BER @ 0.1%		-	-70	dBm
Sensitivity (Multi Slot Packet)	DH5 PRBS9	BER @ 0.1%		-	-70	dBm
C/I Performance	DH1 PRBS9	Co-Channel -60dBm@ BER0.1%		-	11	dB
		C/I @1MHz -60dBm@ BER 0.1%		-	0	
		C/I @2MHz -60dBm@ BER 0.1%		-	-30	
		C/I @≥3MHz -67dBm@ BER0.1%		-	-40	
		C/I @Image -67dBm@ BER 0.1%		-	-9	
		C/I @1MHz Image -67dBm@ BER 0.1%		-	-20	
Blocking Performance	DH1 PRBS9	-10dBm @30M ~2 GHz		-	-67	dBm
		-27dBm @2 ~ 2.4 GHz				
		-27dBm @2.5 ~ 3 GHz				
		-10dBm @3 ~ 12.75 GHz				
Intermodulation Performance	DH1 PRBS9	$\Delta f = +5\text{MHz}$ (CW) $\Delta f = +10\text{MHz}$ (BT) BER @ 0.1%		-	-64	dBm
		$\Delta f = -5\text{MHz}$ (CW) $\Delta f = -10\text{MHz}$ (BT) BER @ 0.1%		-	-64	dBm
Maximum Input Level	DH1 PRBS9	BER @ 0.1%	-20	-		dBm

19) NFC

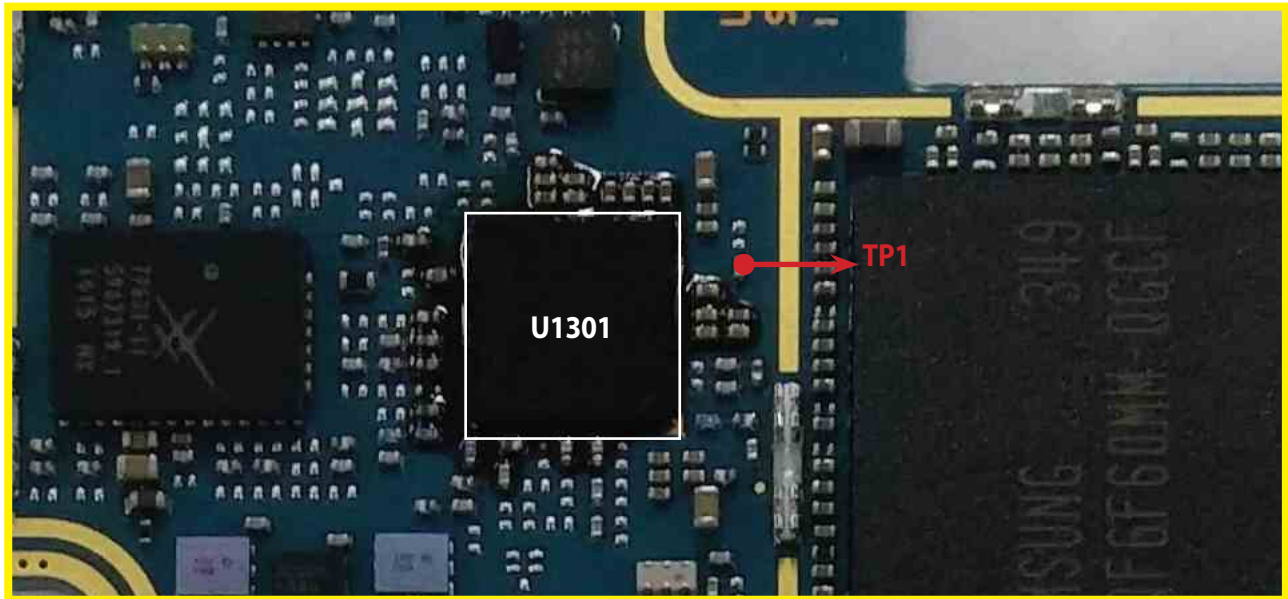
Operate below 3 mode.

- 1) Card mode : Smart Card capability for mobile device(recognition distance : over 20mm)
- 2) Reader mode : Get information from tag(recognition distance : over 20mm)
- 3) Peer-to-peer mode : Device to device communication(recognition distance : over 5mm)

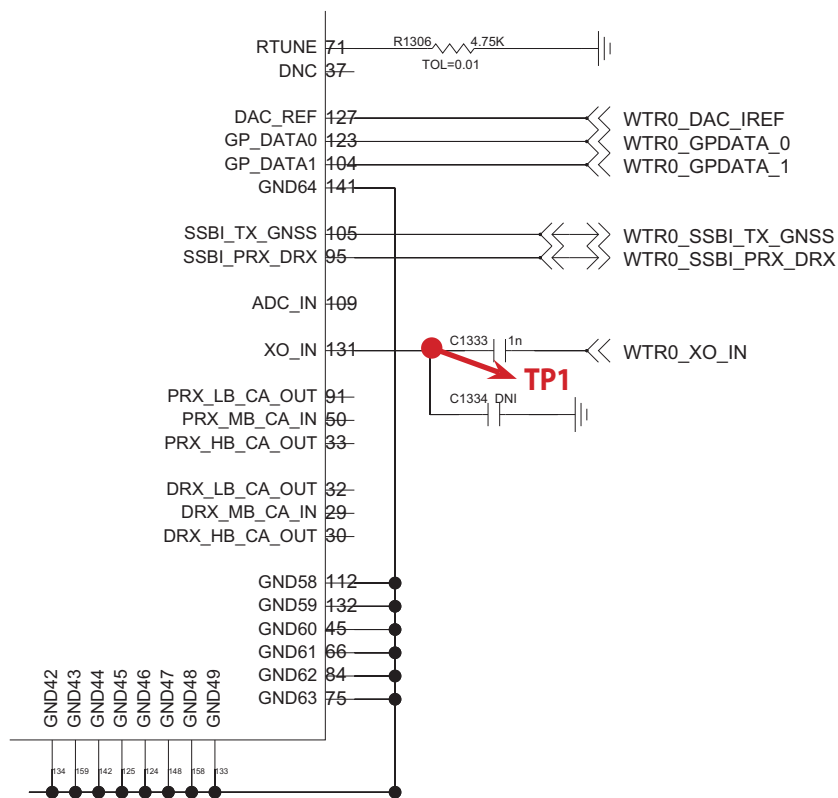
3. TROUBLE SHOOTING

3.1 Checking XO Block

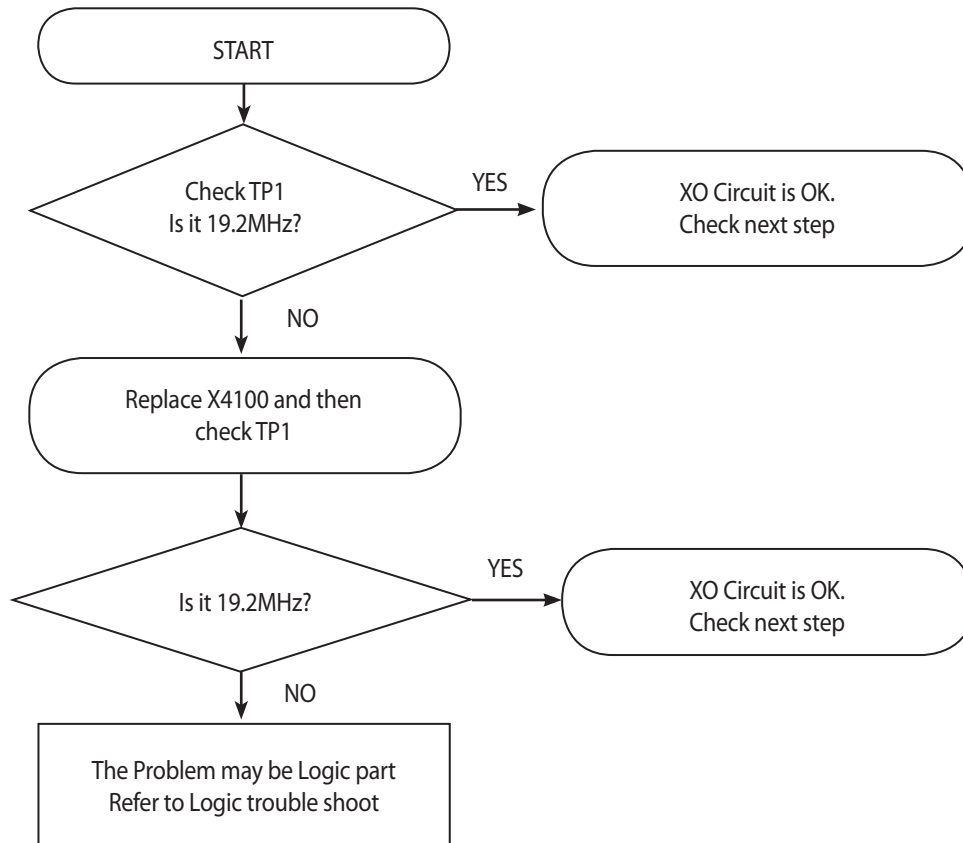
The output frequency(19.2MHz) of XO(X4100) is used as the reference one of WTR1625L and PM8941 internal VCO



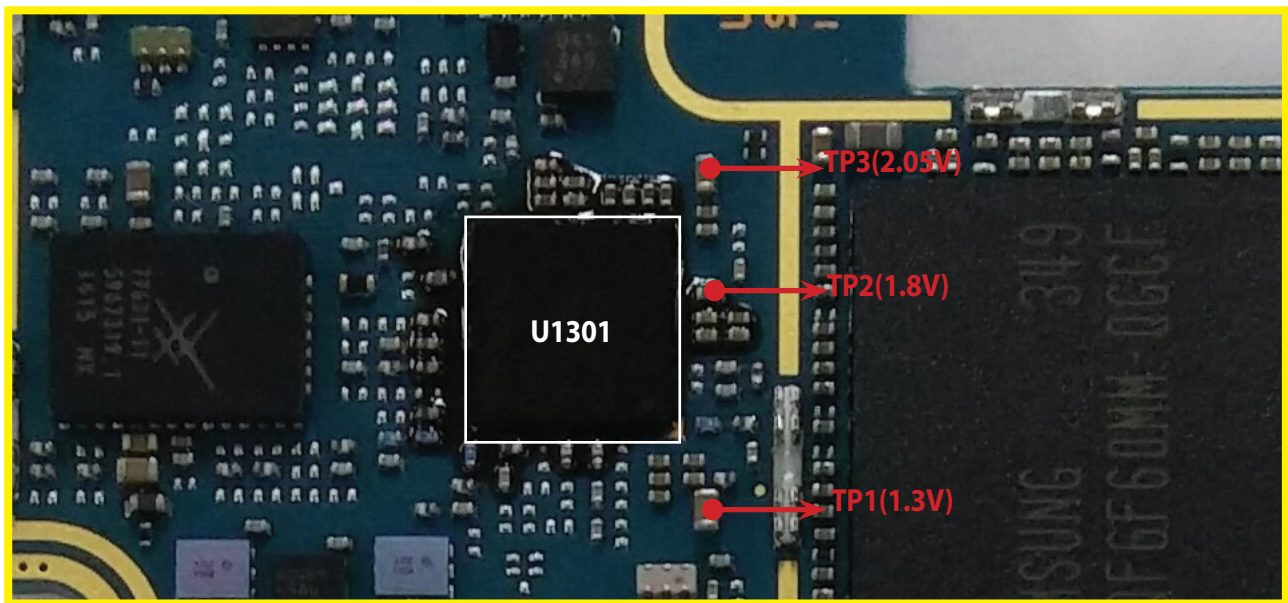
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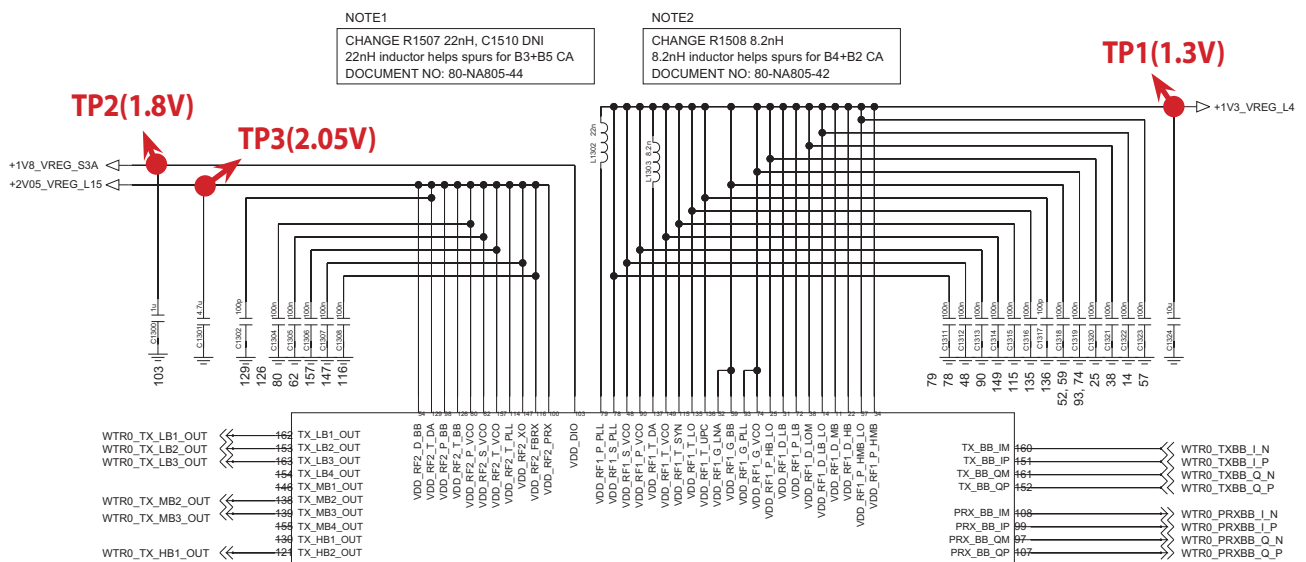
3. TROUBLE SHOOTING



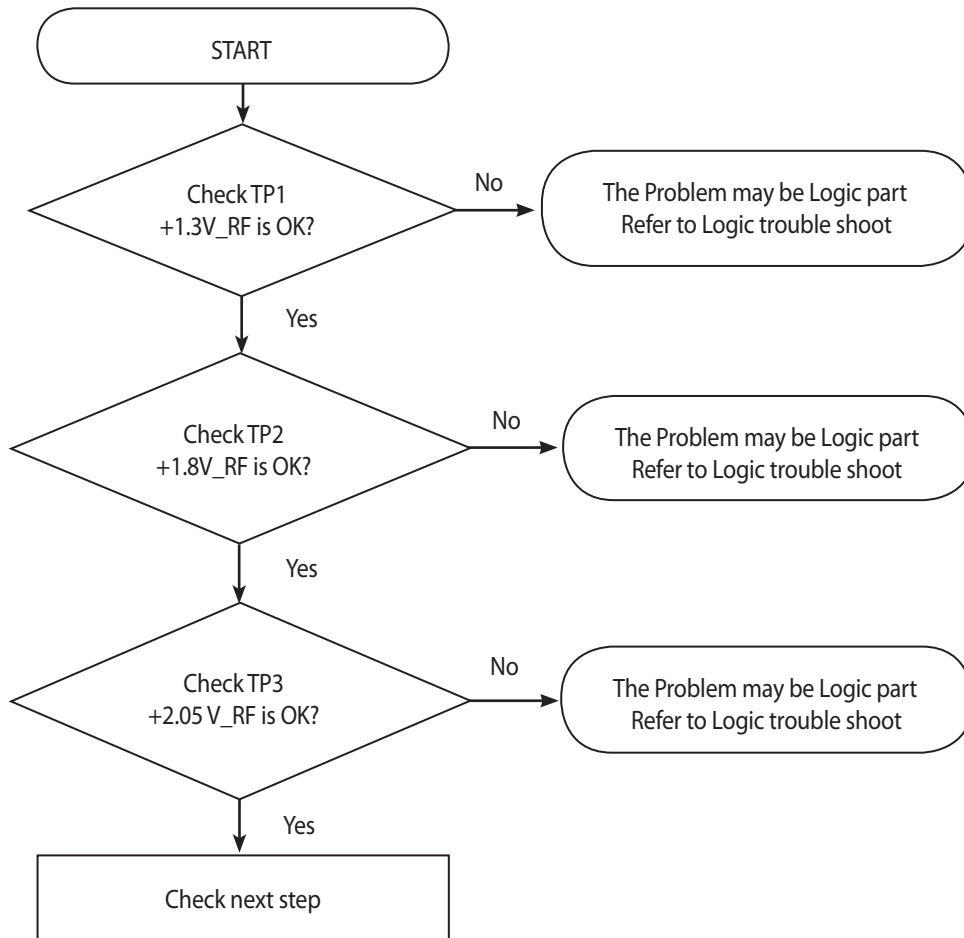
3.2 Checking Transceiver DC Power Supply Circuit Block



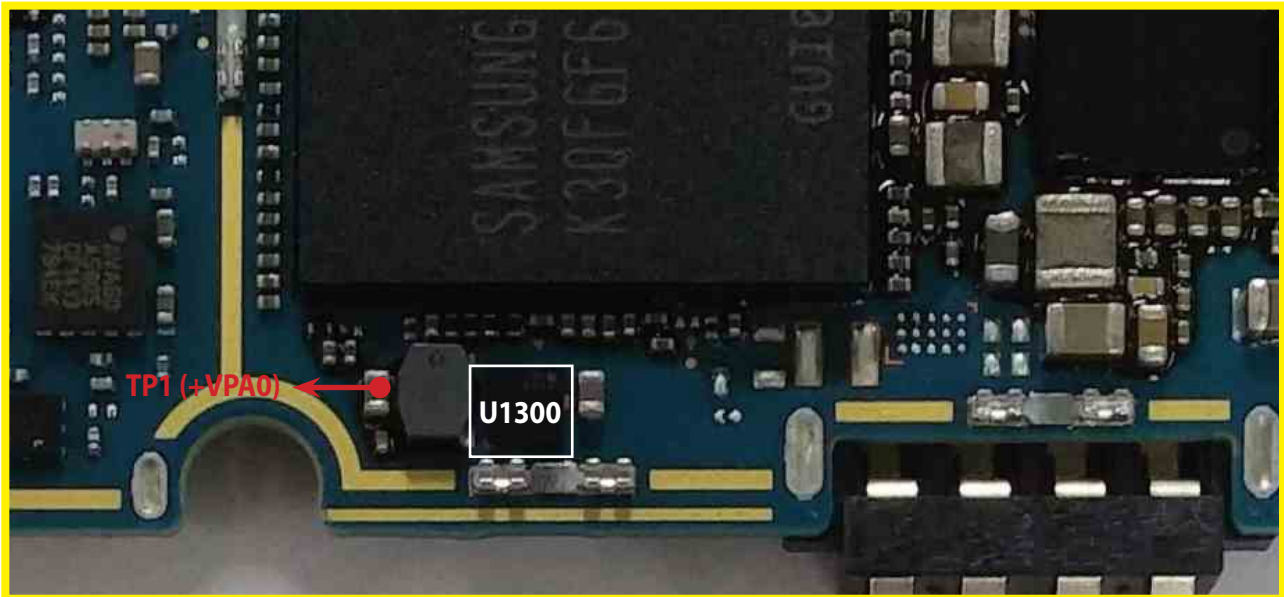
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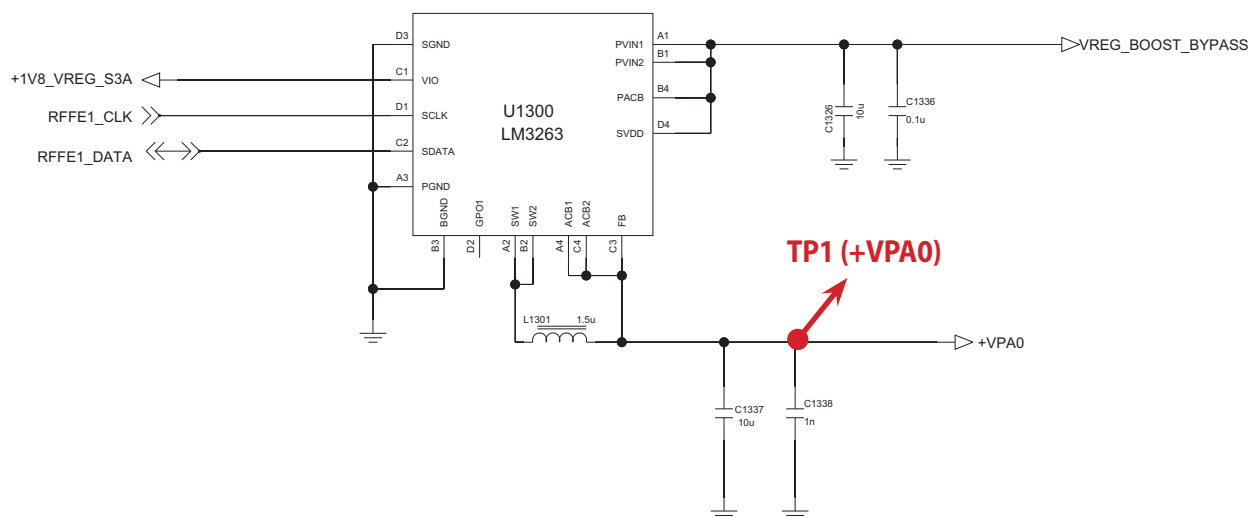
3. TROUBLE SHOOTING

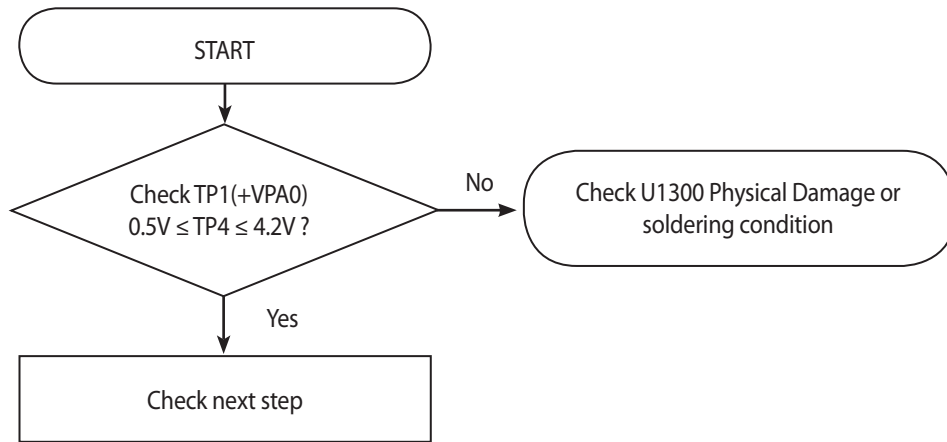


3.3 Checking DC-DC Block



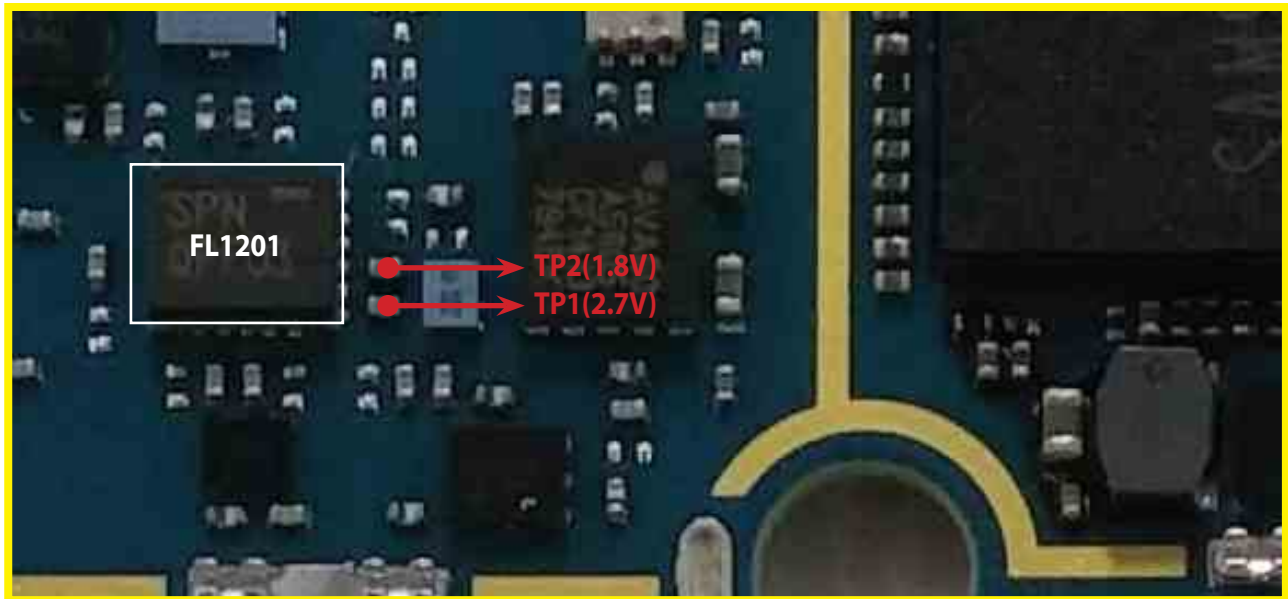
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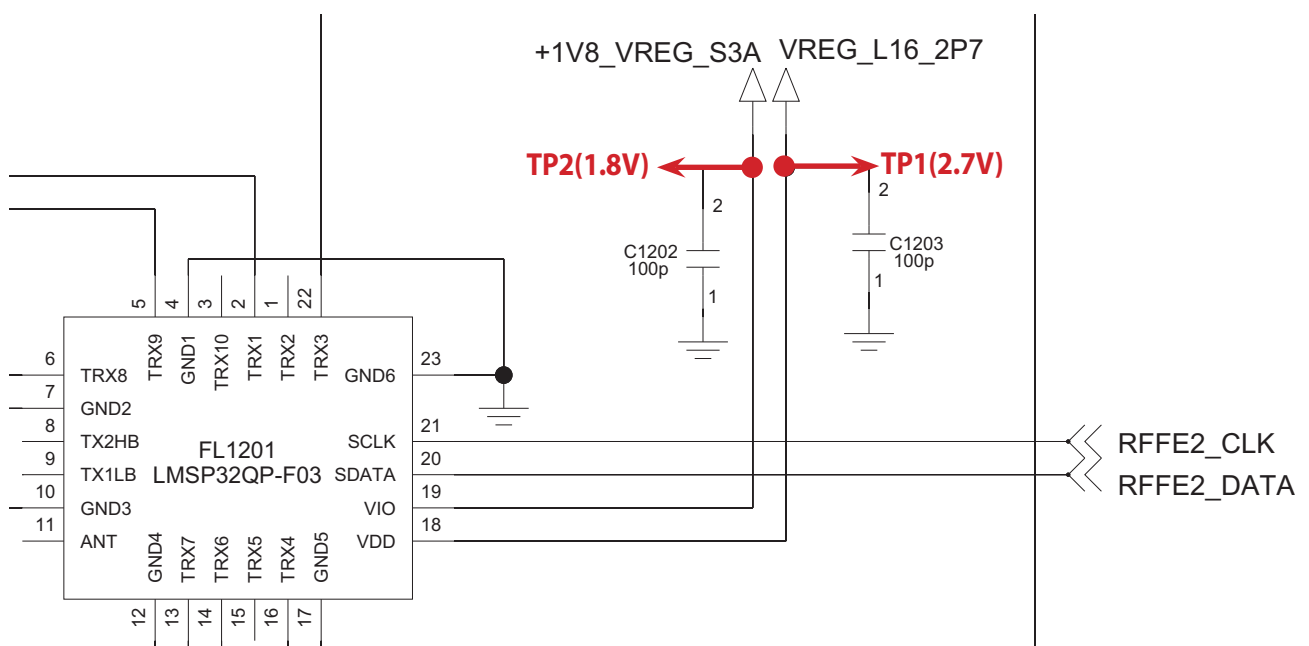


3.4 ASM(Antenna Switch Module) Block

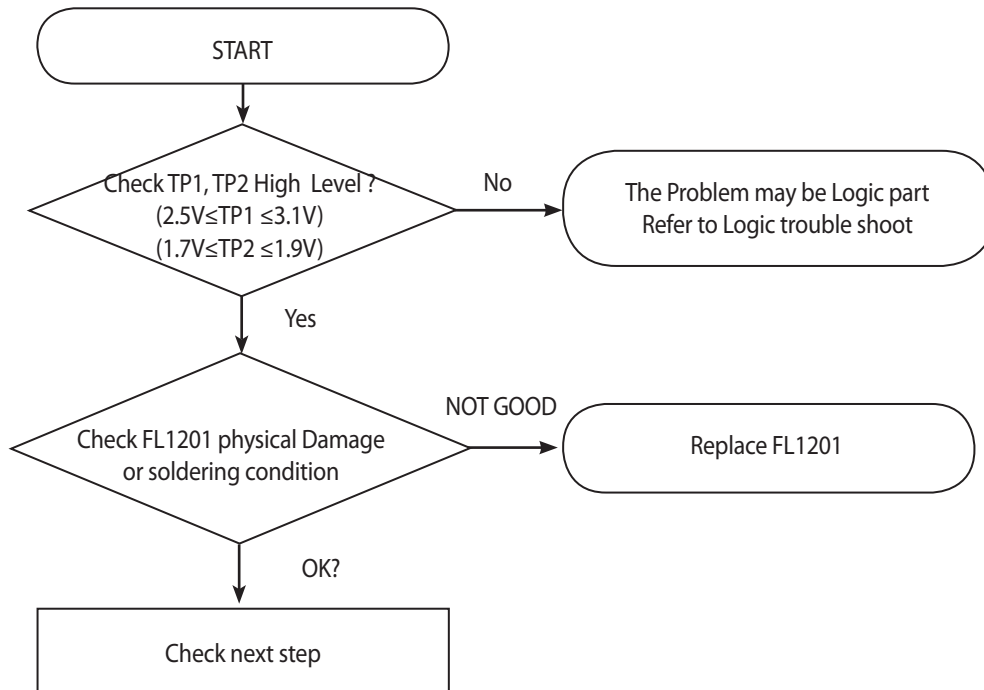
3.4.1 Checking ASM (GSM, W B1/2/5, LTE B25/26, CDMA BC0/1/10) Block



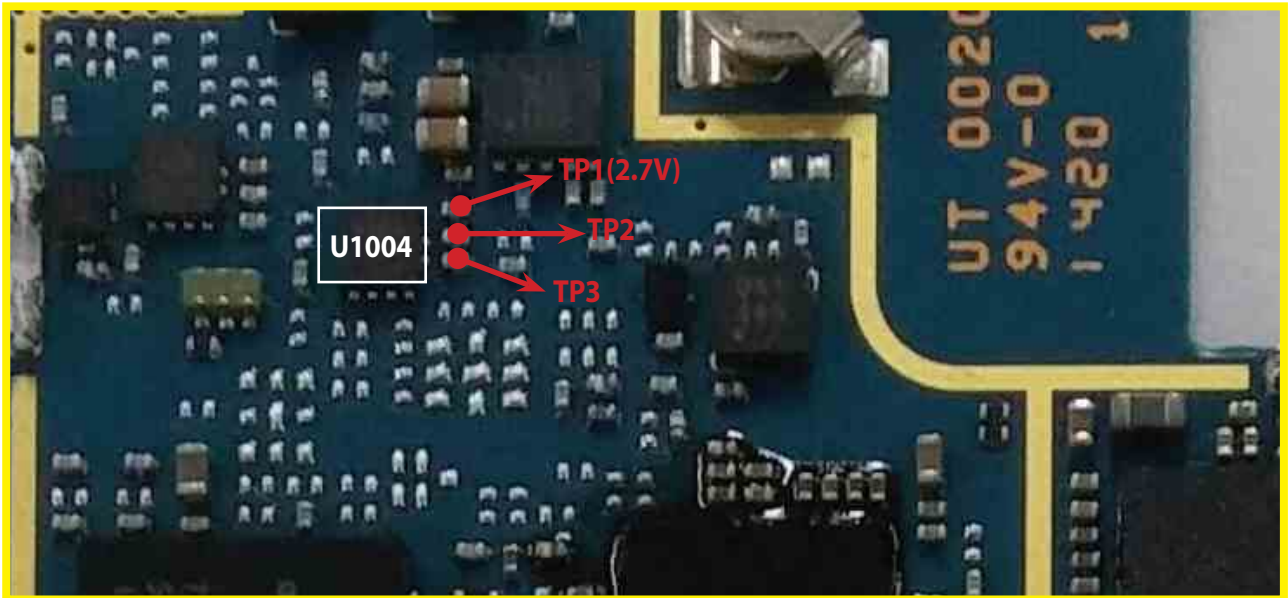
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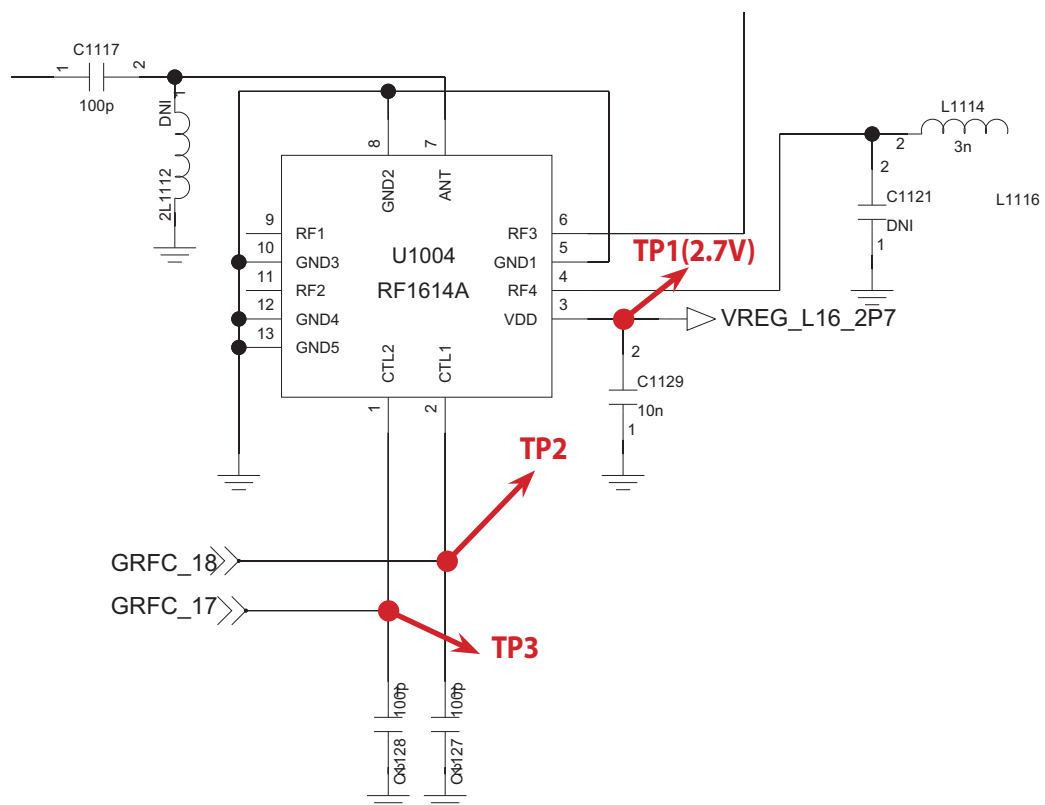
3. TROUBLE SHOOTING



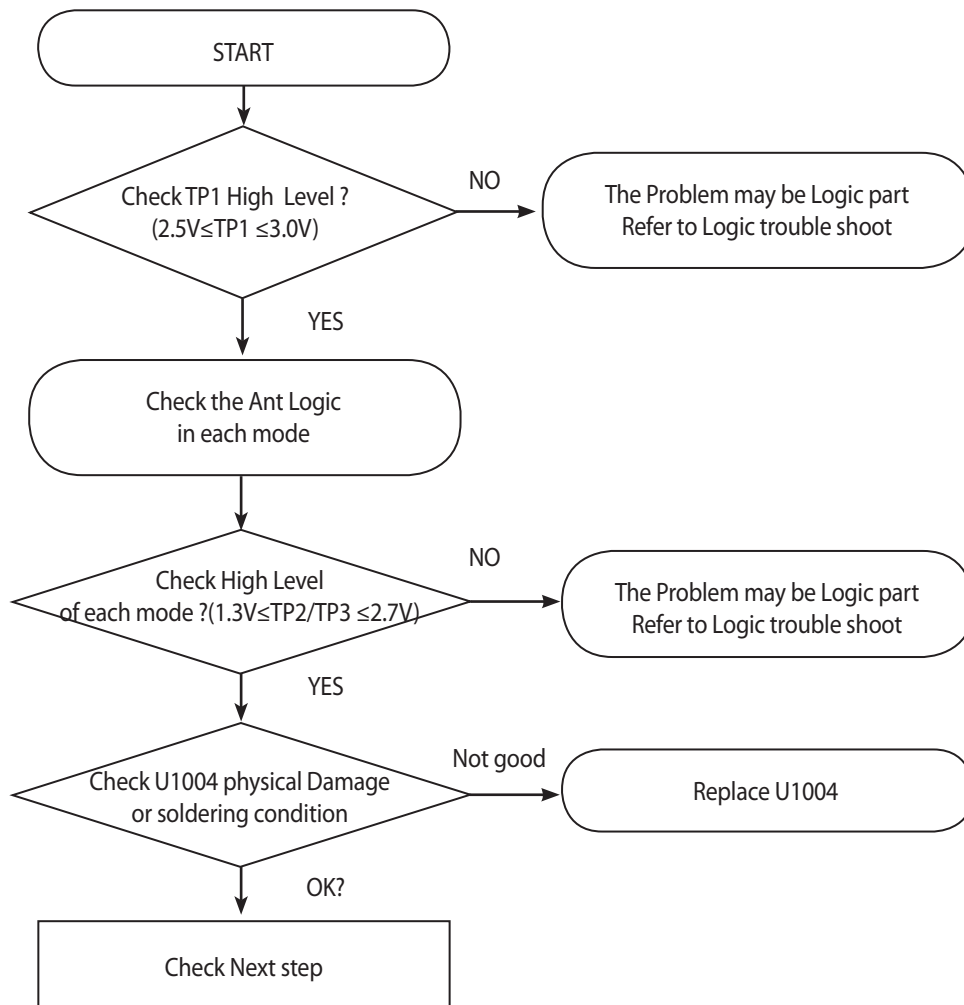
3.4.2 Checking SP4T (LTE B41) Block



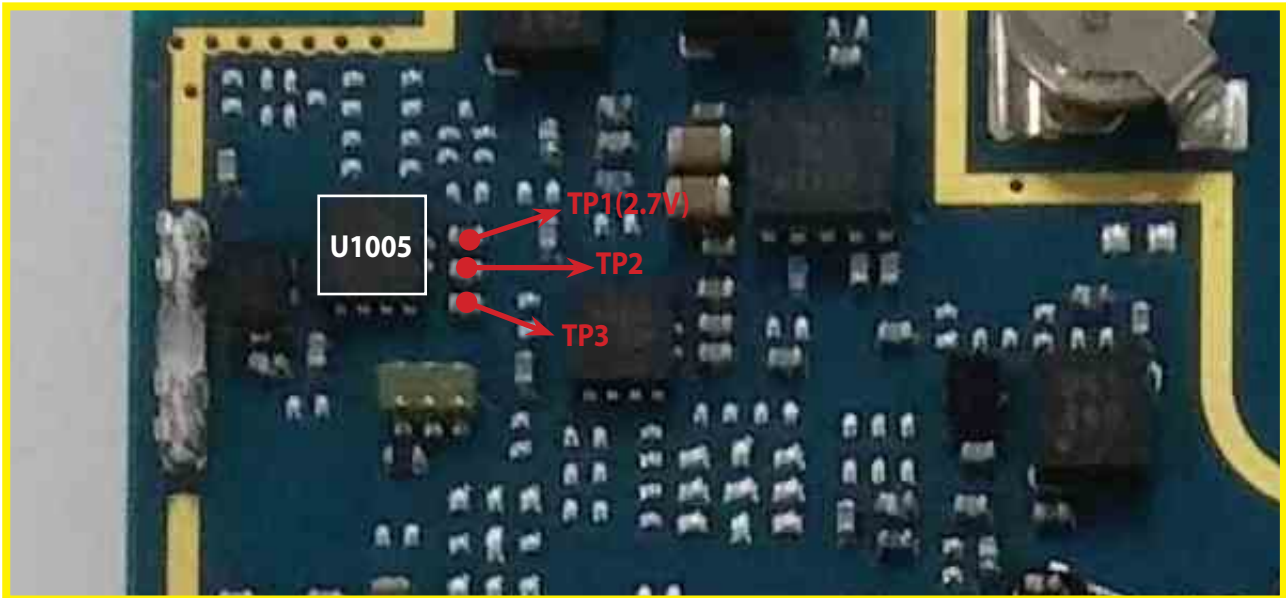
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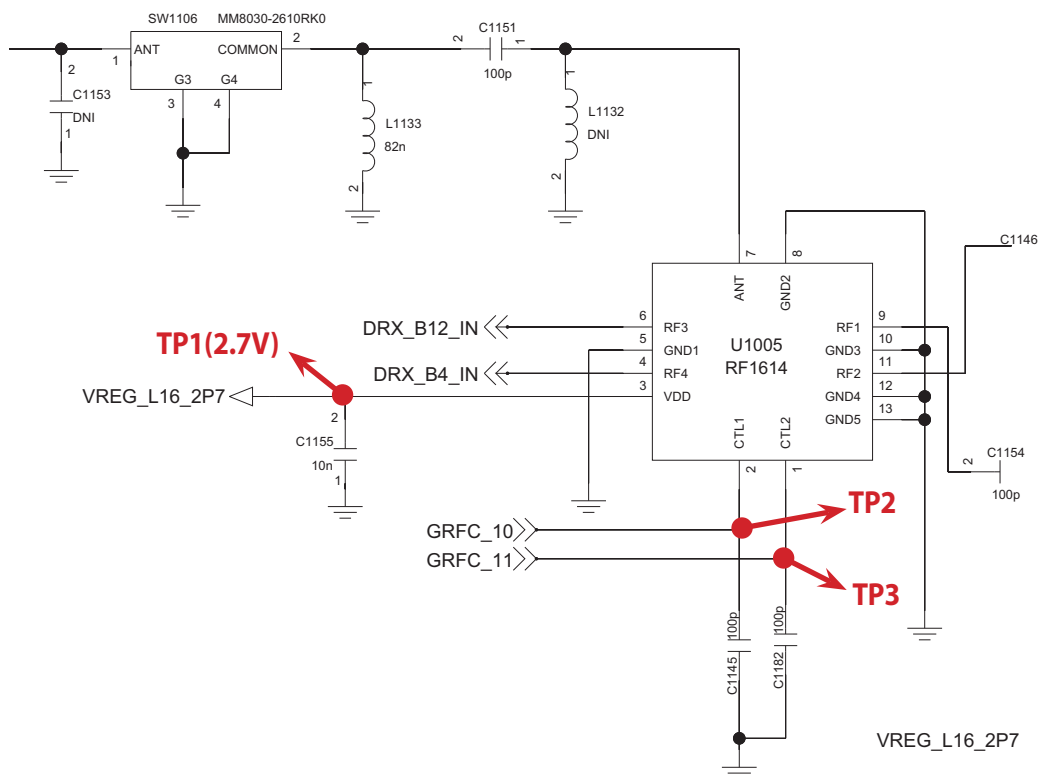
3. TROUBLE SHOOTING



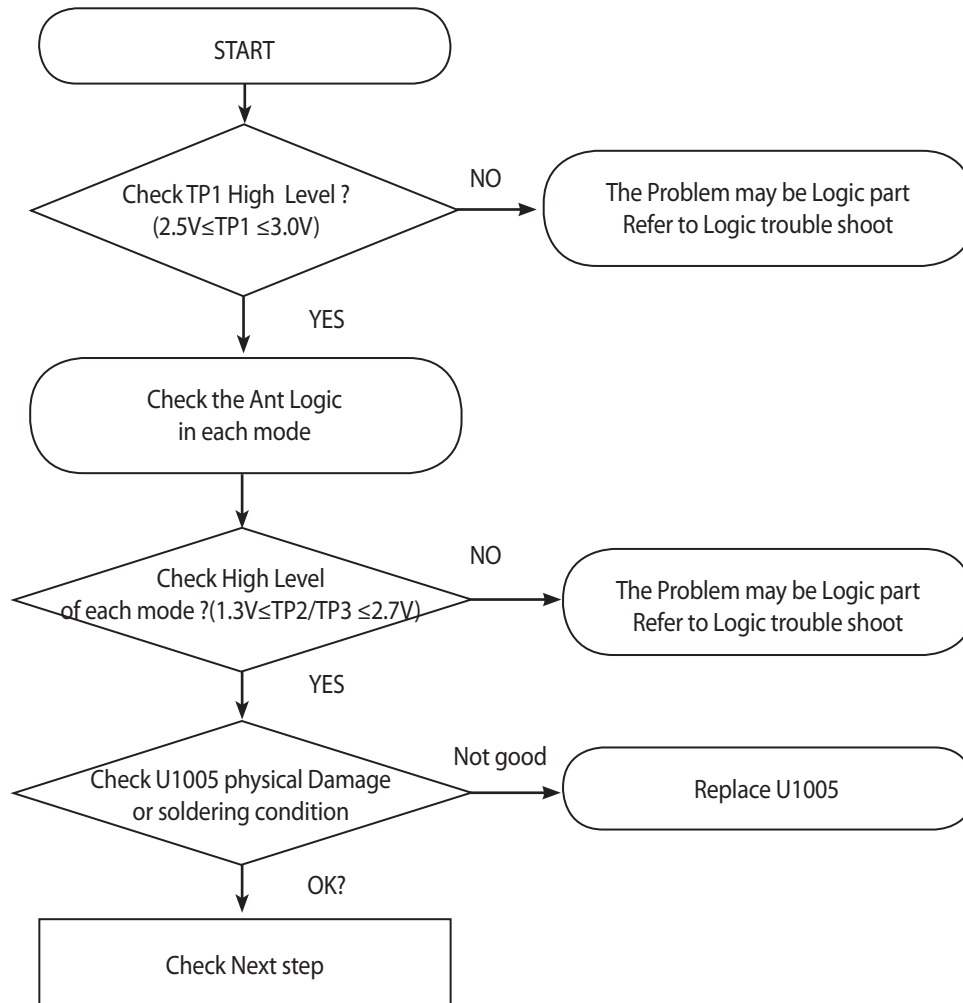
3.4.2 Checking SP4T (DRX) Block



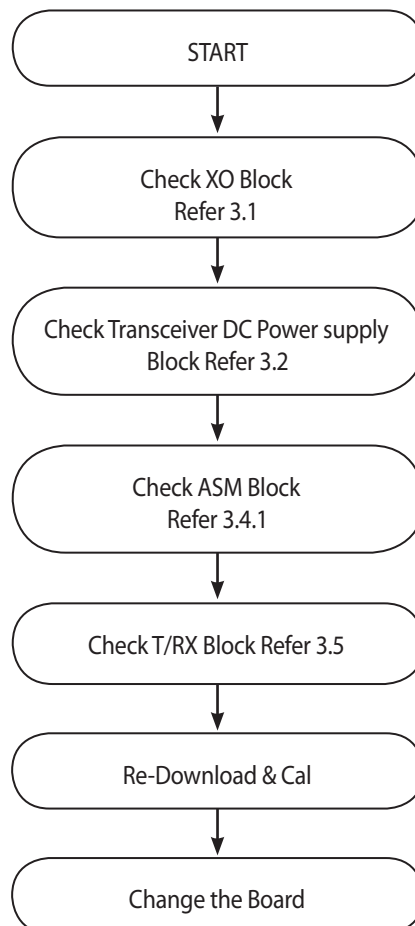
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3. TROUBLE SHOOTING



3.5 GSM RF Part

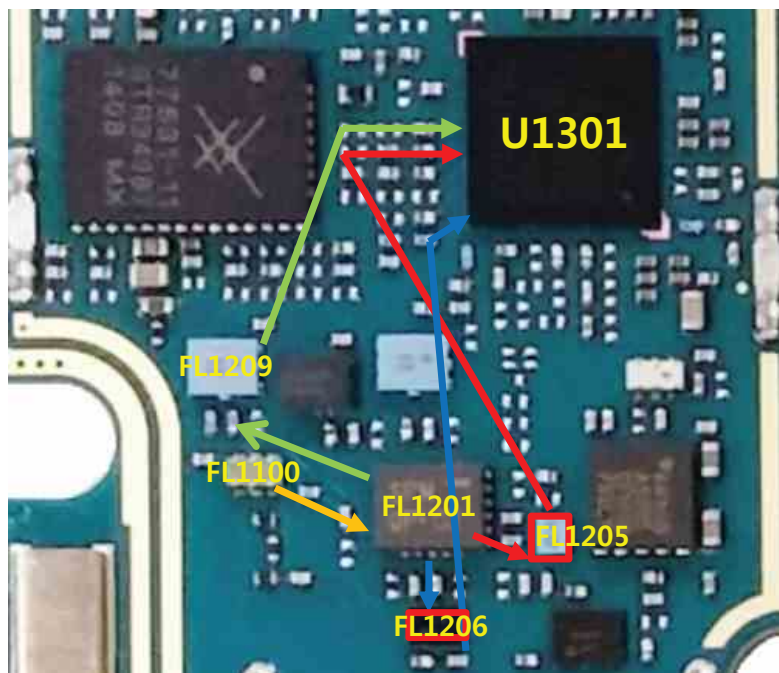


3.5.1 GSM850/900/1800/1900 Rx

GSM850/900/1800/1900 RX RF PATH



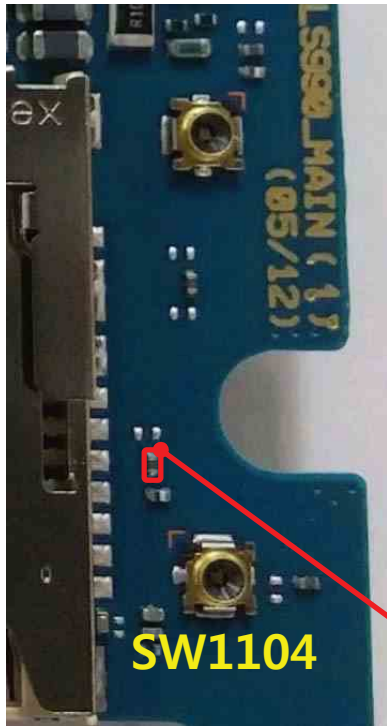
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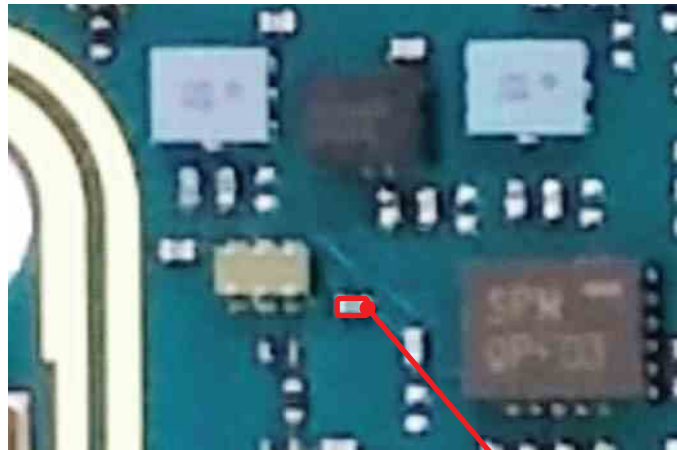
Top

- 1. GSM850 RX PATH
- 2. GSM900 RX PATH
- 3. GSM1800/1900 RX PATH
- 4. COMMON T/RX PATH

3.5.1.1 Checking RF Switch & Coupler

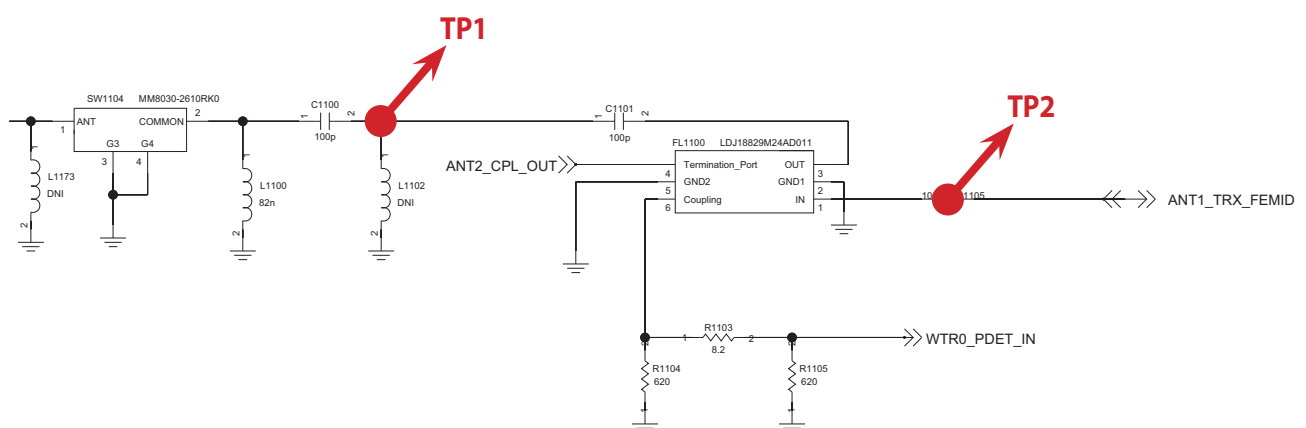


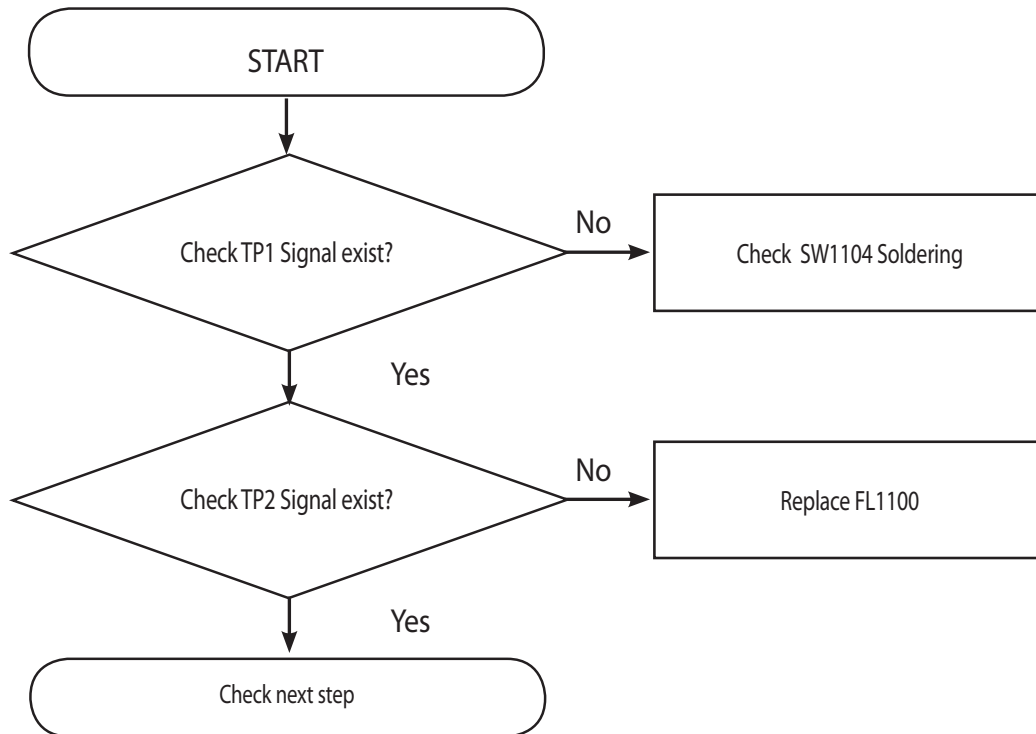
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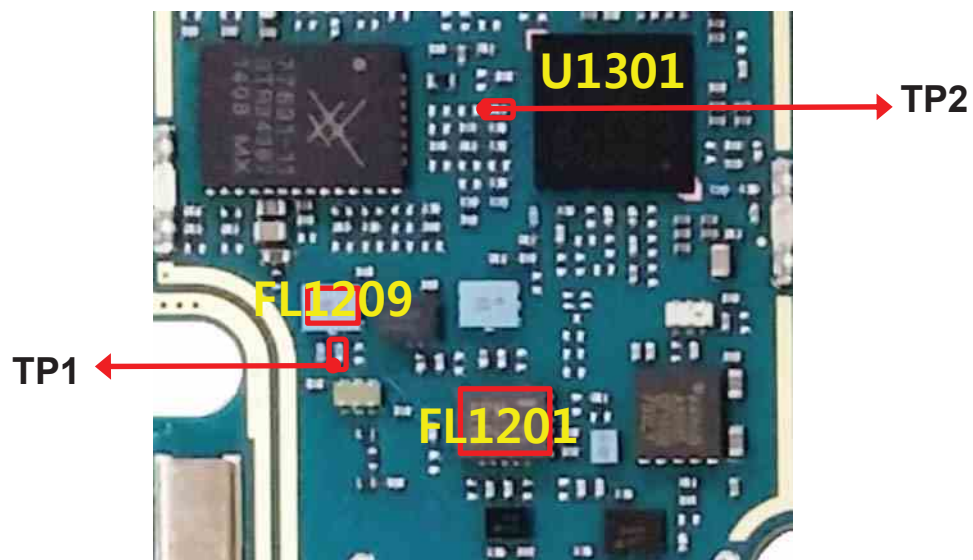
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TP2

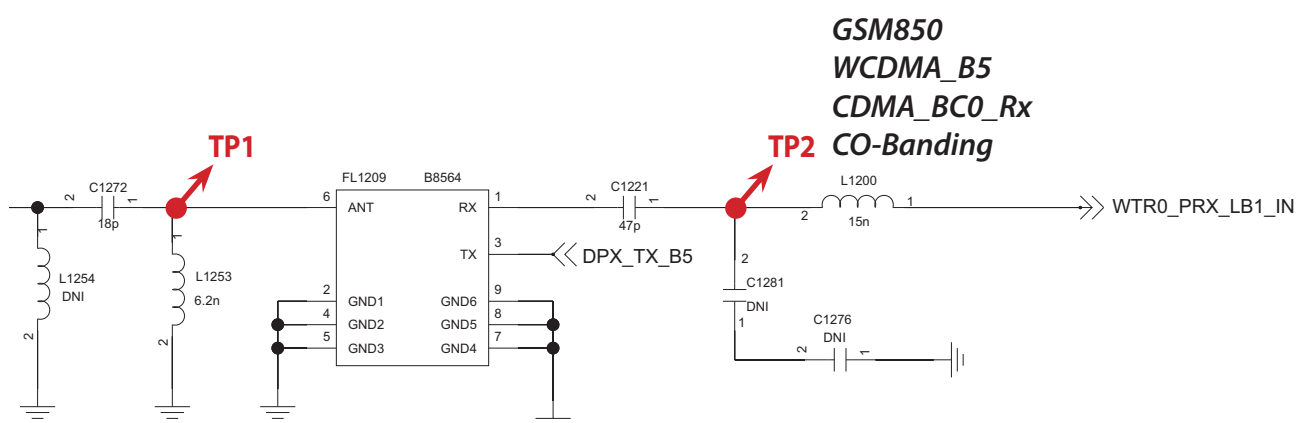




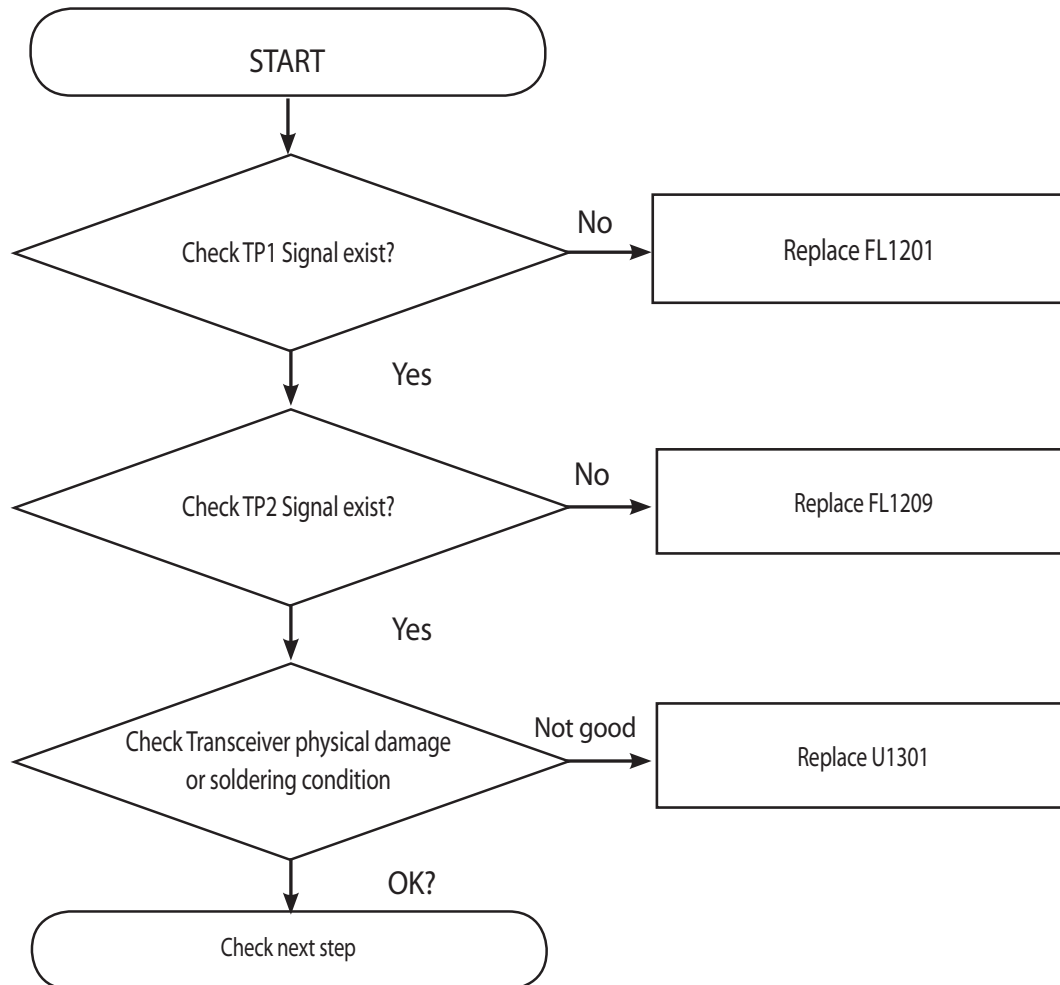
3.5.1.2 Checking RF signal path



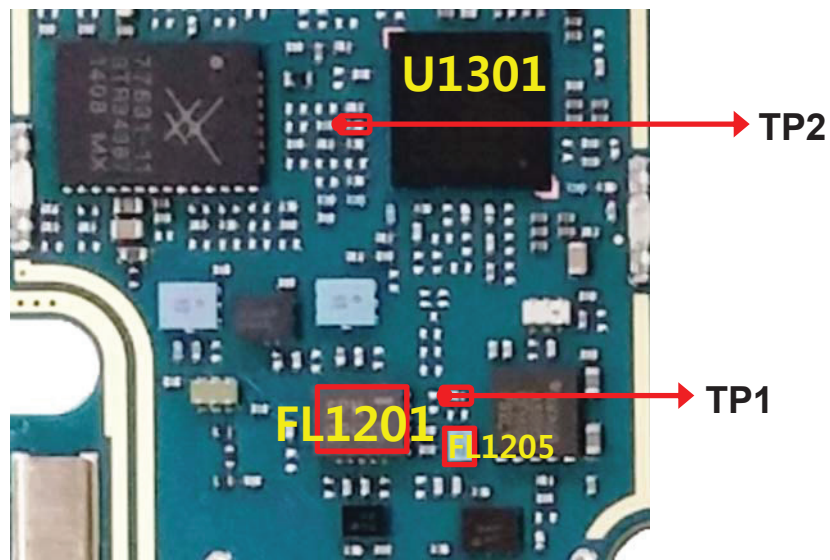
Circuit Diagram



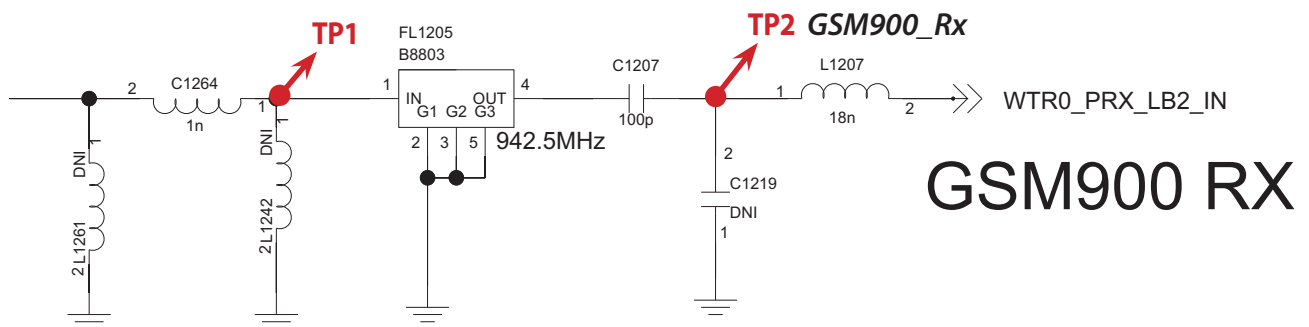
3. TROUBLE SHOOTING



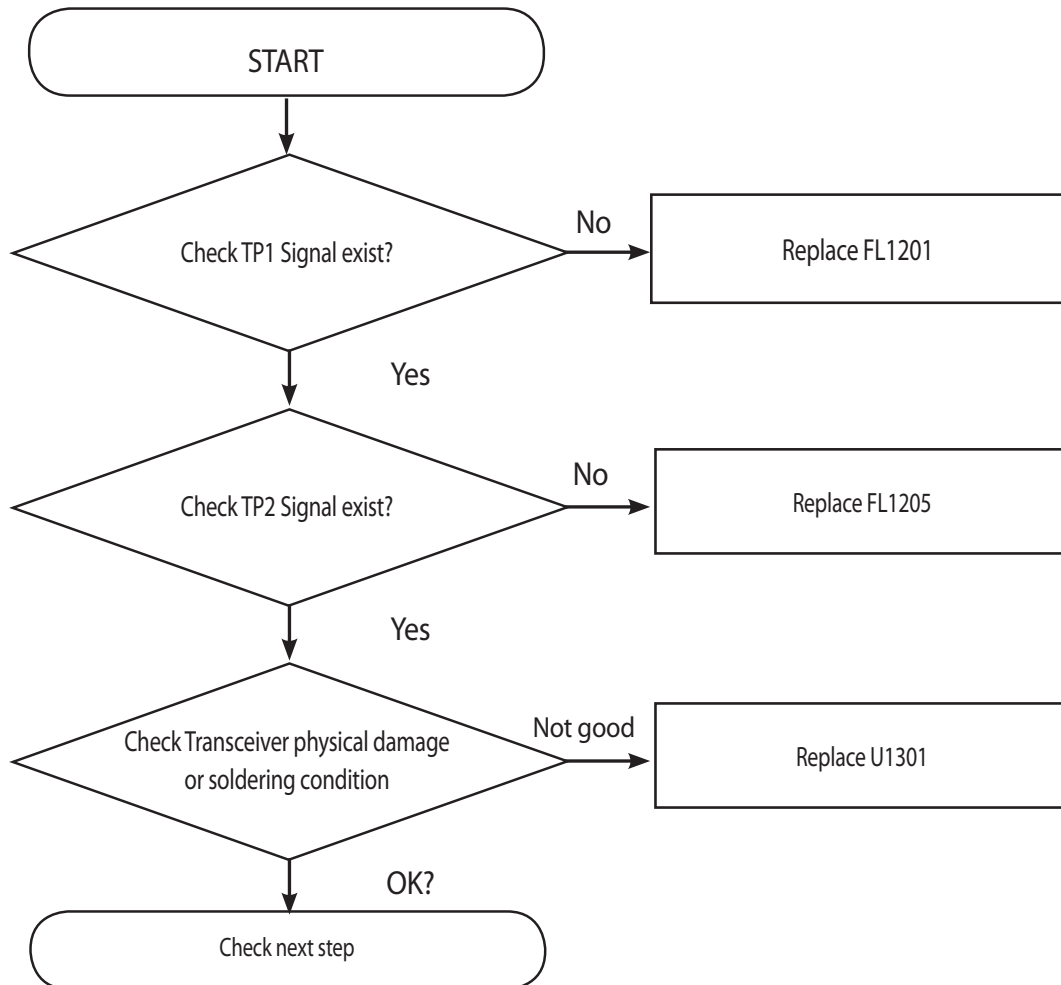
3.5.1.3 Checking RF signal path



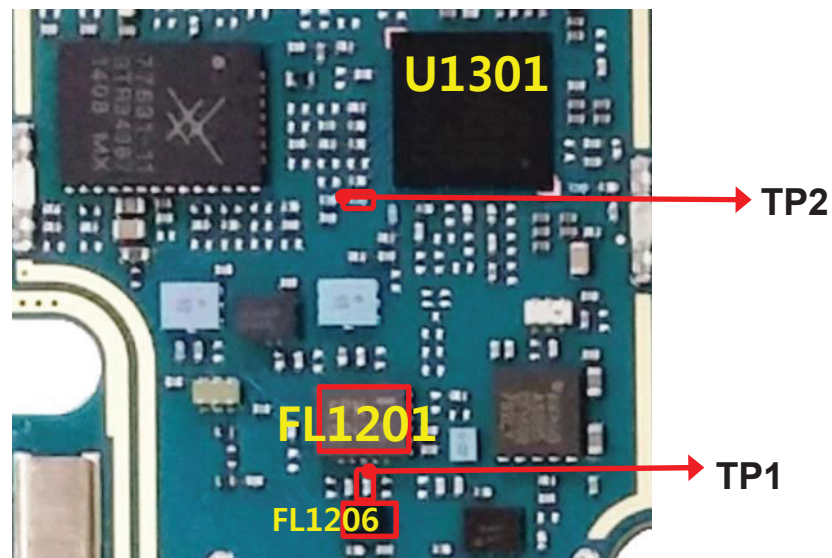
Circuit Diagram



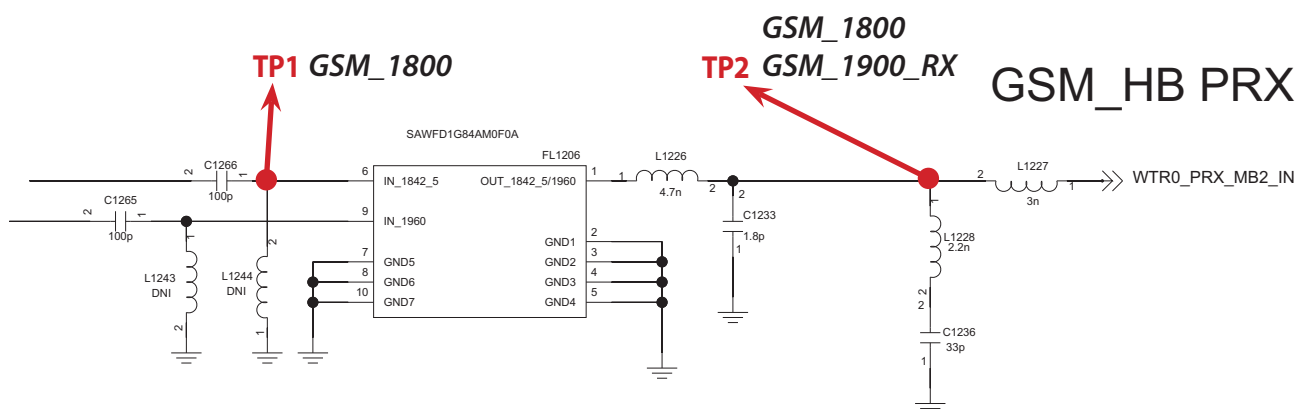
3. TROUBLE SHOOTING



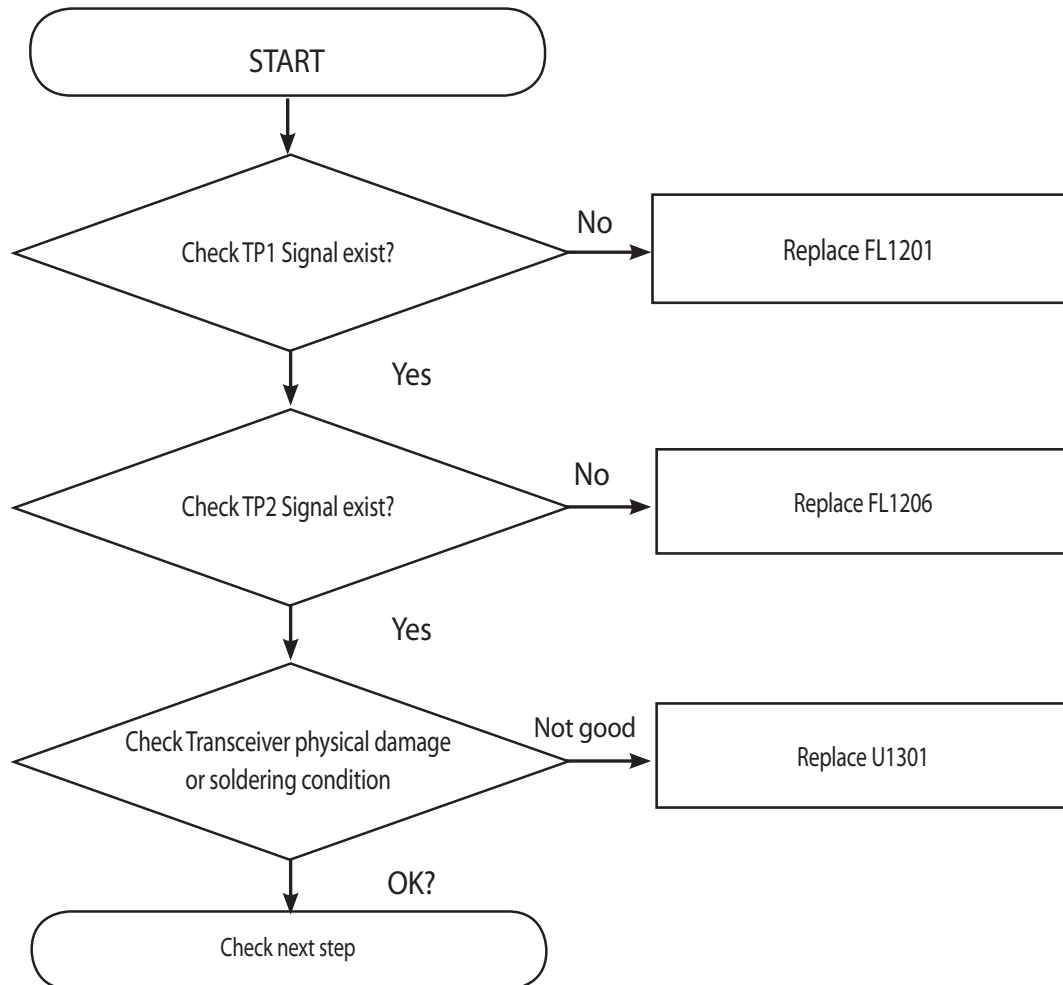
3.5.1.4 Checking RF signal path



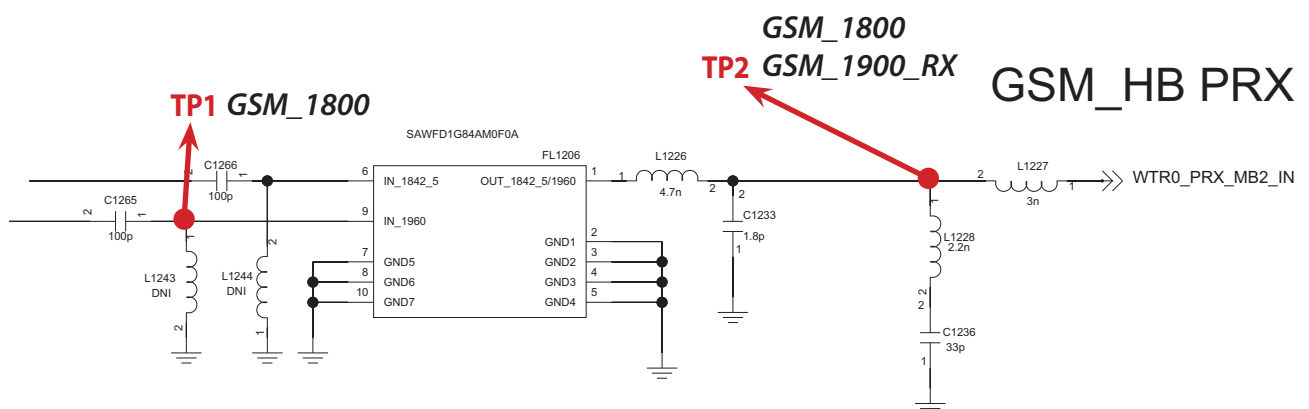
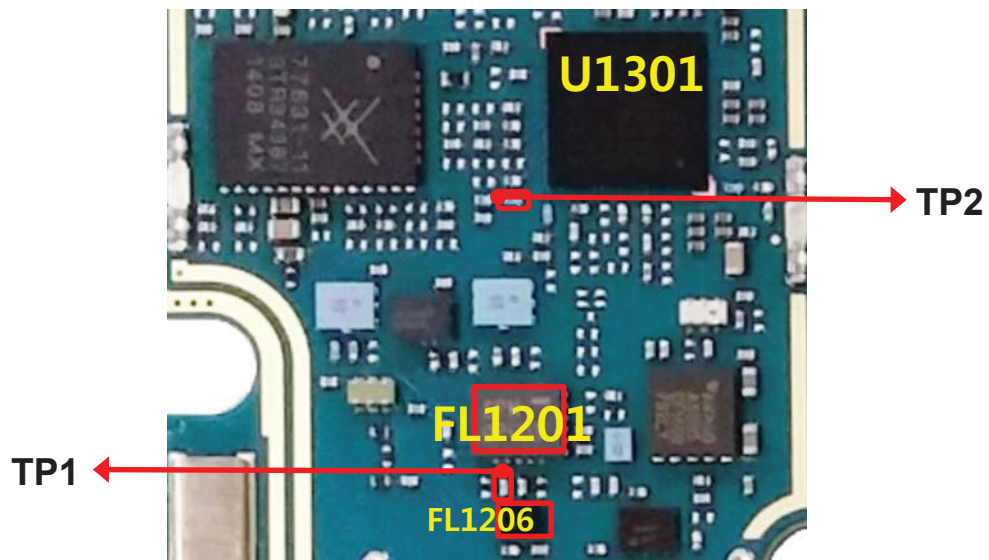
Circuit Diagram



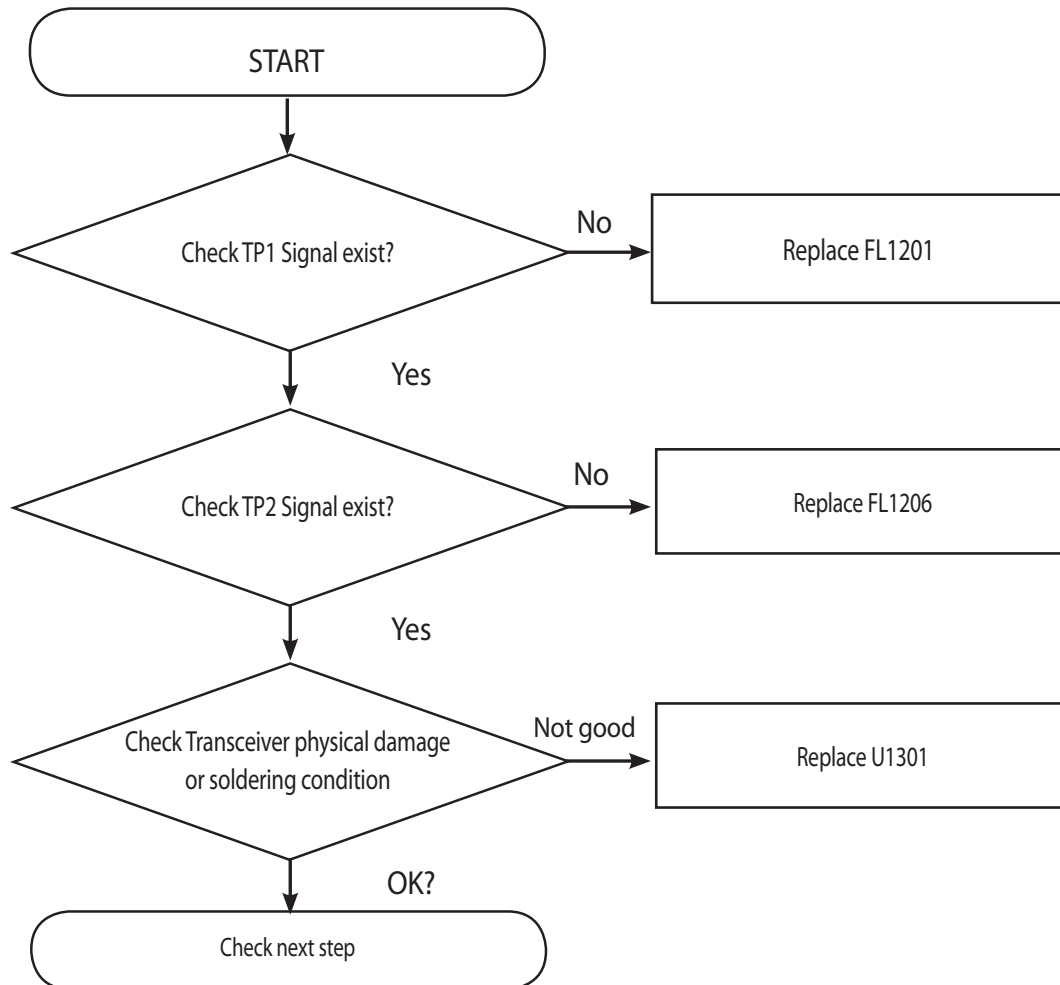
3. TROUBLE SHOOTING



3.5.1.5 Checking RF signal path



3. TROUBLE SHOOTING

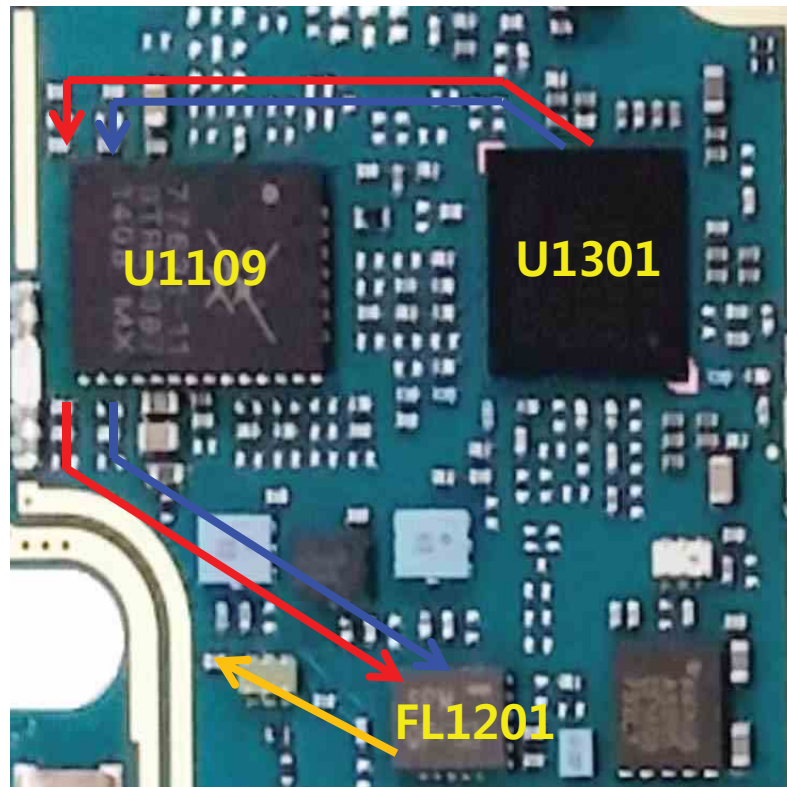


3.5.2 GSM850/900/1800/1900 Tx

GSM850/900/1800/1900 TX RF PATH



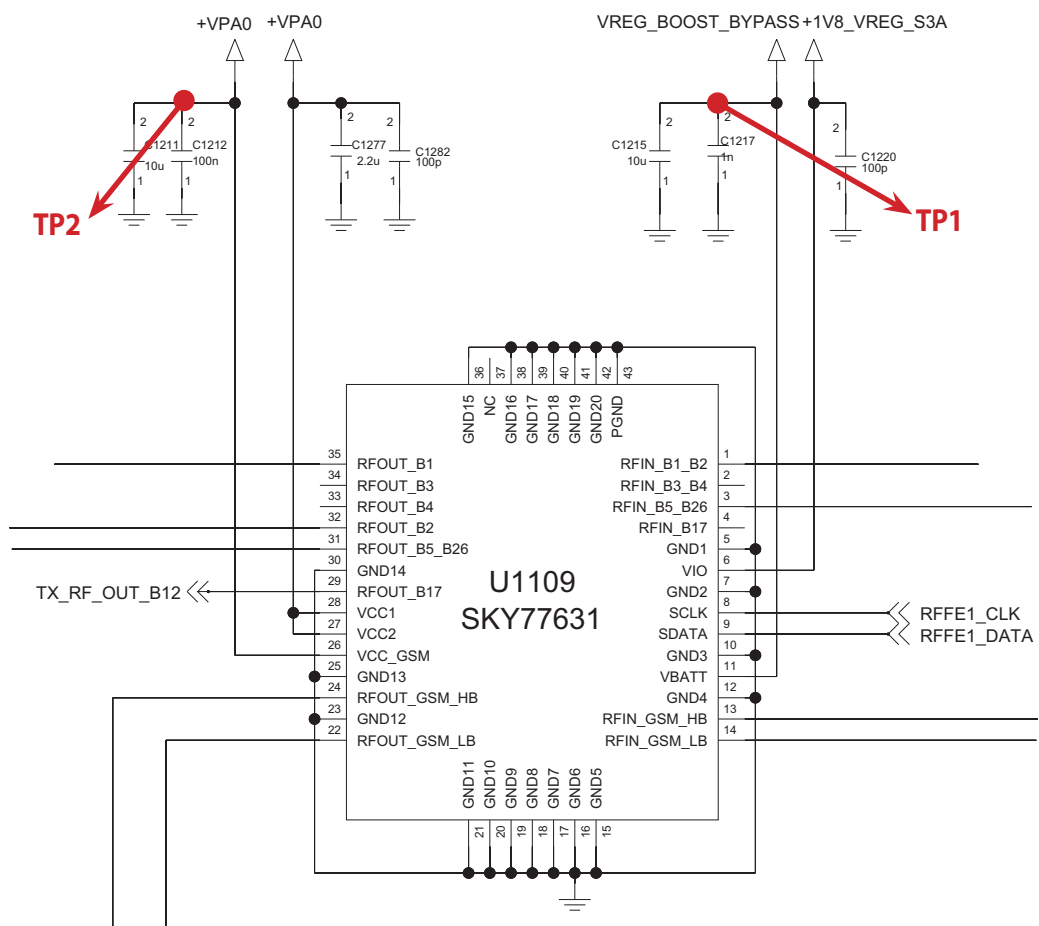
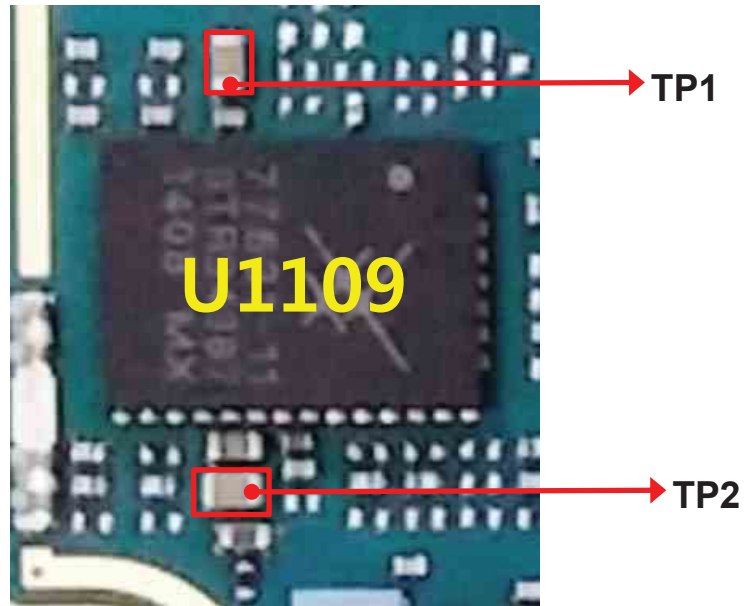
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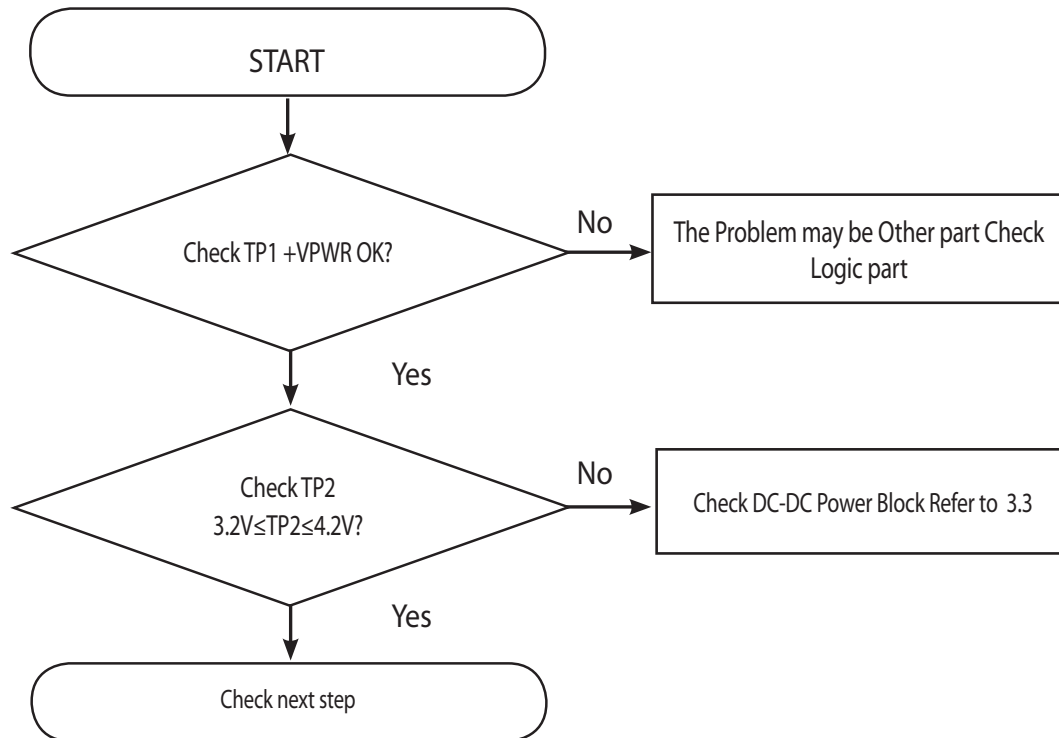


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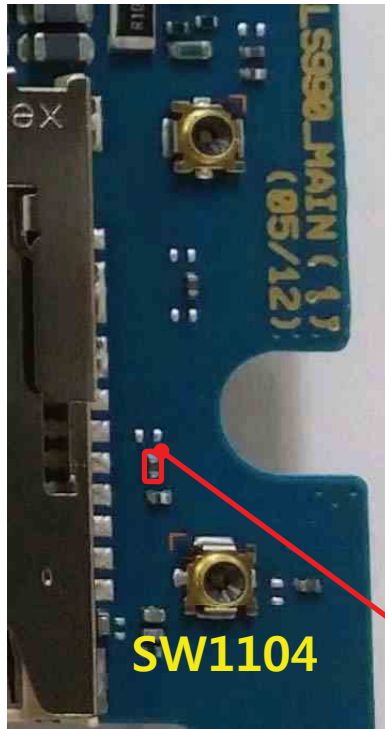
- 1. GSM850/900 TX PATH
- 2. GSM1800/1900 TX PATH
- 3. COMMON T/RX PATH

3.5.2.1 Checking GSM PAM DC Power Circuit

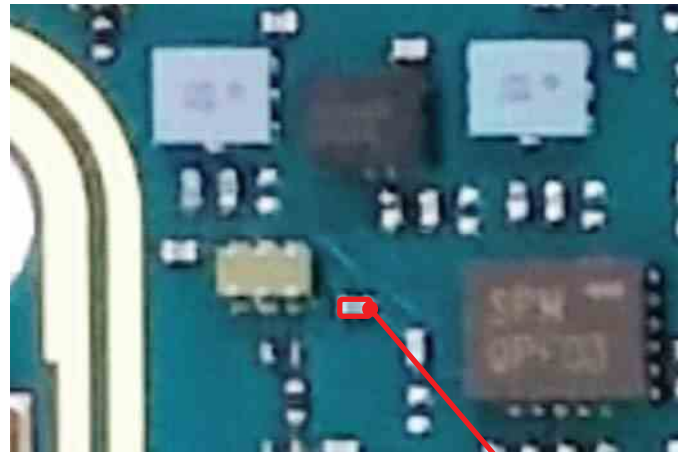




3.5.2.2 Checking RF Switch & Coupler

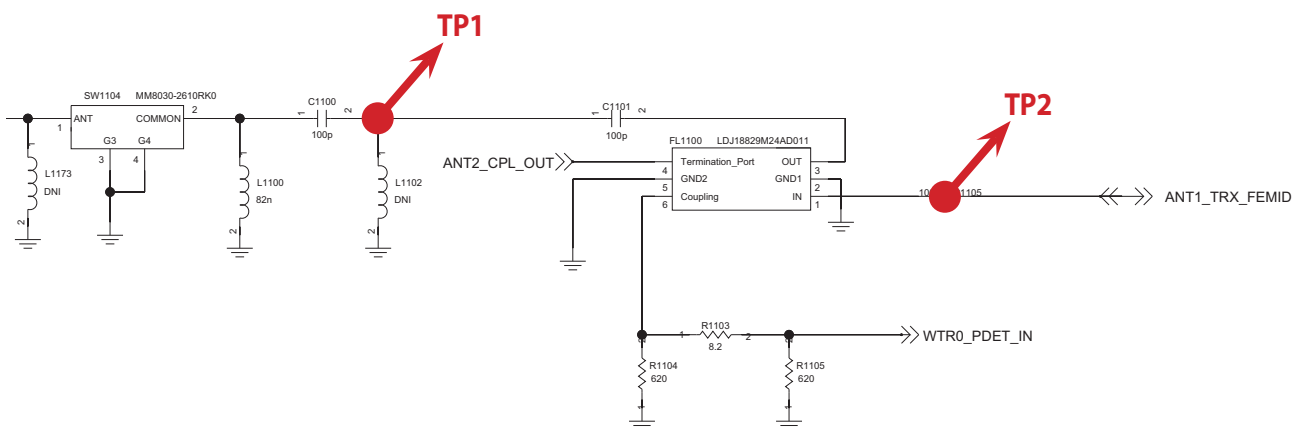


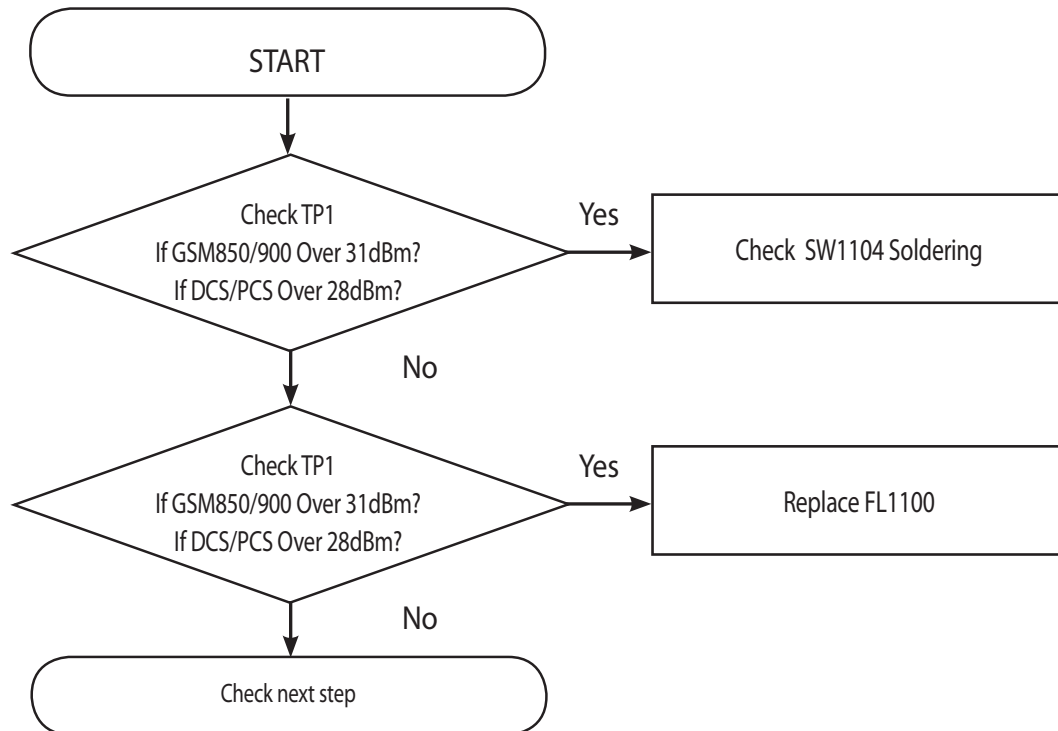
Bottom



Top

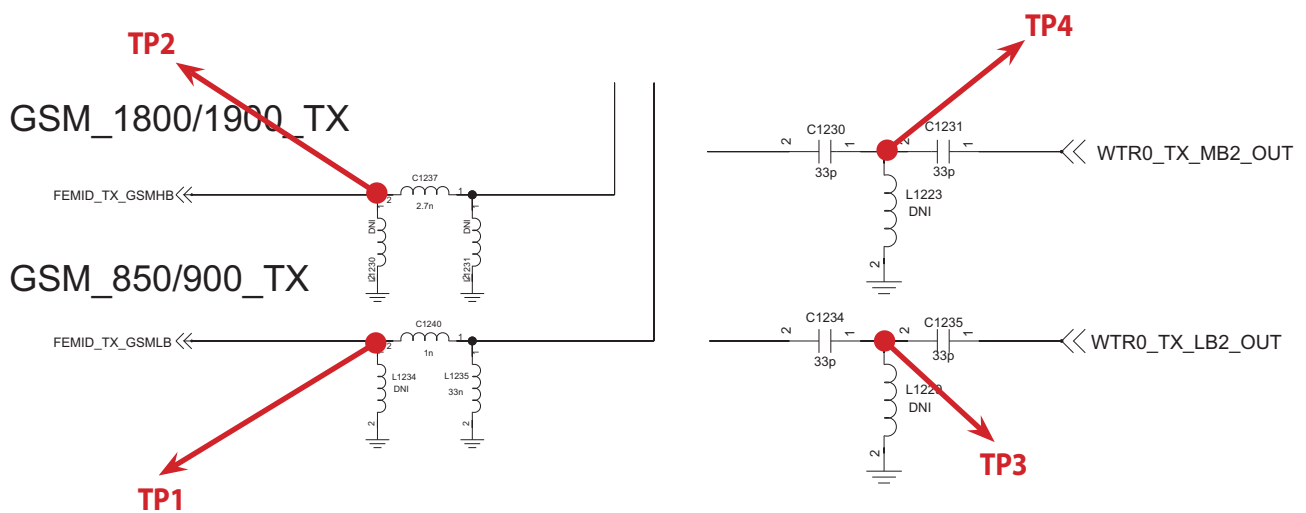
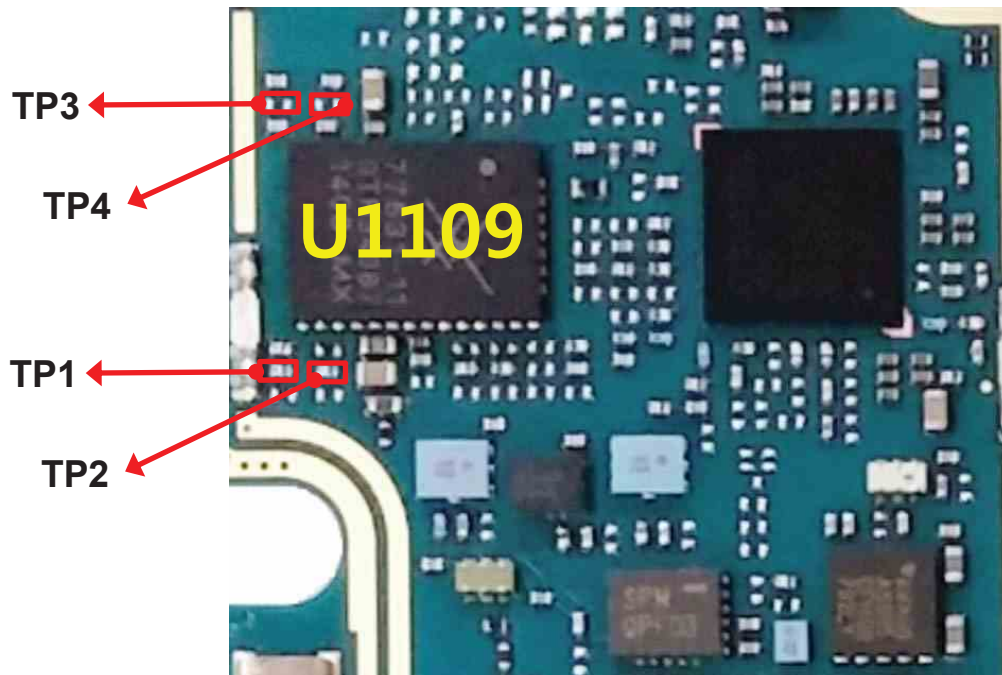
TP2



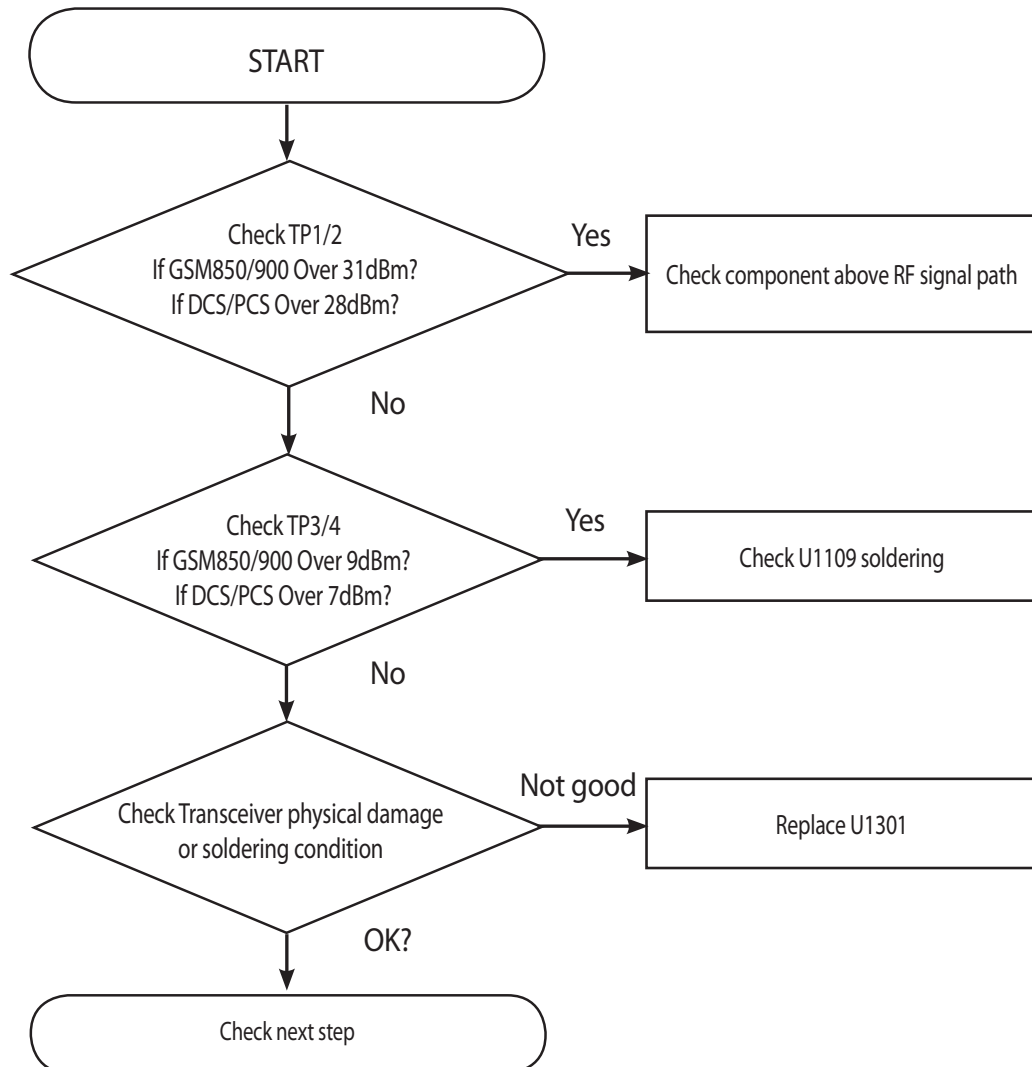


3.5.2.3 Checking RF signal path

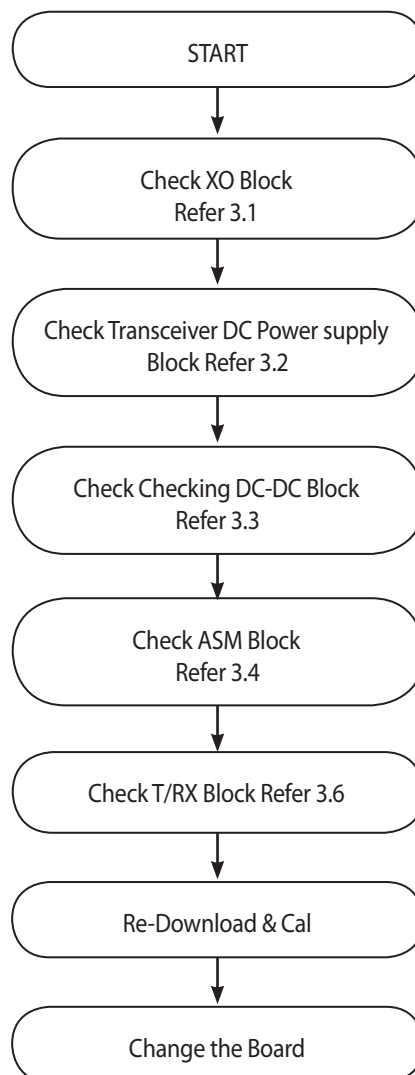
Test Point



3. TROUBLE SHOOTING



3.6 WCDMA RF Part



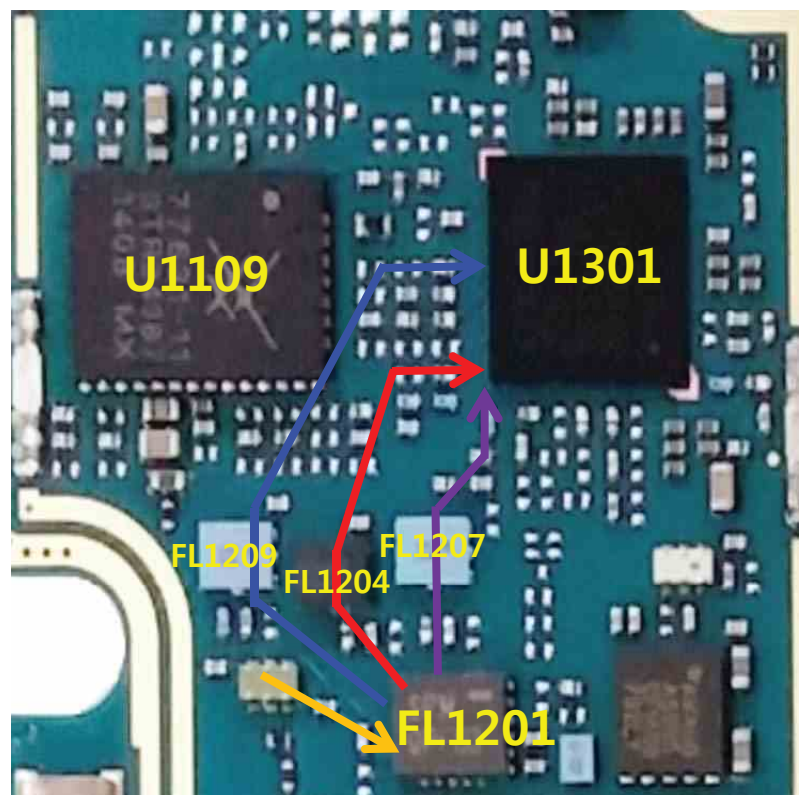
3.6.1 WCDMA B1/2/5/8 Rx

WCDMA B1/2/5/8 RX RF PATH

WCDMA B1/2/5 RX RF PATH



Bottom

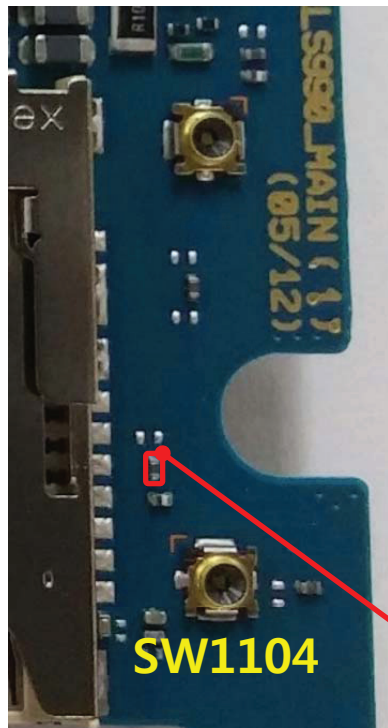


Top

1. WCDMA B1 RX PATH
2. WCDMA B2 RX PATH
3. WCDMA B5 RX PATH
4. WCDMA B8 RX PATH
5. COMMON T/RX PATH

3.6.1.1 Checking RF Switch & Coupler

Test Point

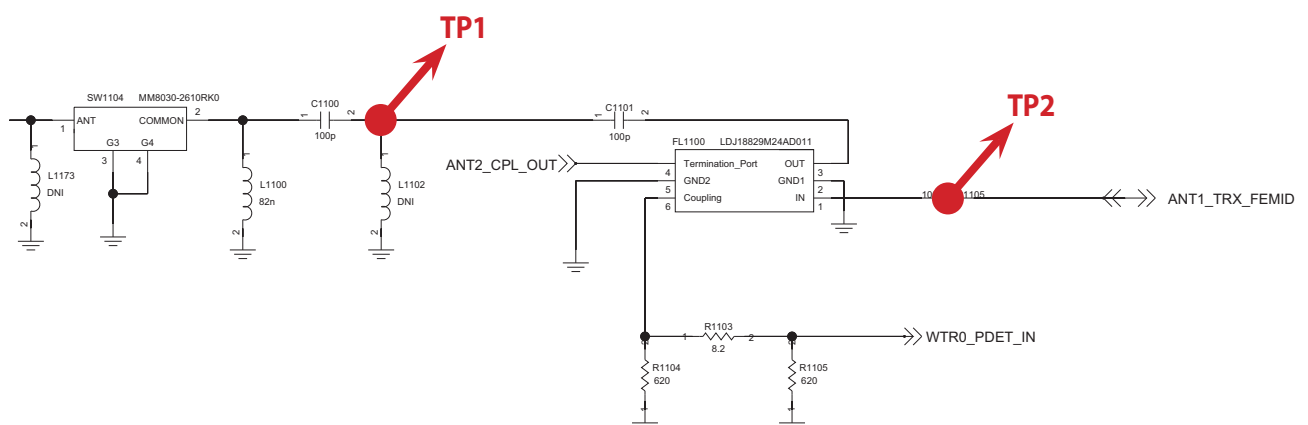


Bottom

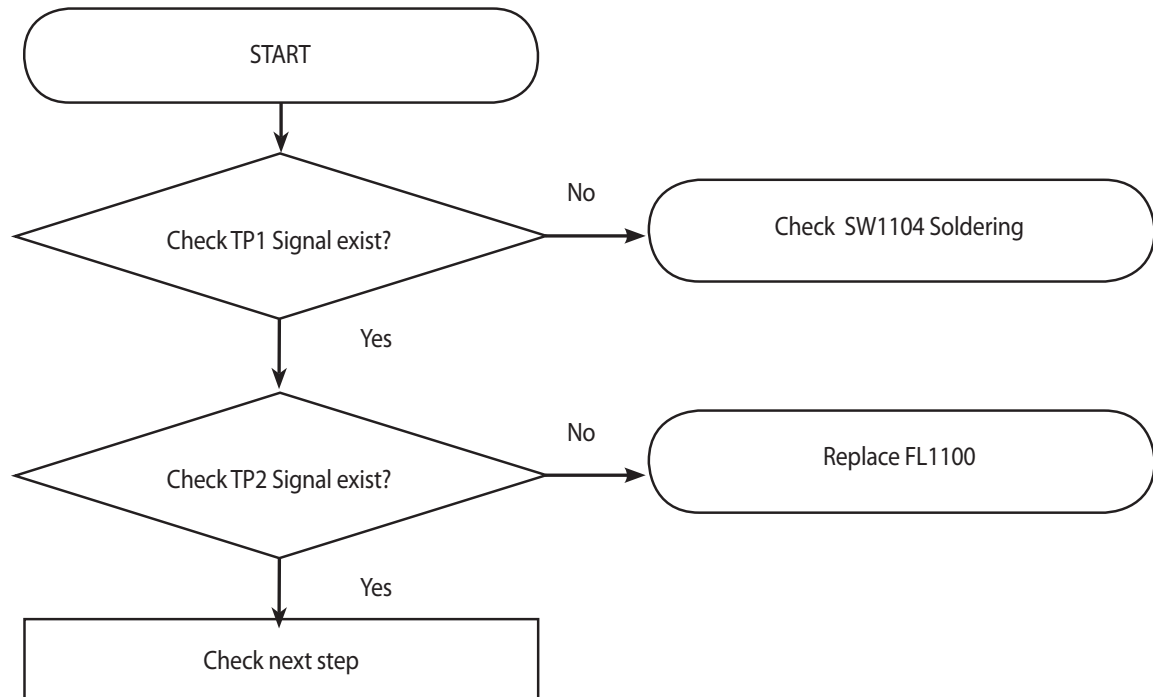


Top

TP2

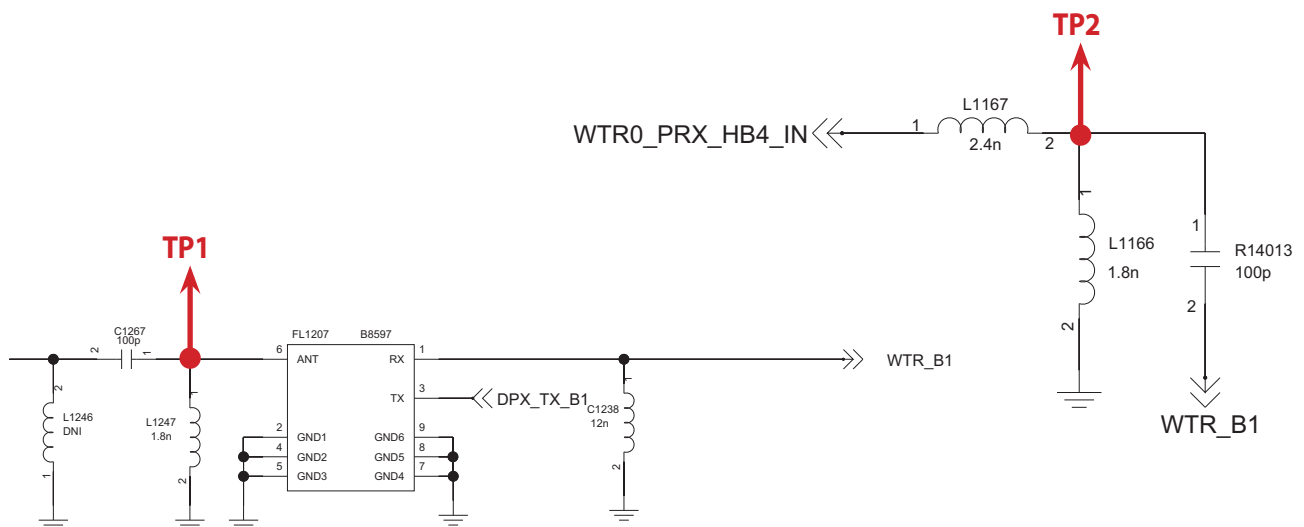
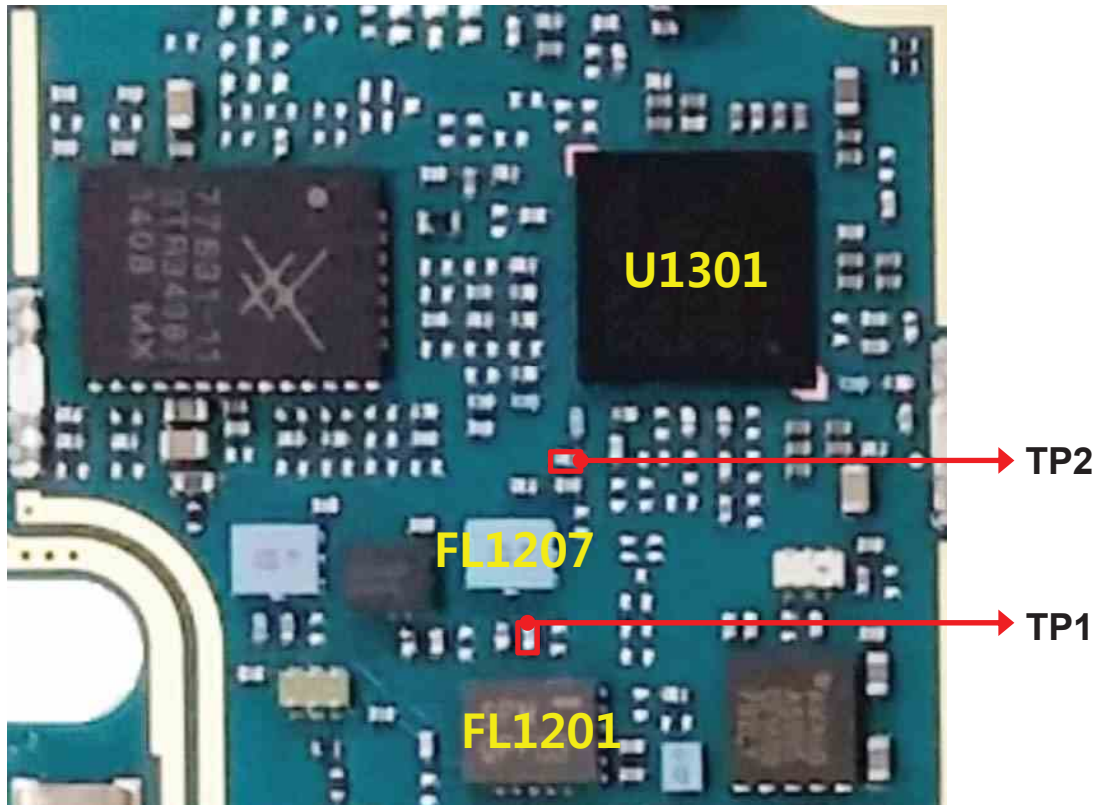


3. TROUBLE SHOOTING

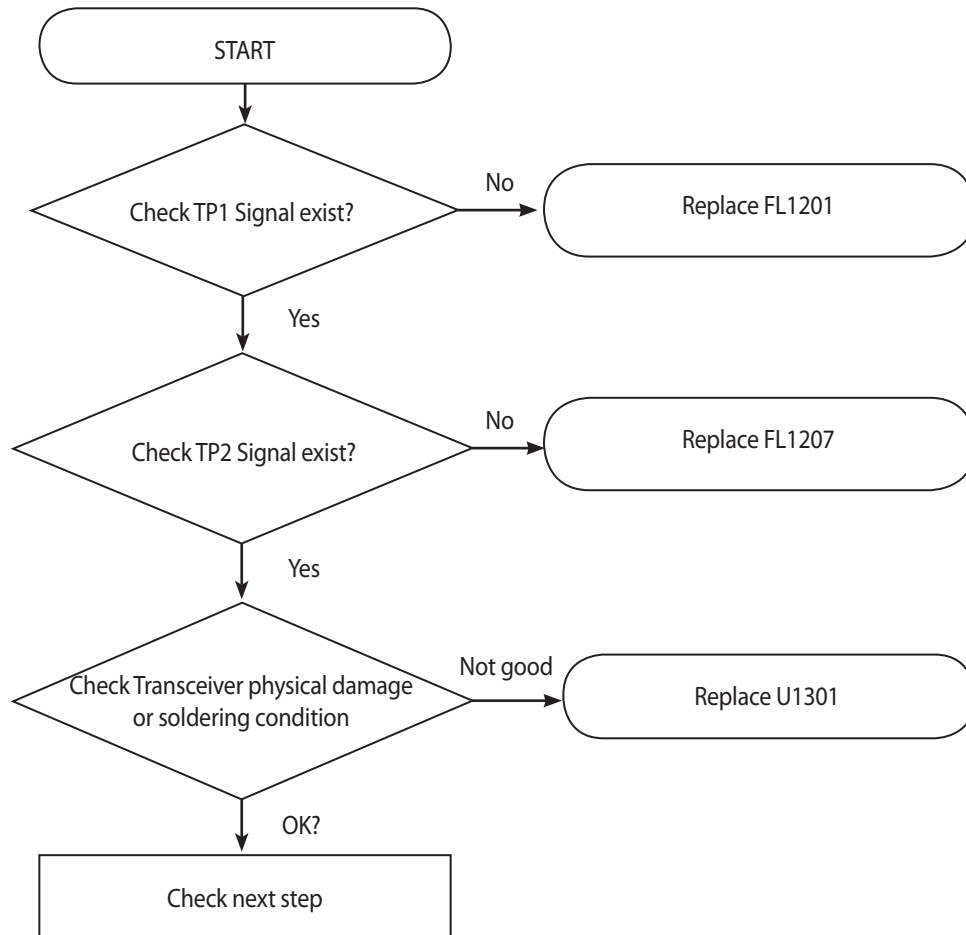


3.6.1.2 Checking RF signal path(B1)

Test Point

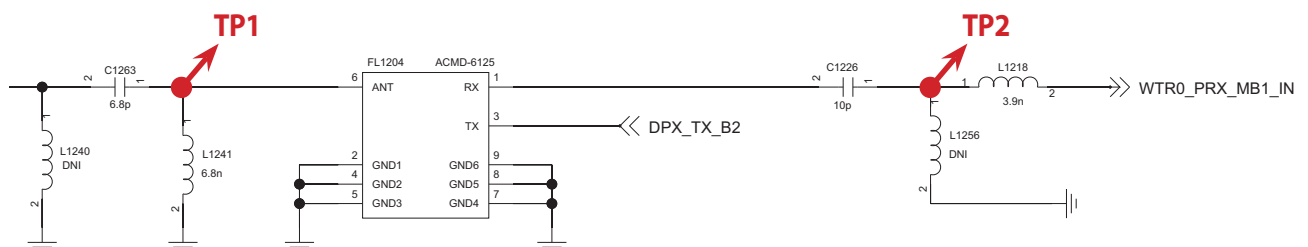
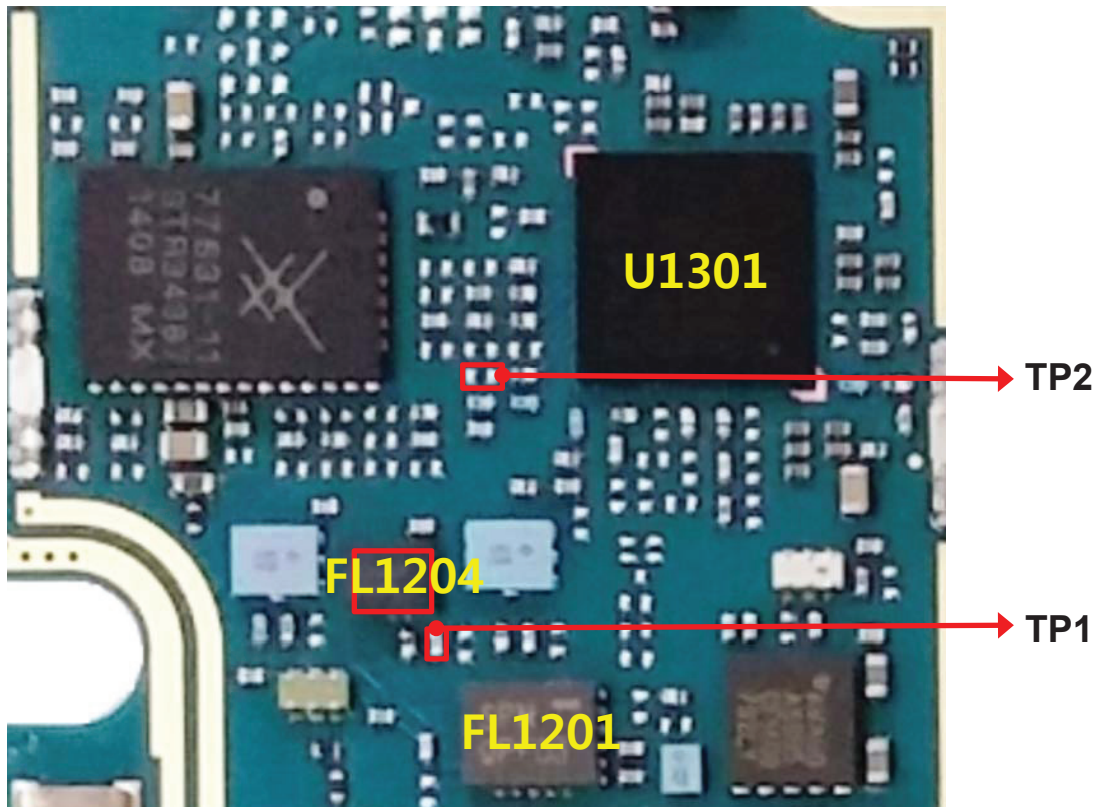


3. TROUBLE SHOOTING

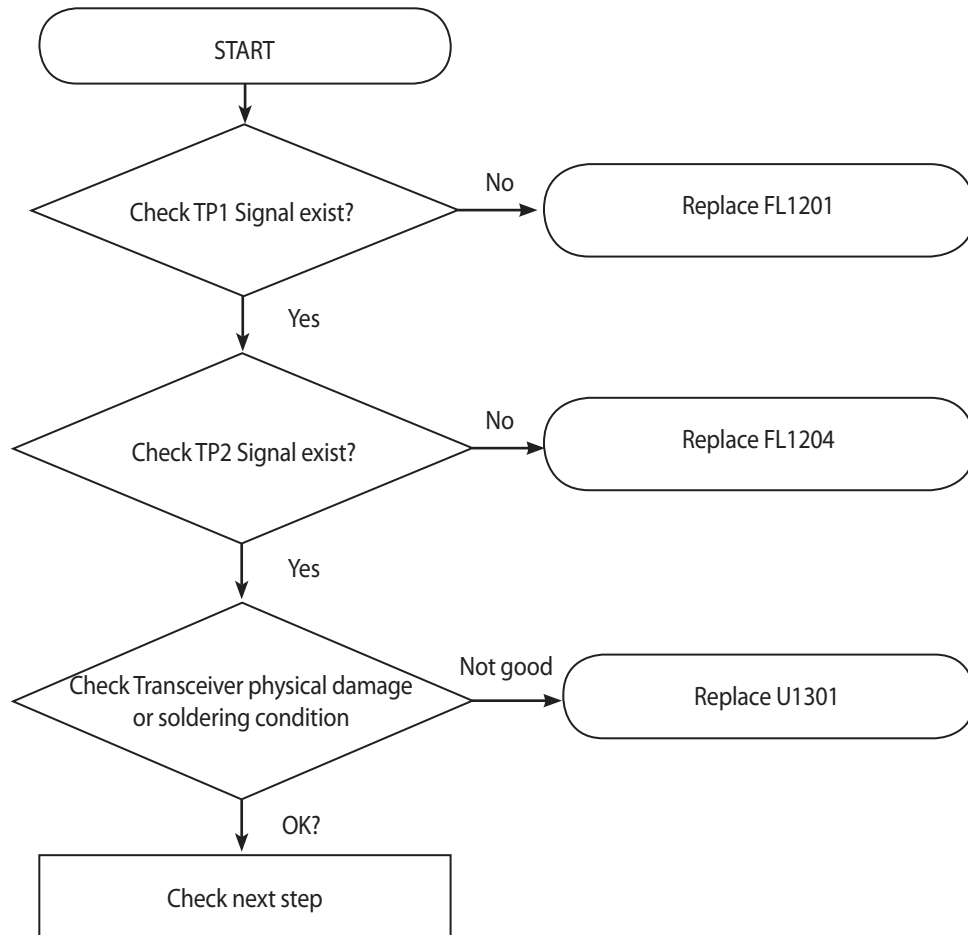


3.6.1.3 Checking RF signal path(B2)

Test Point

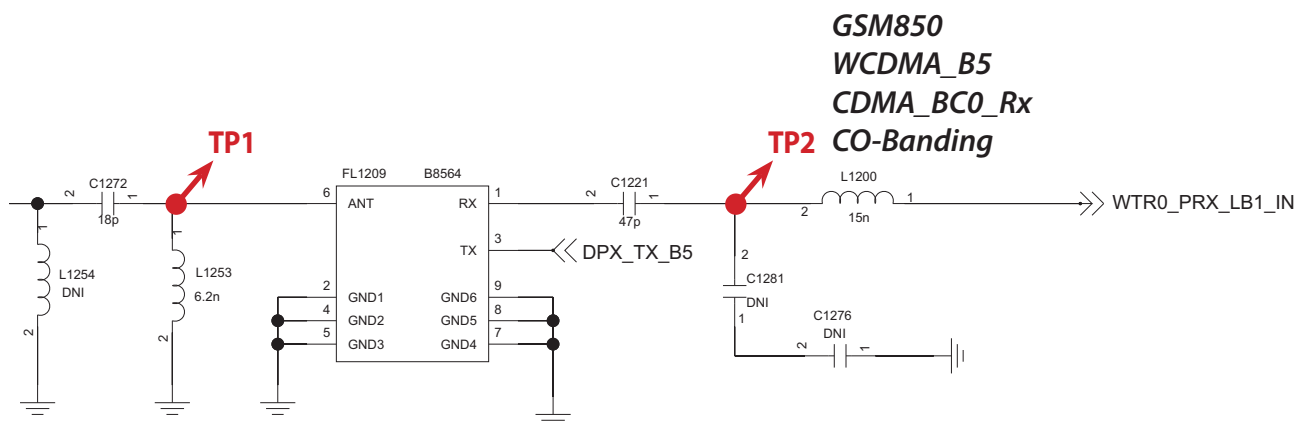
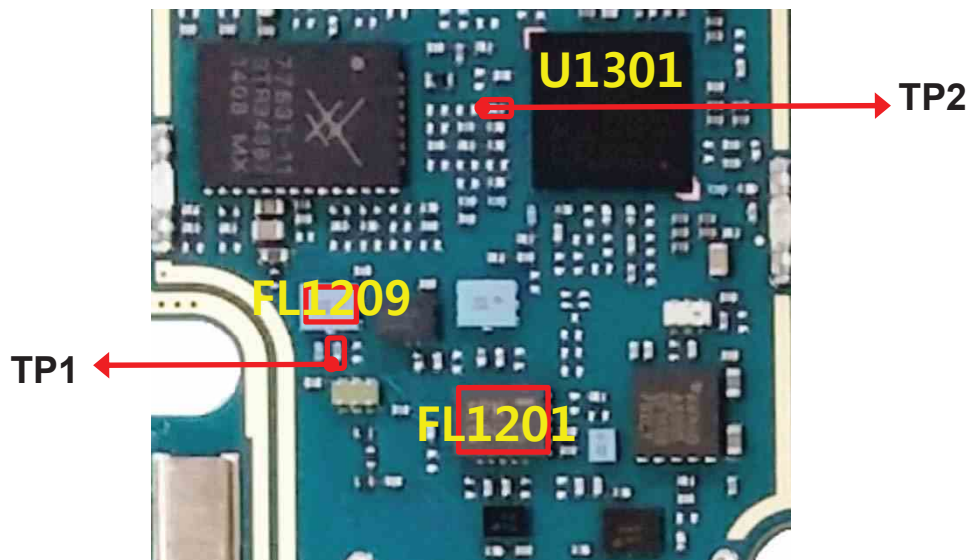


3. TROUBLE SHOOTING

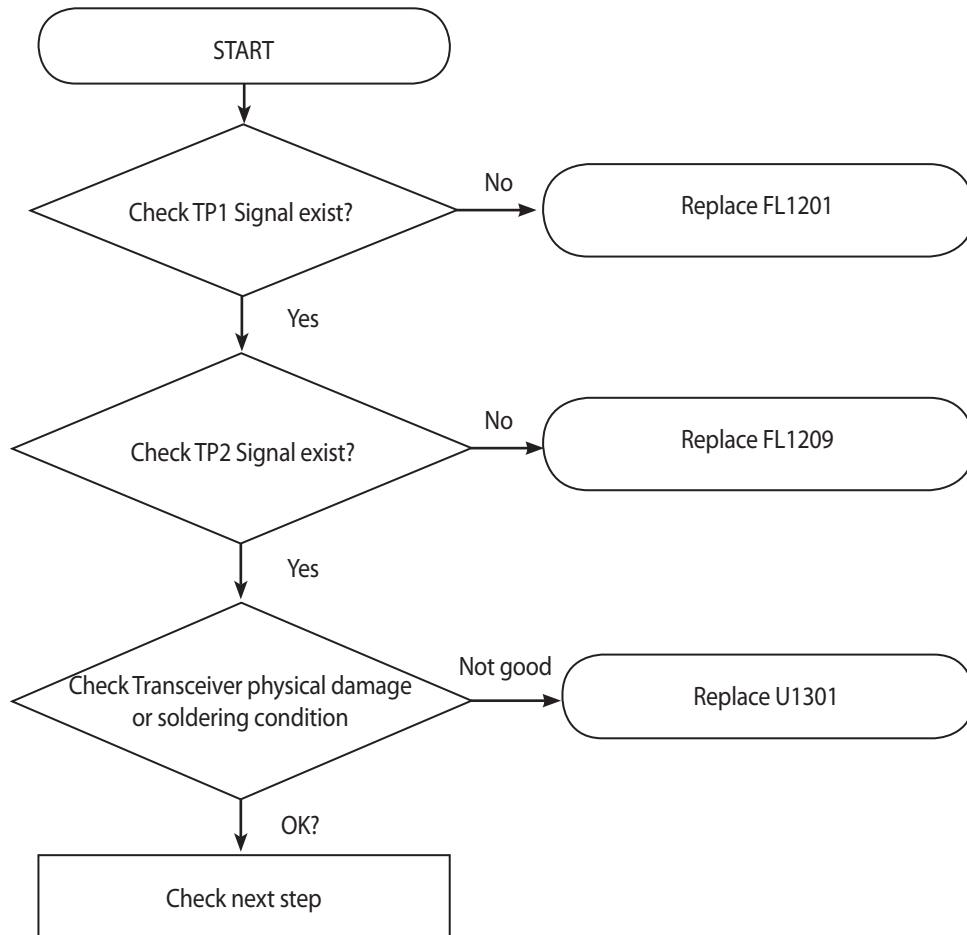


3.6.1.4 Checking RF signal path(B5)

Test Point

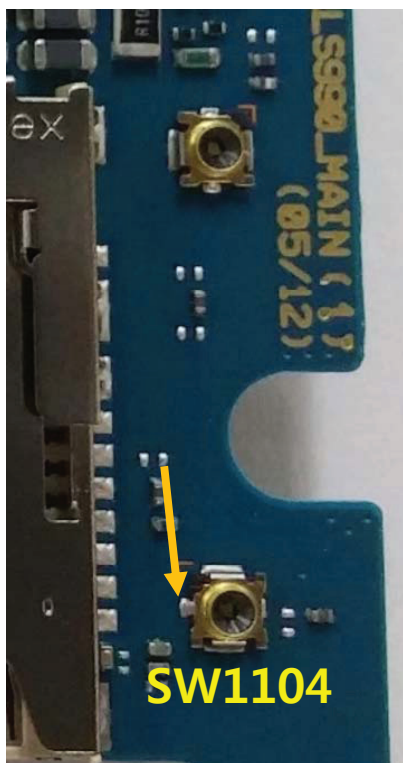


3. TROUBLE SHOOTING

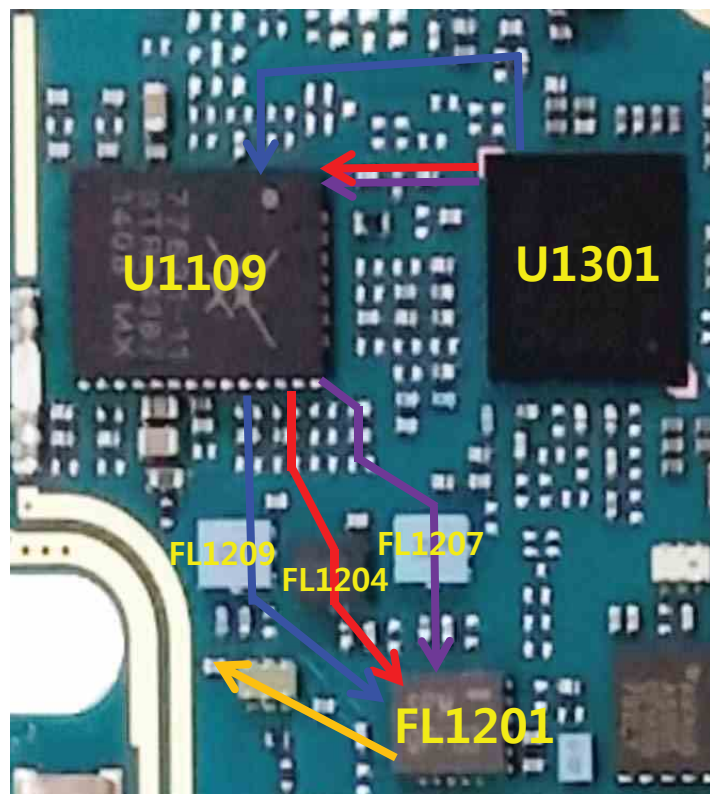


3.6.2 WCDMA B1/2/5 Tx

WCDMA B1/2/5 TX RF PATH



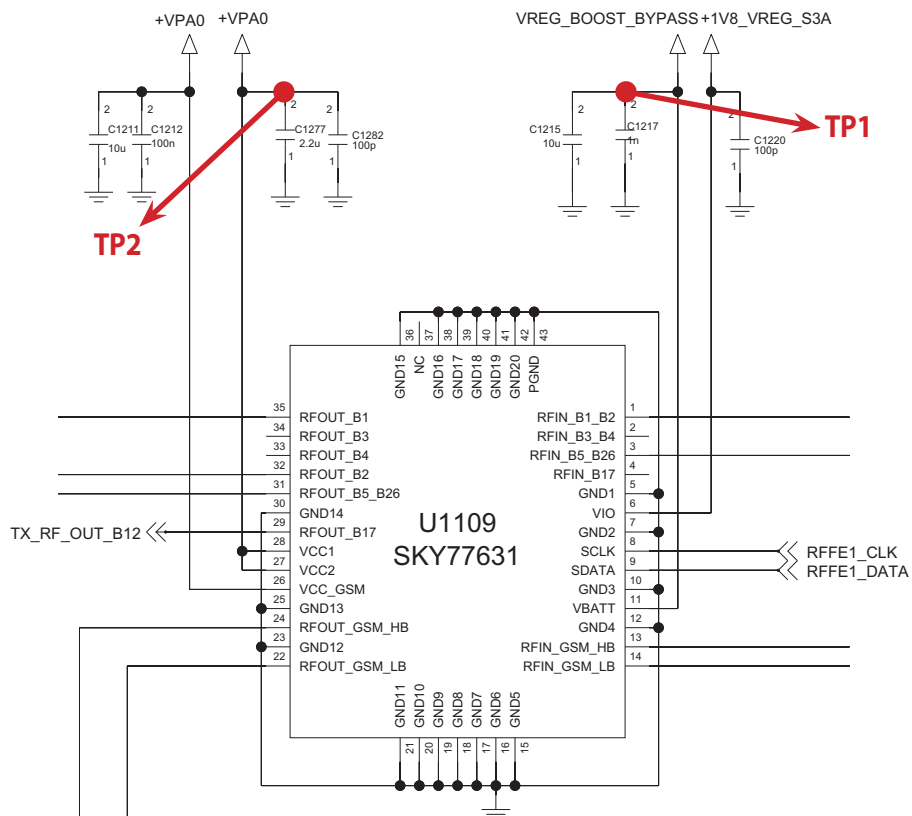
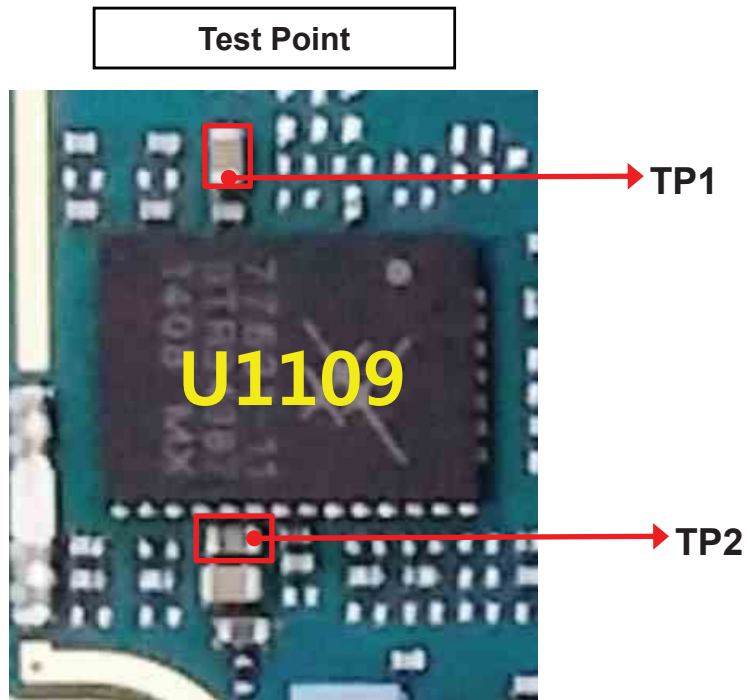
Bottom



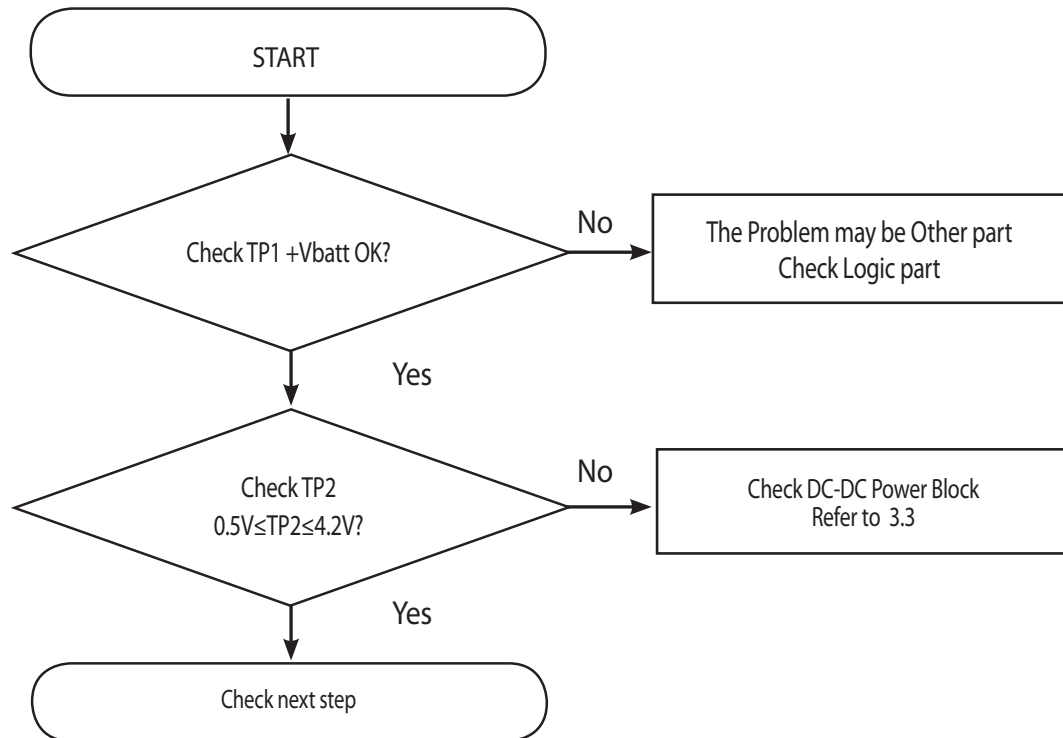
Top

- 1. WCDMA B1 TX PATH
- 2. WCDMA B2 TX PATH
- 3. WCDMA B5 TX PATH
- 4. COMMON T/RX PATH

3.6.2.1 Checking WCDMA PAM DC Power&Control Circuit

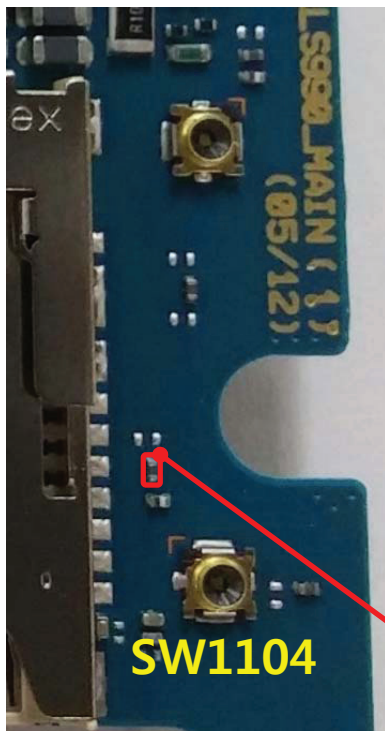


3. TROUBLE SHOOTING



3.6.2.2 Checking RF Switch & Coupler

Test Point

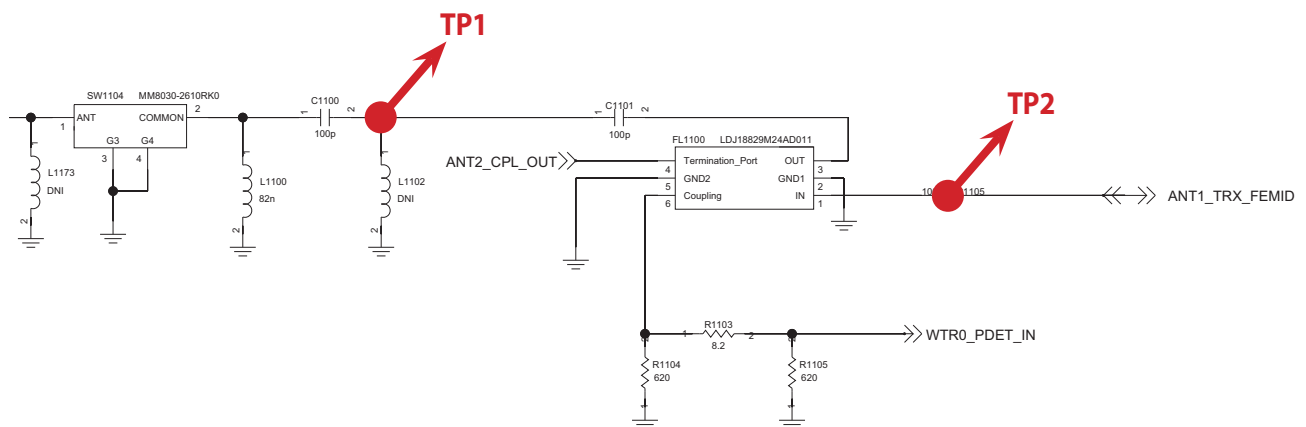


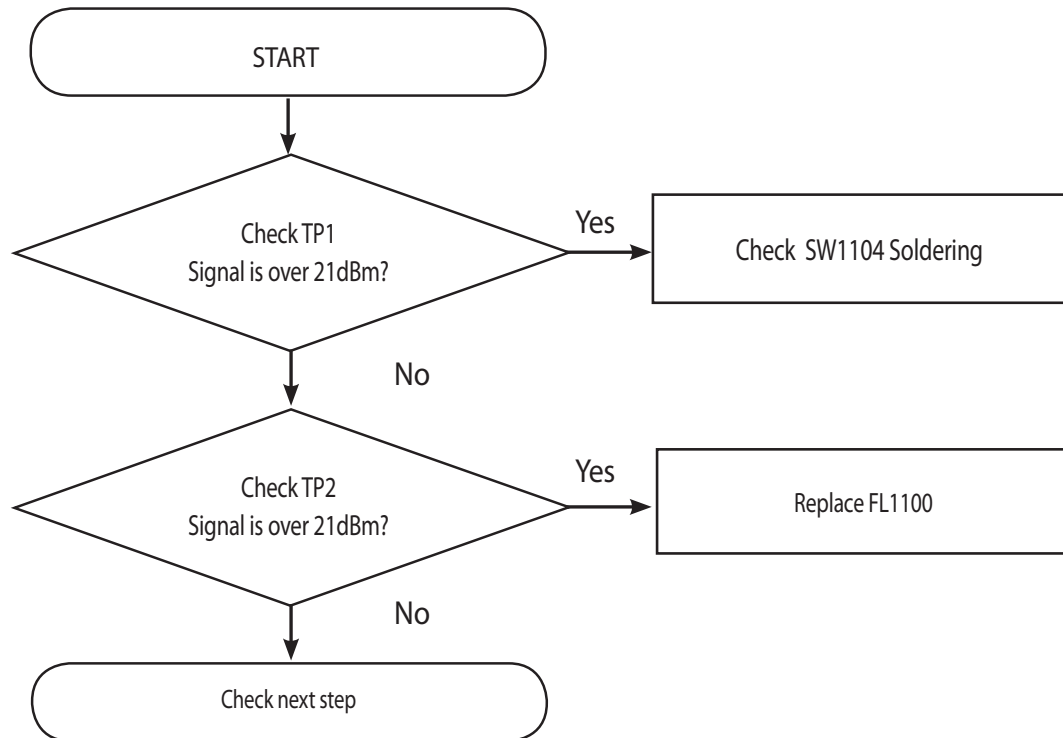
Bottom



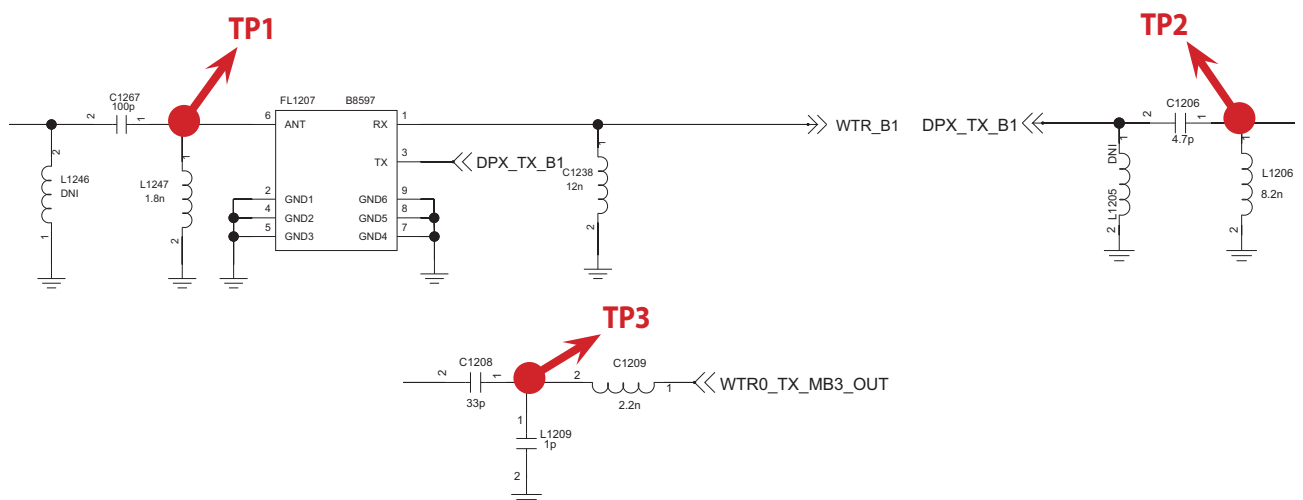
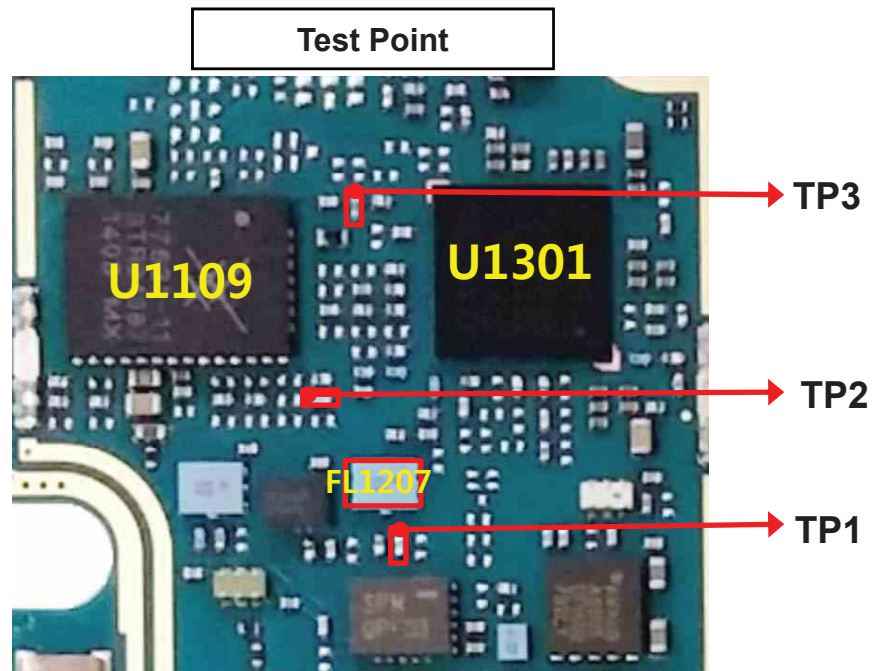
Top

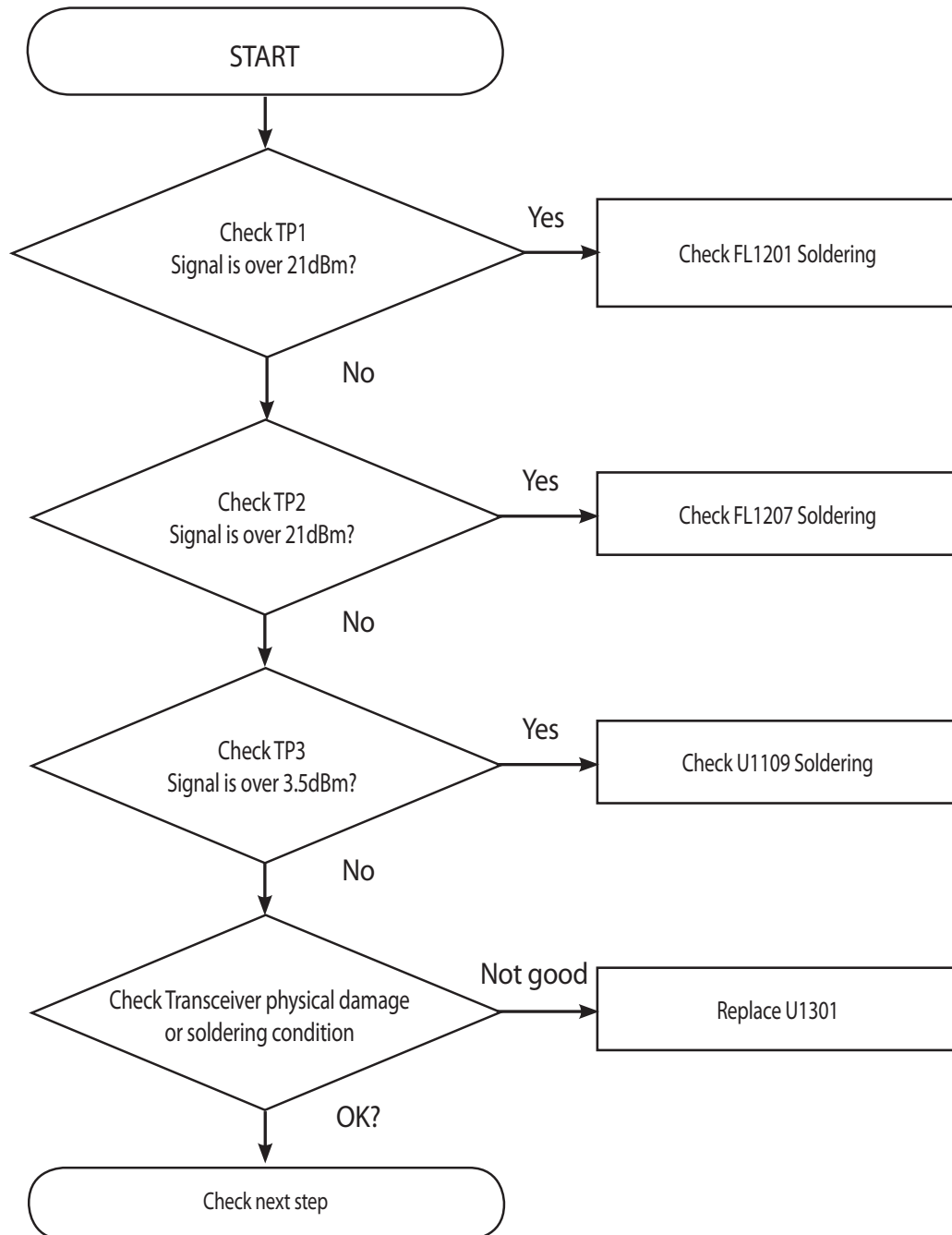
TP2



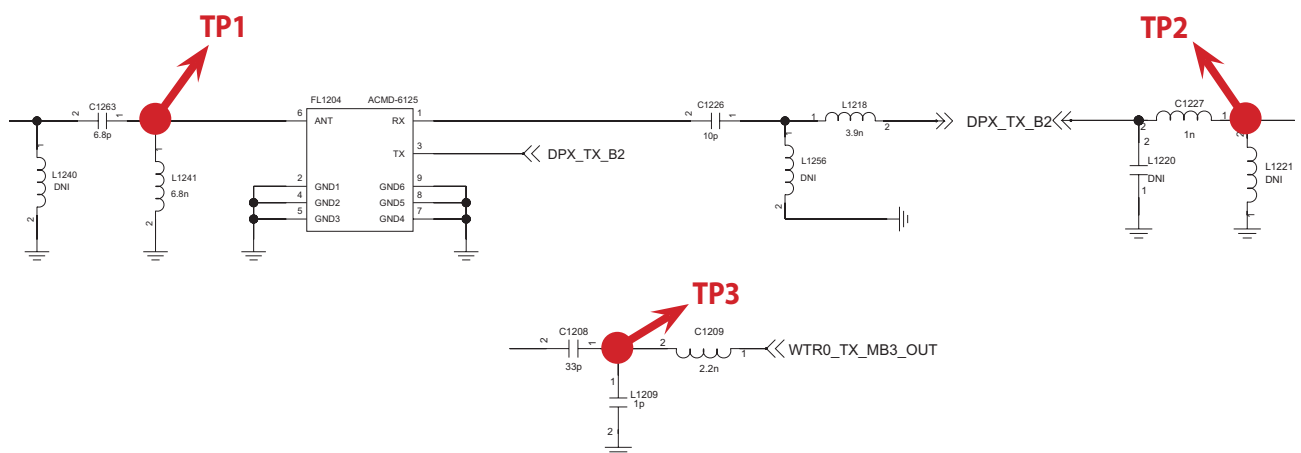
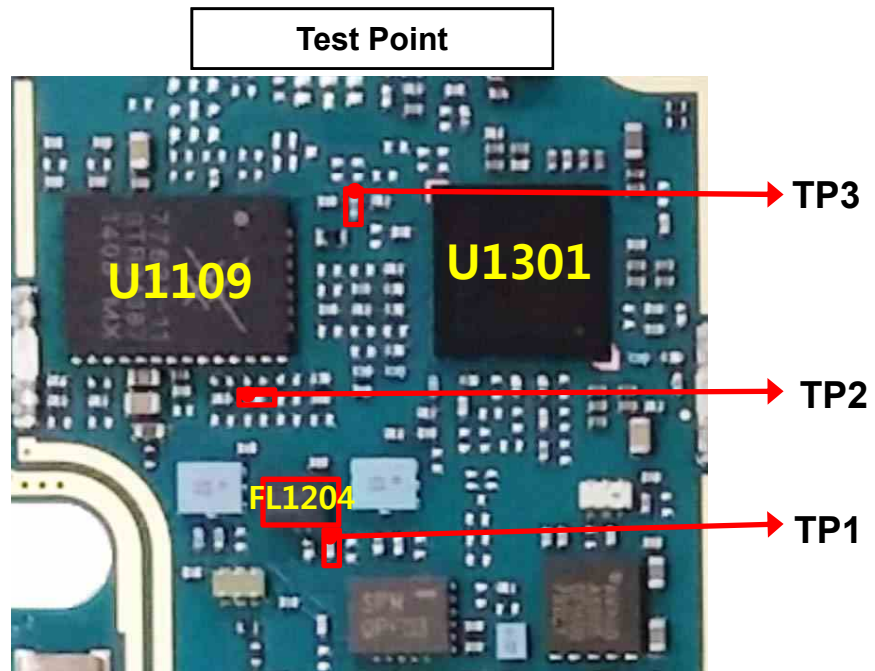


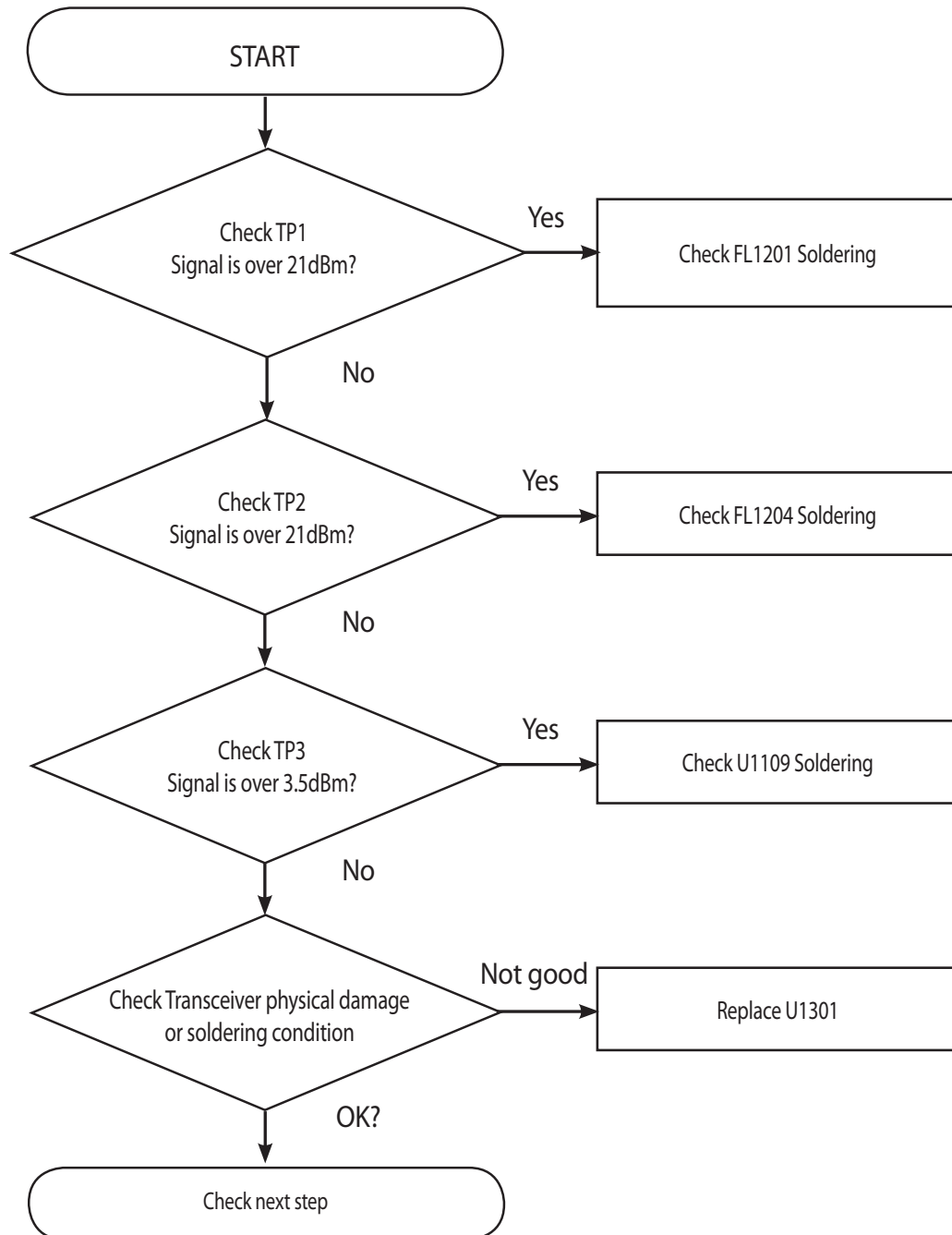
3.6.2.3 Checking RF signal path(B1)



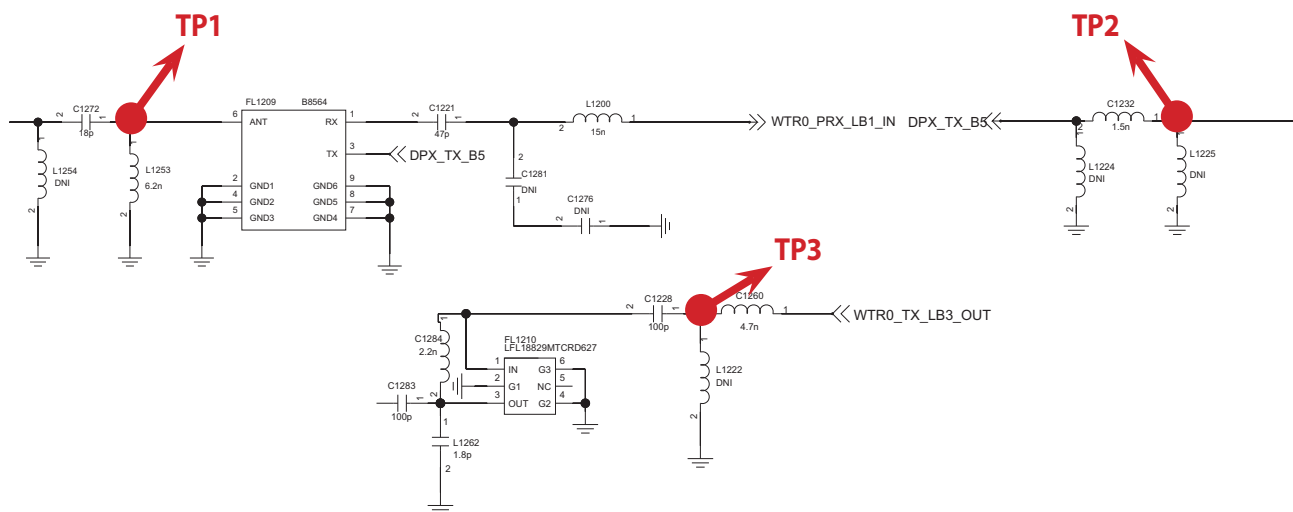
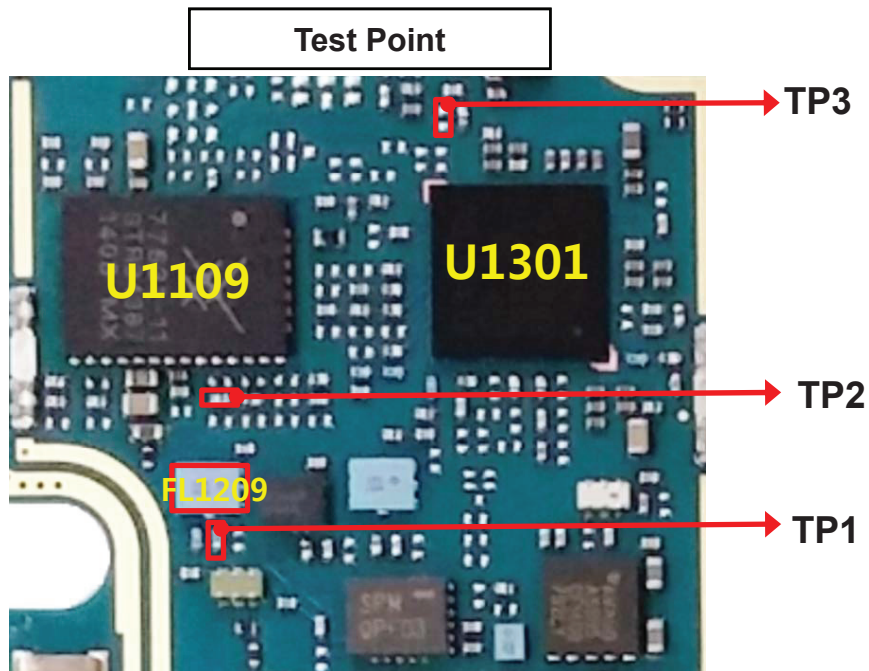


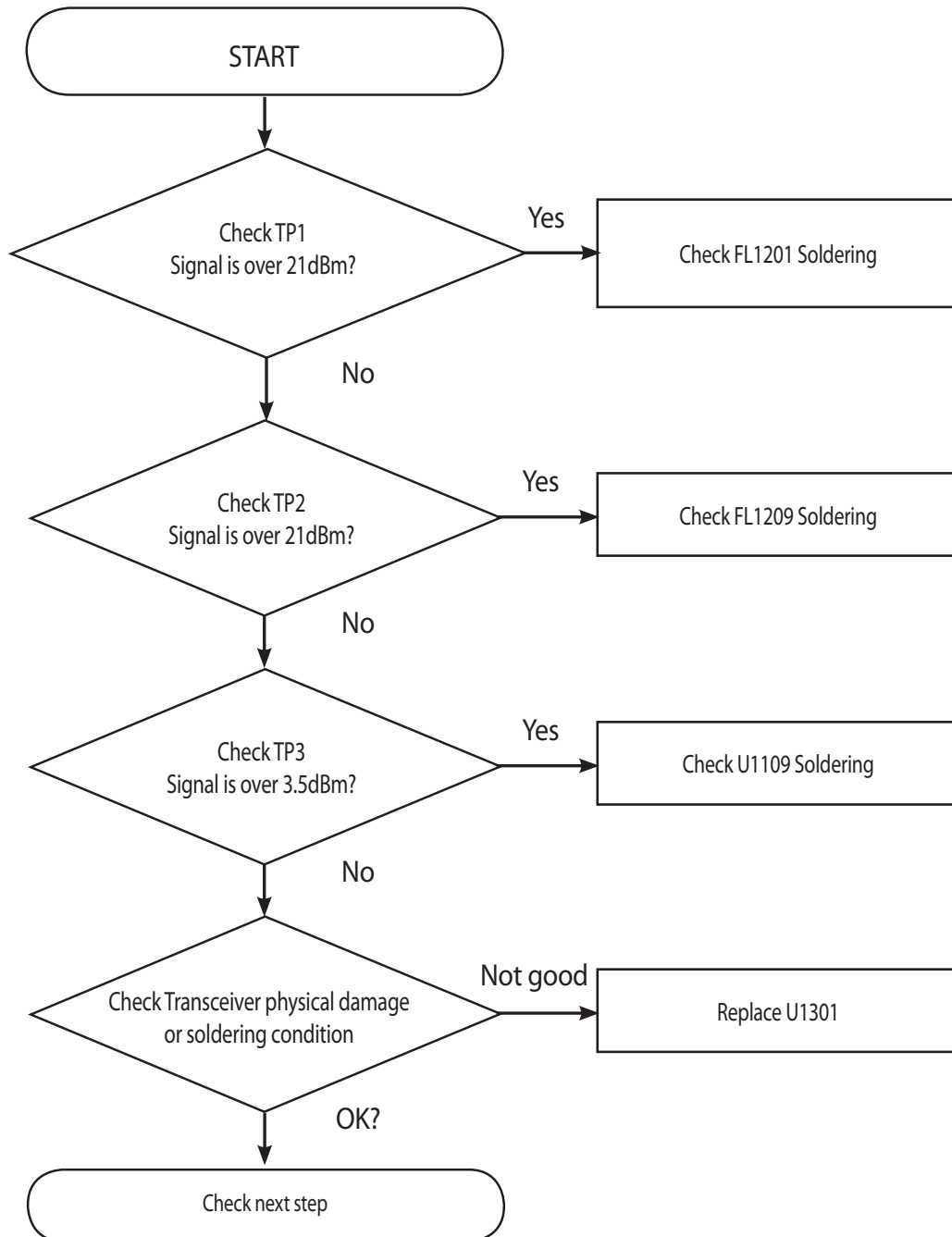
3.6.2.4 Checking RF signal path(B2)



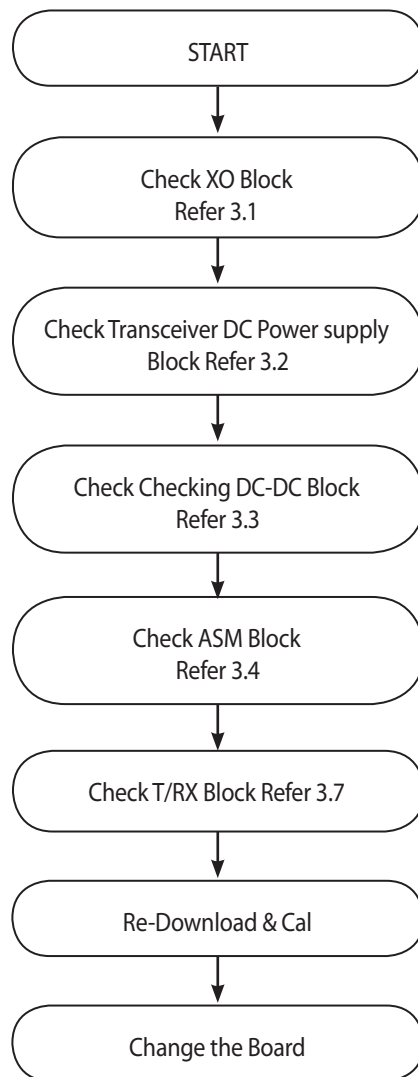


3.6.2.5 Checking RF signal path(B5)



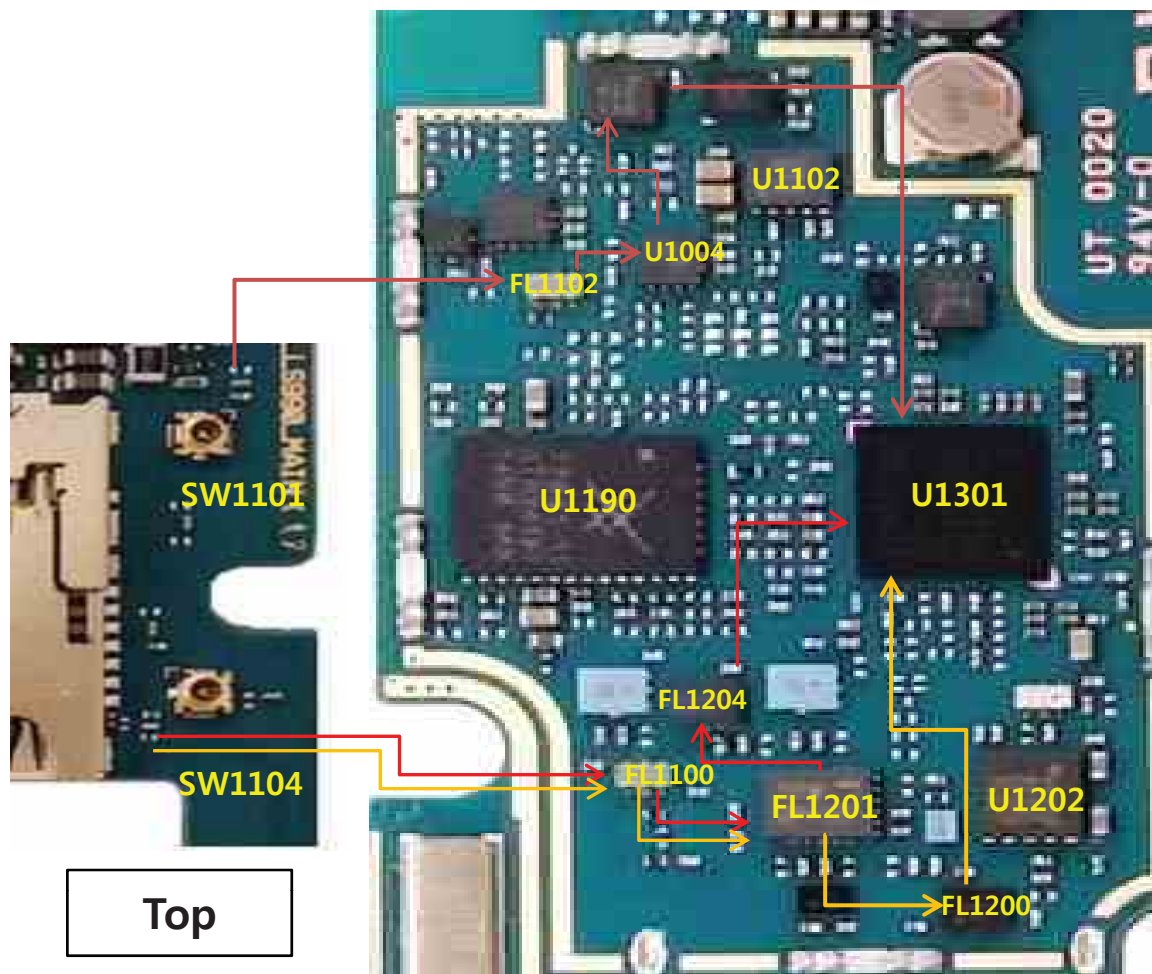


3.7 LTE RF Part



3.7.1 LTE B25/B26/B41 PRX

LTE B25/B26/B41 PRX RF PATH



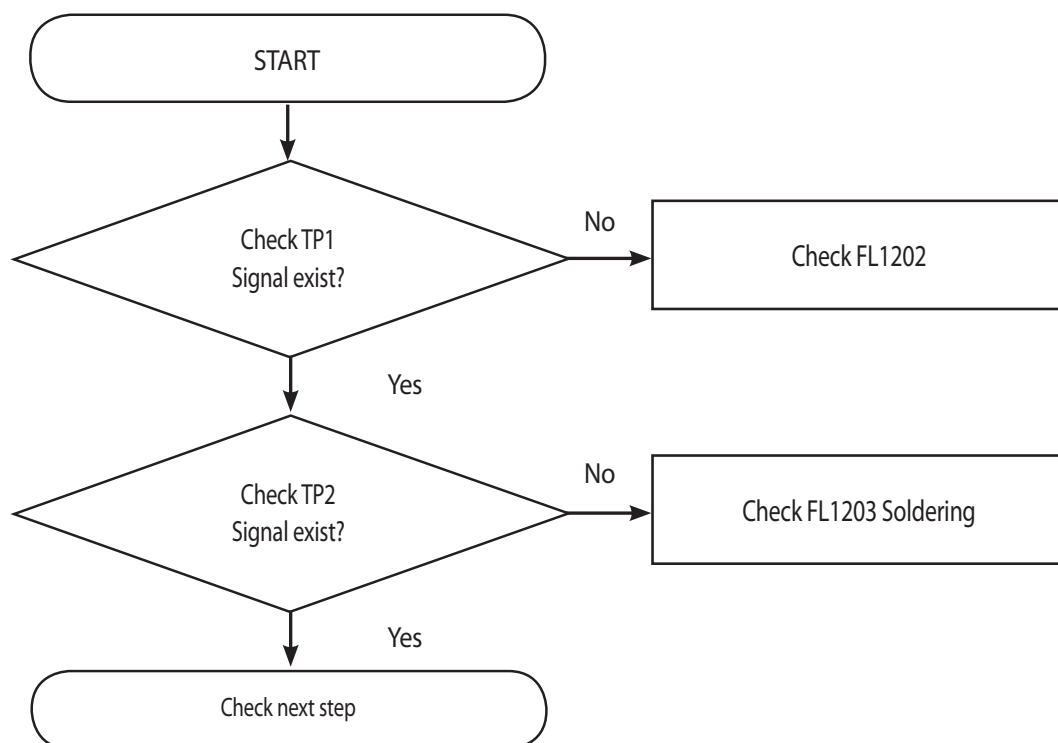
1. LTE B25 / CDMA(BC1)_ PRX PATH
2. LTE B26/ CDMA(BC10)_ PRX PATH
3. LTE B41 PRX PATH

3.7.1.1 Checking LTE Switch (B25,BC1/B26,BC10)

Refer to 3.4.1

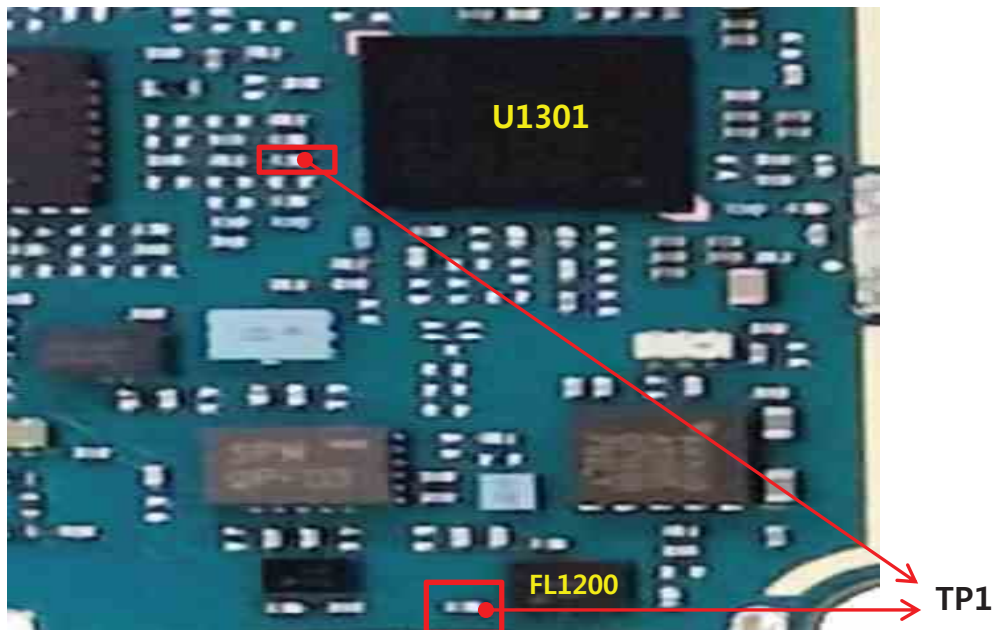
3.7.1.2 Checking RF signal path (B25_BC1)

Refer to 3.6.1.2

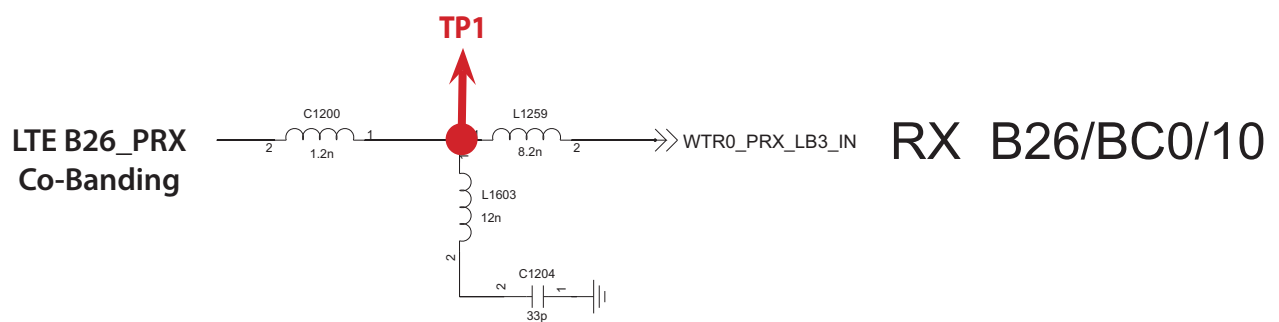


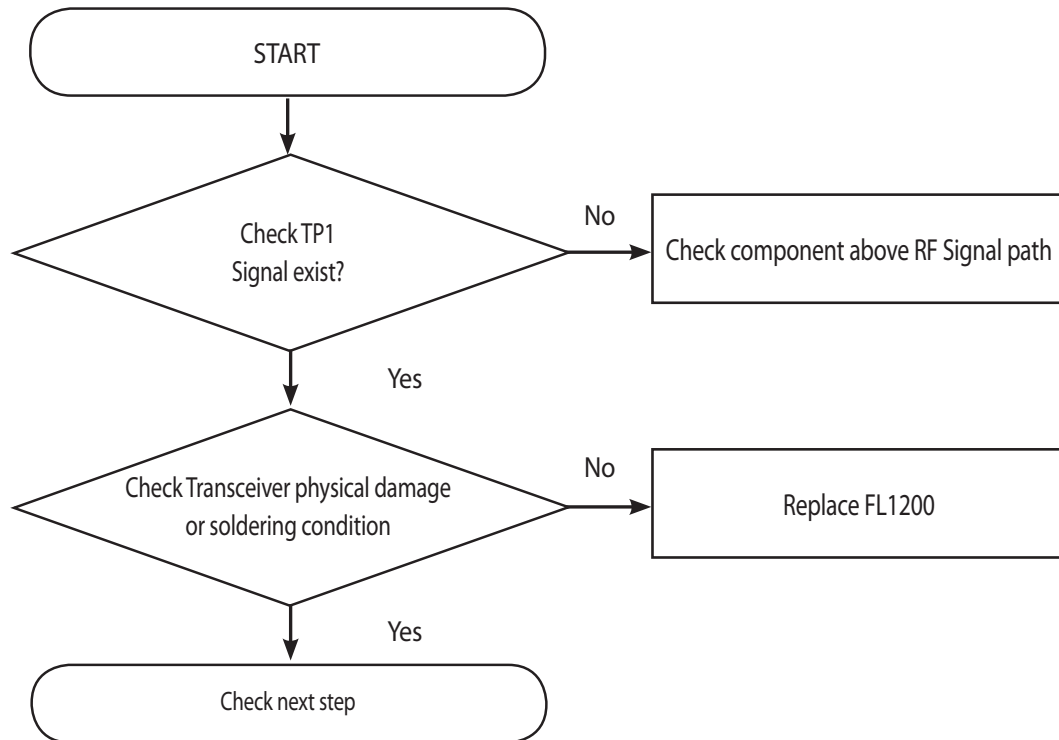
3.7.1.3 Checking RF signal path (B26_BC10)

Test Point



Circuit Diagram

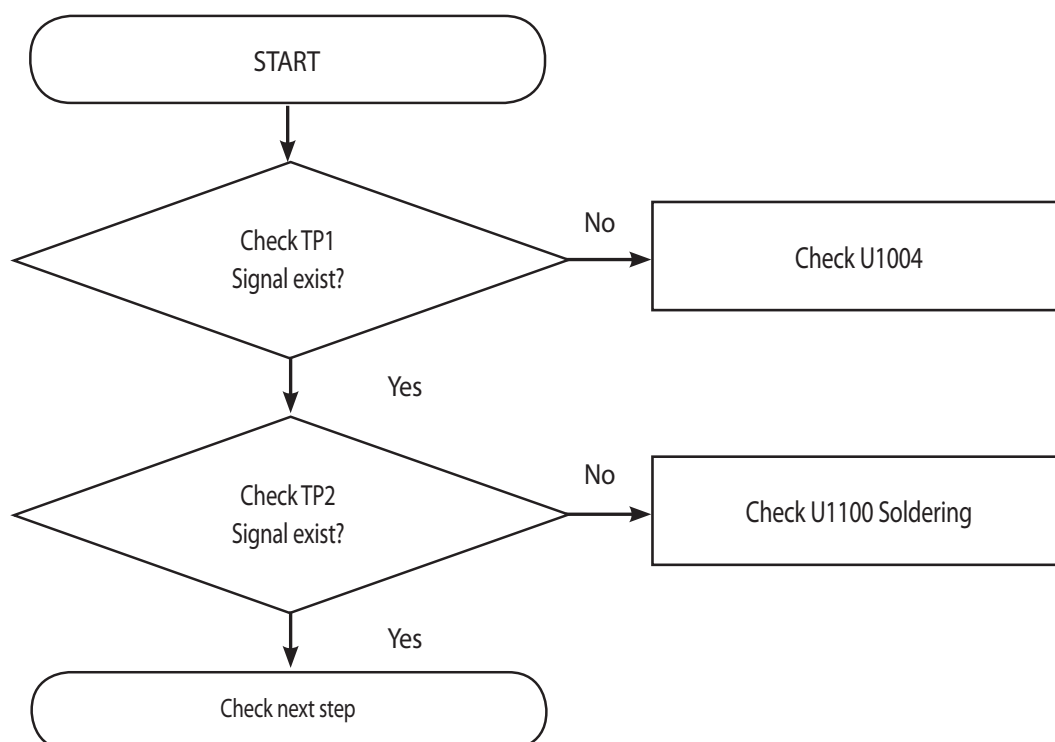
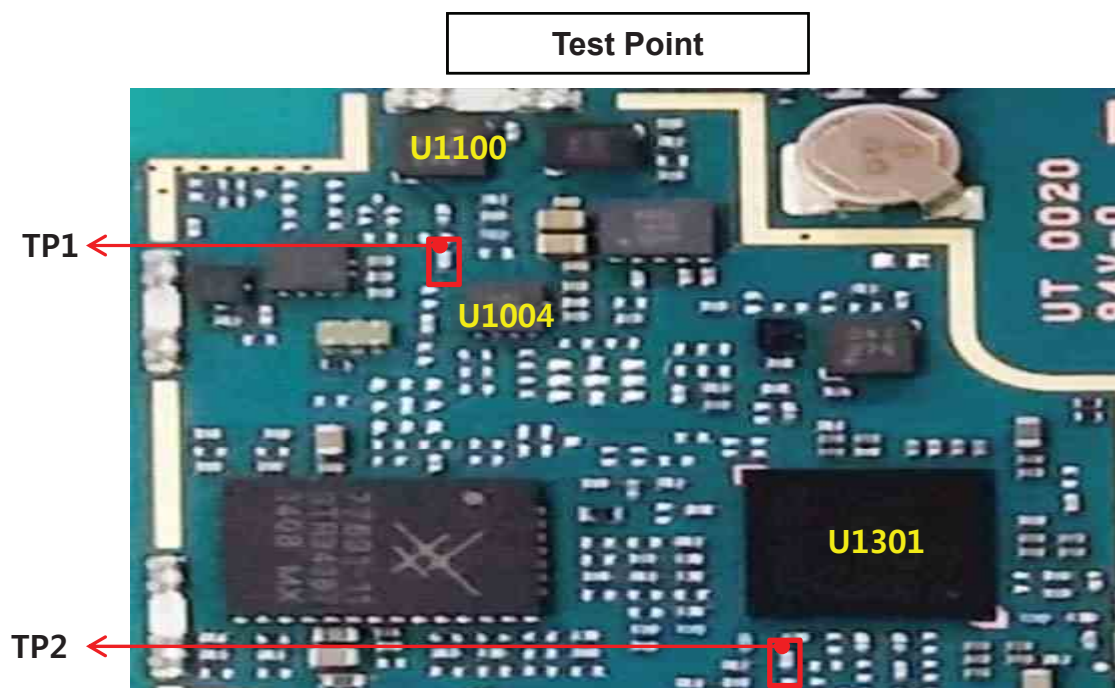




3.7.1.4 Checking RF LTE SP4T switch (B41)

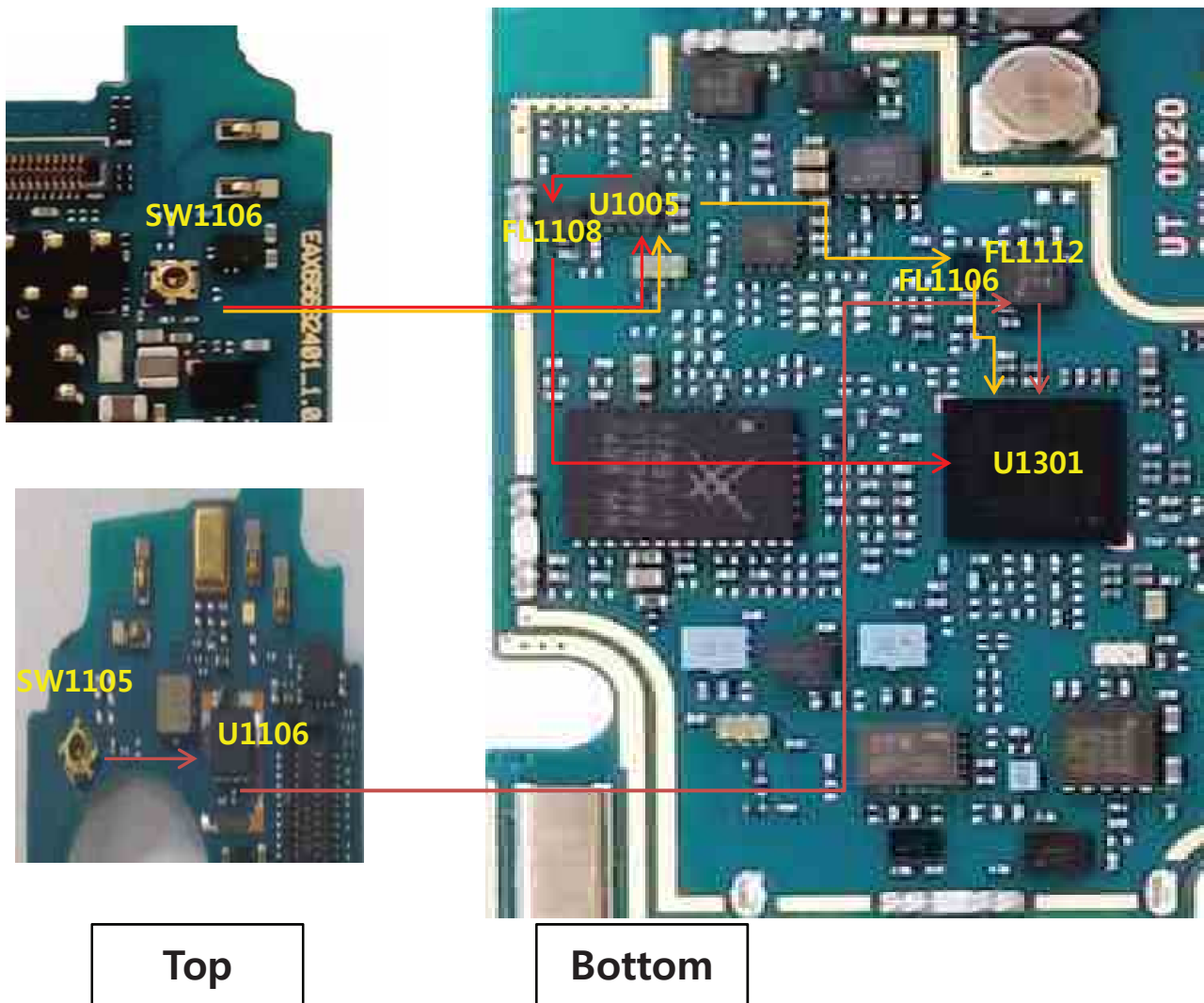
Refer to 3.4.2

3.7.1.5 Checking RF signal path (B41)



3.7.2 LTE B25/B26/B41 DRX

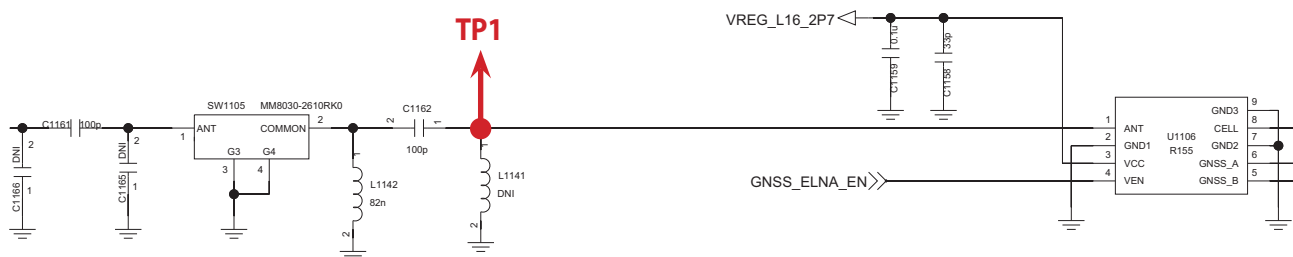
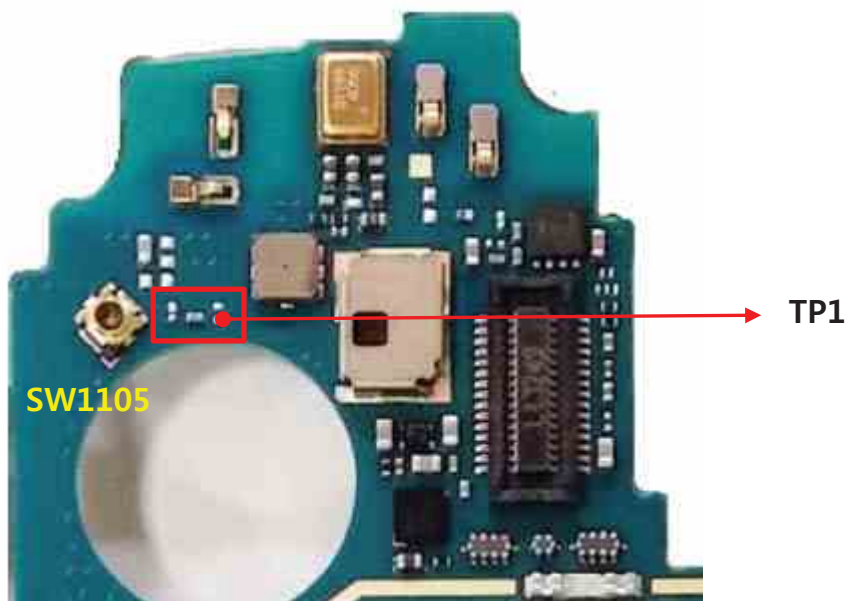
LTE B25/B26/B41 DRX RF PATH

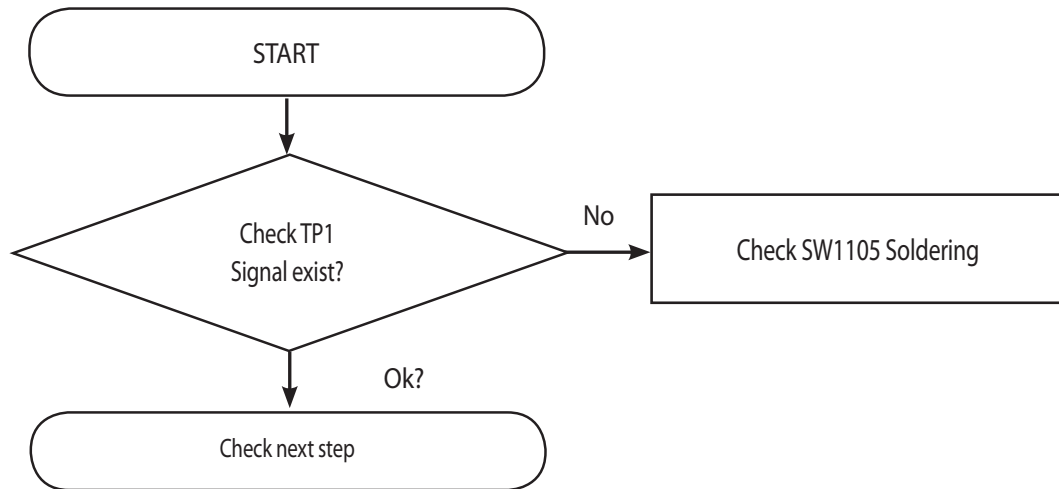


1. *LTE B25/CDMA-BC1 DRX PATH*
2. *LTE B26/CDMA_BC10,BC0 DRX PATH*
3. *LTE B41 PRX PATH*

3.7.2.1 Checking DRX Antenna Switching Module (B41)

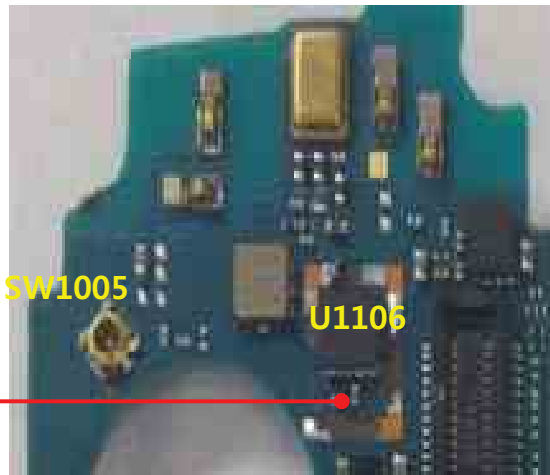
Test Point





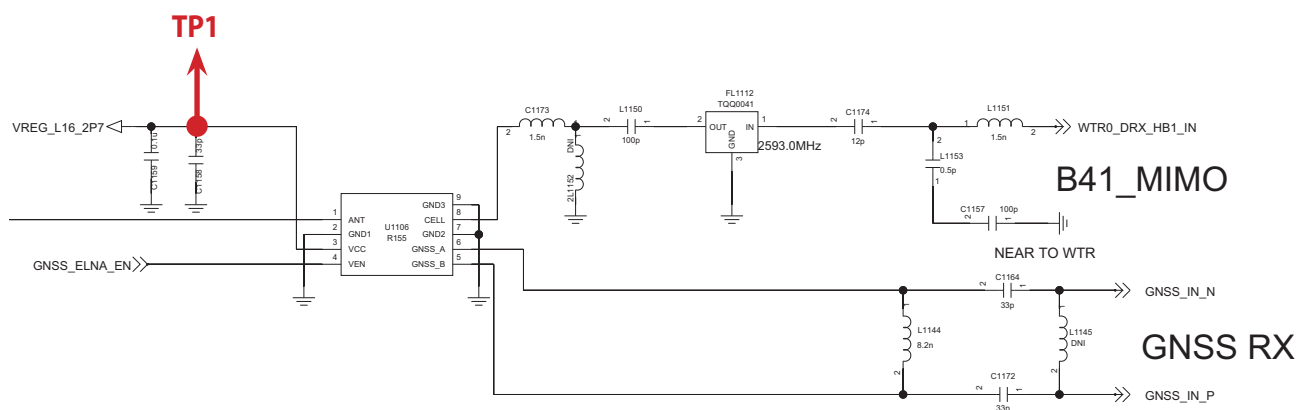
3.7.2.2 Checking SPDT SW DC Power(B41)

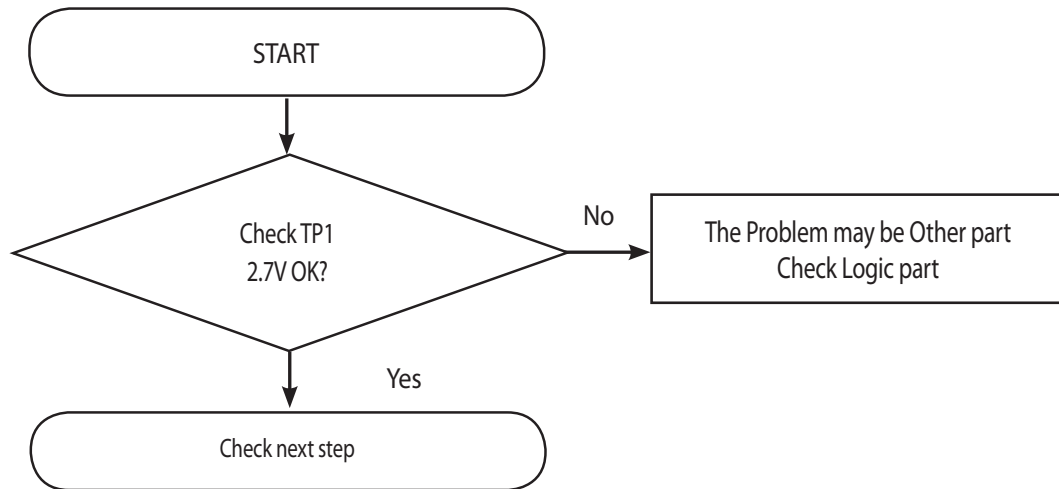
Test Point



TP1

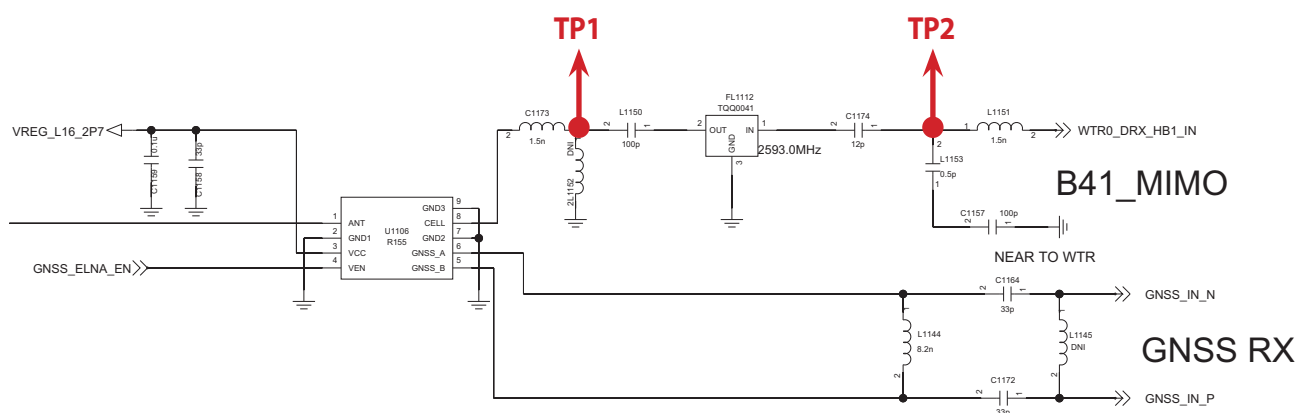
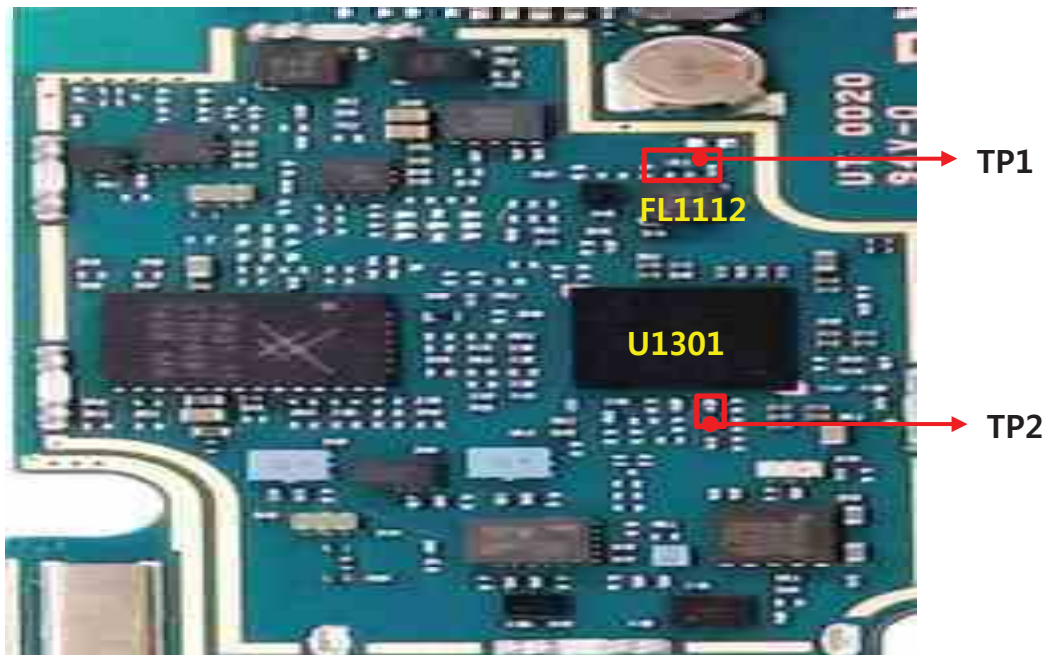
Bottom

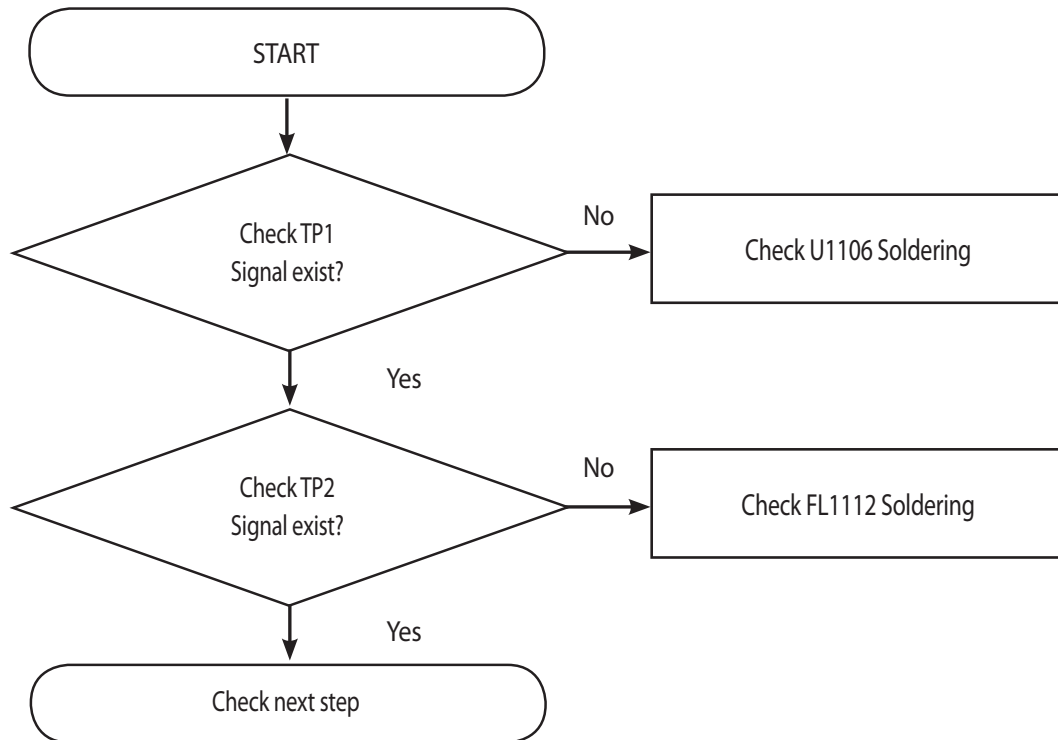




3.7.2.3 Checking Signal Path(B41)

Test Point



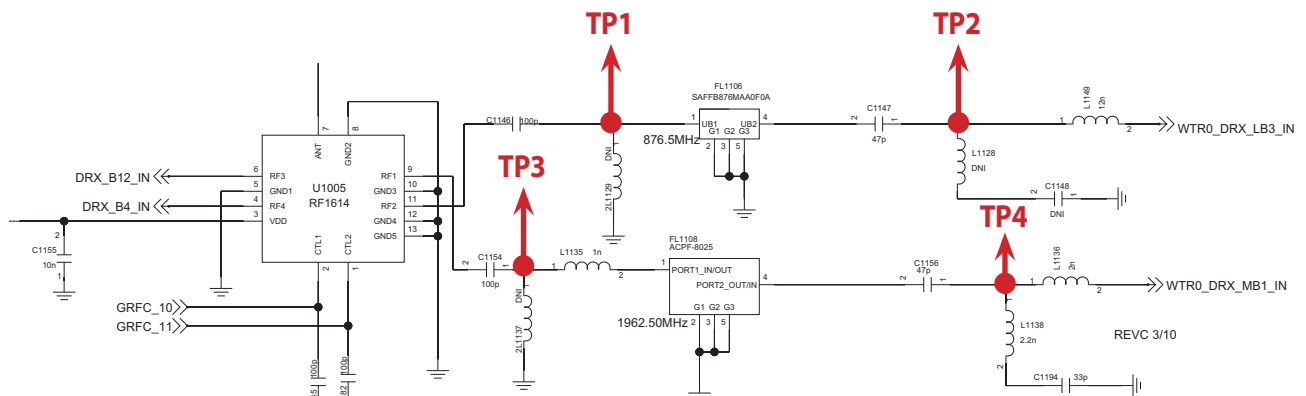
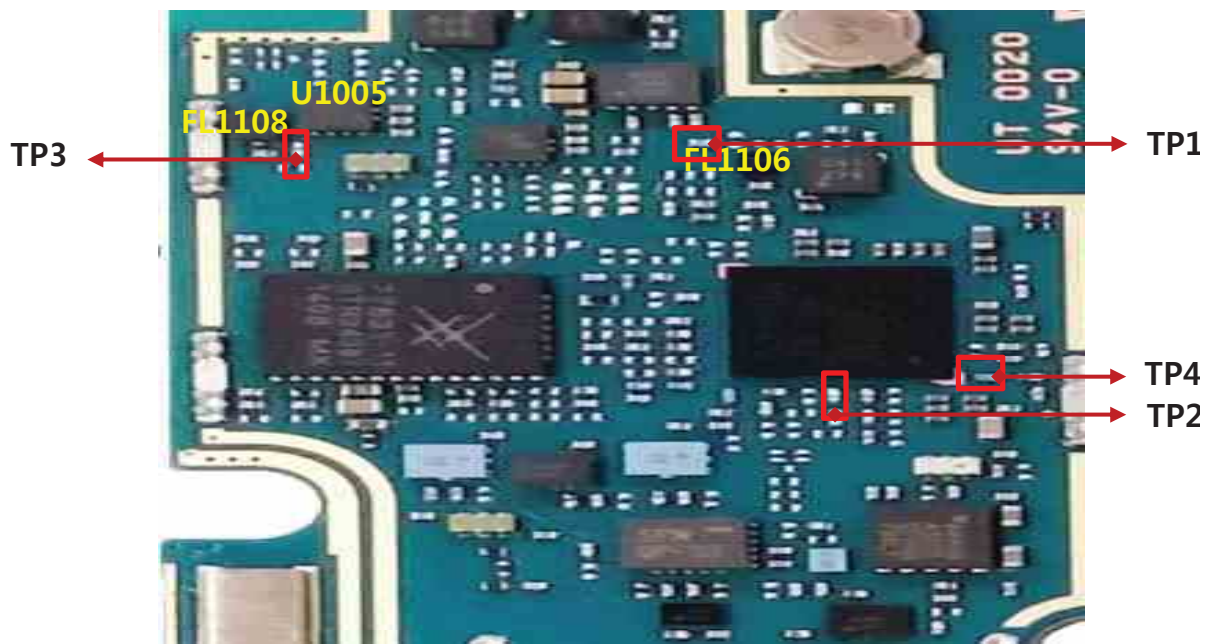


3.7.2.4 Checking RF LTE SP4T switch (B25/BC1_B26/BC10)

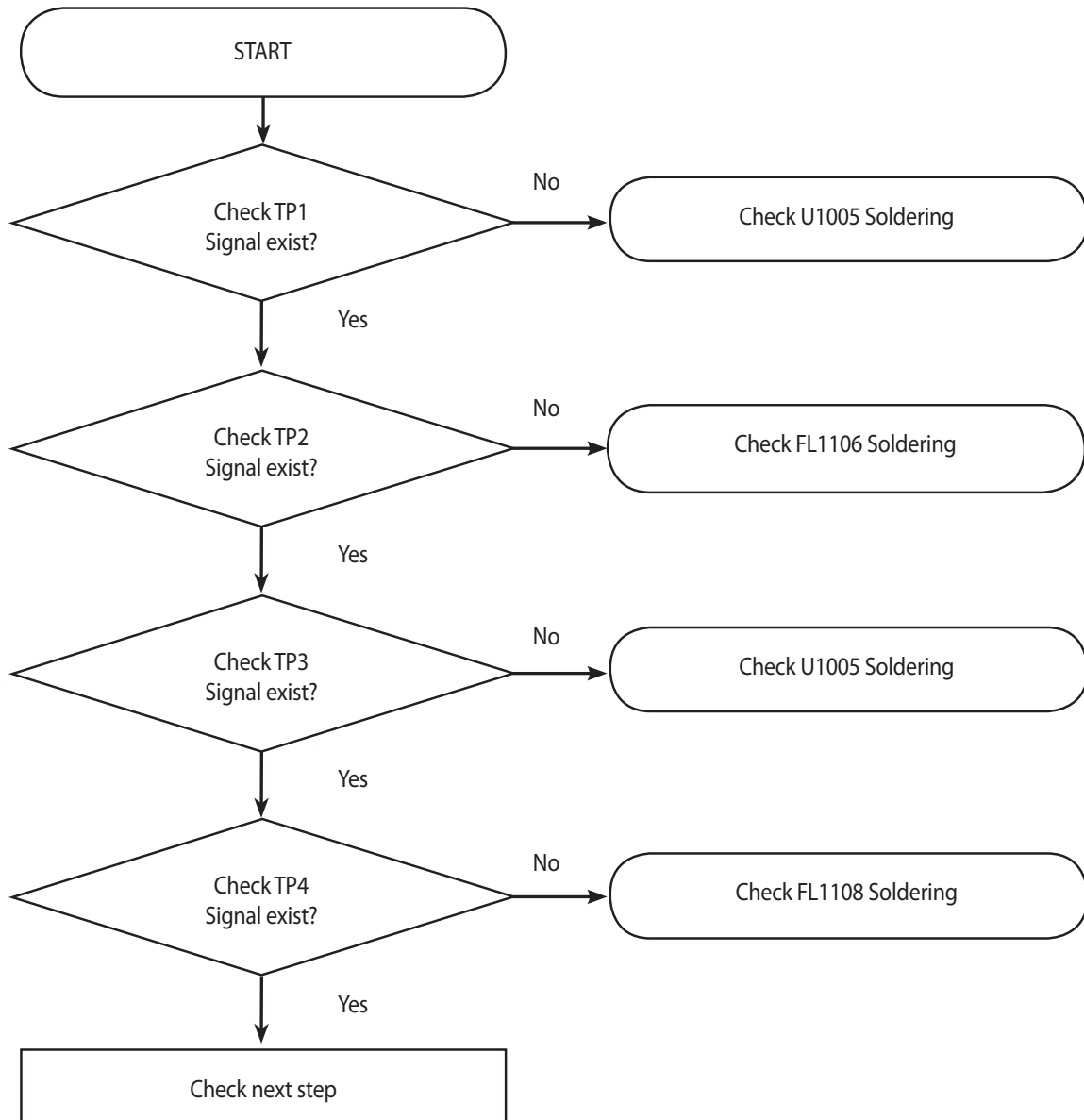
Refer to 3.4.3

3.7.2.5 Checking Signal Path (B25/BC1_B26/BC10/BC0)

Test Point



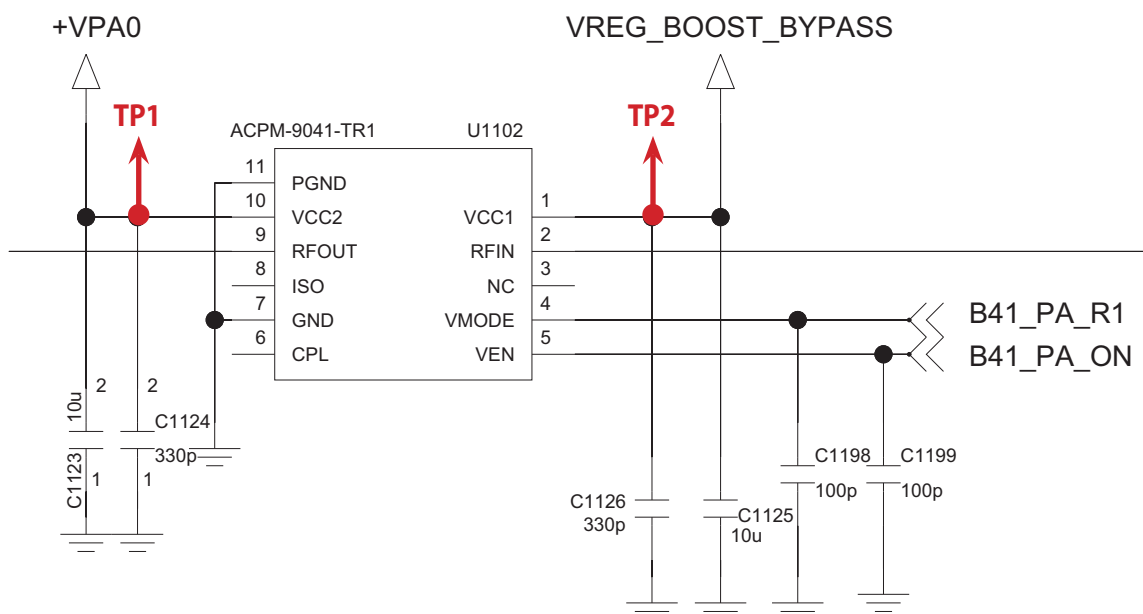
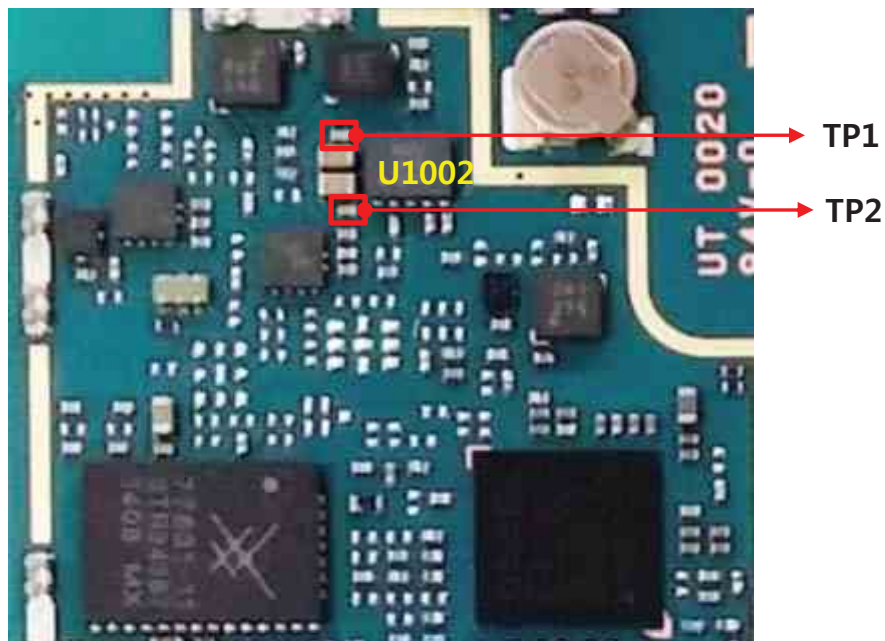
3. TROUBLE SHOOTING

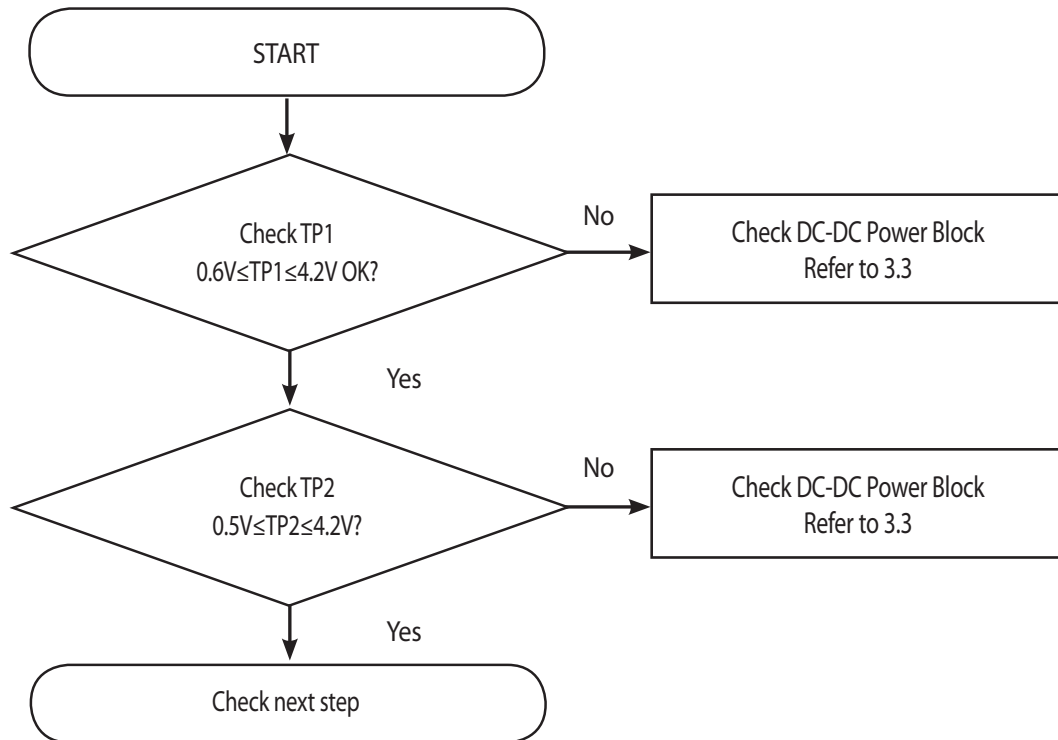


3.7.3 LTE B25/B26/B41 TX

3.7.3.1 Checking LTE PAM DC Power Circuit (B41)

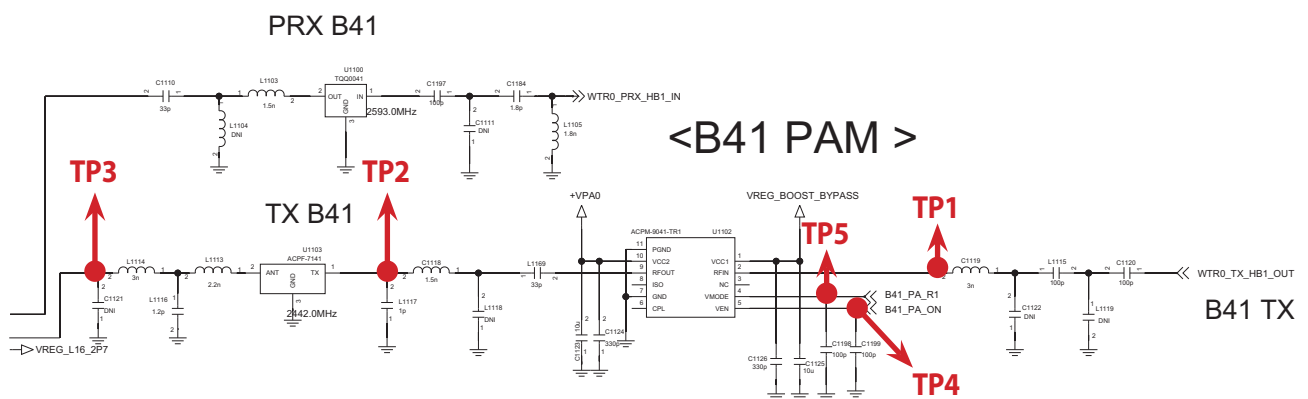
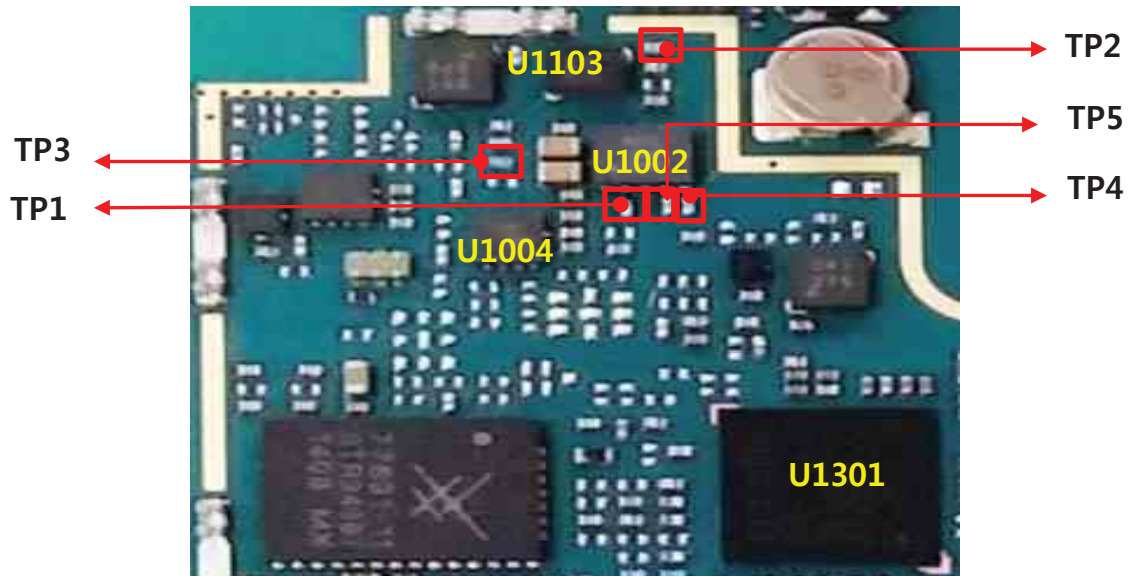
Test Point



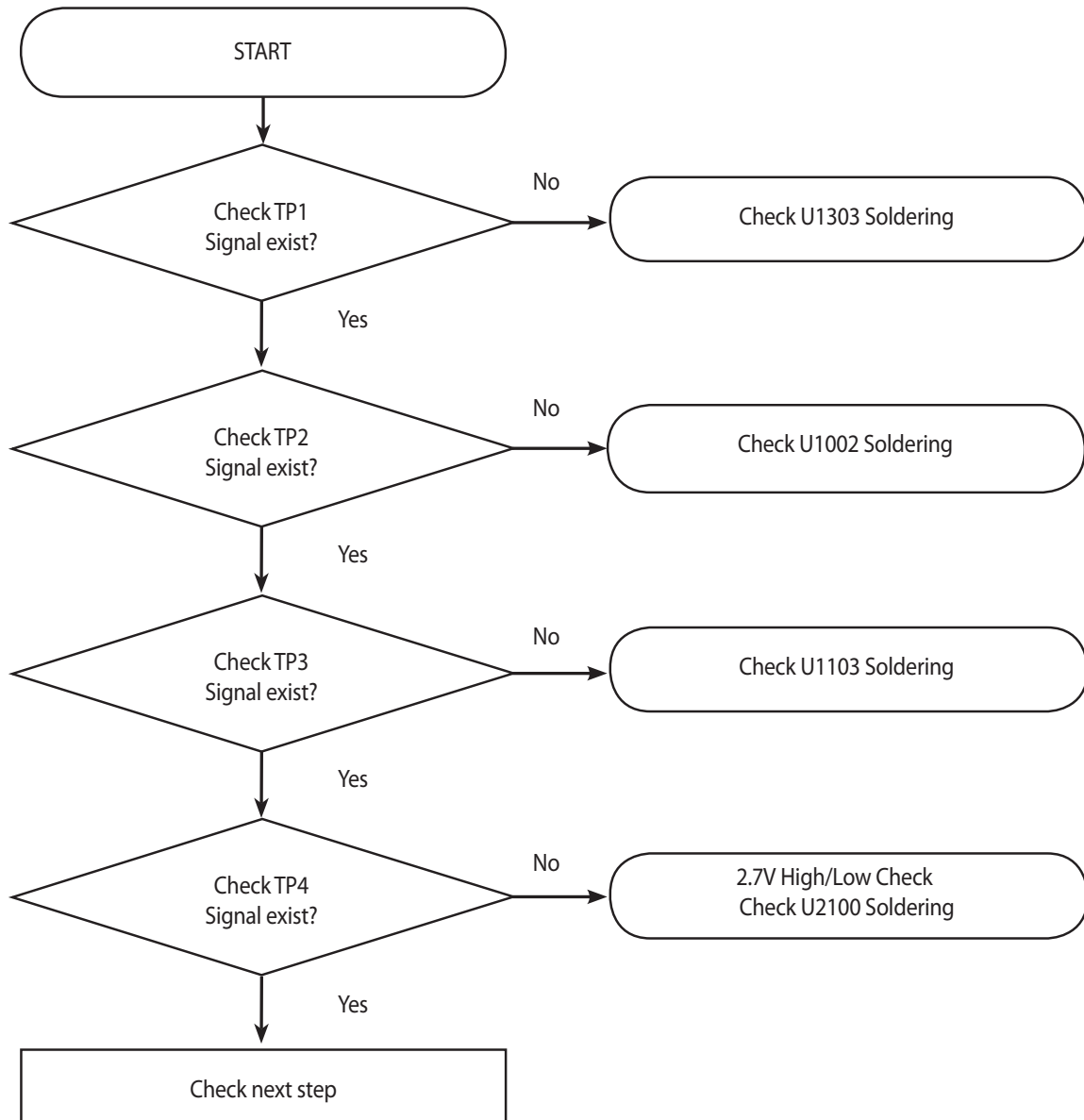


3.7.3.2 Checking RF signal path(B41)

Test Point

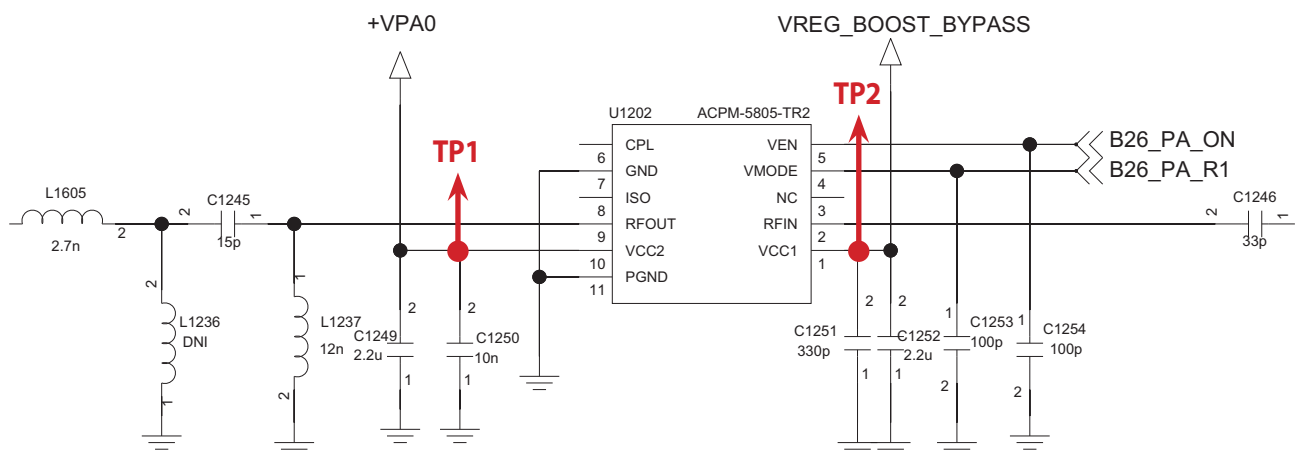
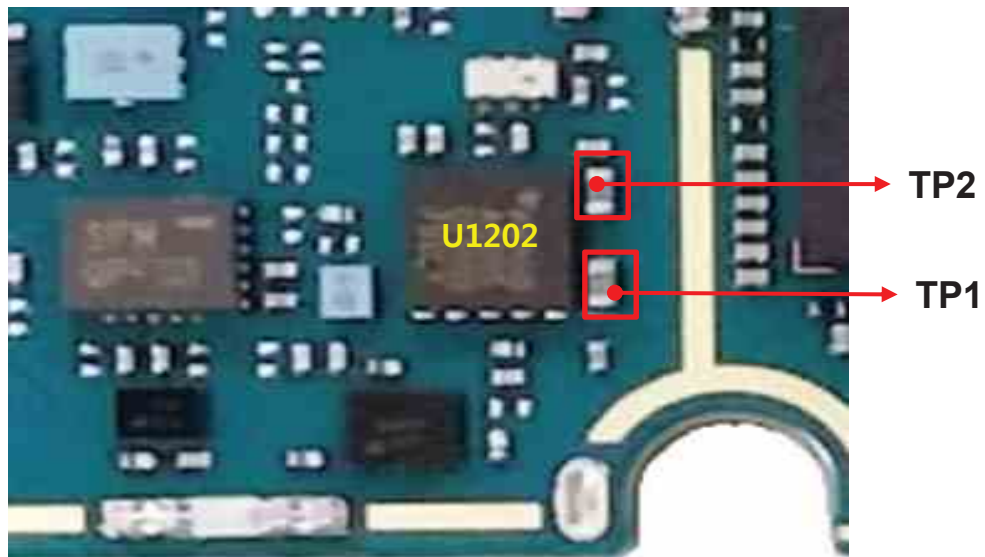


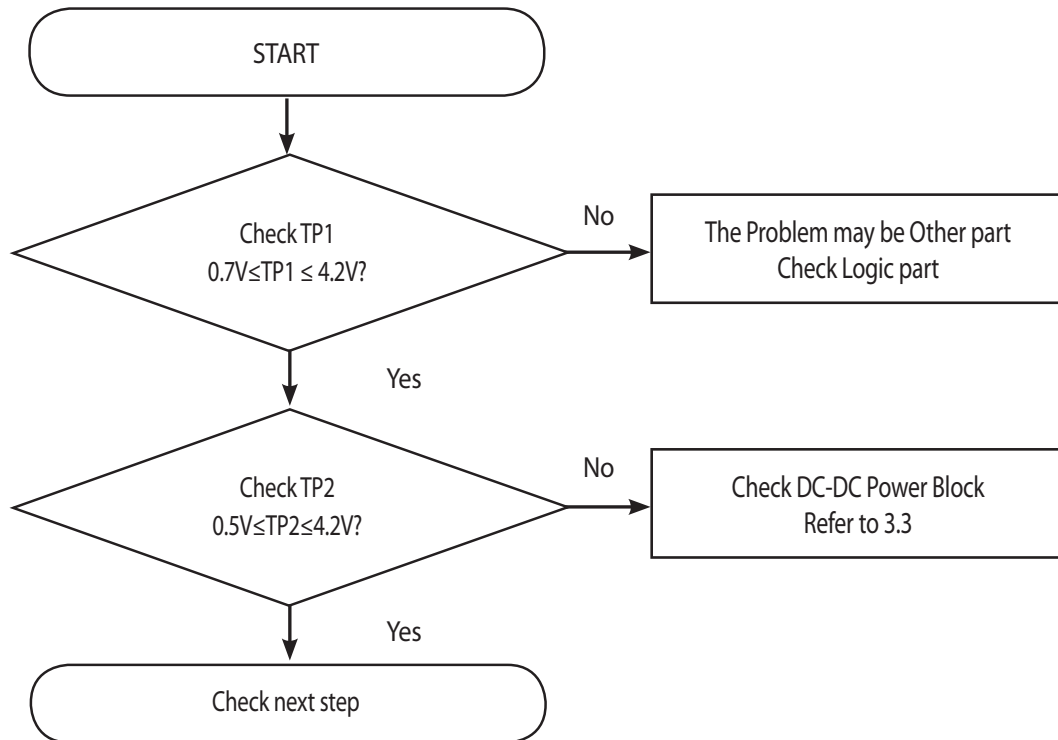
3. TROUBLE SHOOTING



3.7.3.3 Checking LTE PAM DC Power Circuit (B26/BC10)

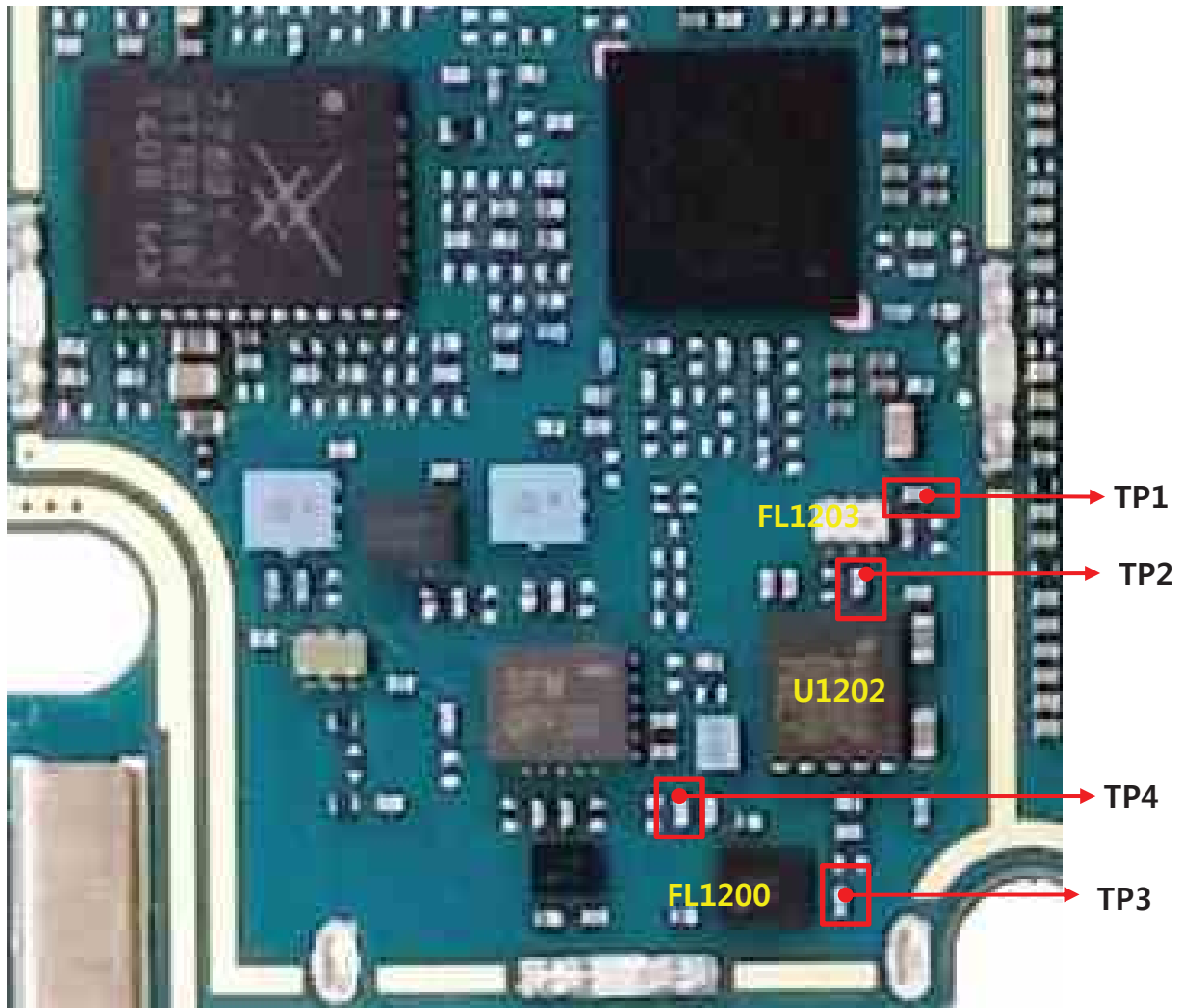
Test Point

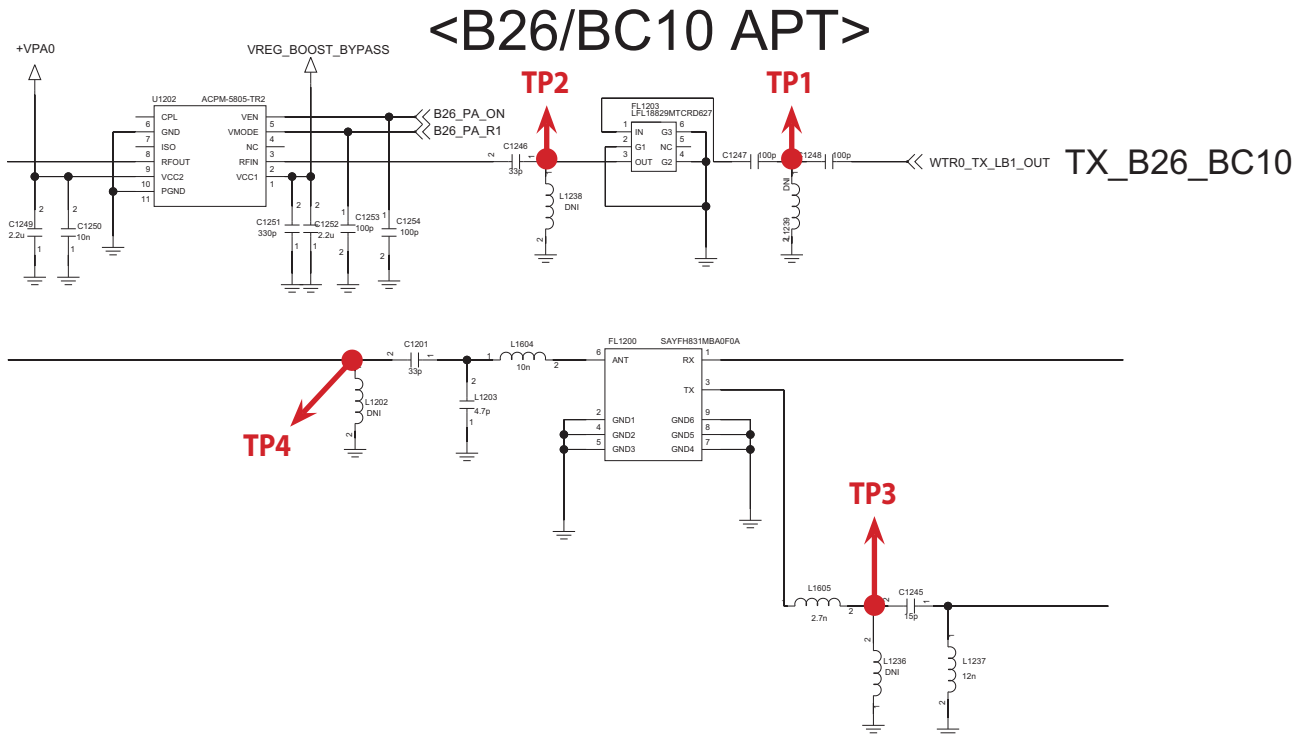


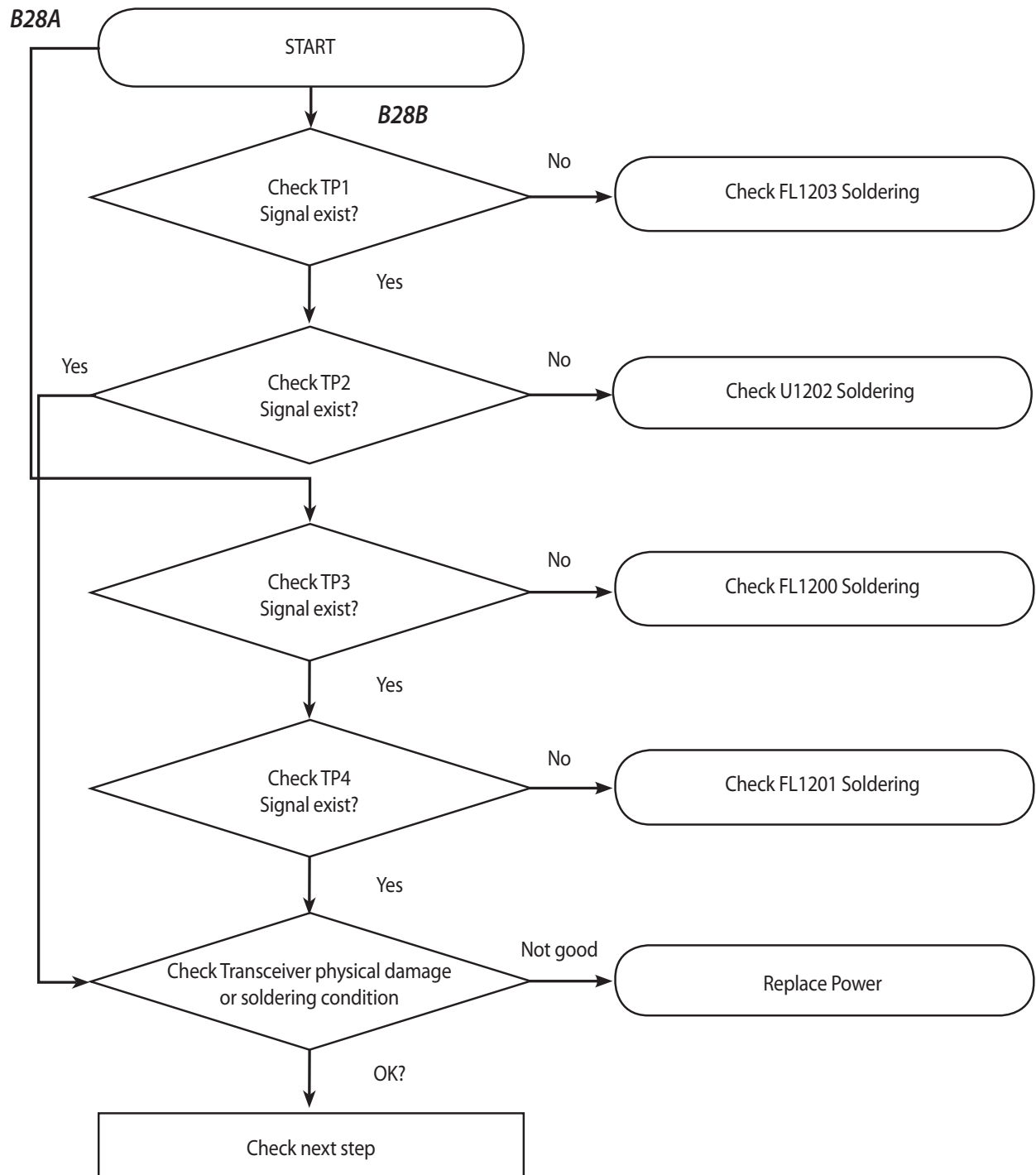


3.7.3.4 Checking RF signal path(B26_BC10)

Test Point







3.7.3.5 Checking LTE PAM DC Power Circuit (B25/BC1)

3.7.3.6 Checking RF signal path (B25/BC1)

Refer to 3.6.1.1 Checking RF Switch & Coupler

Refer to 3.6.2.1 Checking WCDMA PAM DC Power&Control Circuit

Refer to 3.6.2.4 Checking RF signal path(B2) _TX Part

3.8 CDMA RF Part

3.8.1 CDMA BC0/BC1/BC0 RX_TX_DRX

3.8.1.1 (BC1)

Refer to 3.6.1.1 Checking RF Switch & Coupler

Refer to 3.6.1.2 Checking RF signal path(B2) _RX Part

Refer to 3.6.2.1 Checking WCDMA PAM DC Power&Control Circuit

Refer to 3.6.2.4 Checking RF signal path(B2) _TX Part

Refer to 3.7.2.5 Checking Signal Path (B25/BC1_B26/BC10/BC0)_DRX

3.8.1.2 (BC0)

Refer to 3.6.1.1 Checking RF Switch & Coupler

Refer to 3.6.2.4 Checking RF signal path(B5)_RX

Refer to 3.6.2.1 Checking WCDMA PAM DC Power&Control Circuit_TX

Refer to 3.6.2.4 Checking RF signal path_TX (B5)

Refer to 3.7.2.5 Checking Signal Path (B25/BC1_B26/BC10/BC0)_DRX

Refer to 3.7.1.3 Checking RF signal path RX_(B26_BC10)

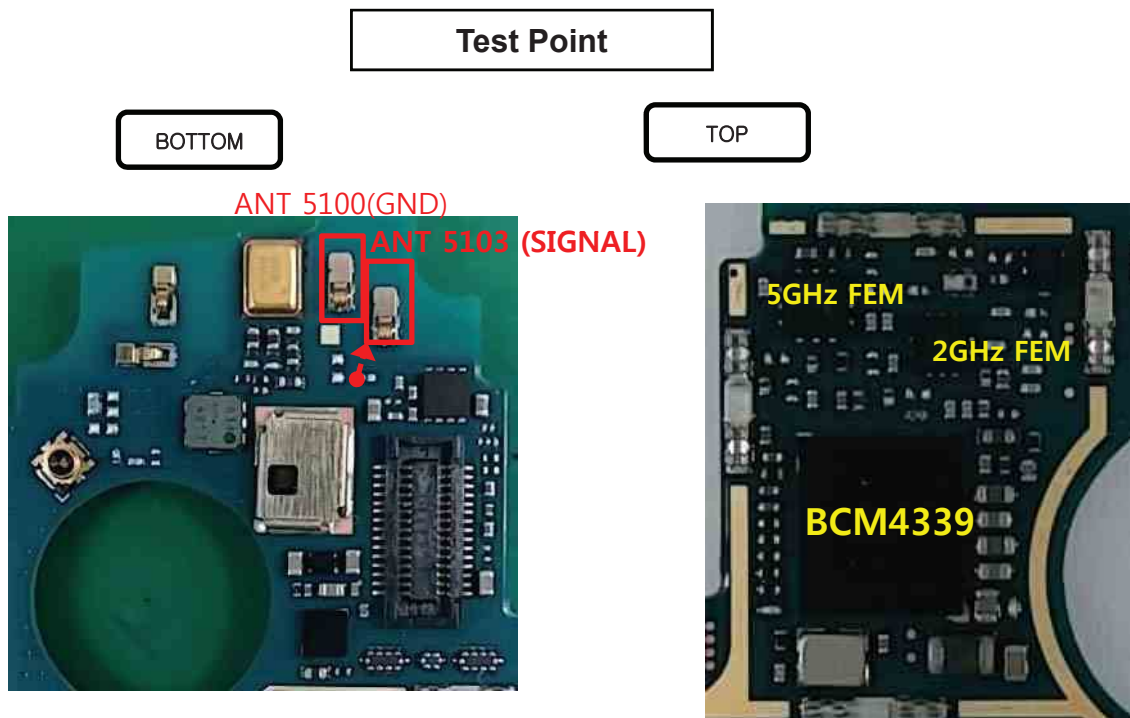
Refer to 3.7.2.5 Checking Signal Path DRX (B25/BC1_B26/BC10/BC0)

Refer to 3.7.3.3 Checking LTE PAM DC Power Circuit _TX(B26/BC10)

Refer to 3.7.3.4 Checking RF signal path_TX(B26_BC10)

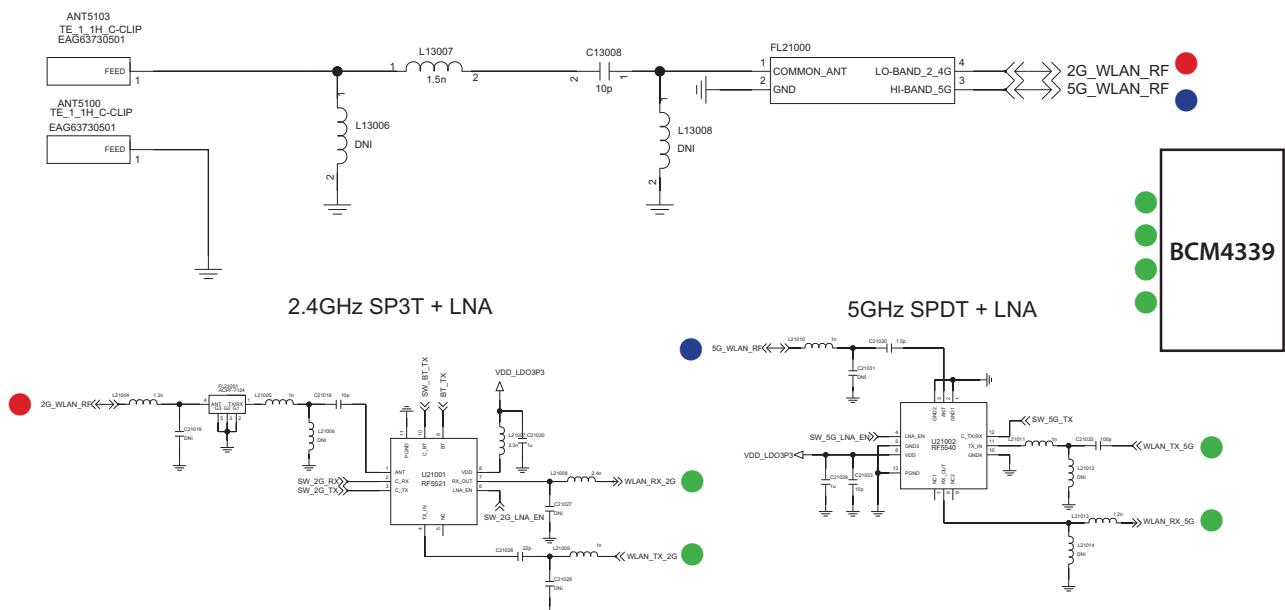
3.9 BT&WIFI&FM PART

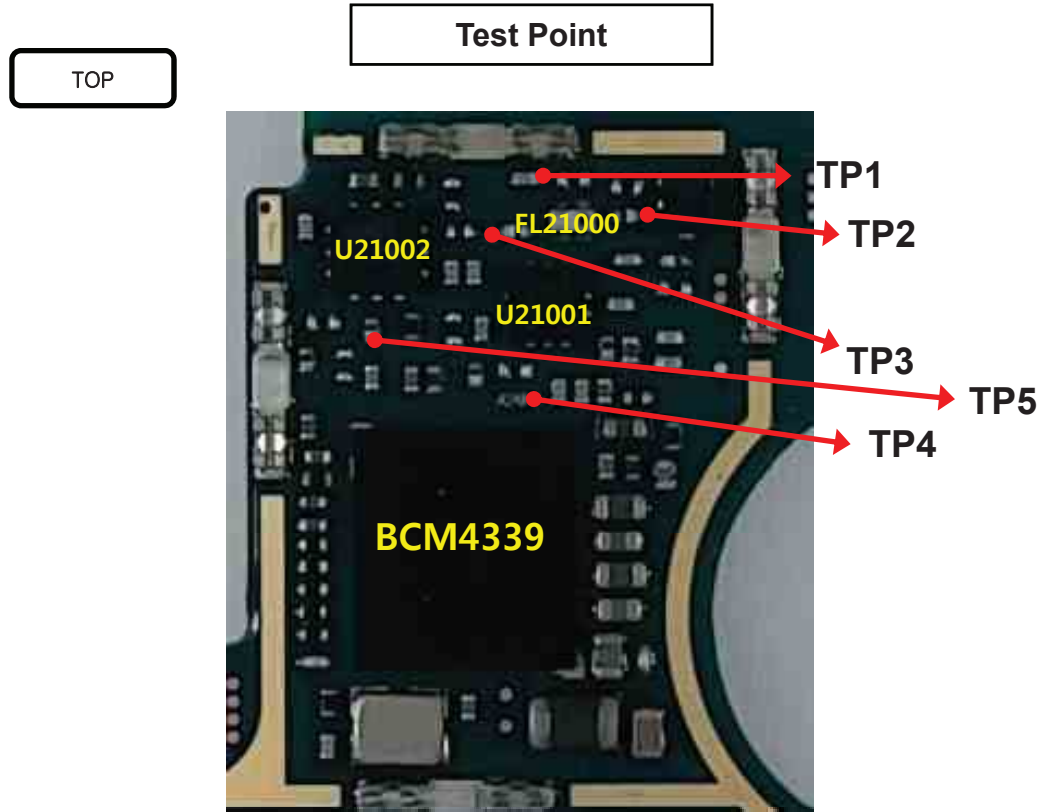
3.9.1 BT&WIFI



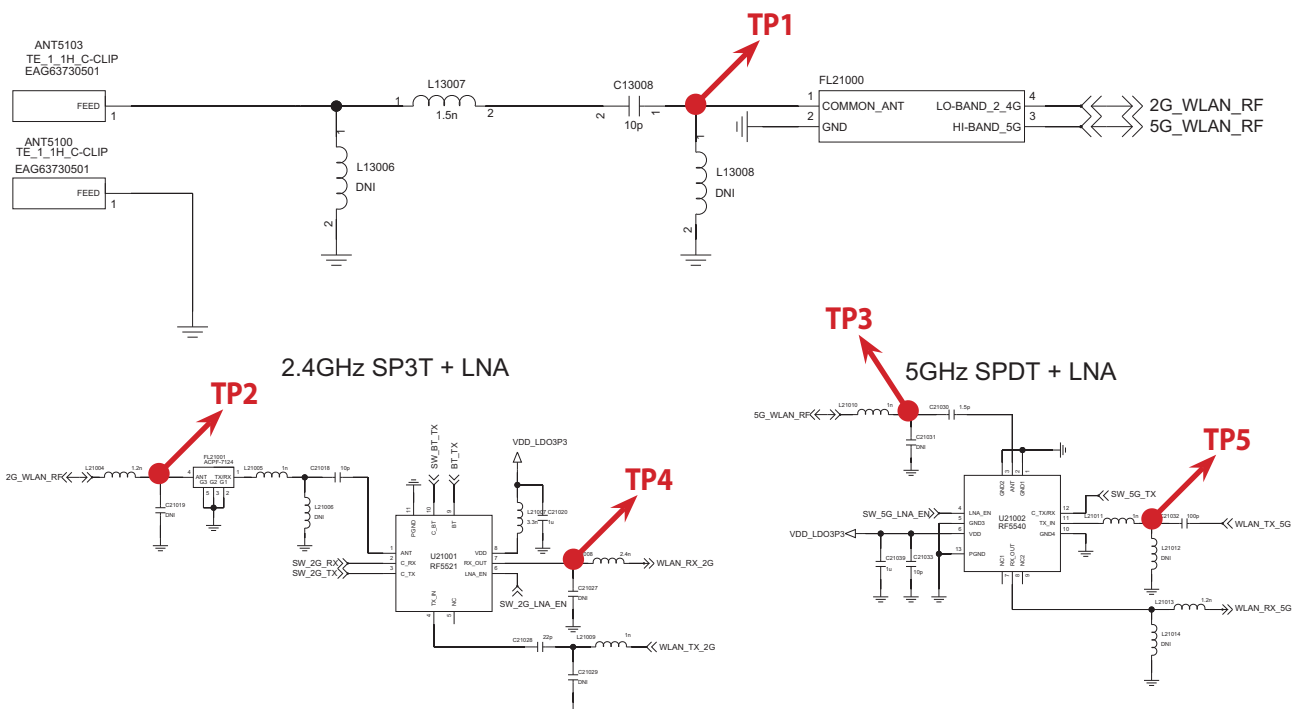
Diplexer(2.4G/5G BPF)

● ● ● : Connection line

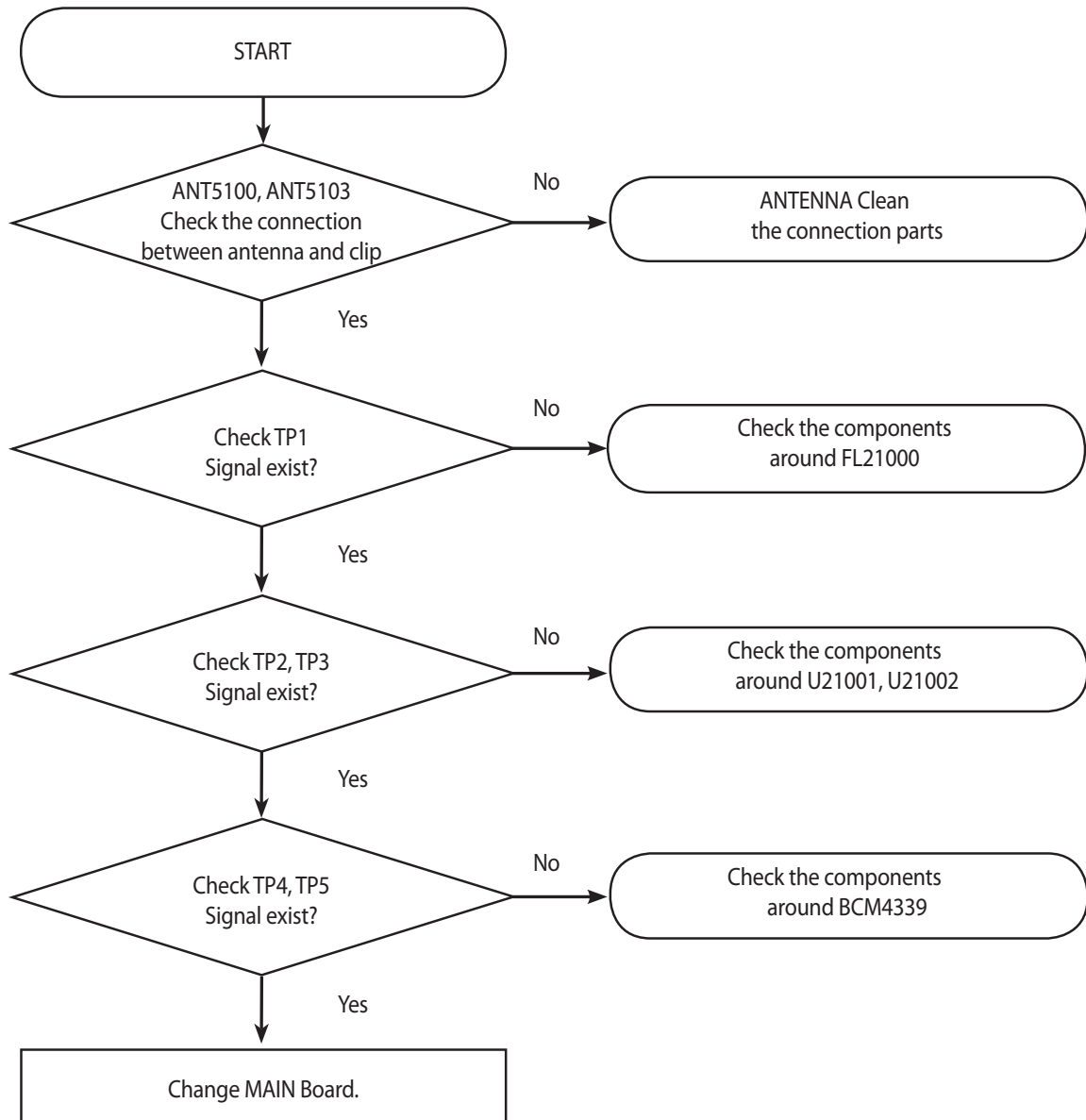




Diplexer(2.4G/5G BPF)



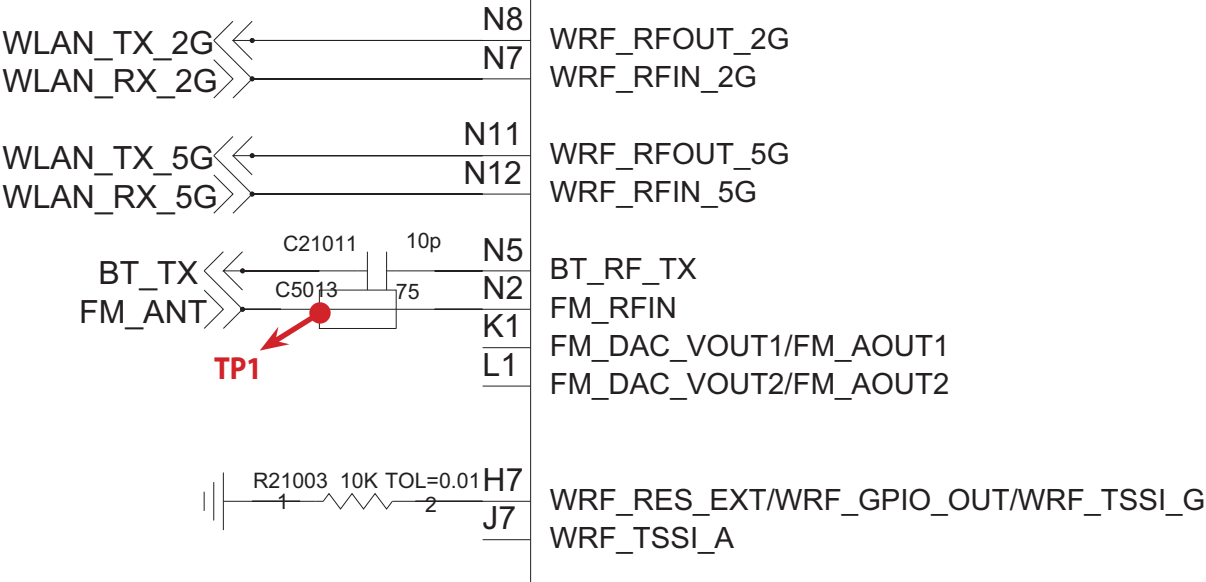
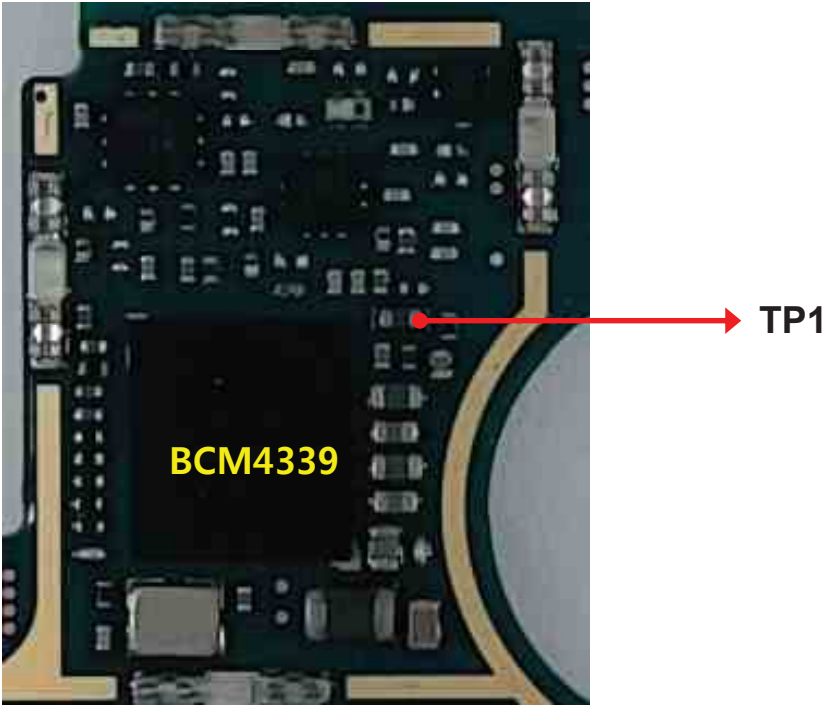
3. TROUBLE SHOOTING

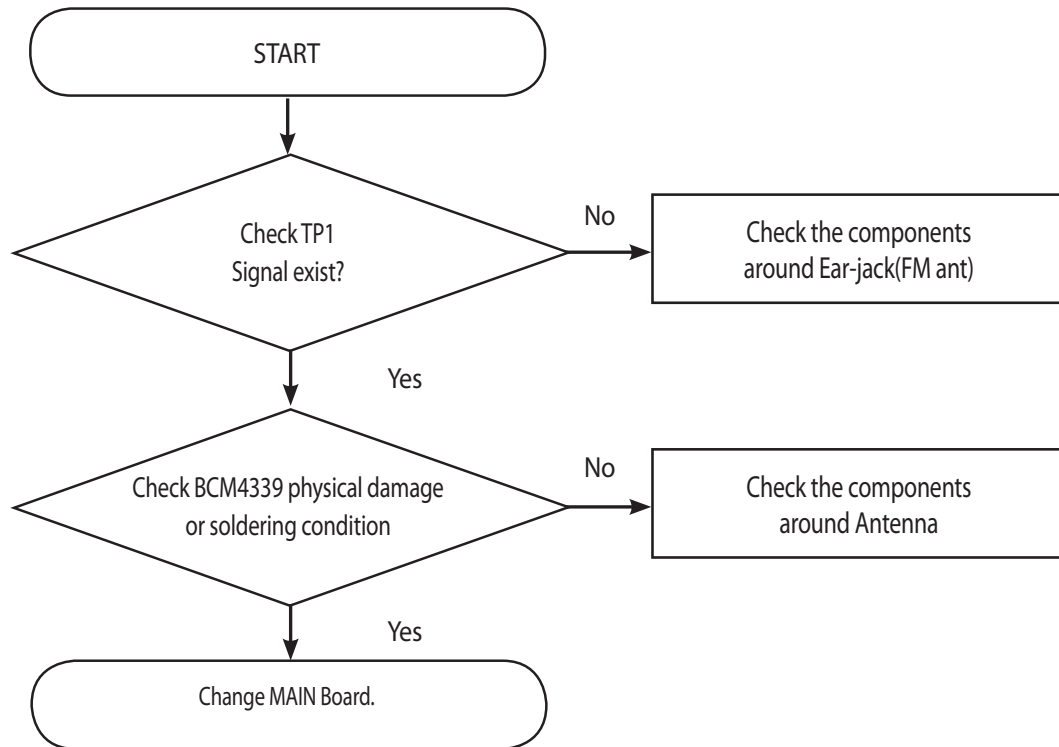


3.9.2 FM

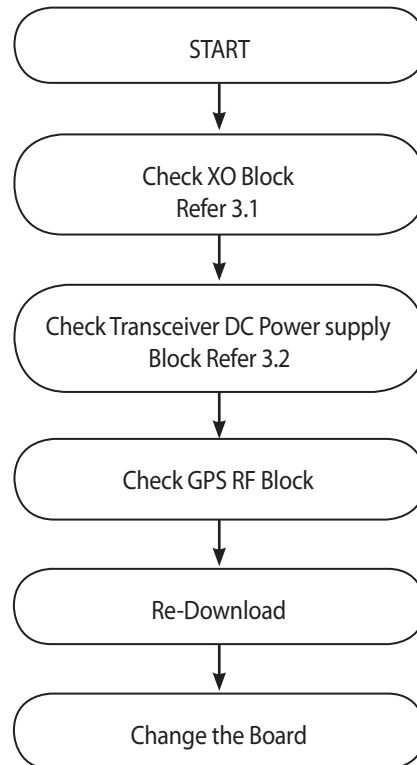
Test Point

TOP

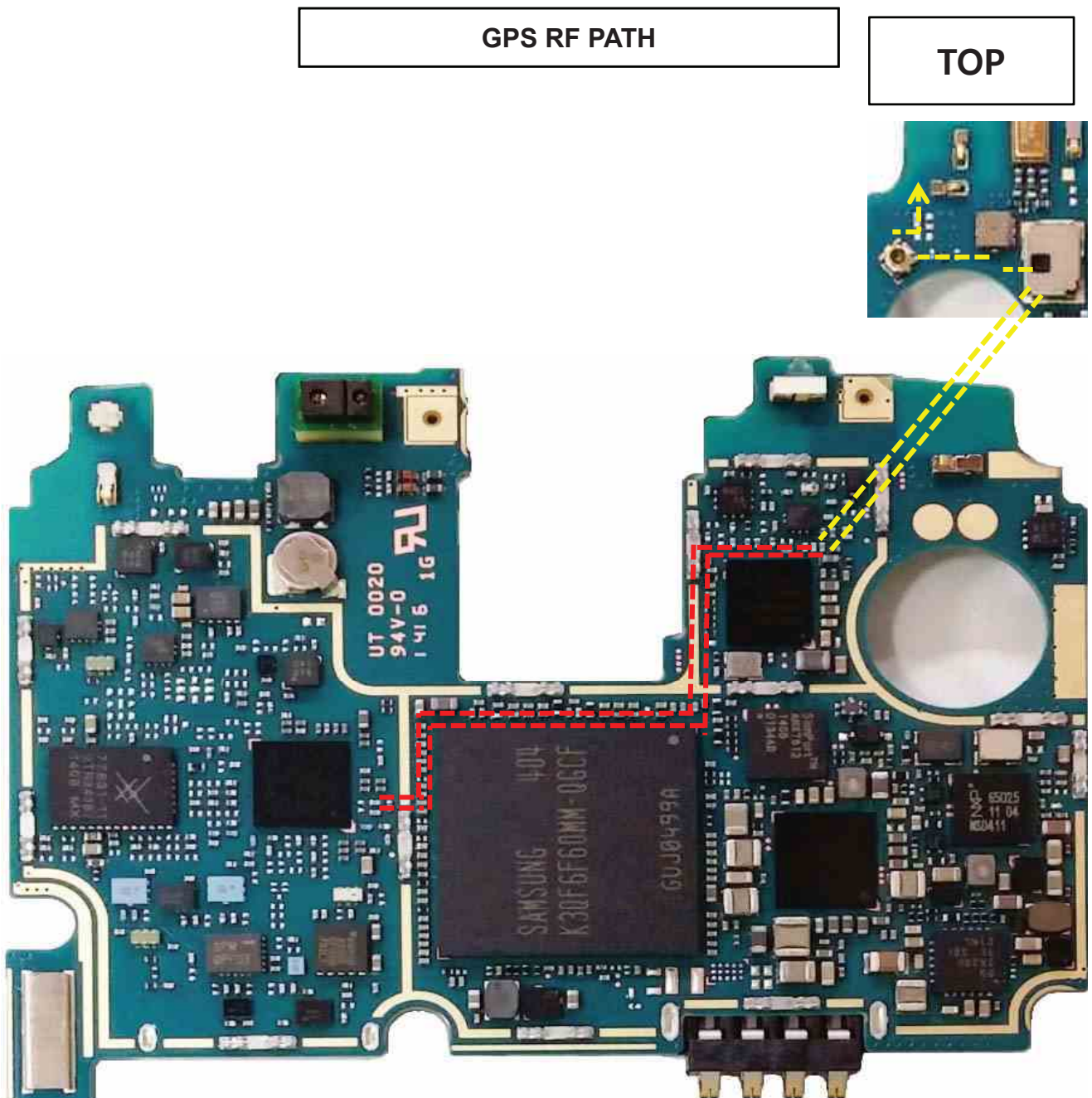




3.10 GPS Part

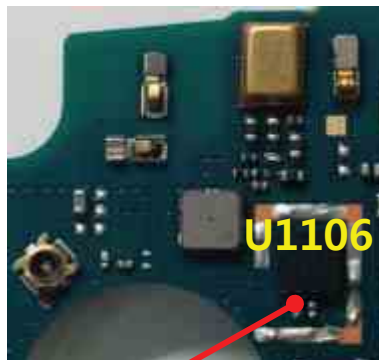


3. TROUBLE SHOOTING



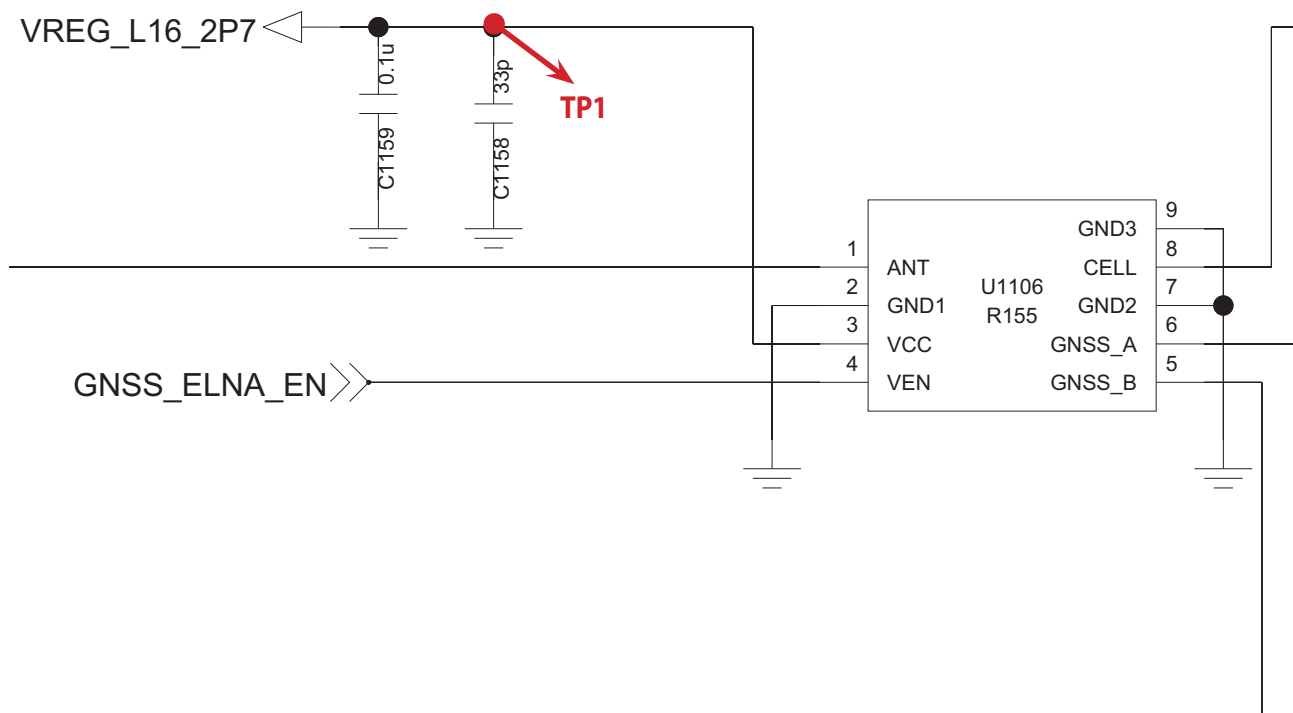
3.10.1 Checking GPS LNA DC Power Circuit

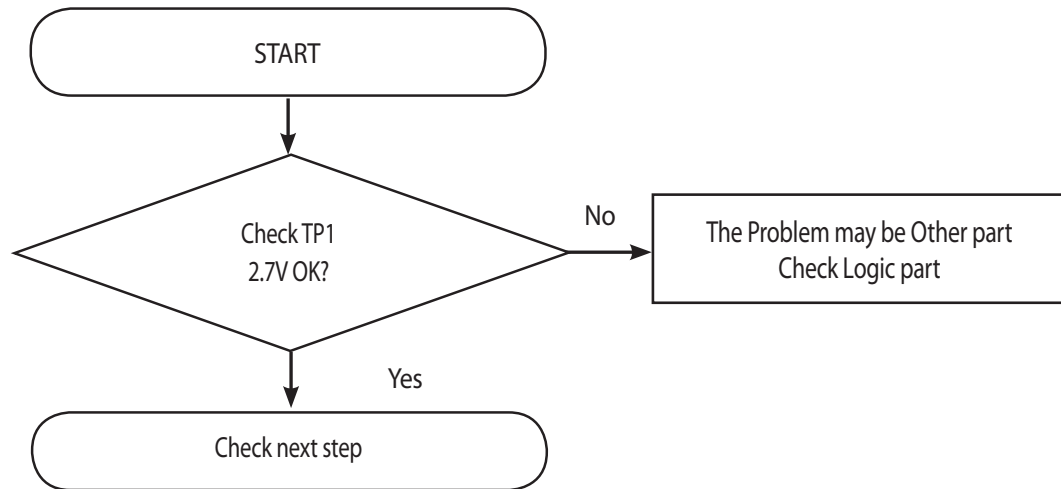
Test Point



TP1

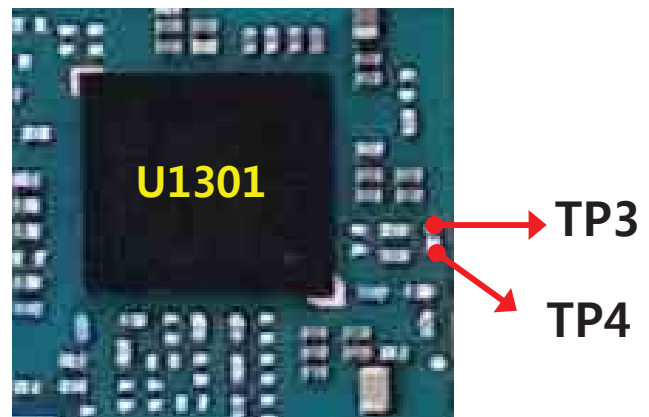
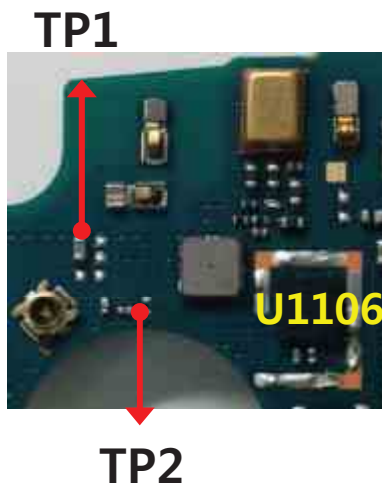
Bottom



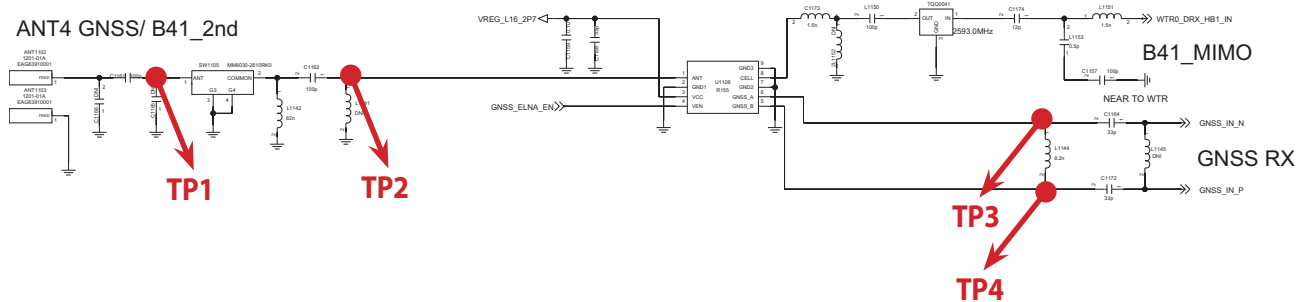


3.10.2 Checking GPS RF Signal path

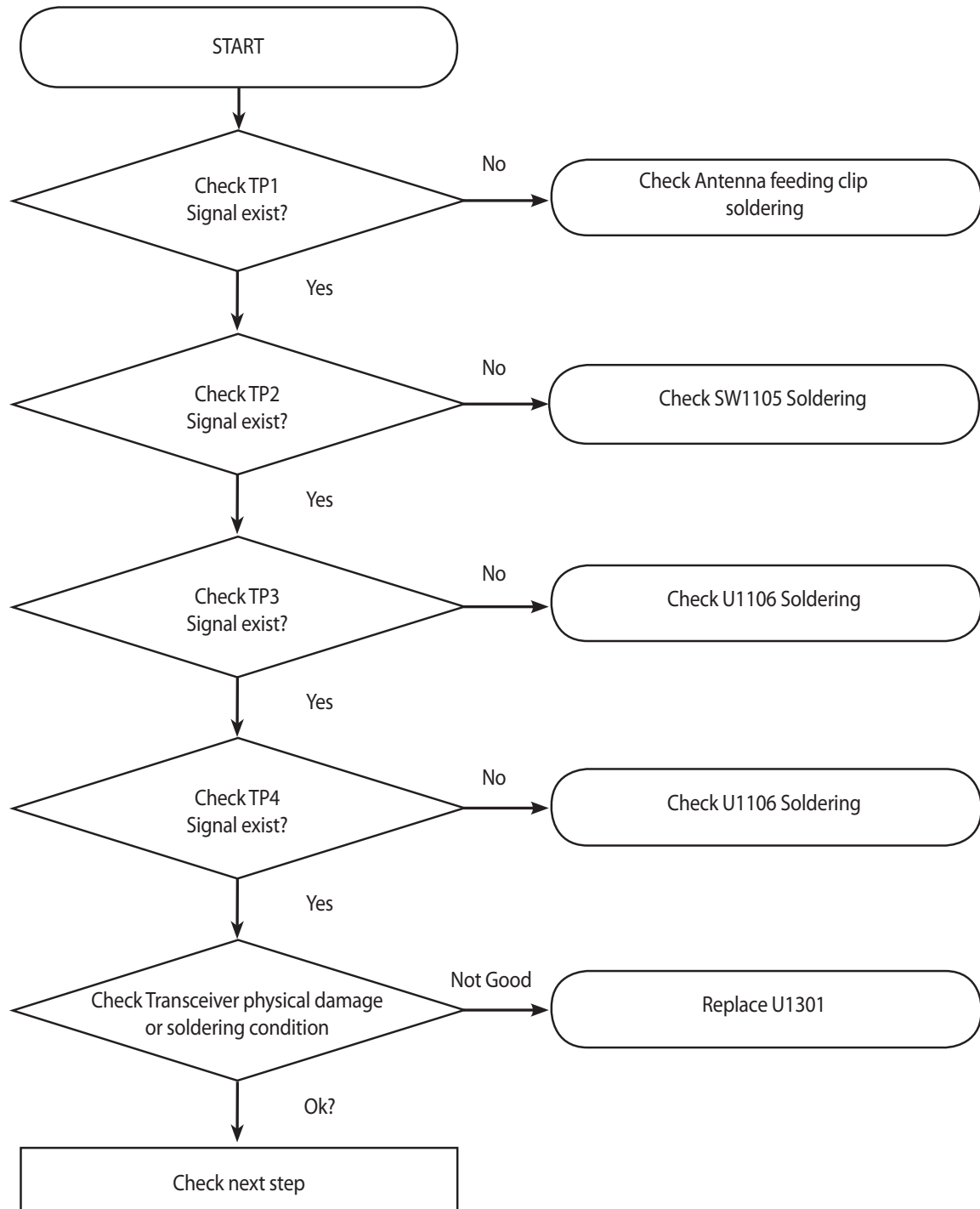
Test Point



Bottom

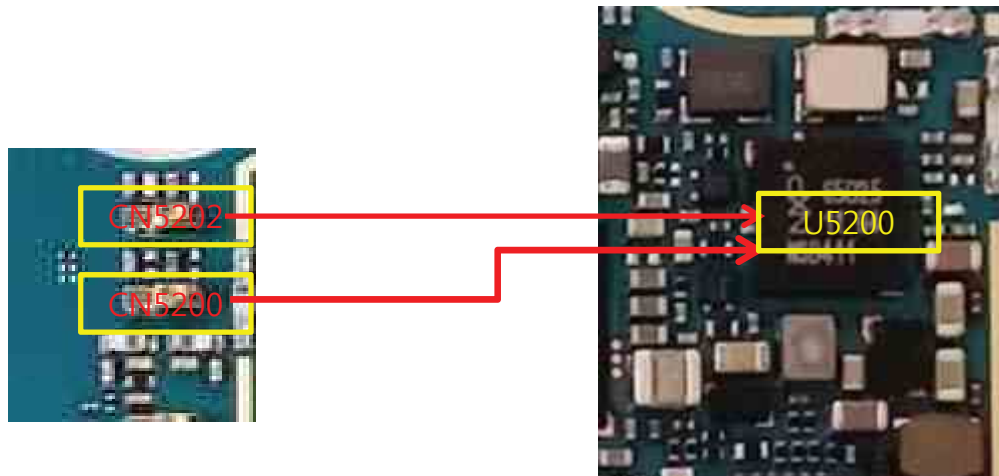


3. TROUBLE SHOOTING



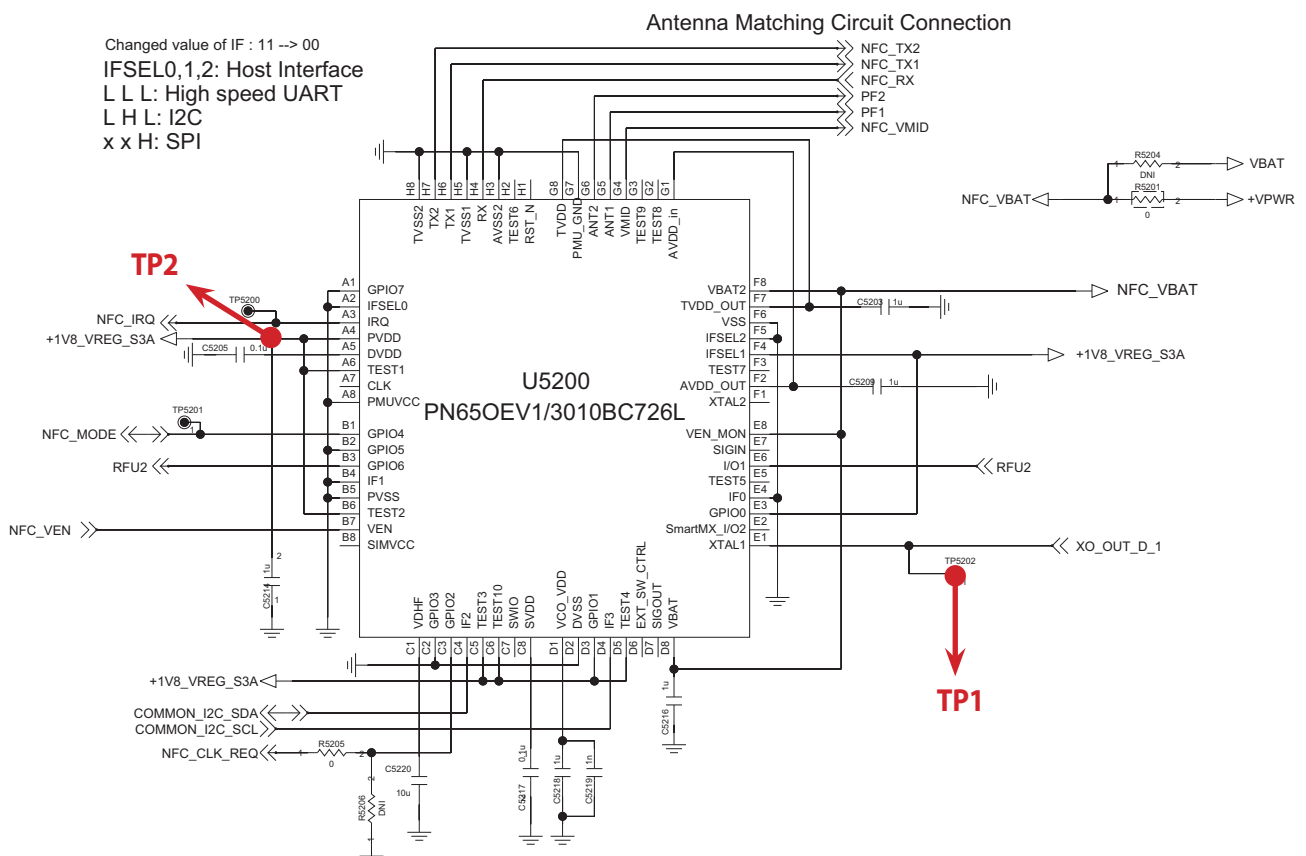
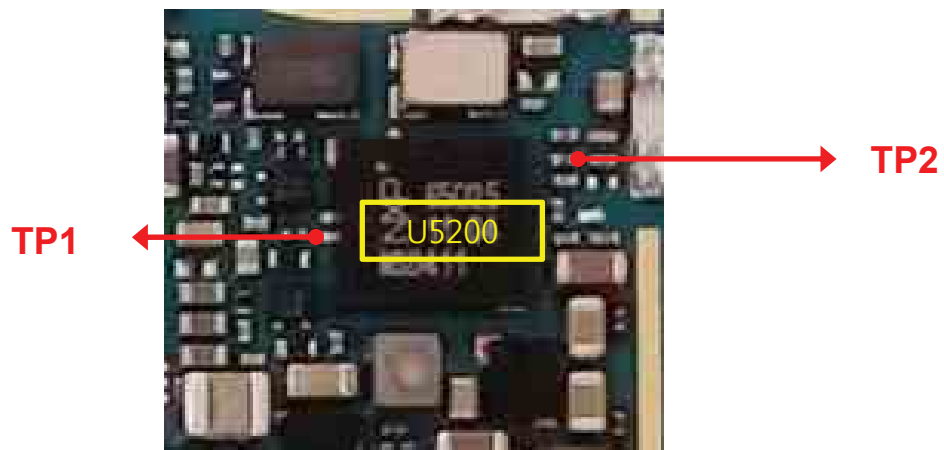
3.11 NFC Part

NFC RF PATH

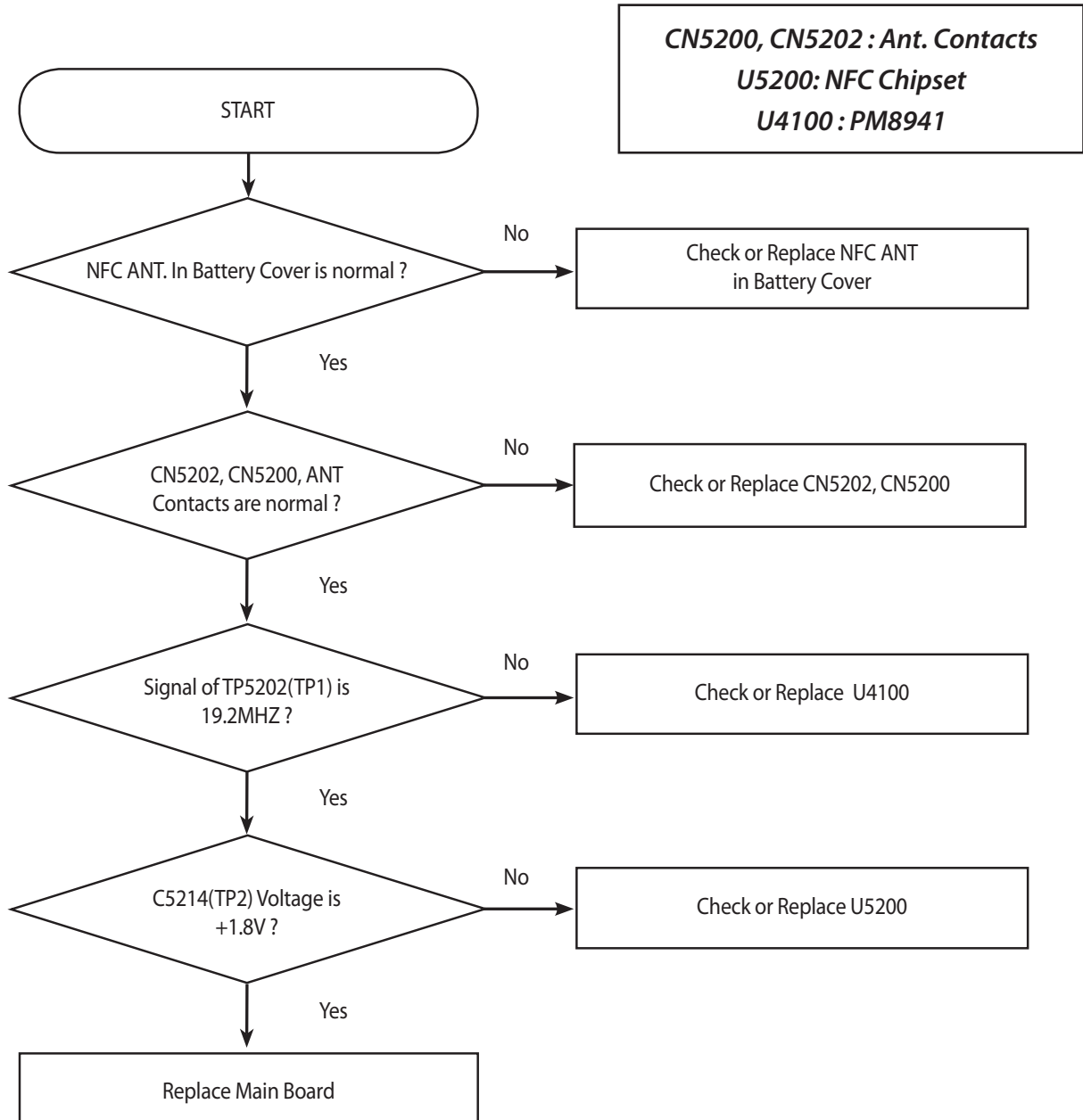


3. TROUBLE SHOOTING

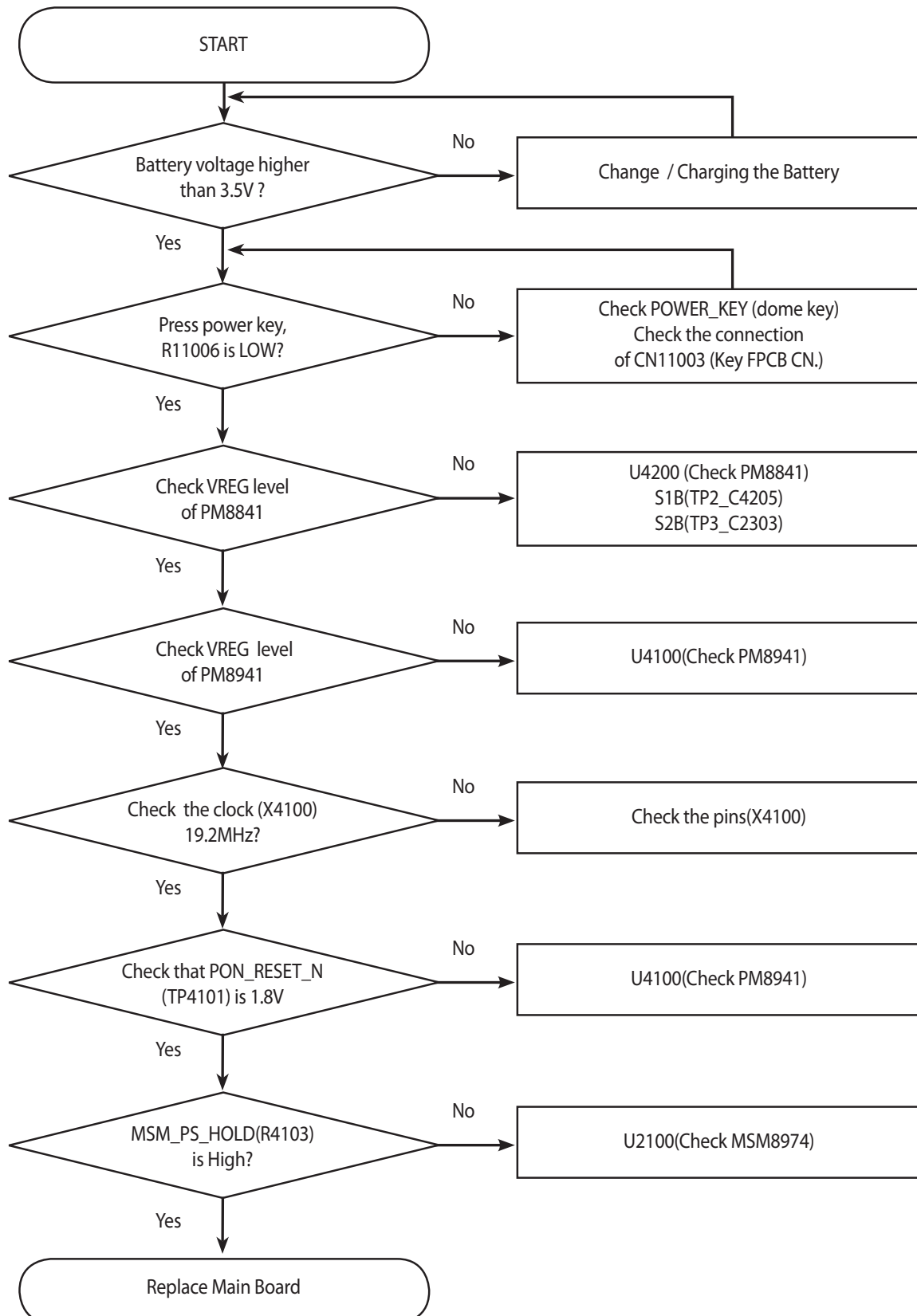
Test Point



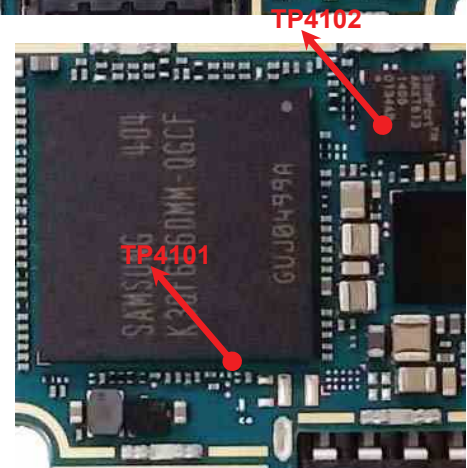
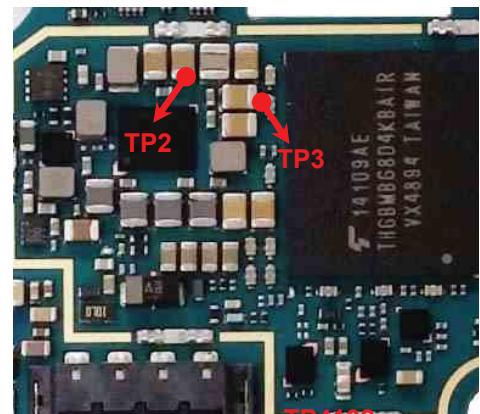
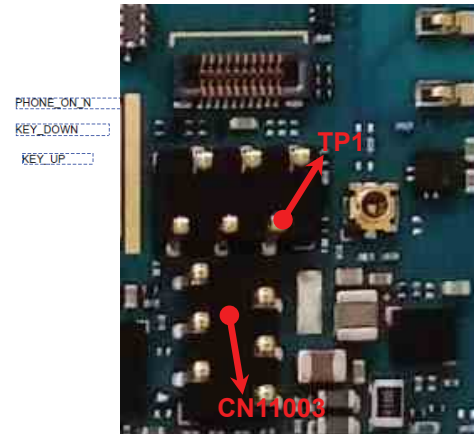
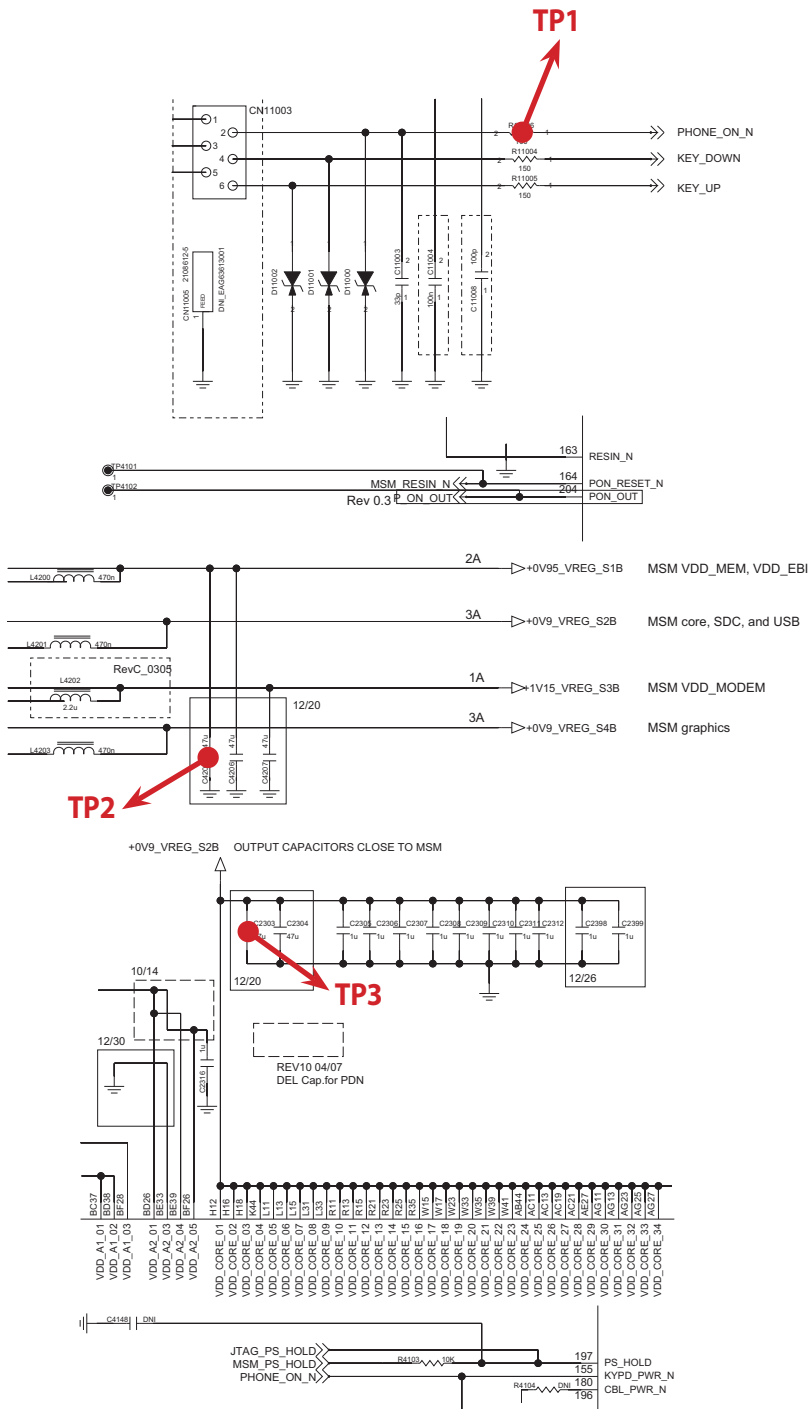
3. TROUBLE SHOOTING



3.12 Power & Charger

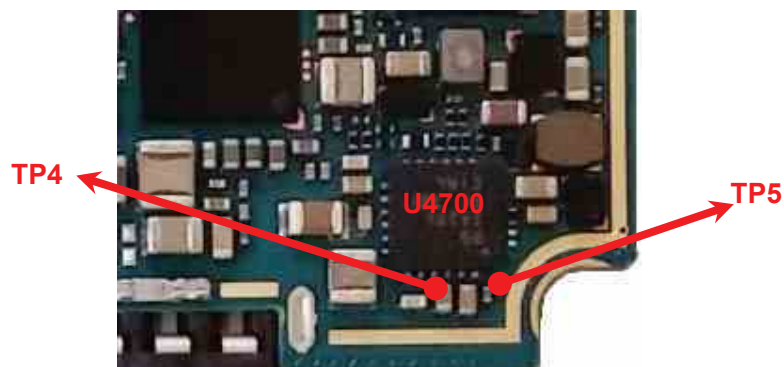
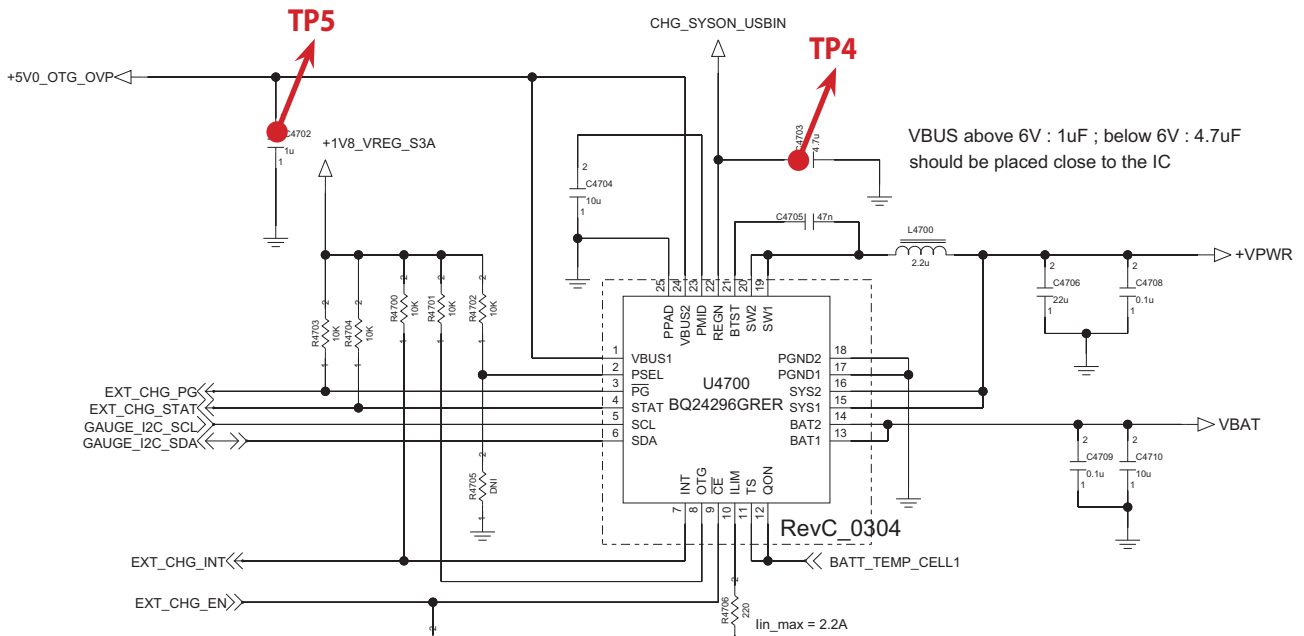


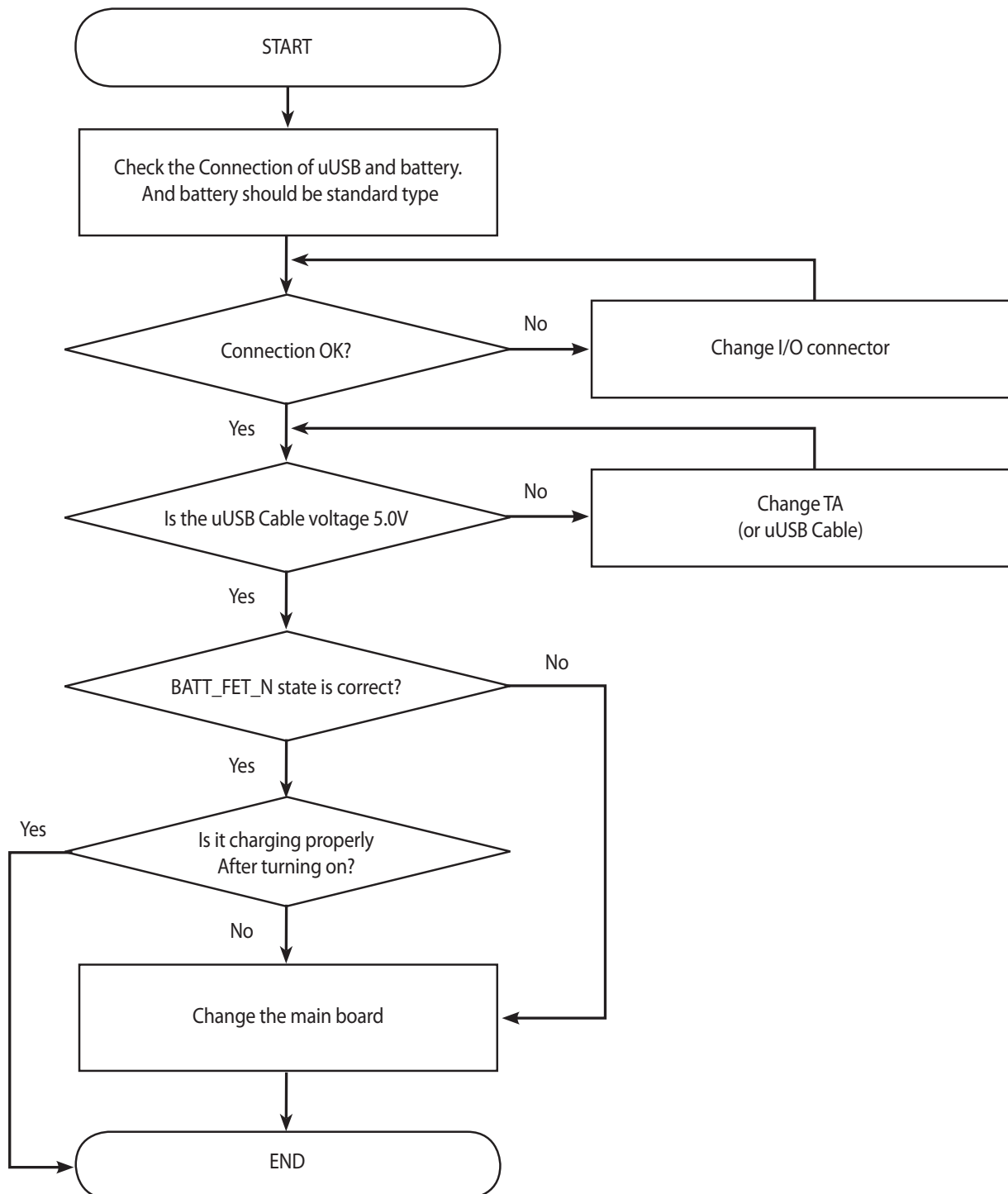
3. TROUBLE SHOOTING



3. TROUBLE SHOOTING

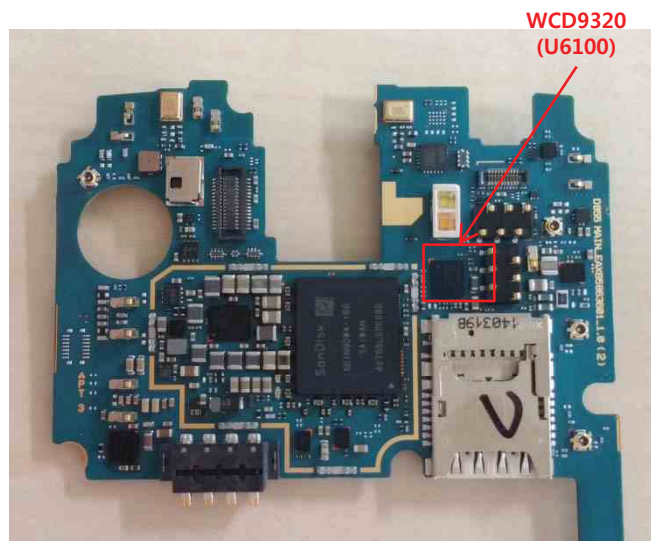
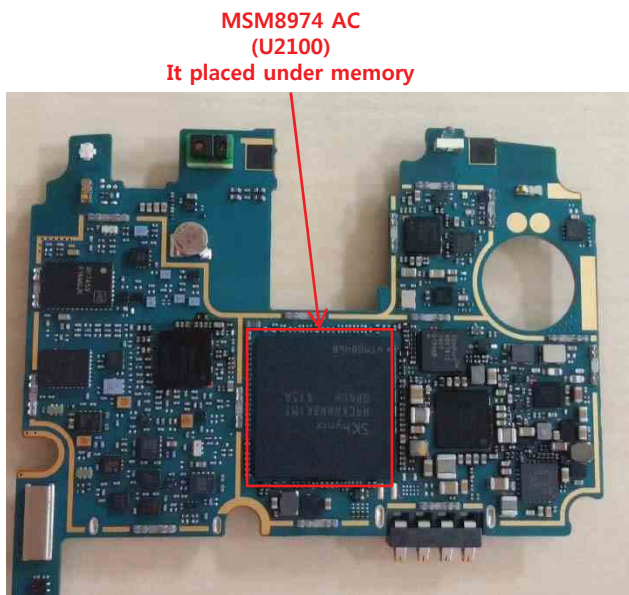
- LS990 uses BQ24296 (External charger IC) for charging control.
- Charging Procedure
 - Supply the VBUS_USB_IN_PM through a TA or u-USB cable
 - BQ24296 controls the charging current
 - Charging current flows into the main battery
- Troubleshooting Check Point
 - Connection of the TA or u-USB cable (OVP U4750)
 - Charging current path (Charger IC U4700)
 - Main battery
- During charging operation TP signal goes low and when stop charging goes high.



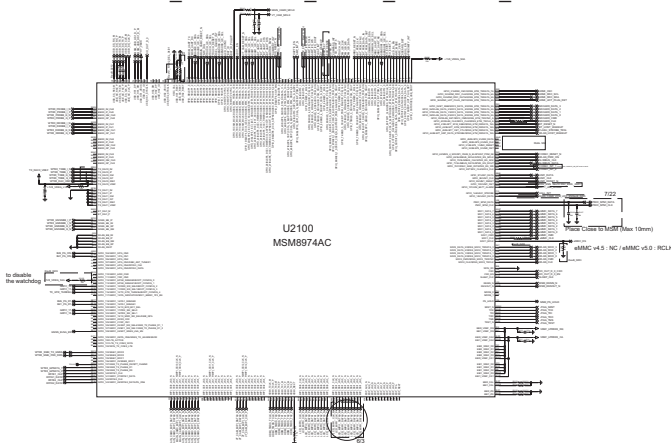


3.13 Audio

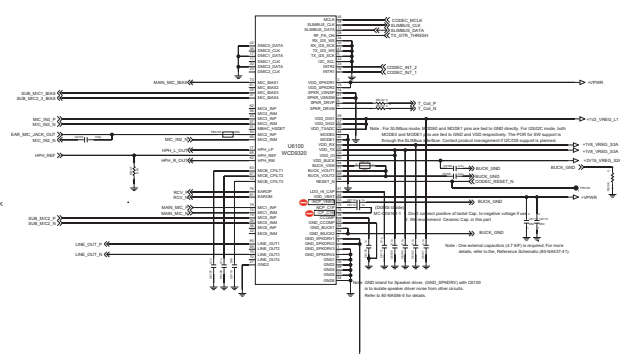
3.13.1 Overview (PCB & Circuit design)



<2-1-2-2-1_MSM8974AB_Control_GPIO>Rev_0.1

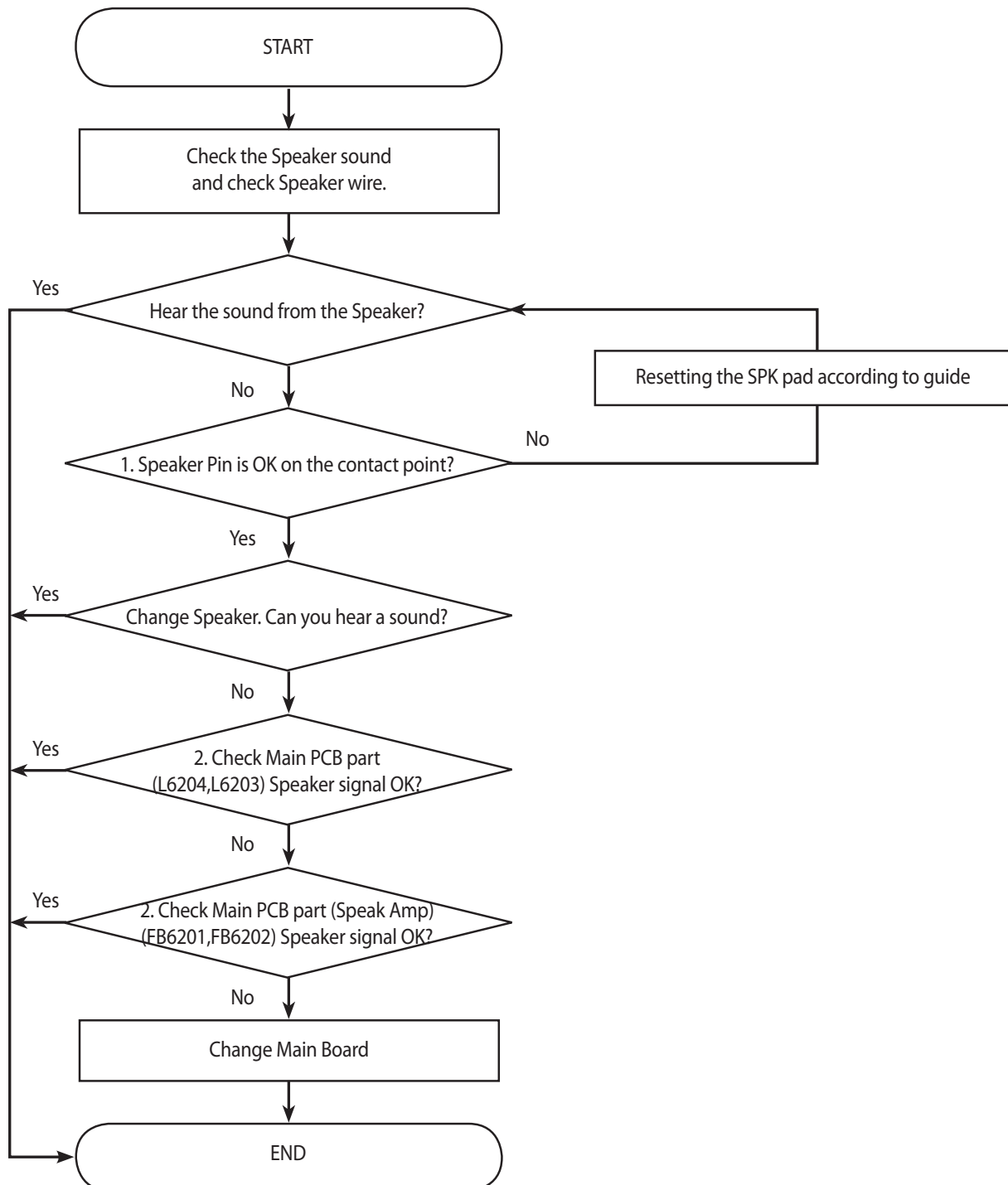


<6-1-1-5_WCD9320>REV_0.3

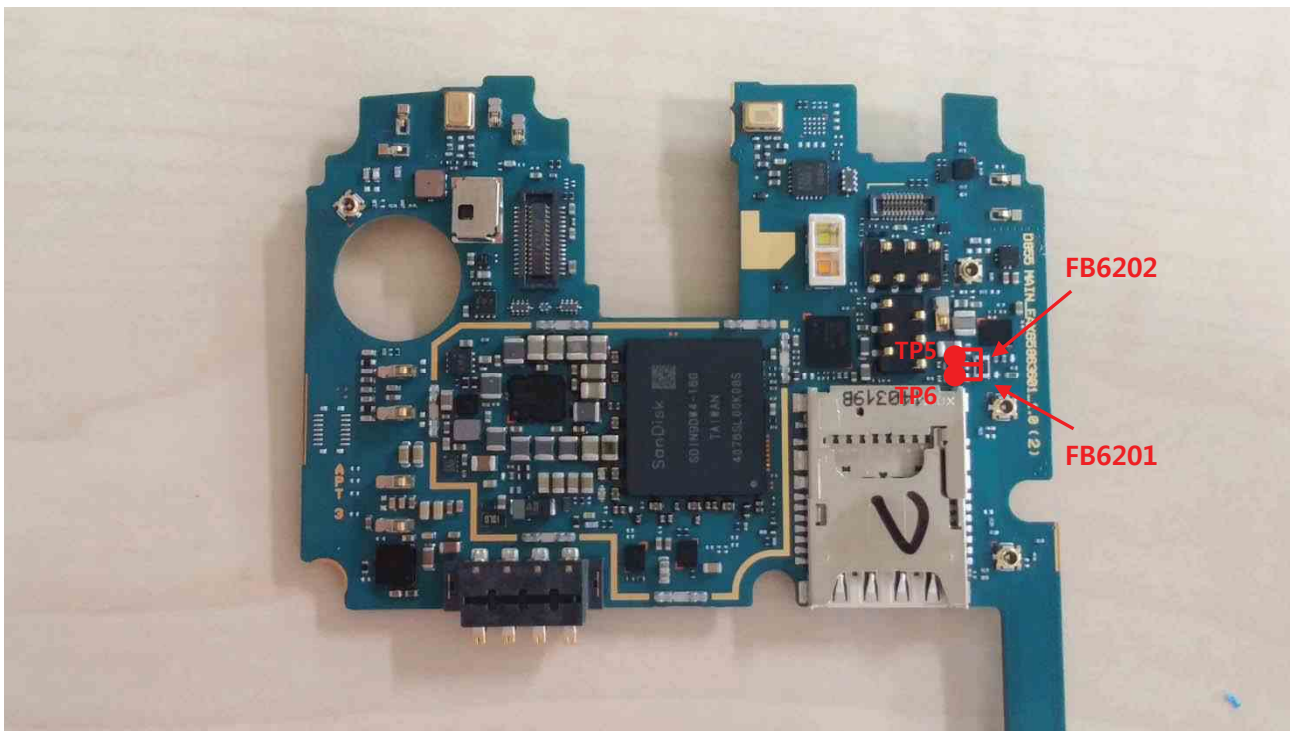
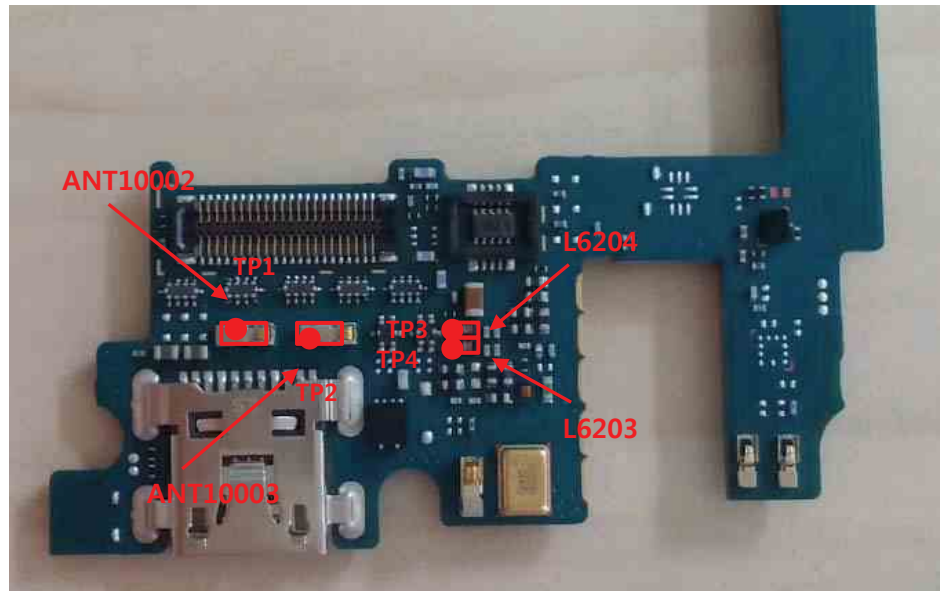


3.13.2 Speaker Trouble Shooting

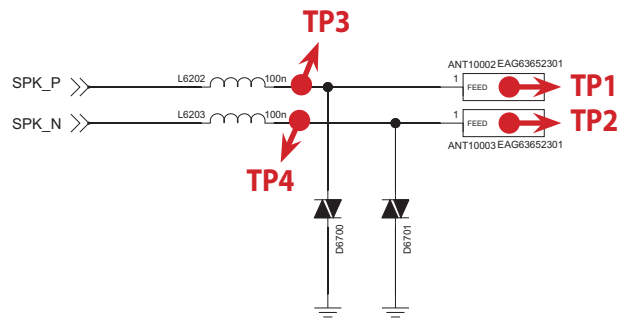
It's trouble shooting guide for no sound case.
 Speaker control signals are generate by MSM8974(U2100), WCD9320(U6100)
 Speak Boost_Amp (U6200)



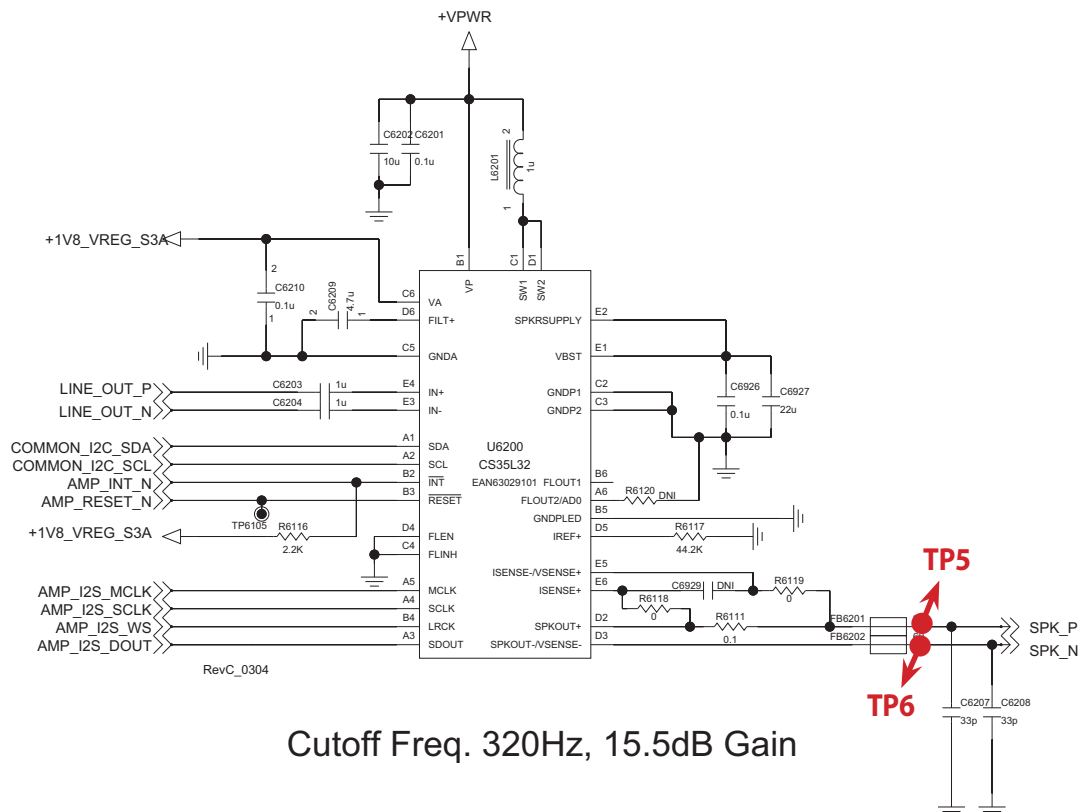
3. TROUBLE SHOOTING



< 6-7-1_Speaker > Rev_0.4



SPK BOOST AMP



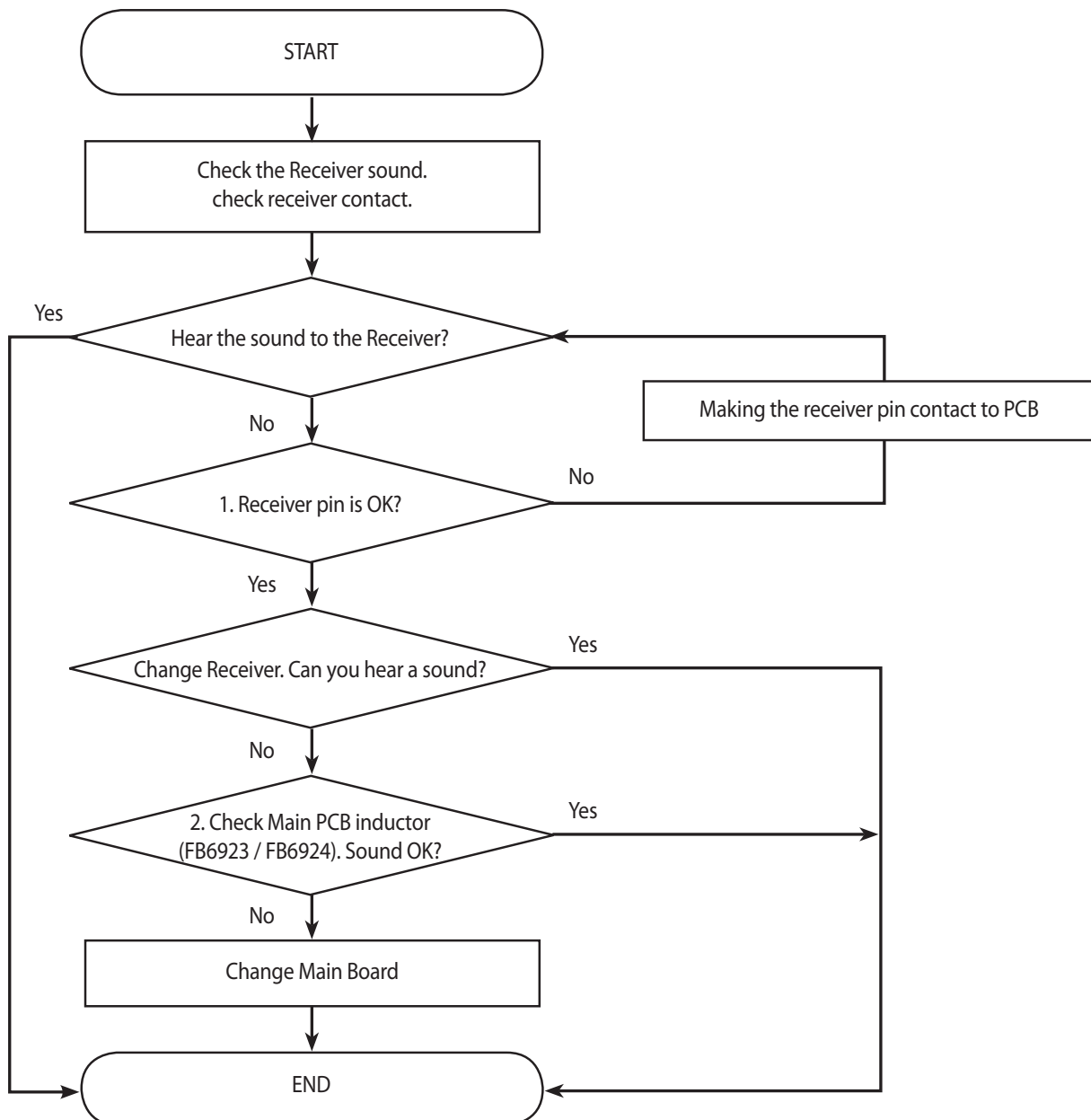
1. Speaker contact is OK on the contact point?
check the ANT10002_TP1, ANT10003_TP2
2. Check PCB part (L6204_TP3, L6203_TP4). Speaker sound OK?
3. Check PCB part (FB6201_TP5, FB6202_TP6). Speaker sound OK?

3.13.3 Receiver Trouble Shooting

Receiver control signals are generate by MSM8974(U2100), WCD9320(U6100)

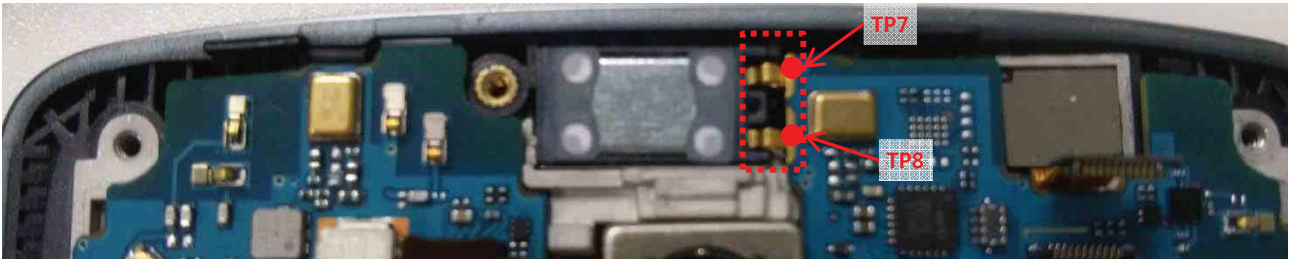
Voice Receiving path as below :

MSM8974 → WCD9320 → EAROP/EARON → Receiver

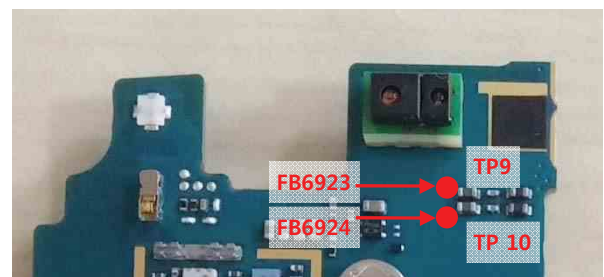
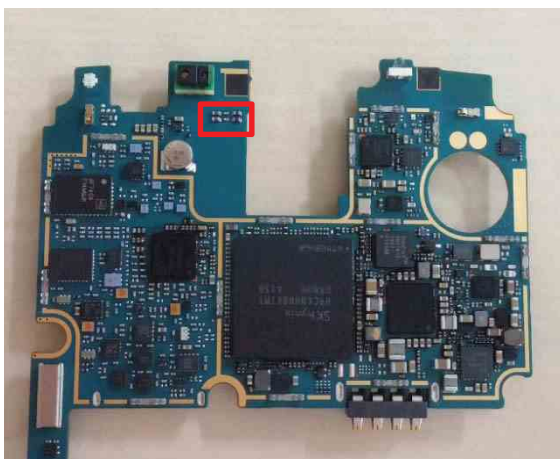
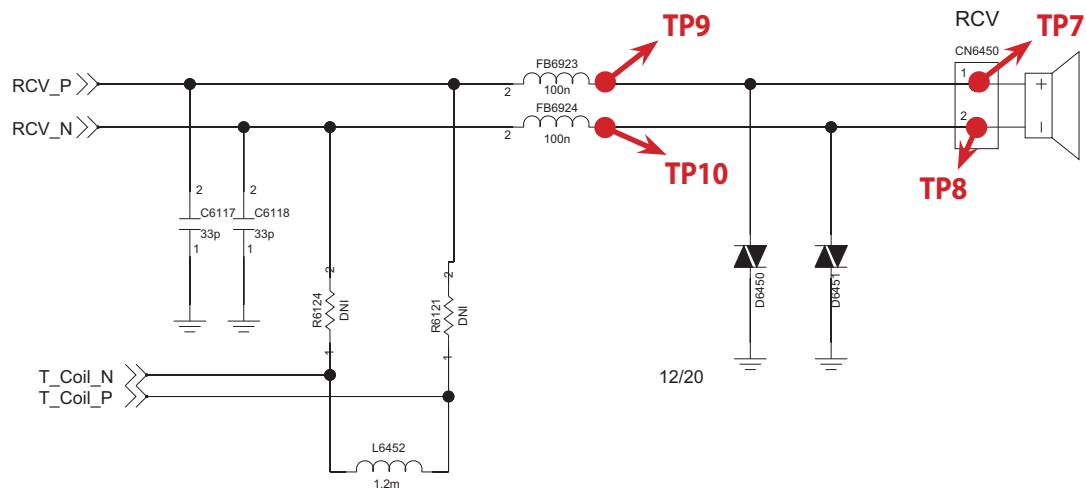


3. TROUBLE SHOOTING

1. Receiver pin is OK? Checking the receiver pin and PCB are contacted well.



2. Check Main PCB inductor (FB6923/ FB6924).

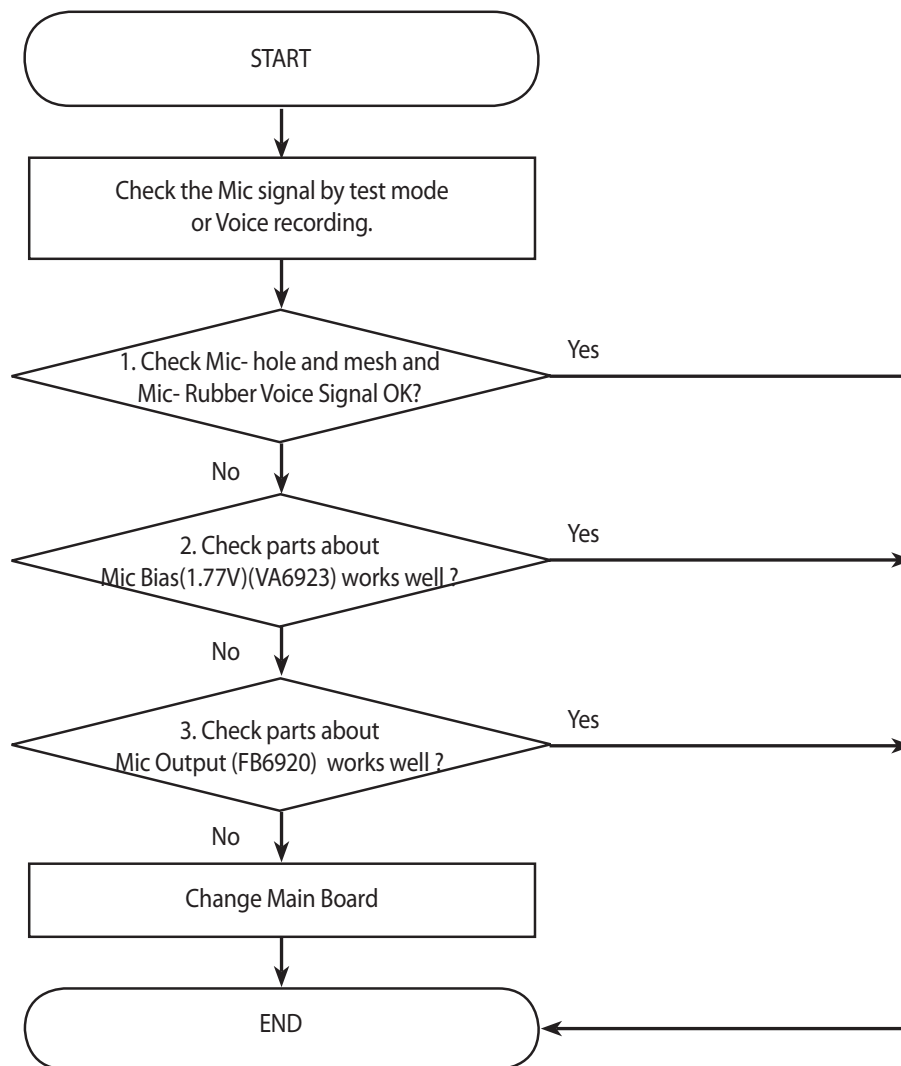


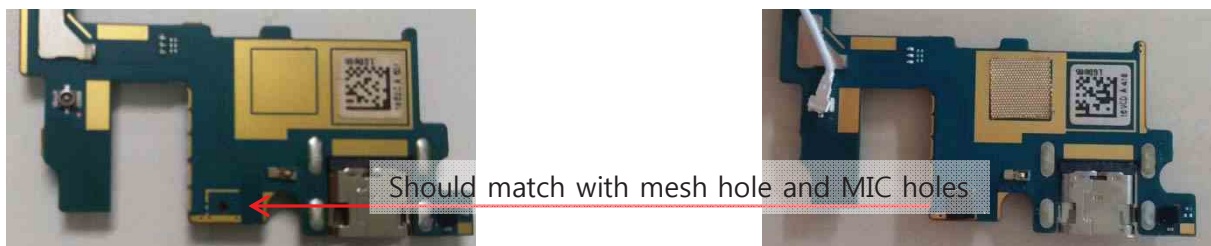
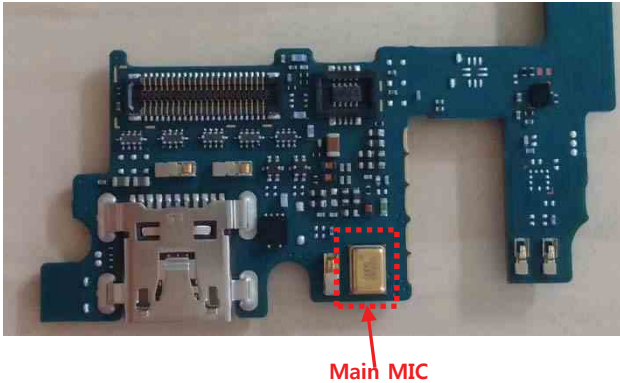
In case of No Receiver sound, not only Receiver fault but MIC fault.
You must check the voice recording test first. After voice recording, if you could hear the recorded sound, It is nor MIC fault And receiver path or parts is something wrong.

3.13.4 Main MIC Trouble Shooting

A Main MIC is located at the bottom of PCB

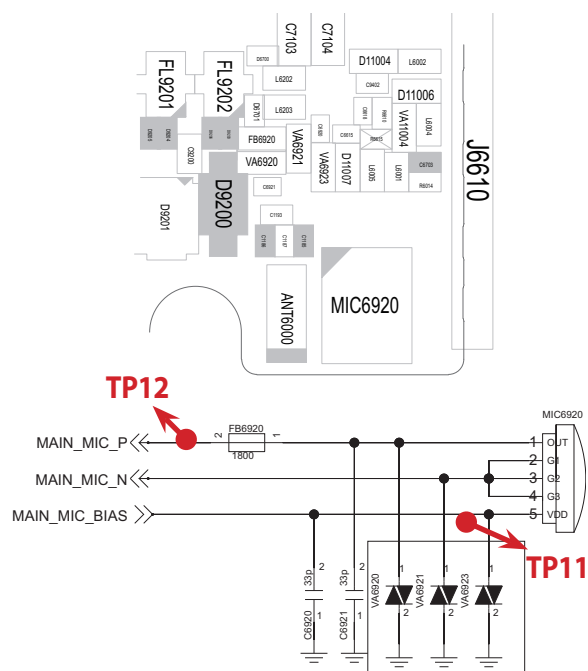
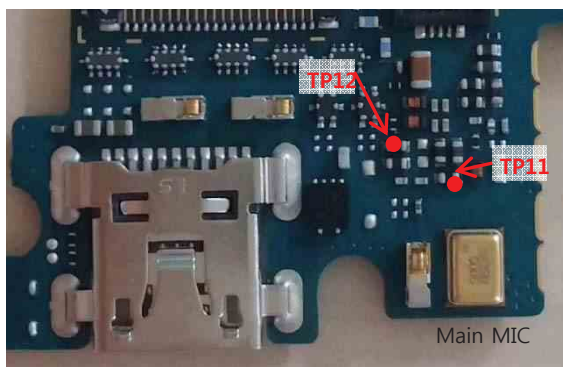
It operates in case of voice call (handset), voice recording, camcorder recording



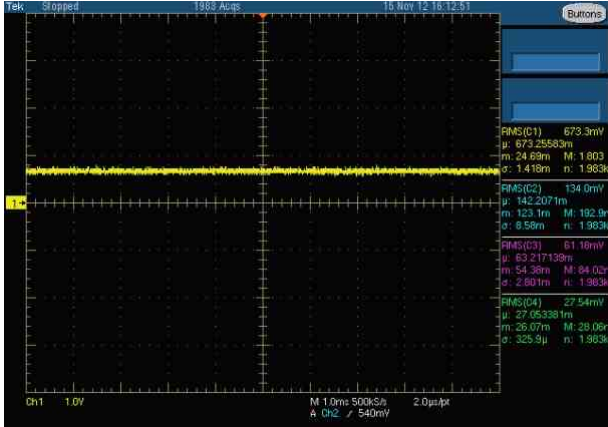
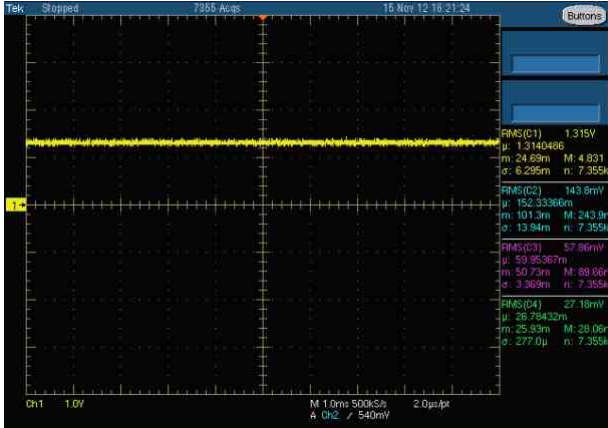


3. Check parts about Mic Output (FB6920_TP12) works well ?

- waveform moves with oscilloscope

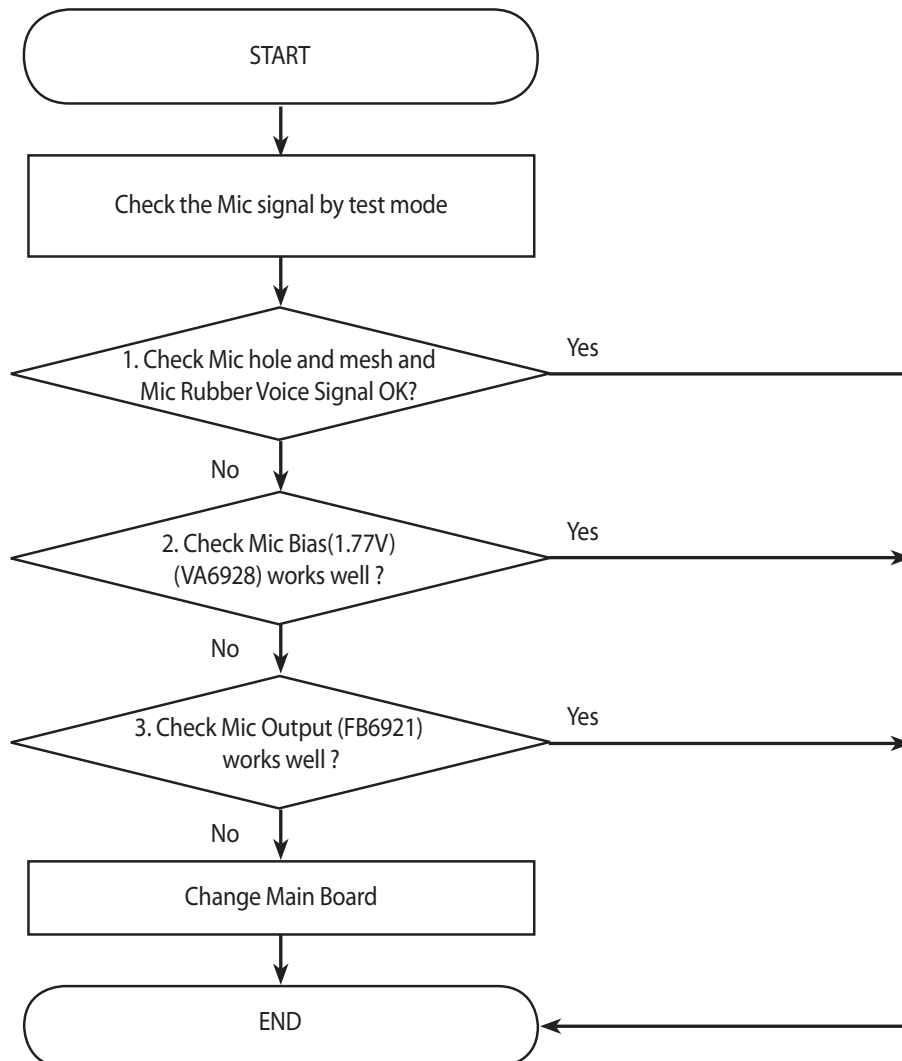


waveform with oscilloscope

Normal MIC Out Signal (Idle)		About 0.7V (Flat)
Normal MIC Out Signal (Voice Input)		Moves(Dynamic)
Malfunction MIC Out Signal (Idle and Voice Input)		1.31V (fixed)

3.13.5 Sub MIC 1 Trouble Shooting

A sub MIC1 (called MIC 2 during Loopback Test) is located beside receiver.
It operates in case of camcorder recording



- Sub MIC 1

MIC Rubber

Main PCB

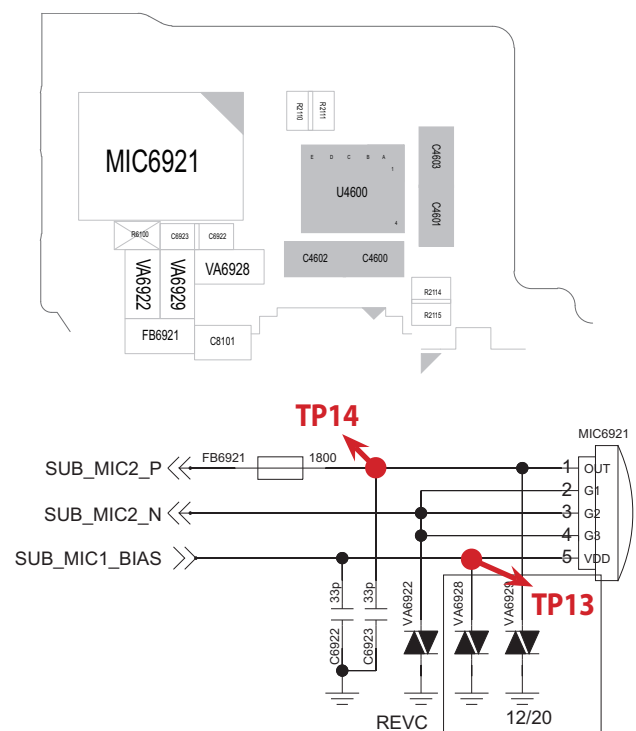
Front Assembly

2

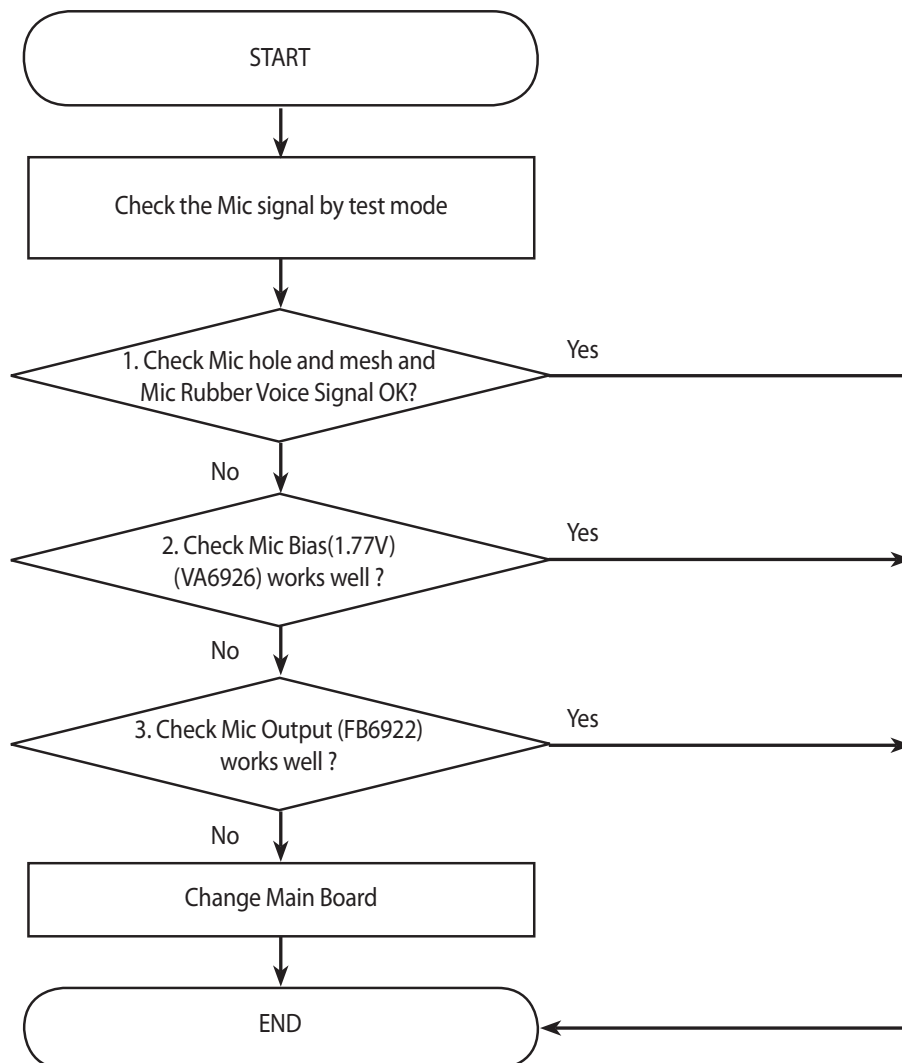
MIC Mesh

Should match with mesh hole and MIC holes

-

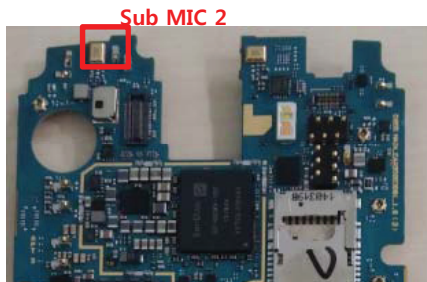


A sub MIC2 (called MIC 3 during Loopback Test) is located at the top of PCB
It operates in case of voice call (Speaker phone).



3. TROUBLE SHOOTING

1. Check Mic hole and mesh Mash on PCB and Mic Rubber in Front Ass'y.
Voice Signal OK?



Main PCB



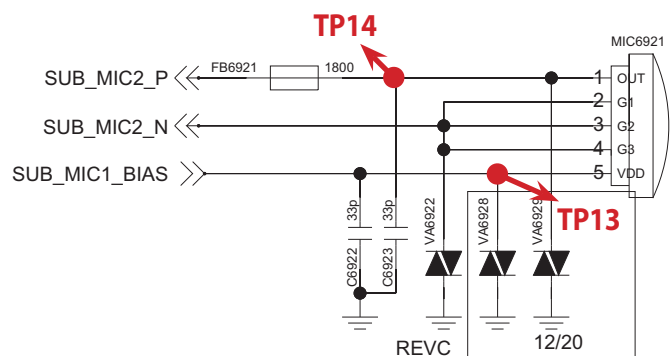
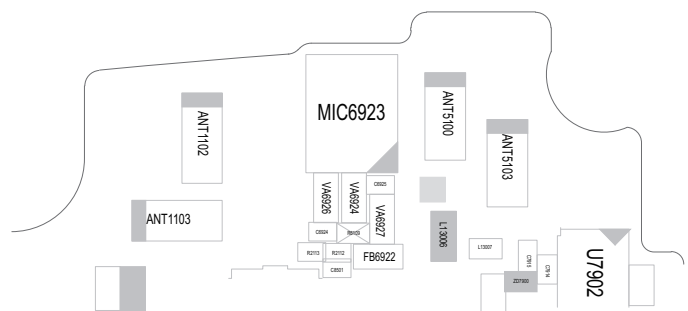
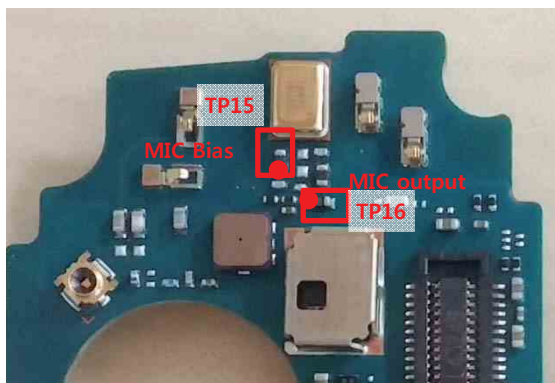
Front Assembly



MIC Mesh

Should match with mesh hole and MIC holes

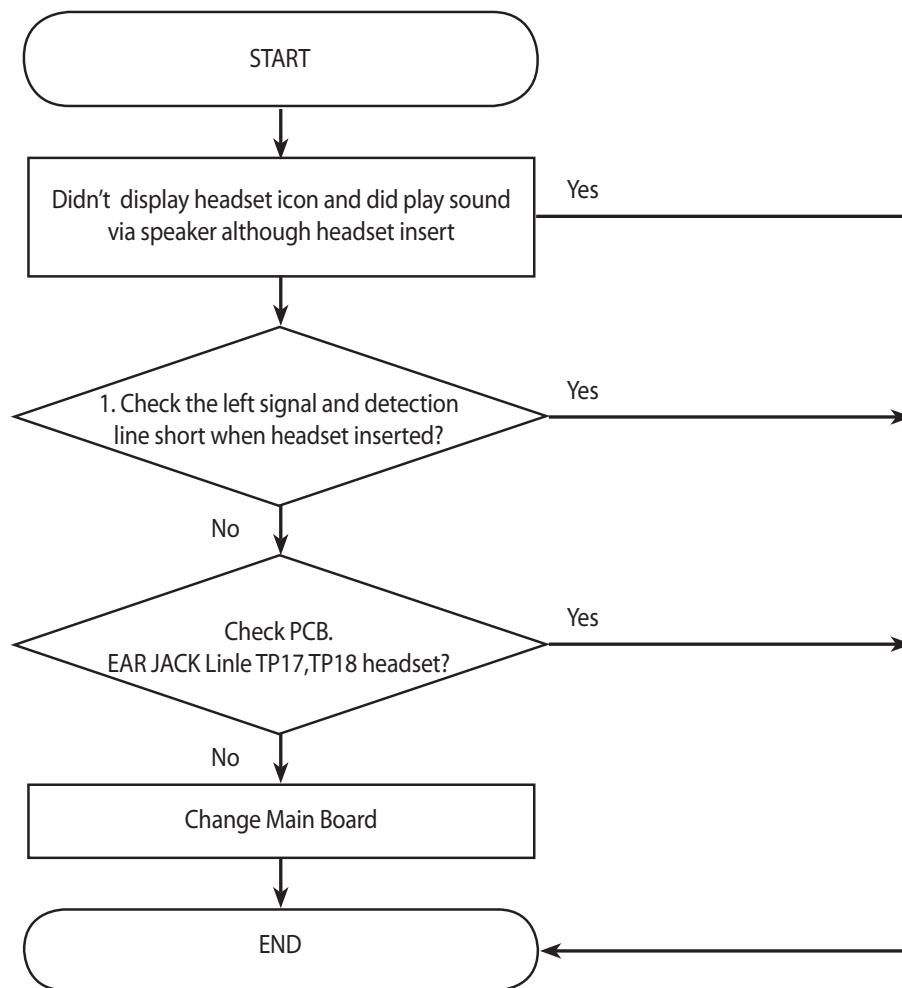
2. Check parts about Mic Bias(VA6926_TP15) works well (about 1.77V) ?
3. Check parts about Mic Output (FB6622_TP16) works well ?
– waveform moves



3.13.6 Ear-MIC Trouble Shooting

Ear MIC control signals are generate by MSM8974AC(U2100), WCD9320(U6100)

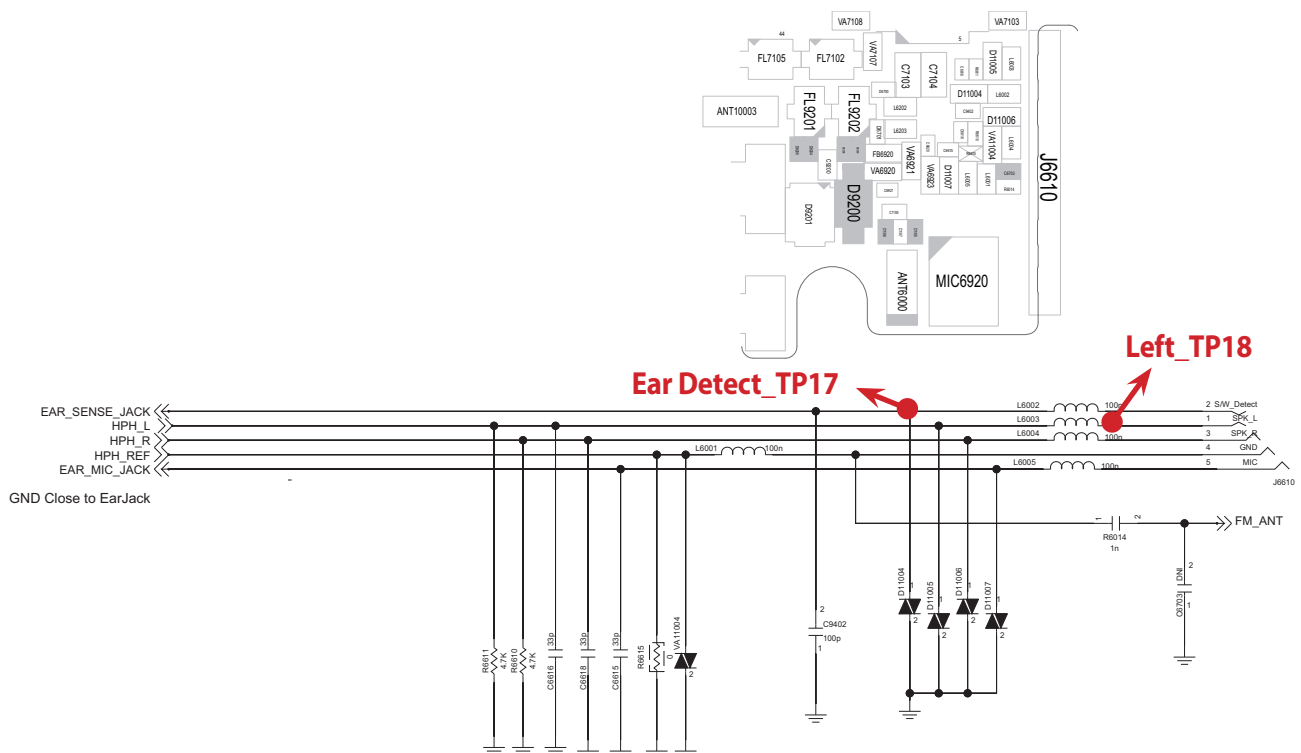
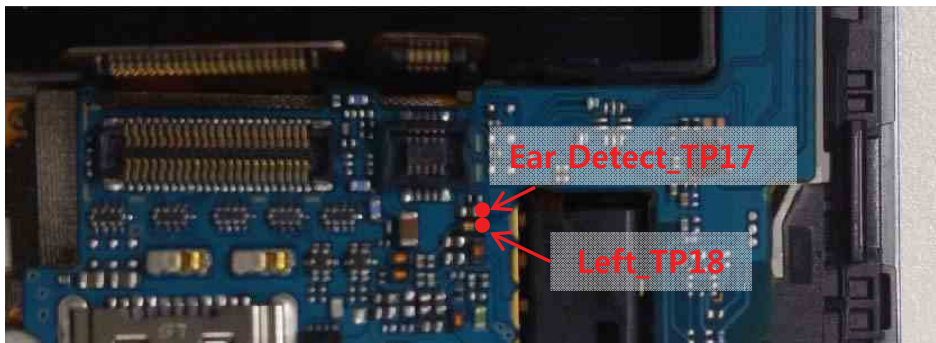
Case A. Unable detection of headset insert.



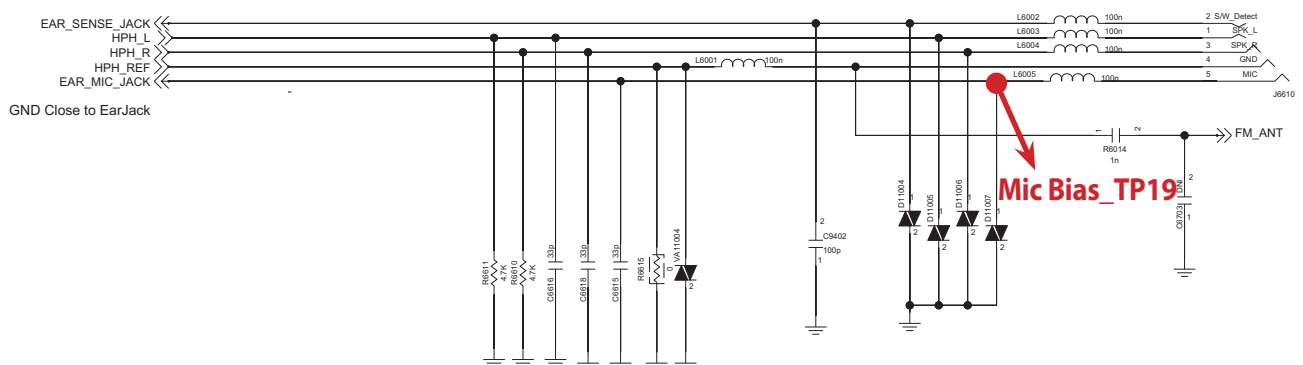
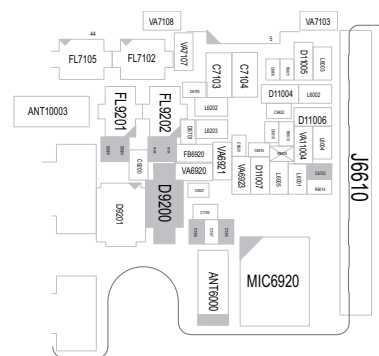
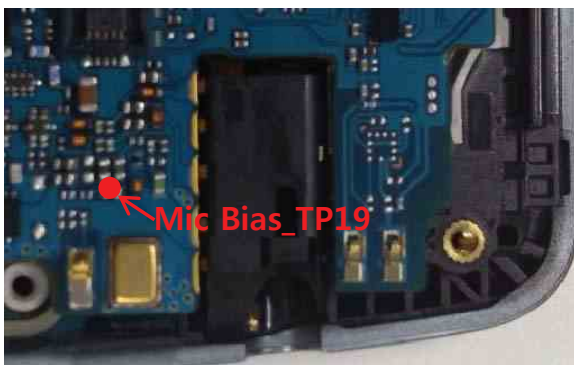
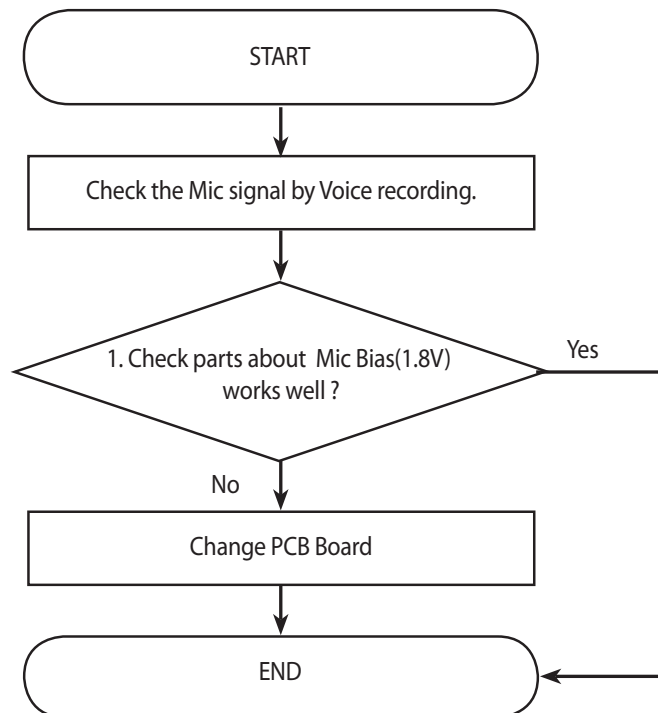
Ear MIC control signals are generate by MSM8974AC(U2100), WCD9320(U6100)

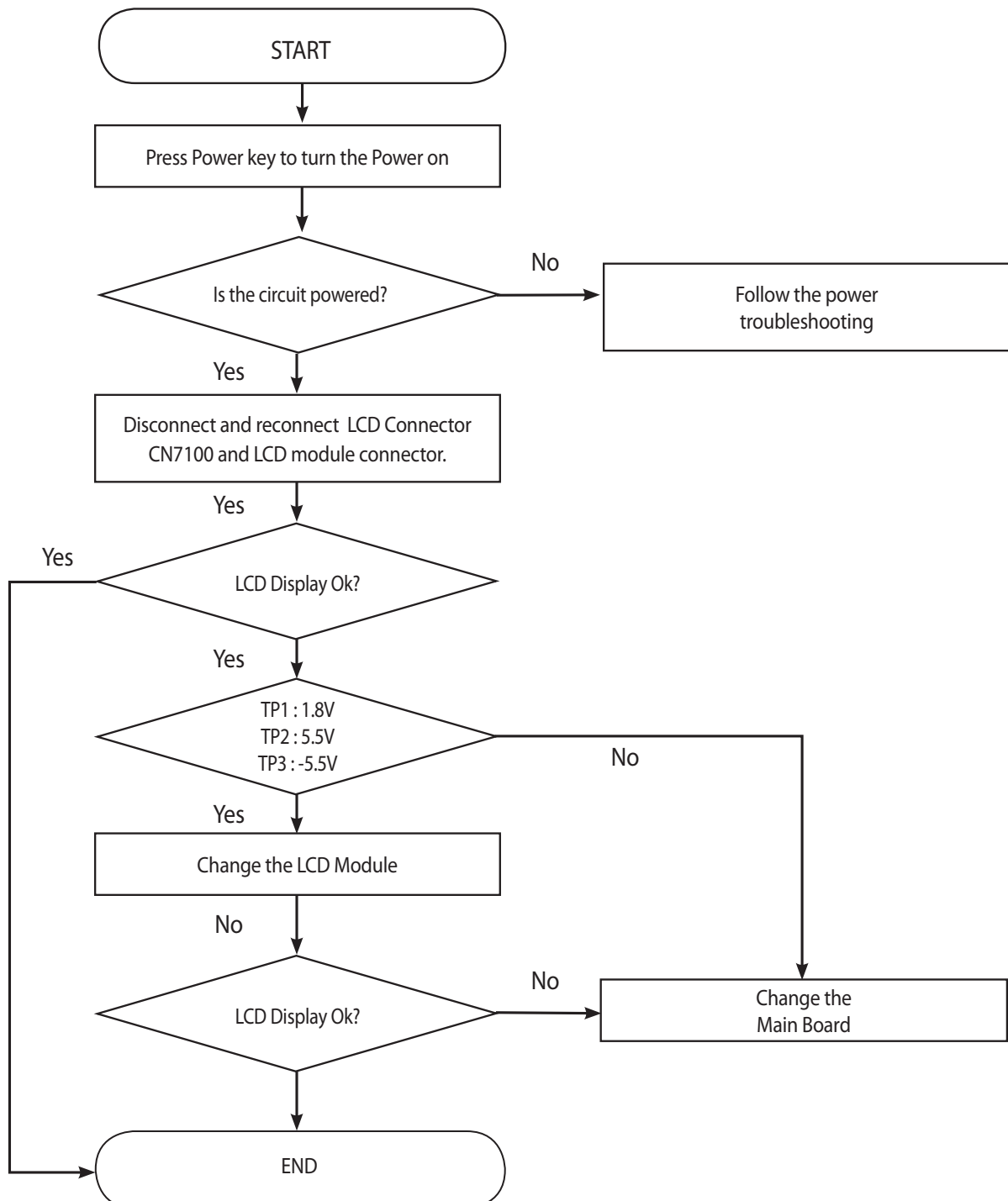
Case A. Unable detection of headset insert.

1. Check the left signal(HPH_L)(FB6611 TP17) and detection line(EAR_SENSE_JACK)(D11004 TP18) are shorted when headset inserted. See PCB Contact terminal also.

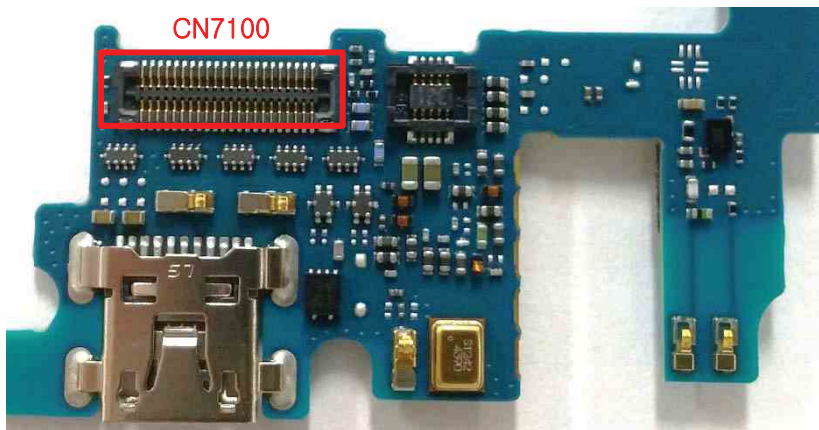
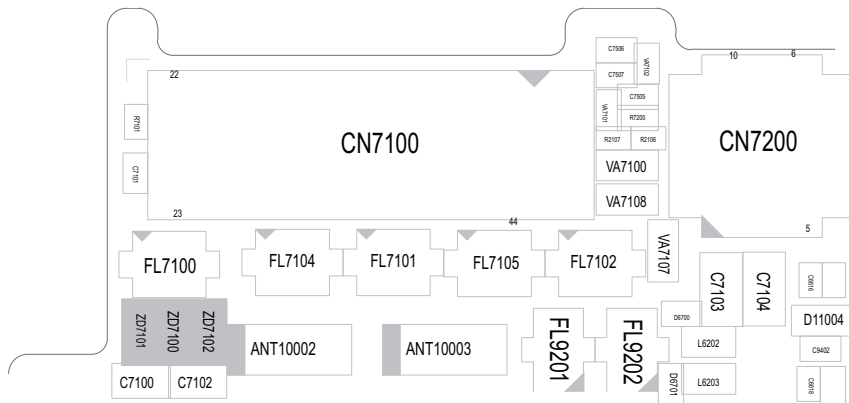


Case B. No Sound from Ear-mic

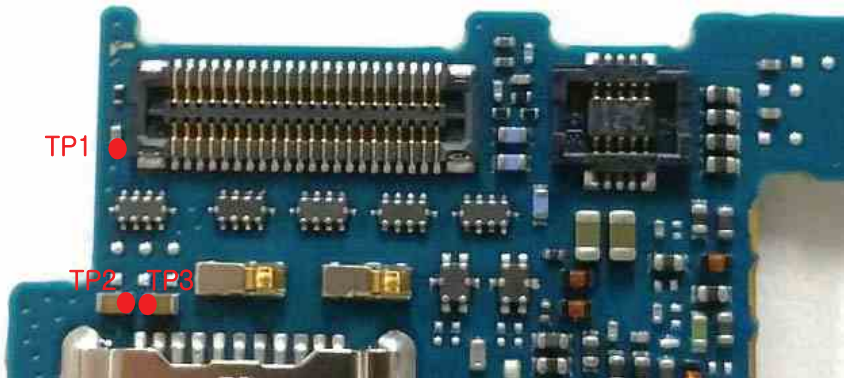
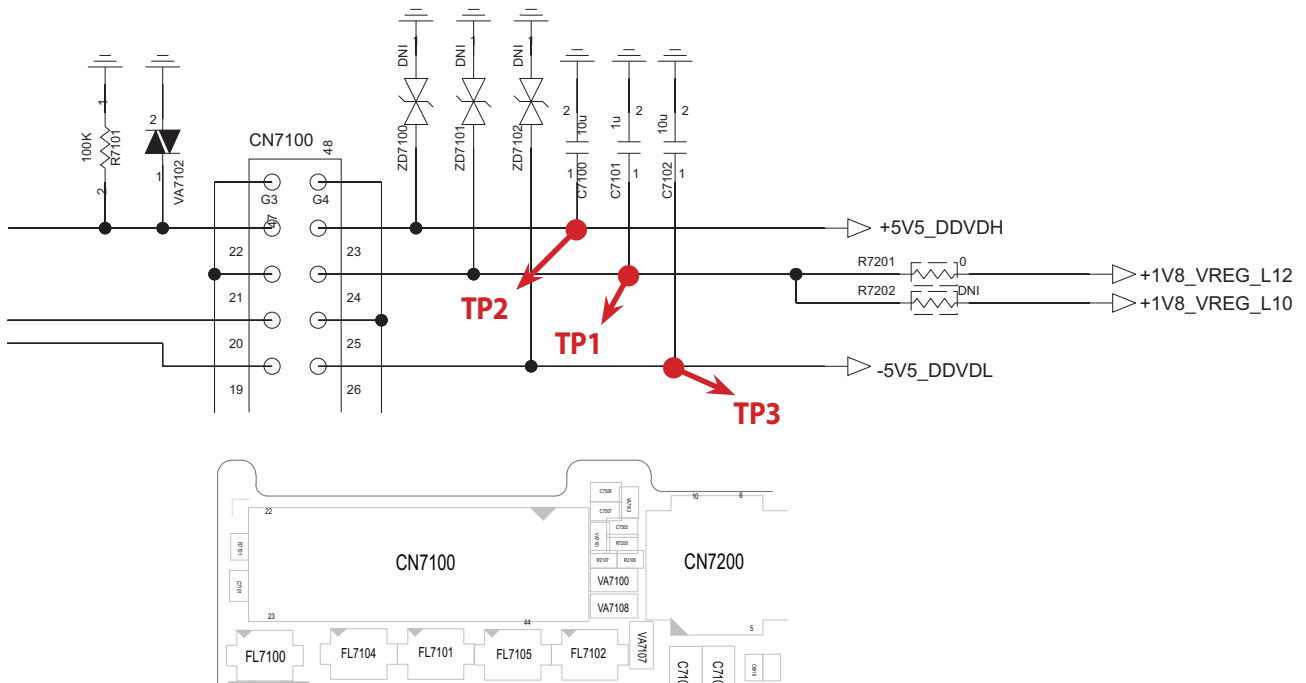




3.14 LCD



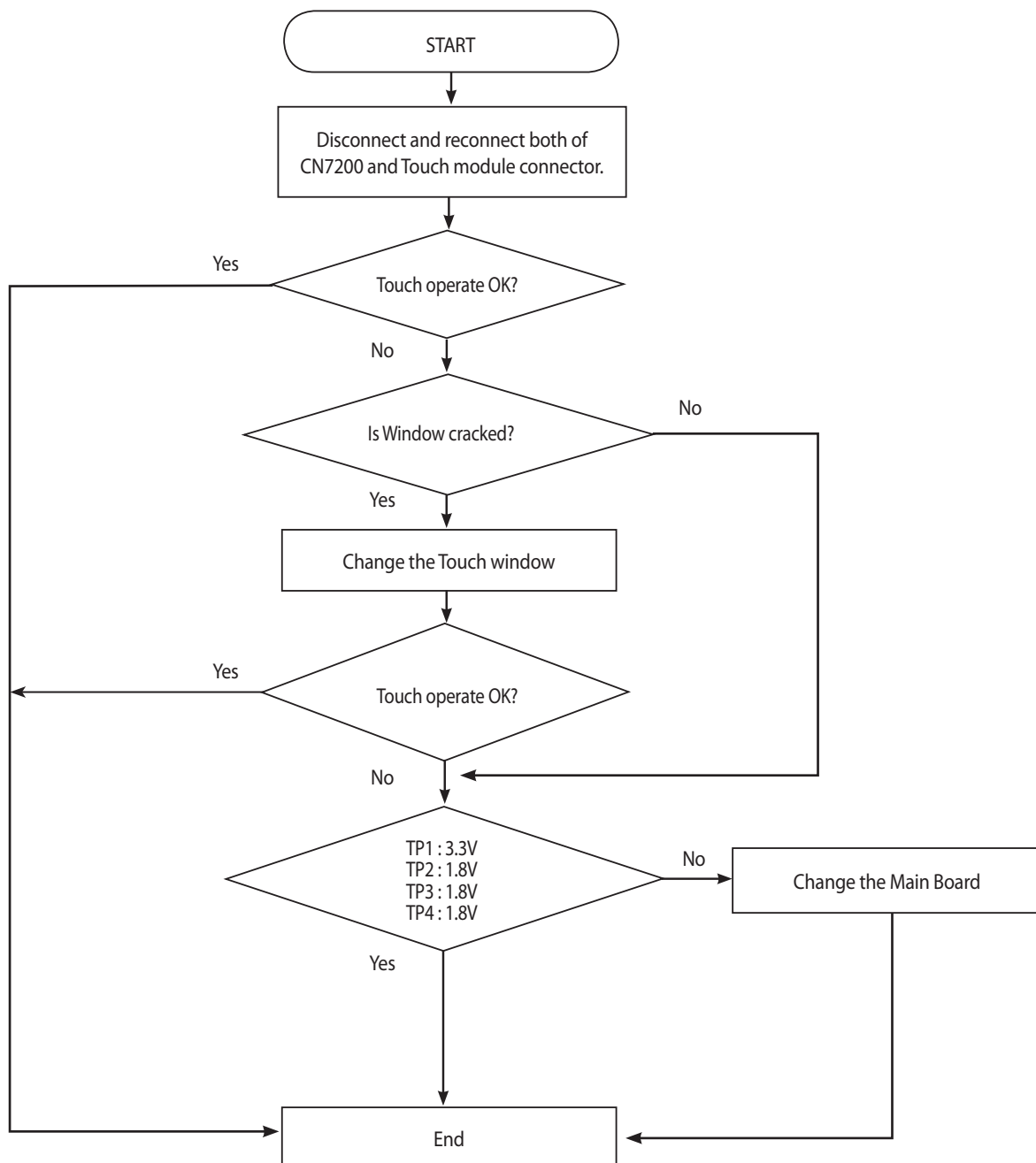
3. TROUBLE SHOOTING



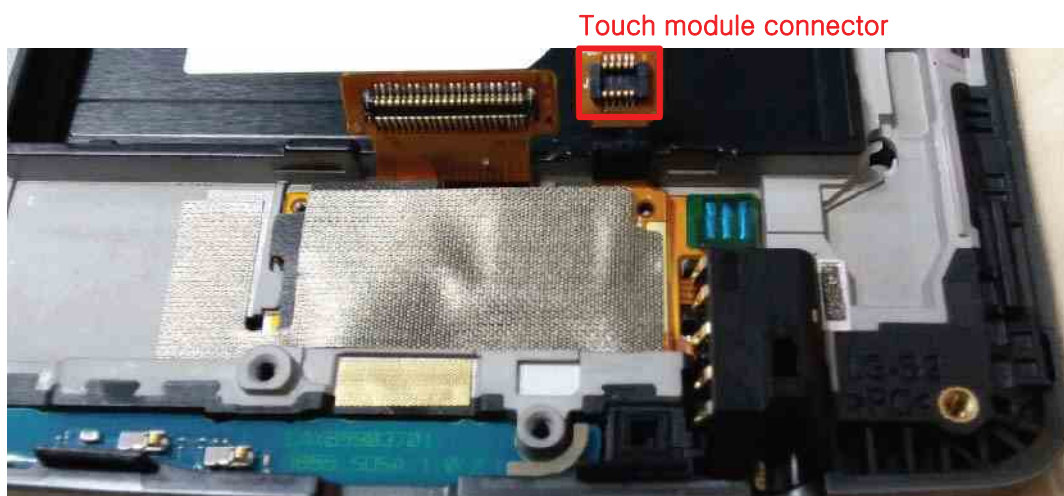
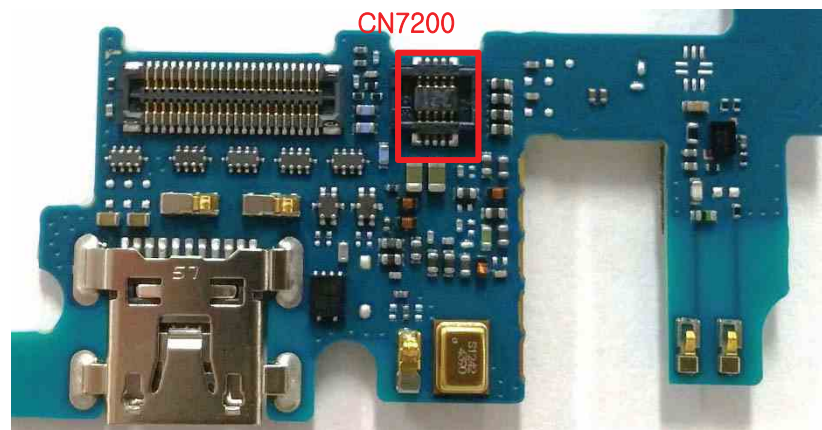
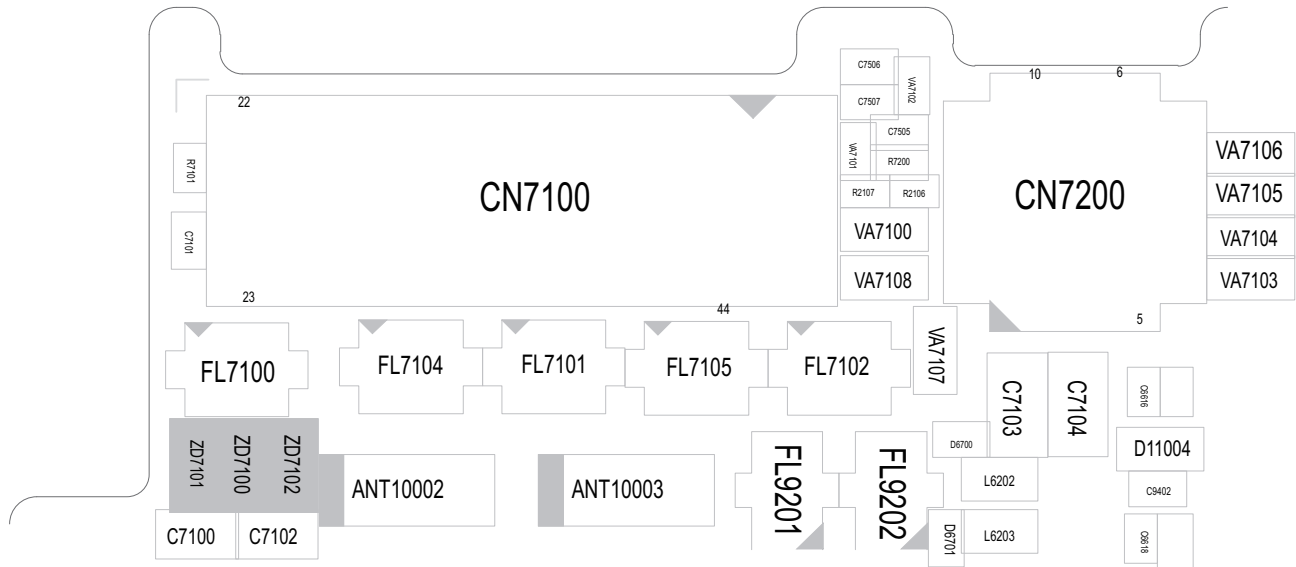
3.15 Touch

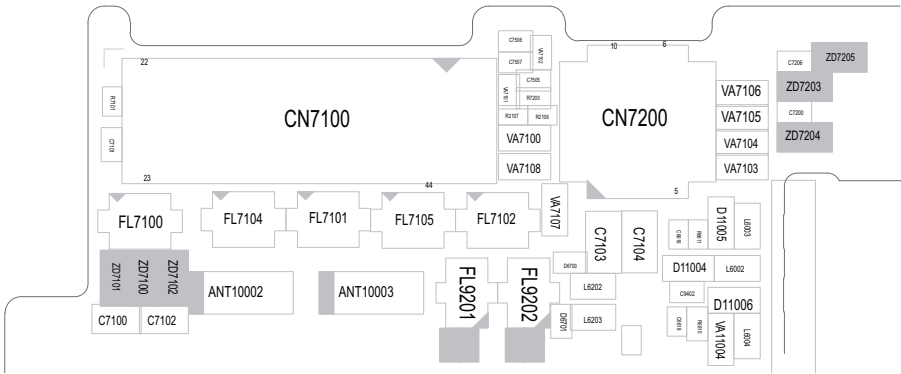
Touch control signals are generated by touch sensor.

Those signal's path are touch sensor → MSM8974

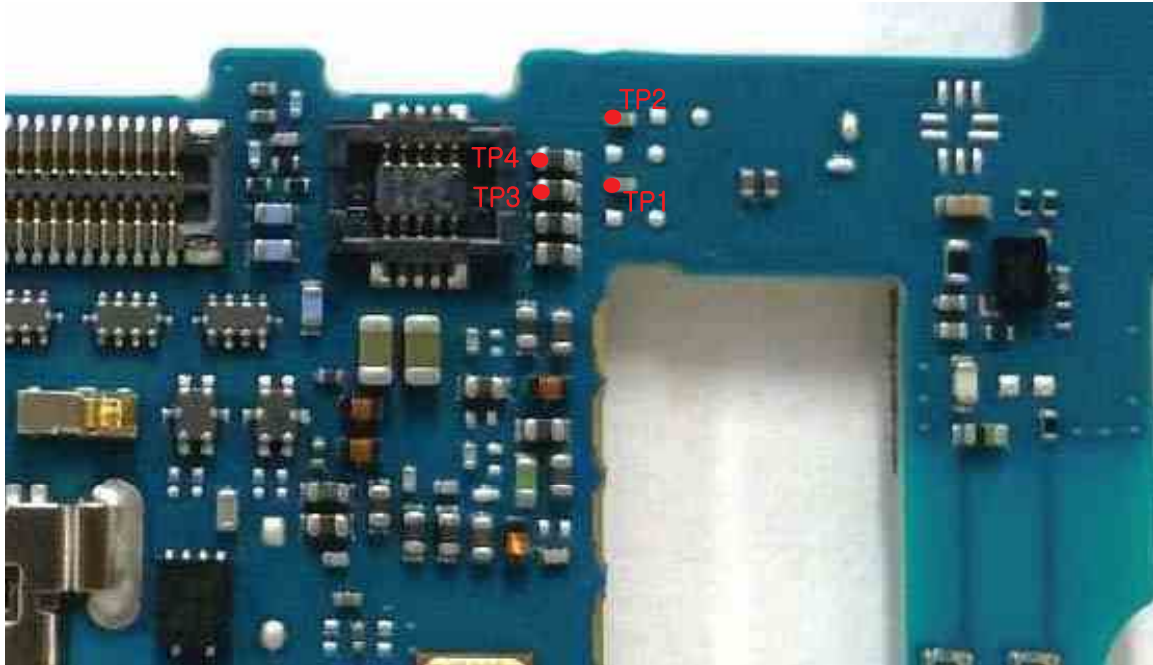


3. TROUBLE SHOOTING



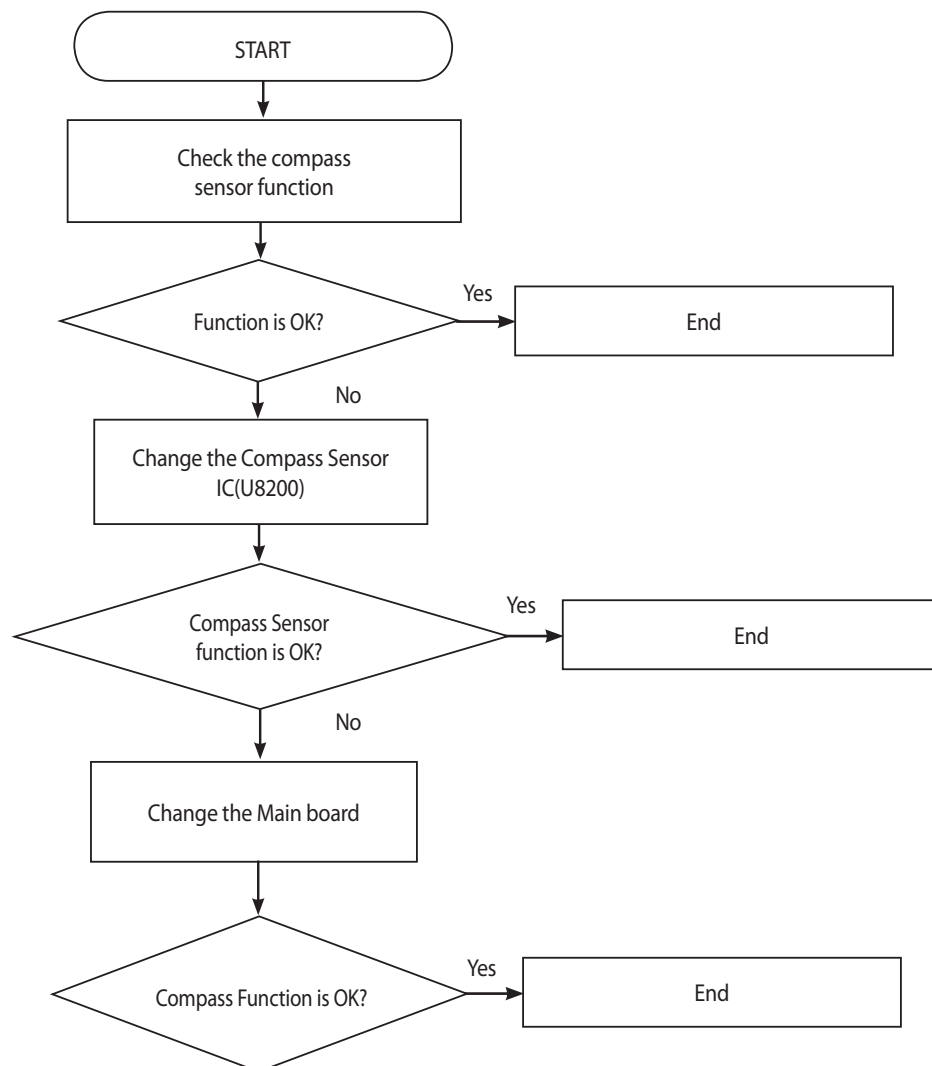


3. TROUBLE SHOOTING

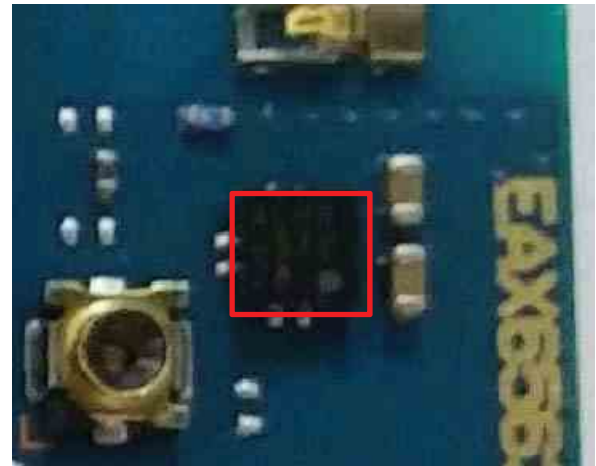
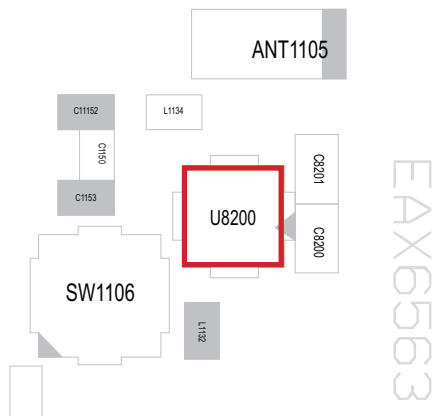
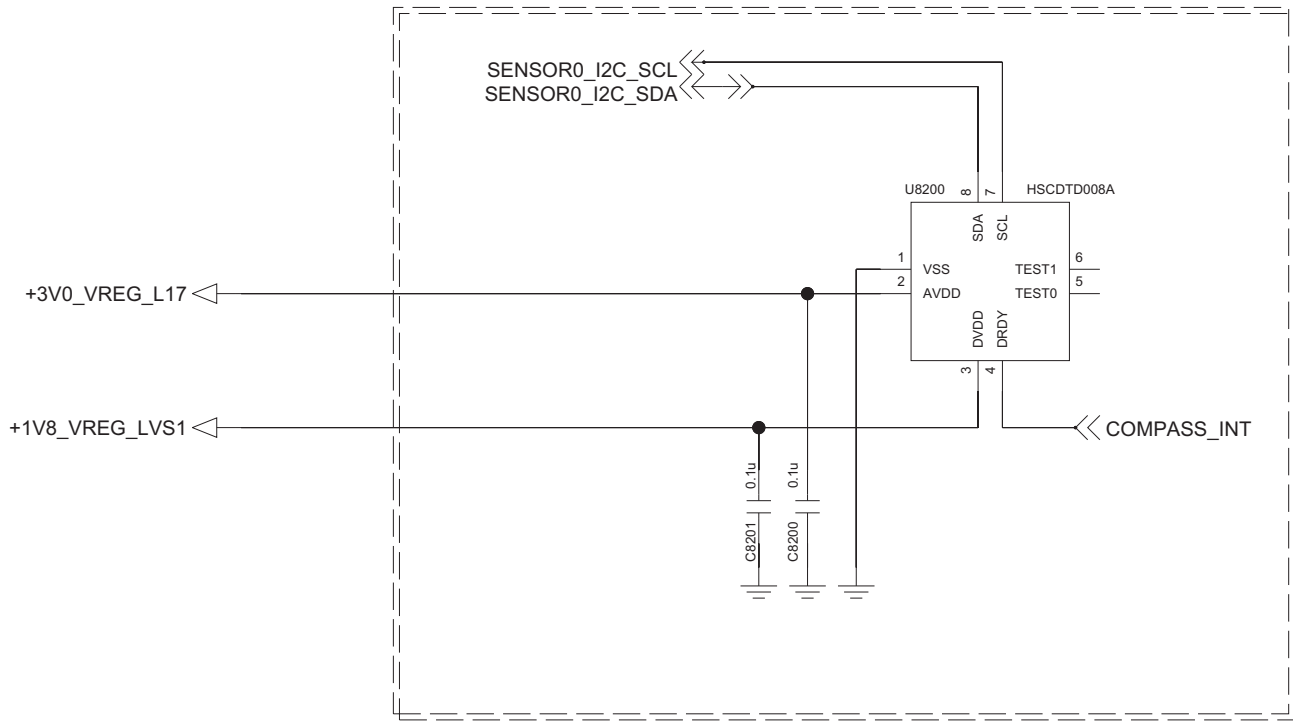


3.16 Sensor

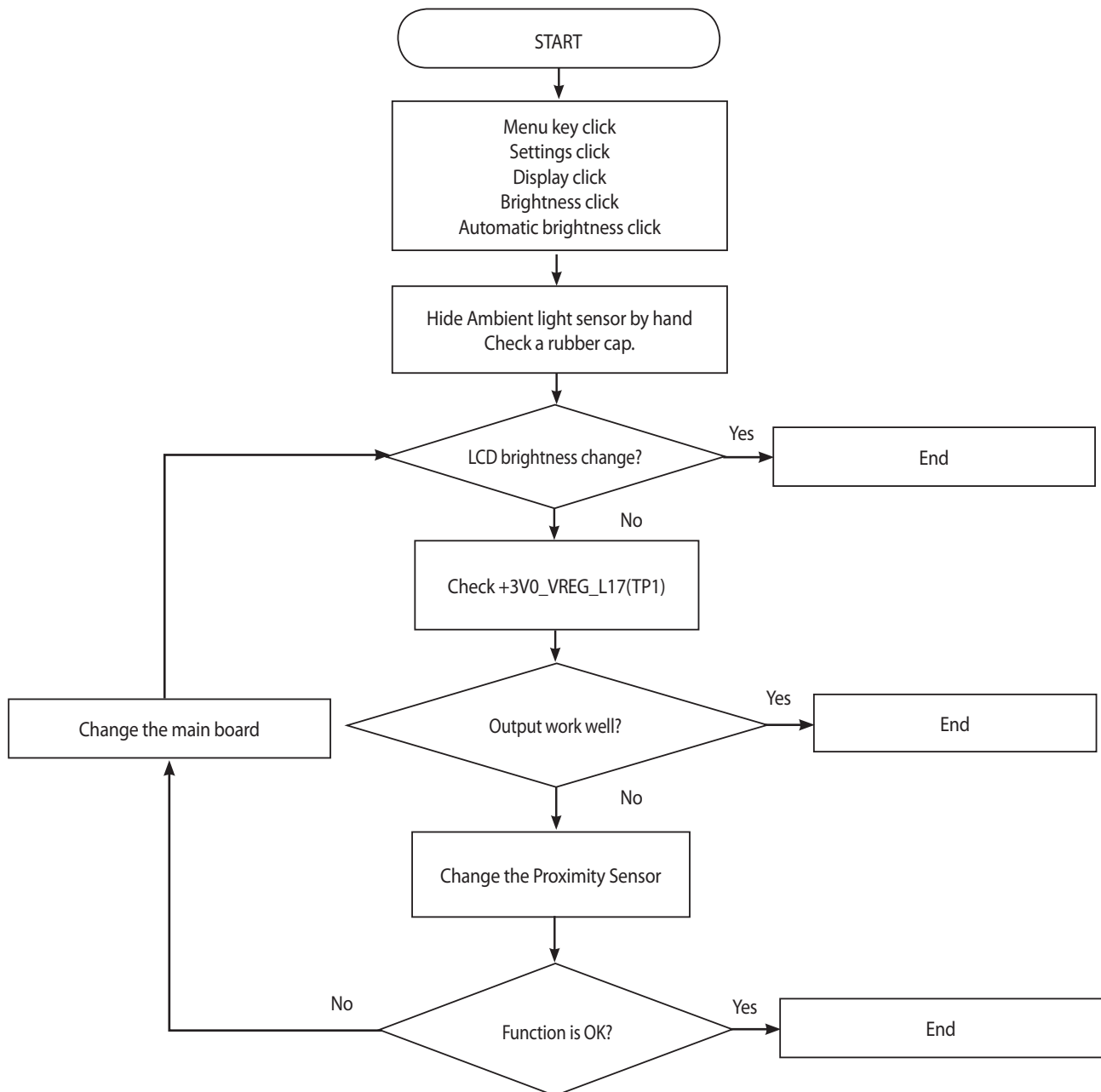
3.16.1 Compass + Motion Sensor

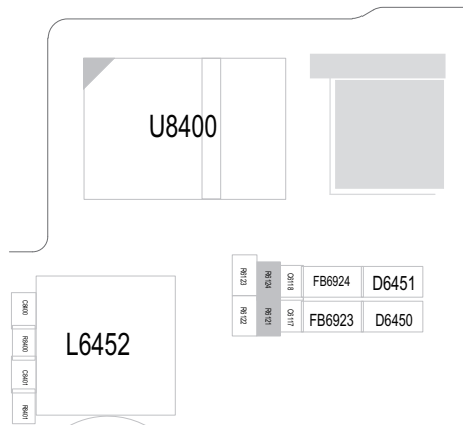
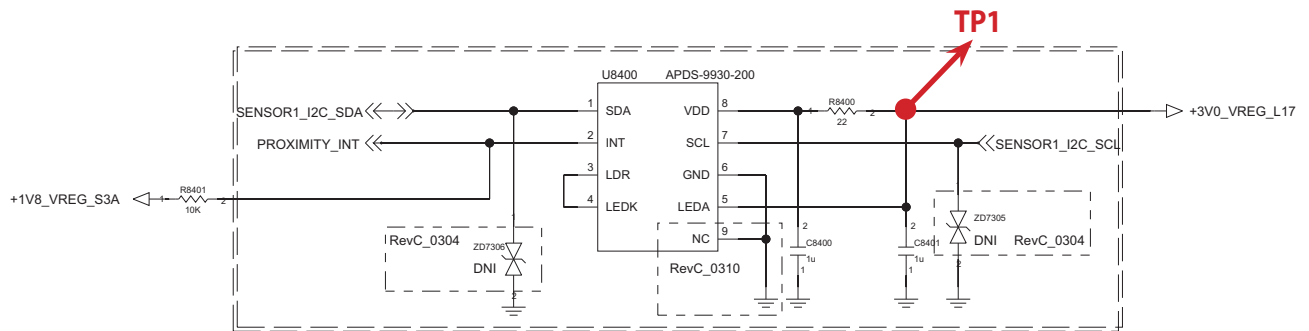


3. TROUBLE SHOOTING



3.16.2 Ambient light sensor



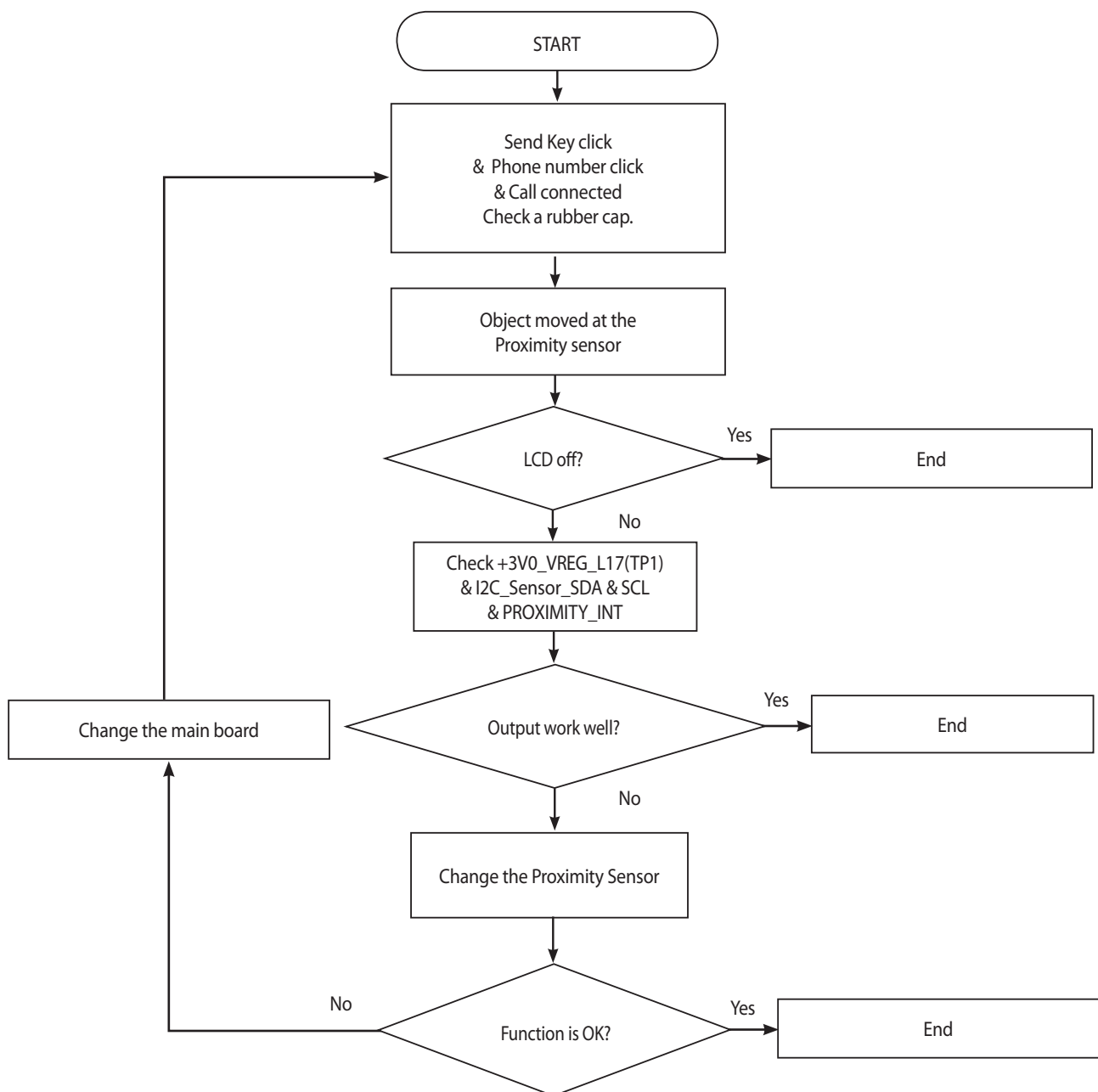


3.16.3 Proximity sensor

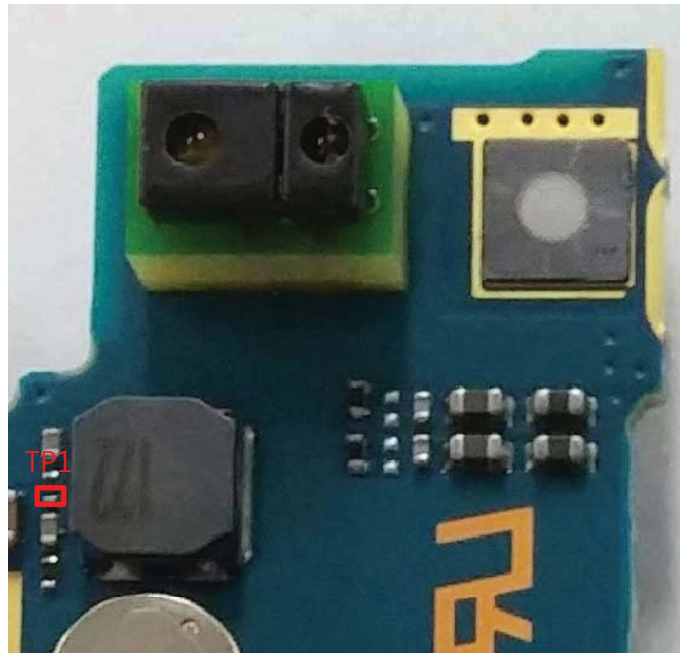
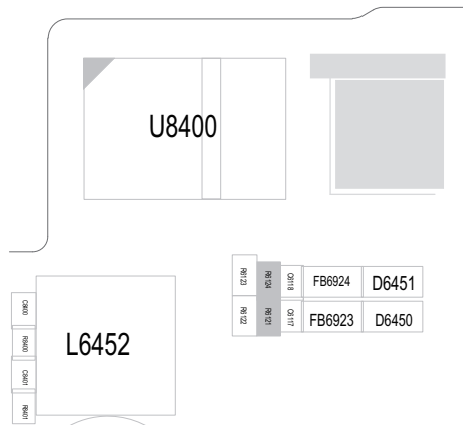
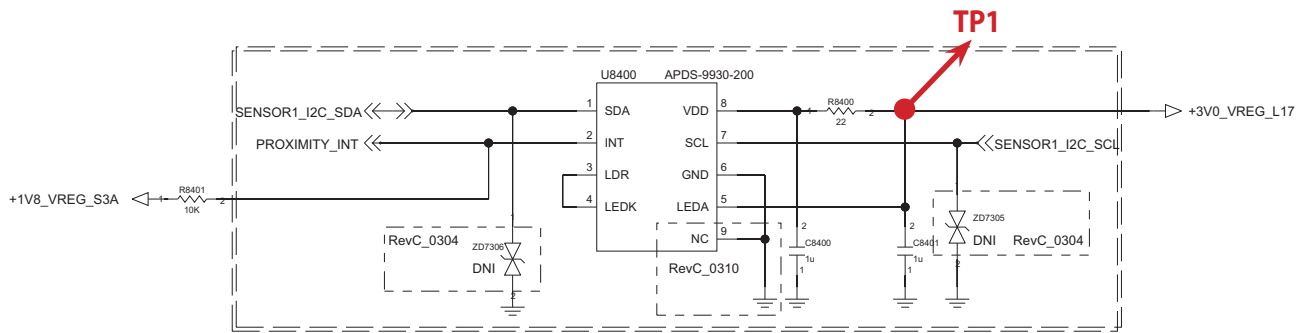
APDS-9130-200 is Proximity sensor.

Proximity Sensor is worked as below:

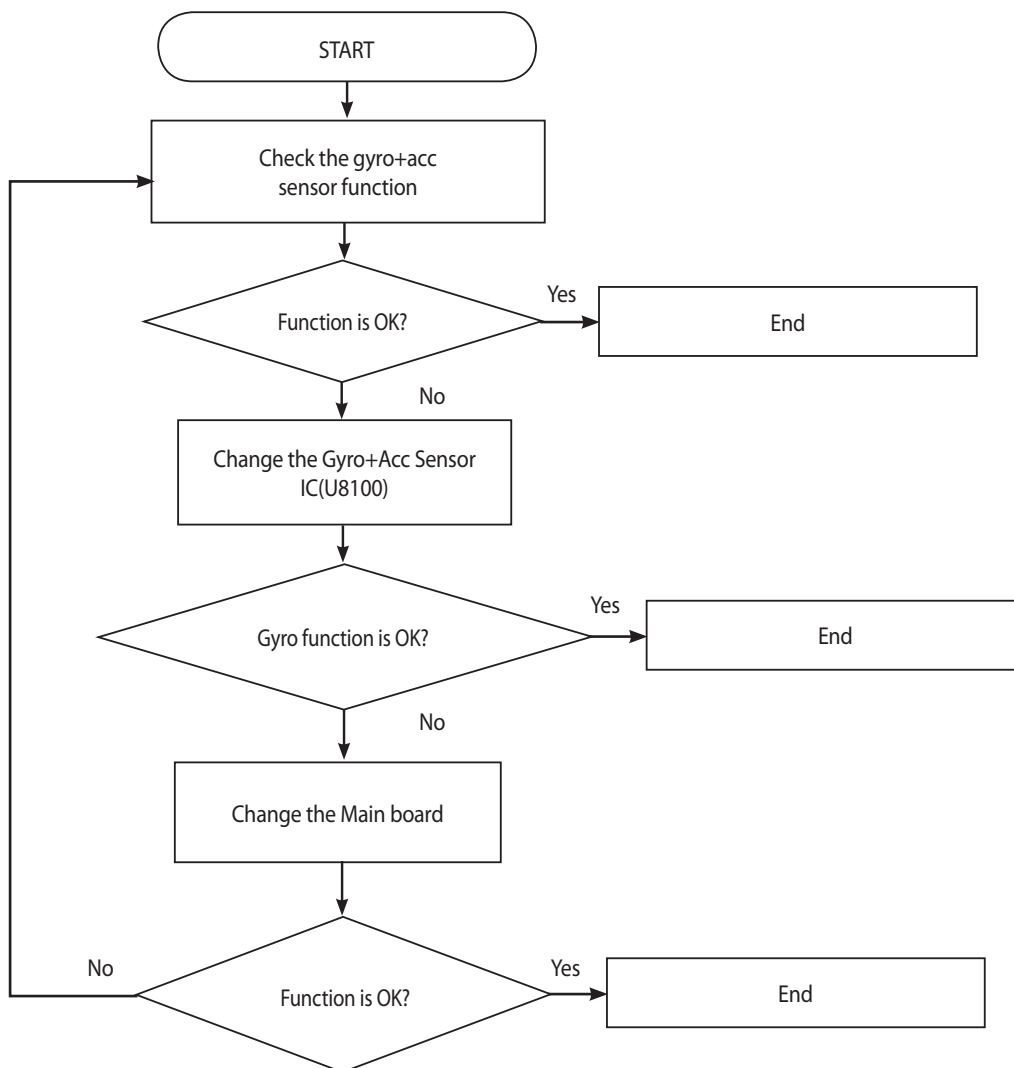
Send Key click → Phone number click → Call connected → Object moved at the sensor → Control the screen's on/off operation automatically



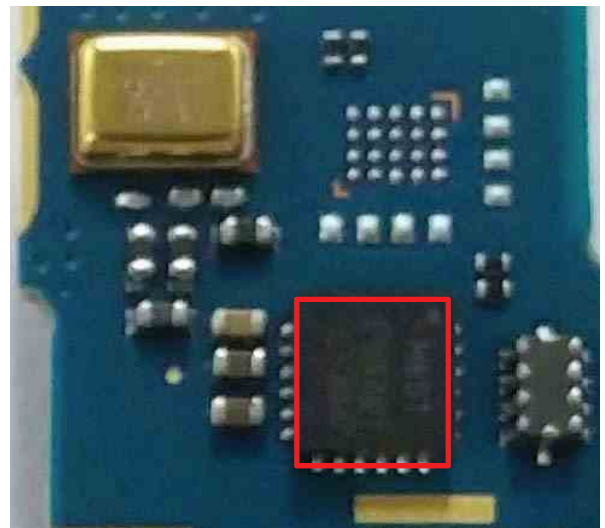
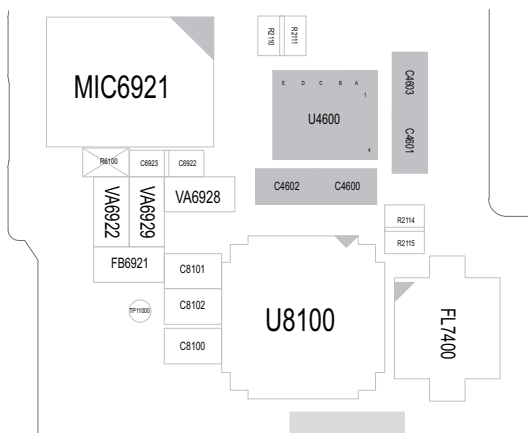
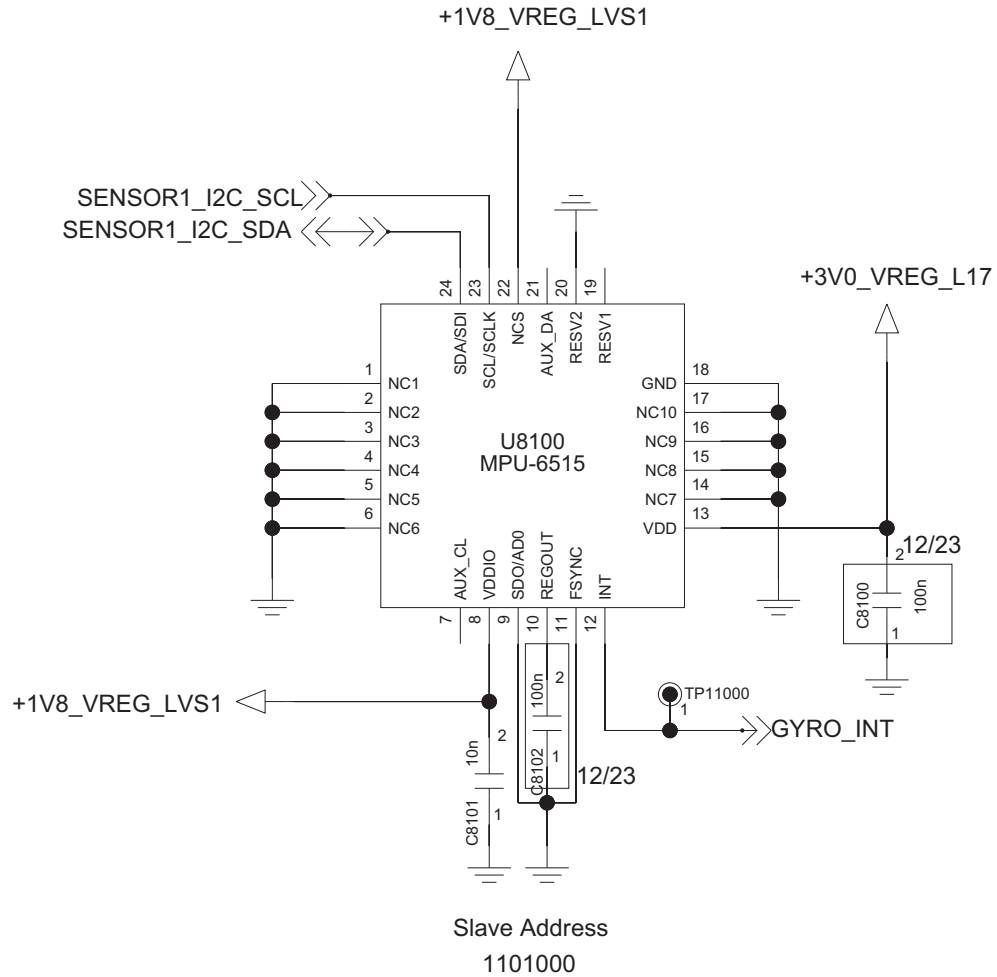
3. TROUBLE SHOOTING



3.16.4 Gyroscope + Accelometer sensor



3. TROUBLE SHOOTING

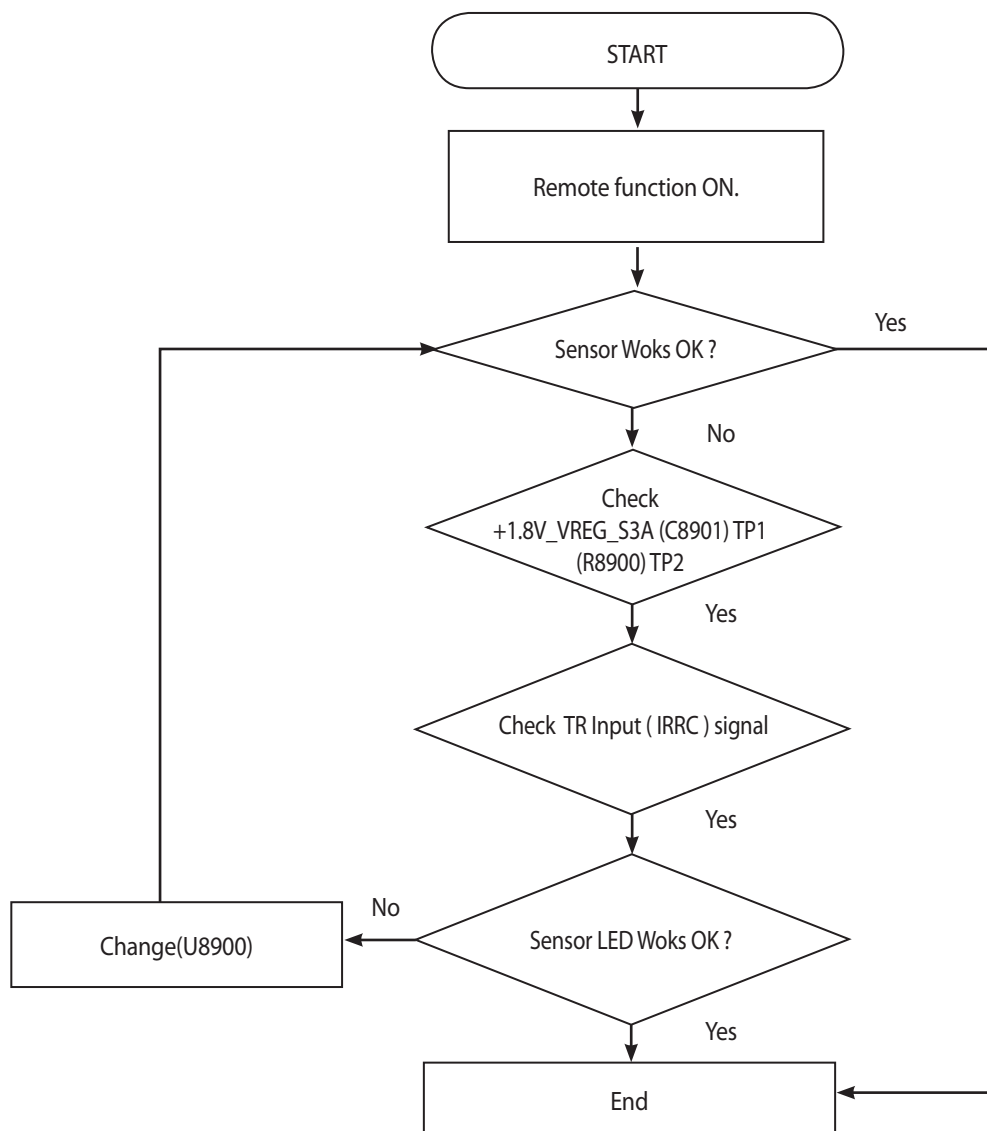


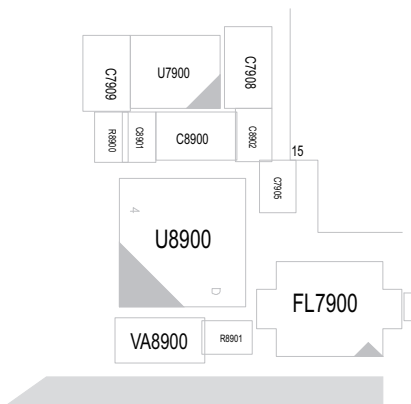
3.17 Remote Sensor (IRRC Sensor)

The sequence of QuickRemote Operation Path

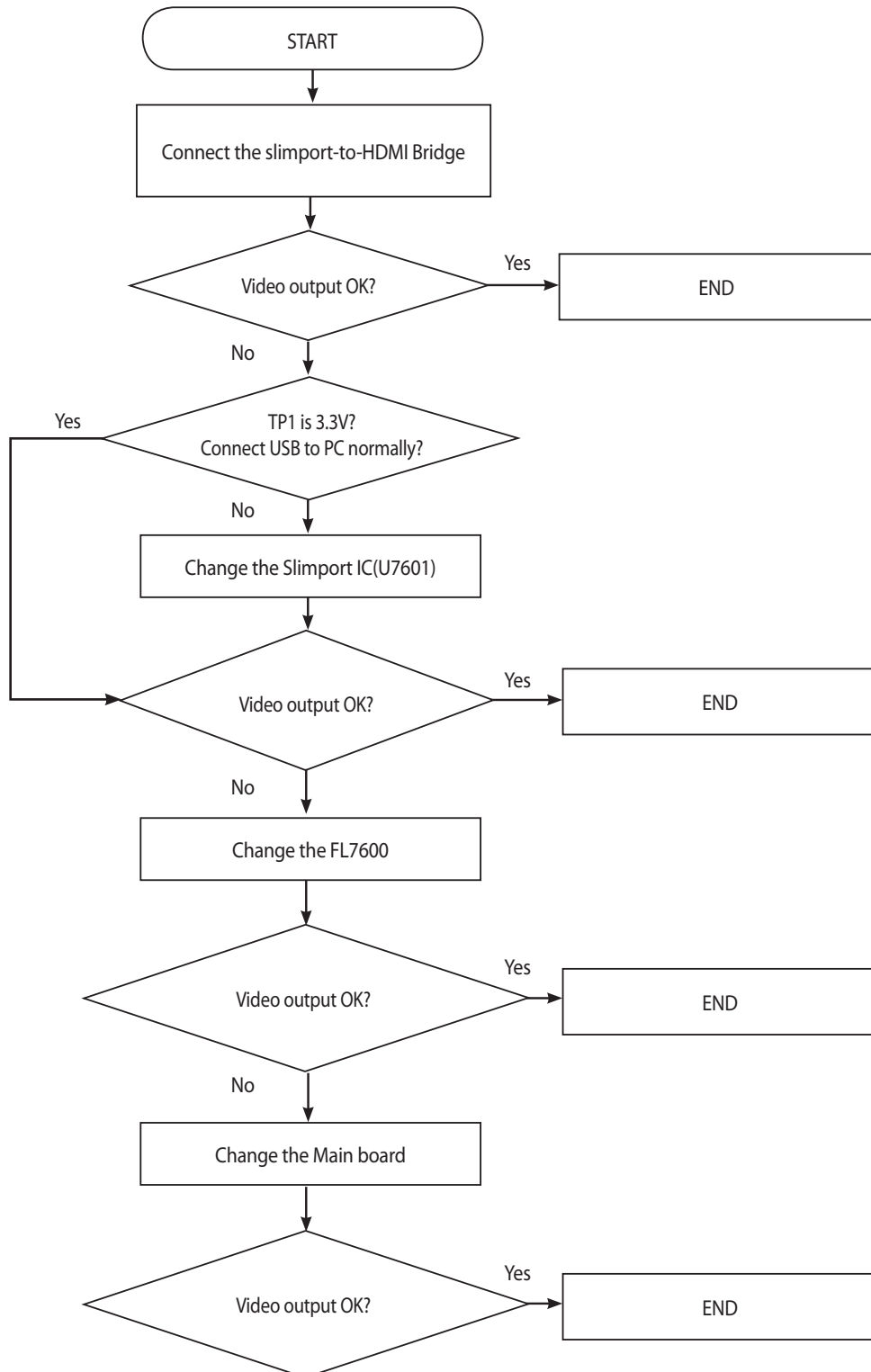
Proximity Sensor is worked as below:

QuickRemote function Check → Menu Remote on → TV Setting on

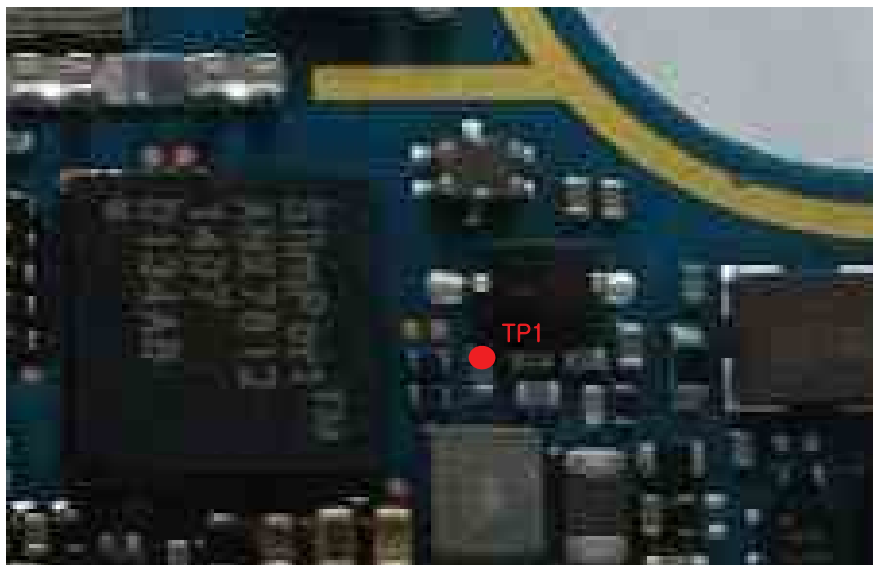
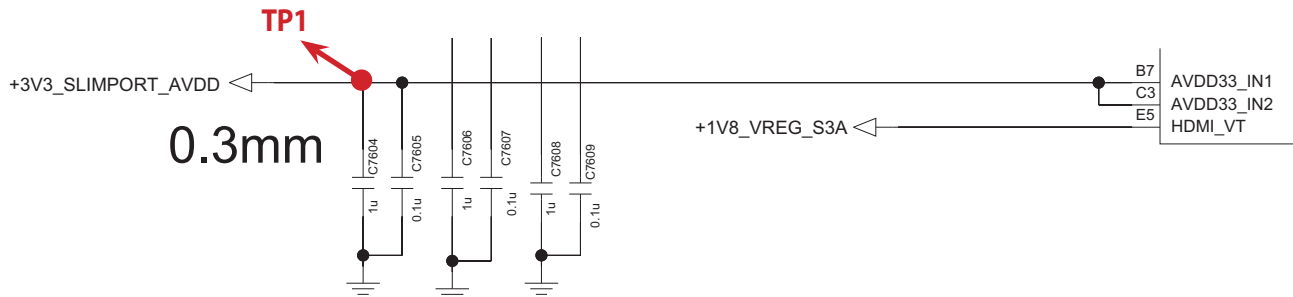




3.18 USB / Slimport

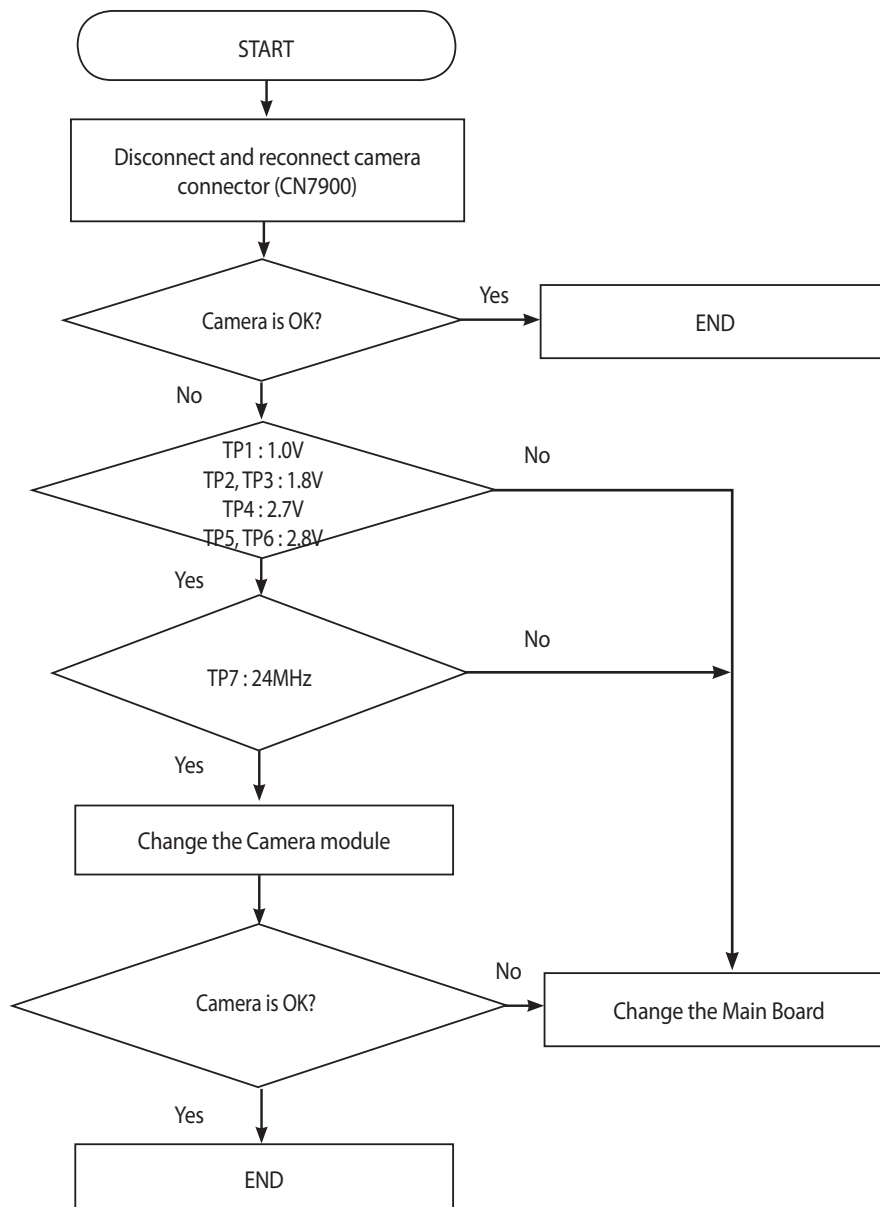


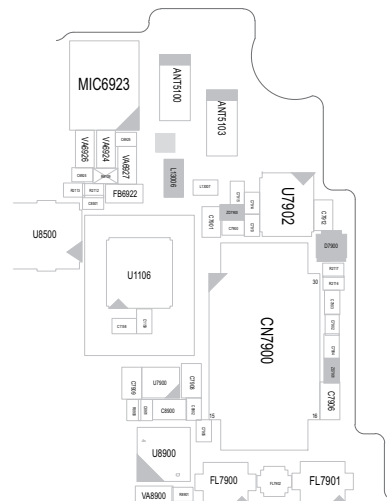
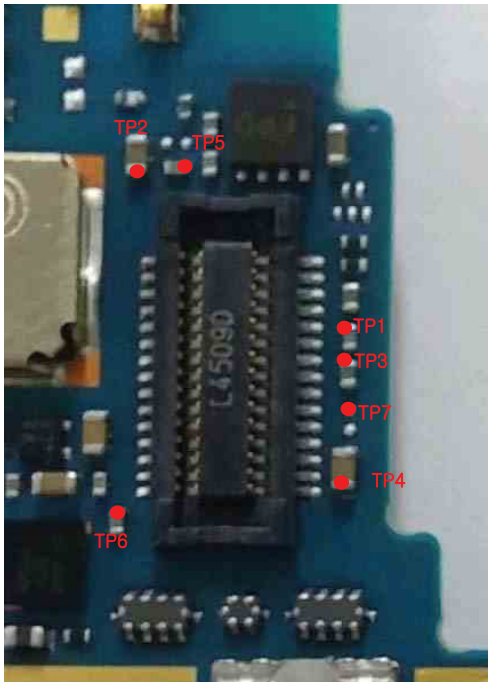
3. TROUBLE SHOOTING



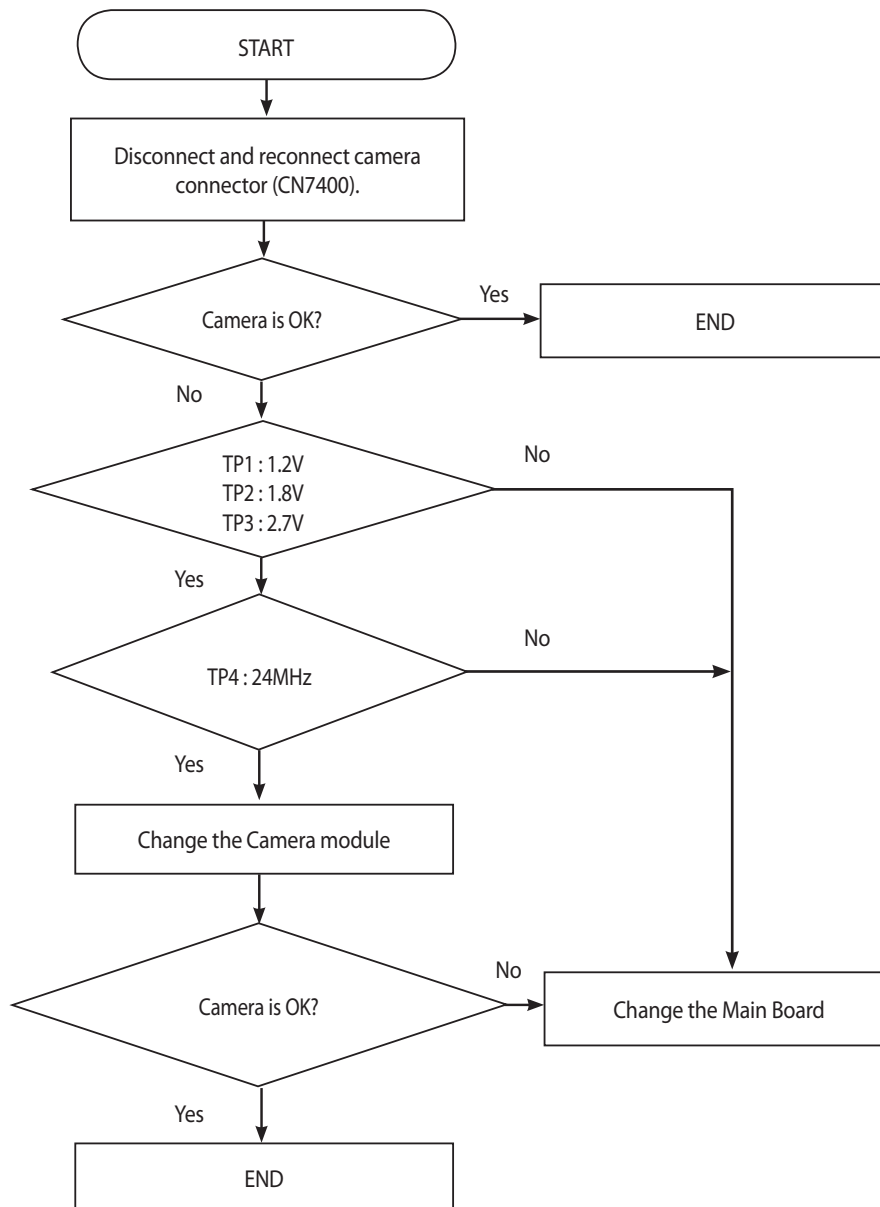
3.19 CAMERA

3.19.1 MAIN CAMERA





3.19.2 VT CAMERA



4. DOWNLOAD

This document is draft version, so the contents will be updated after registering the final one.

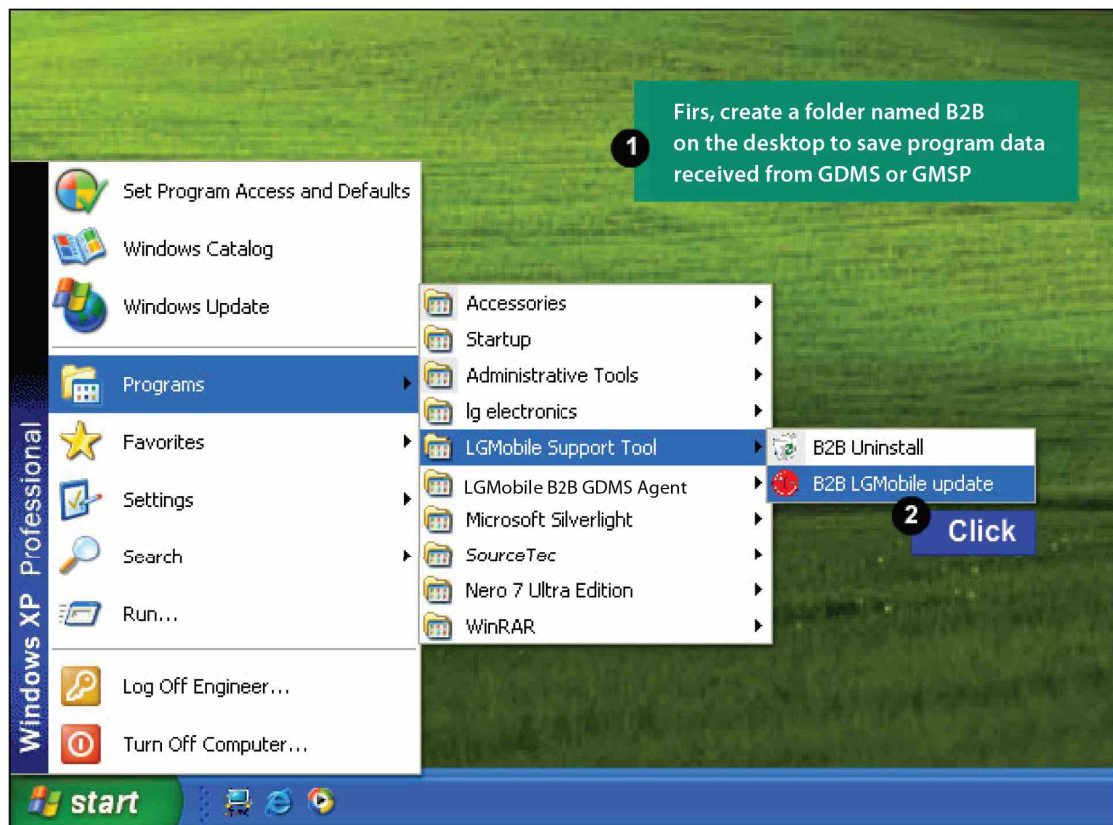
TOOL INFORMATION

TOOL VERSION	DLL NAME	USB DRIVER

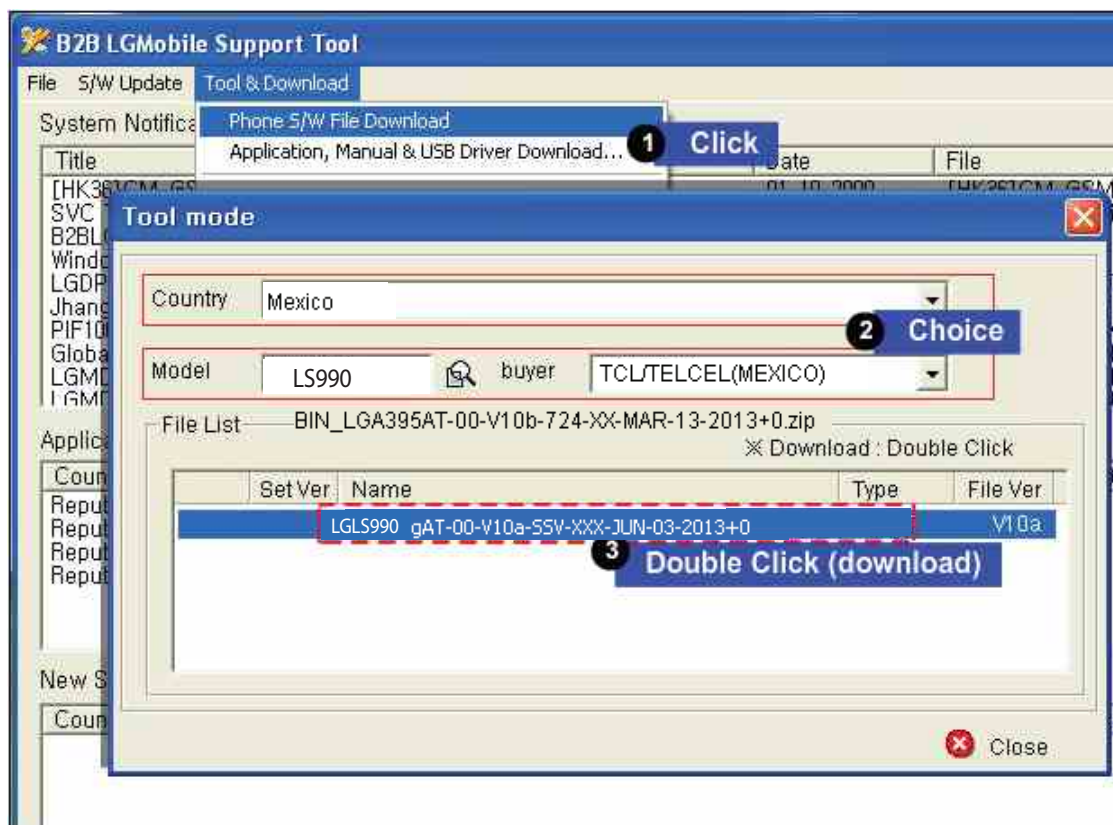
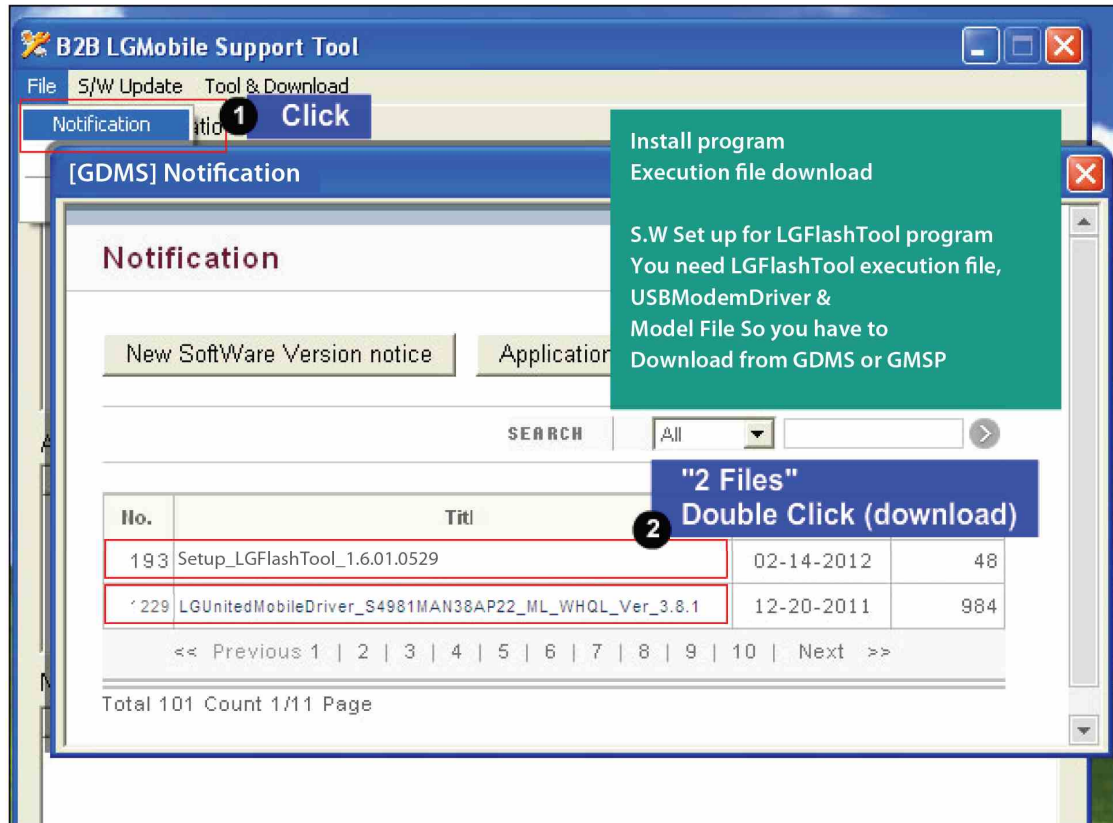
Please Check the Version to "B2B"

H/W

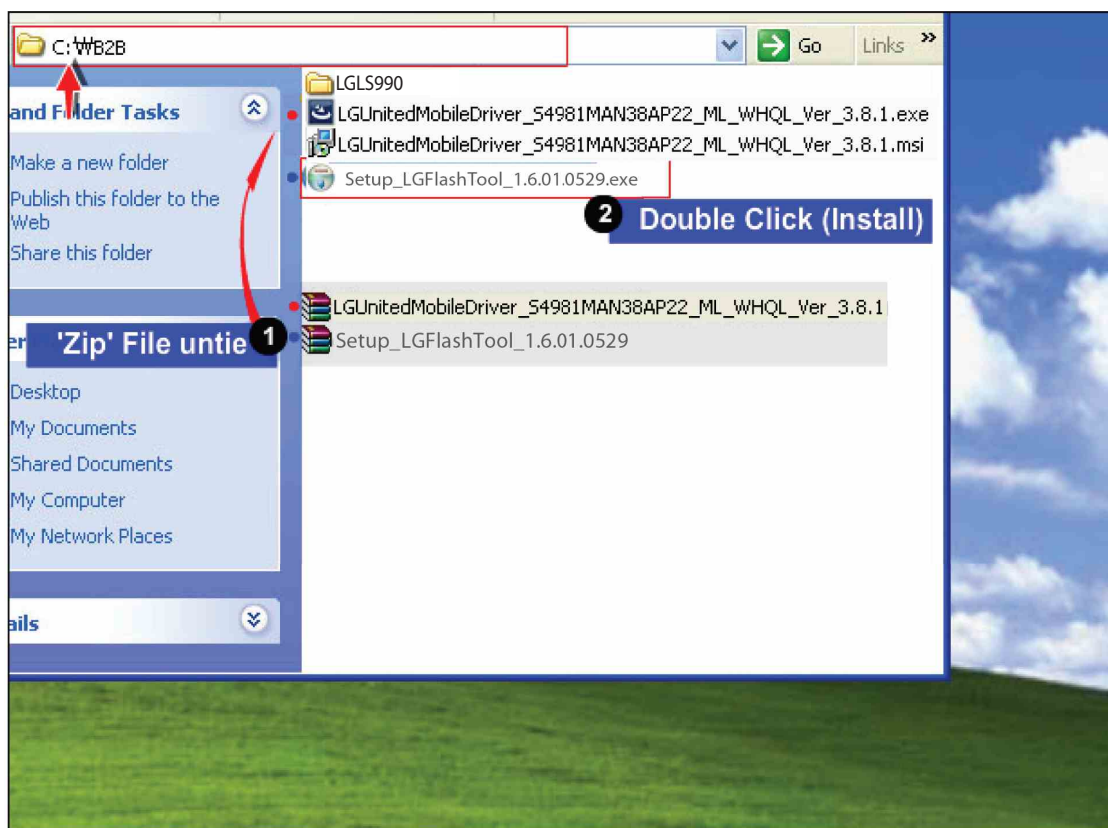
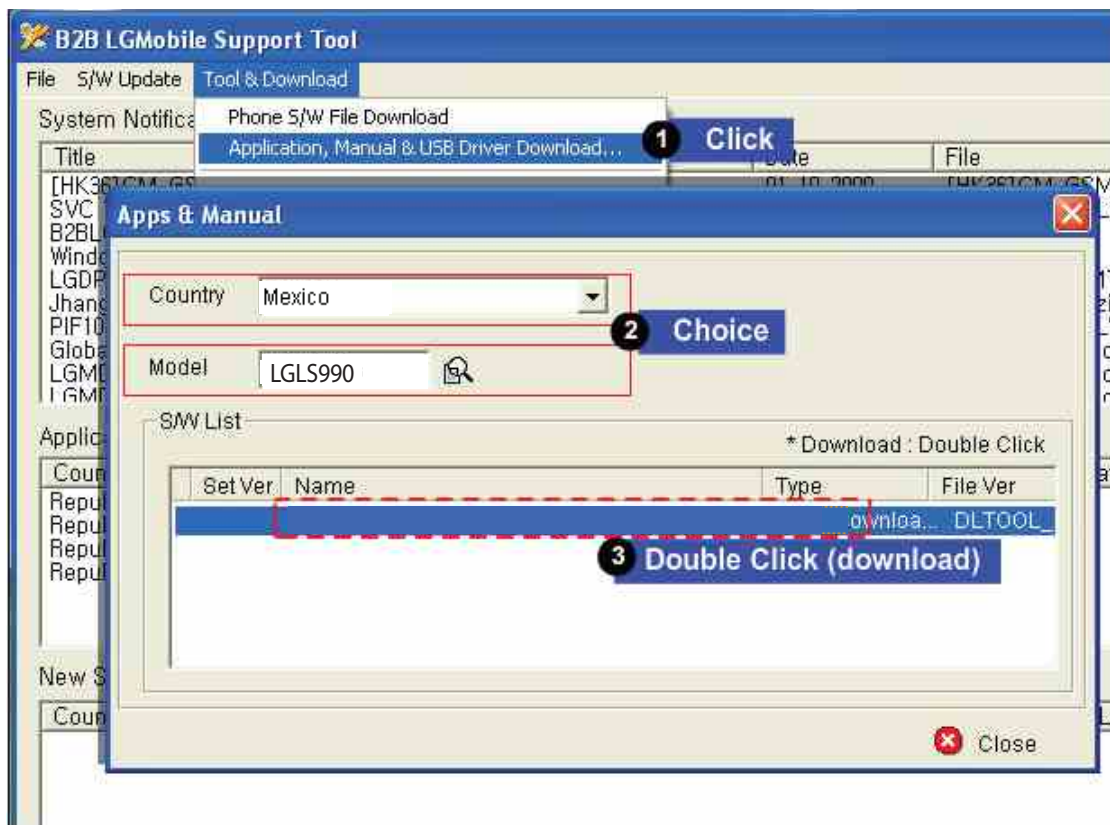
	Name	Part No.	SW
D/L Cable			



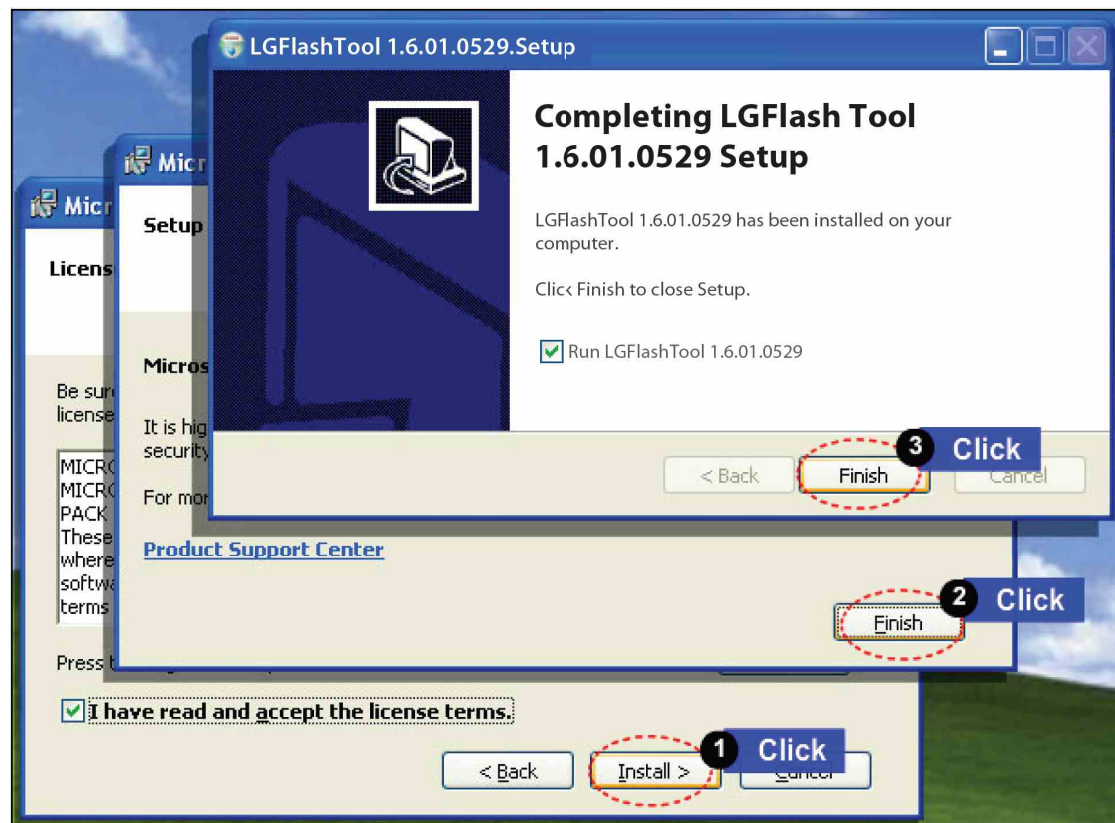
4. DOWNLOAD



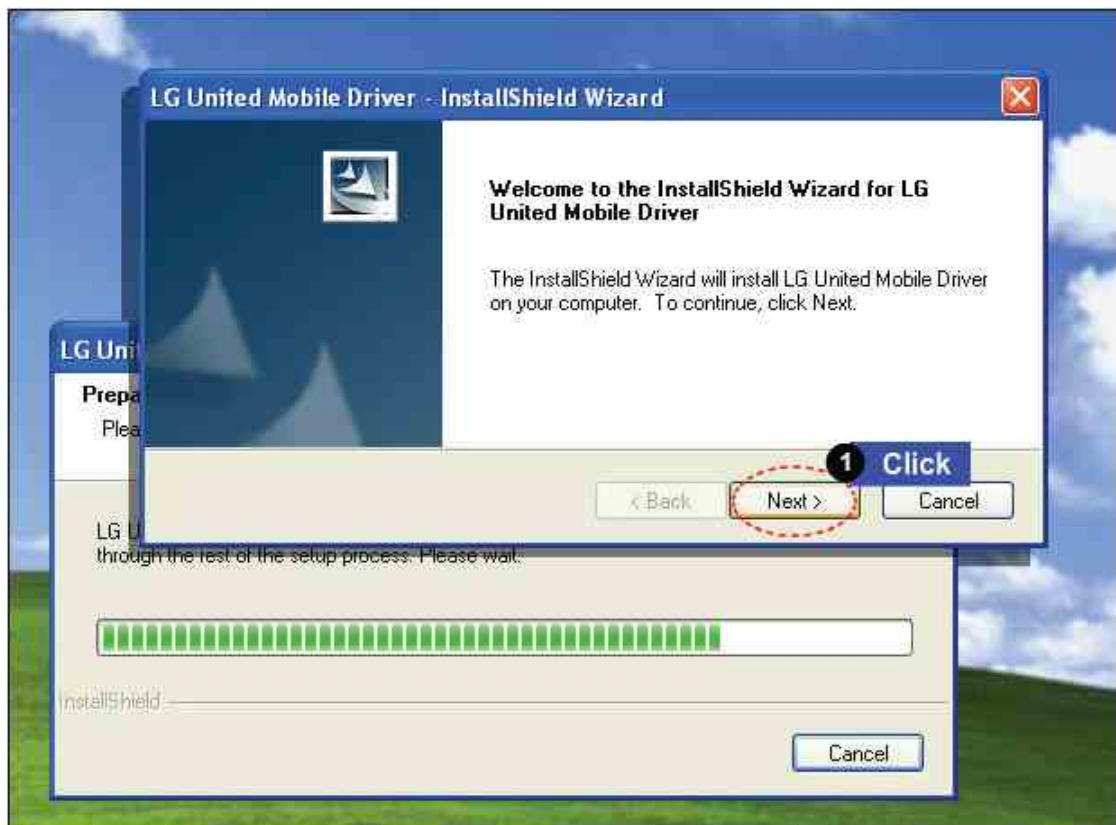
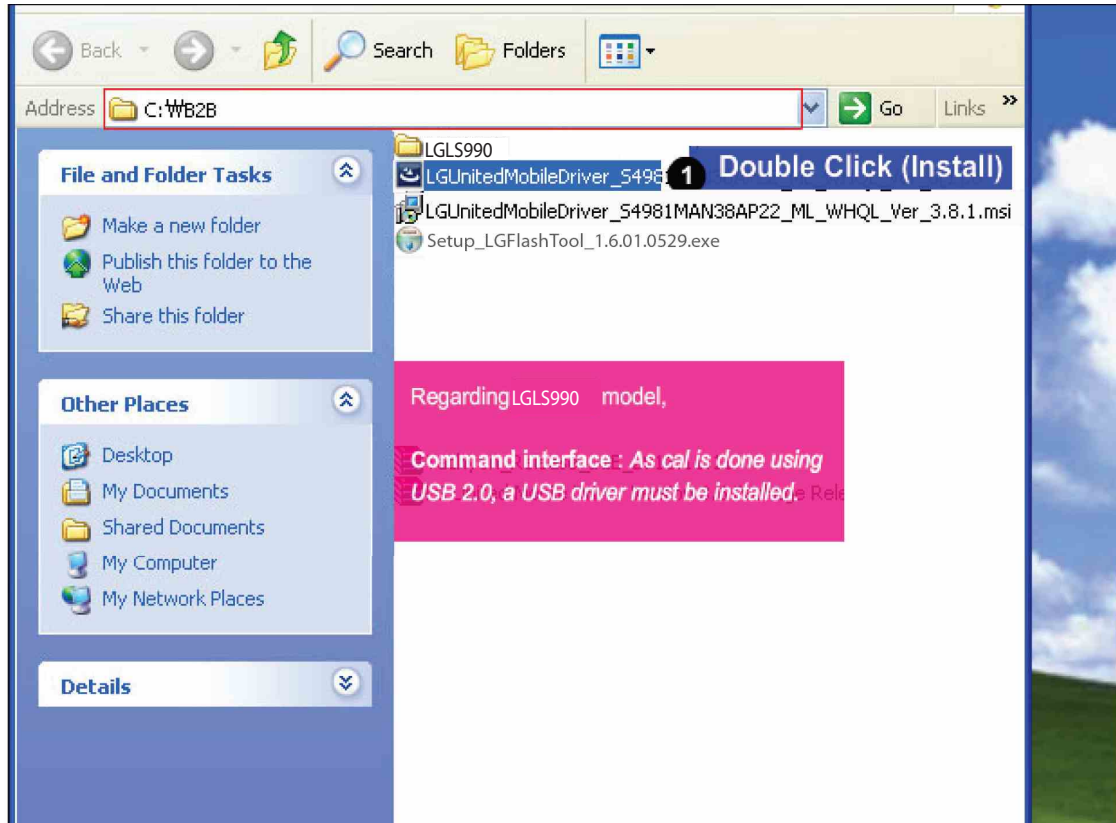
4. DOWNLOAD



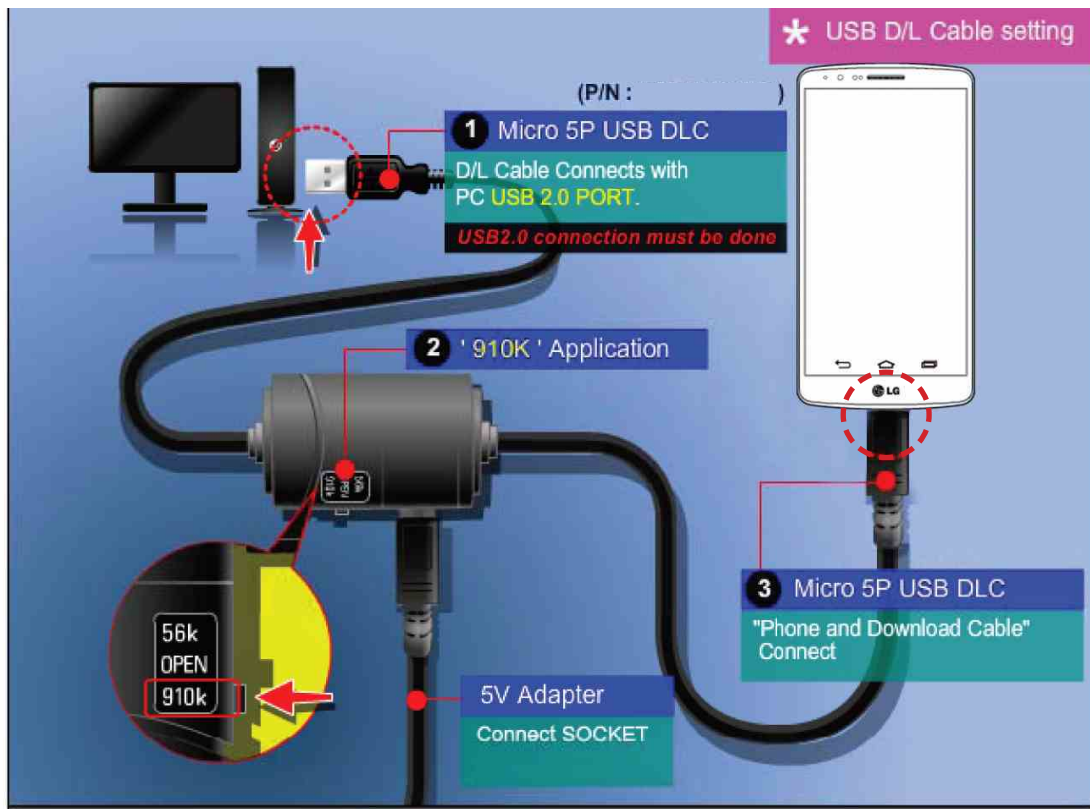
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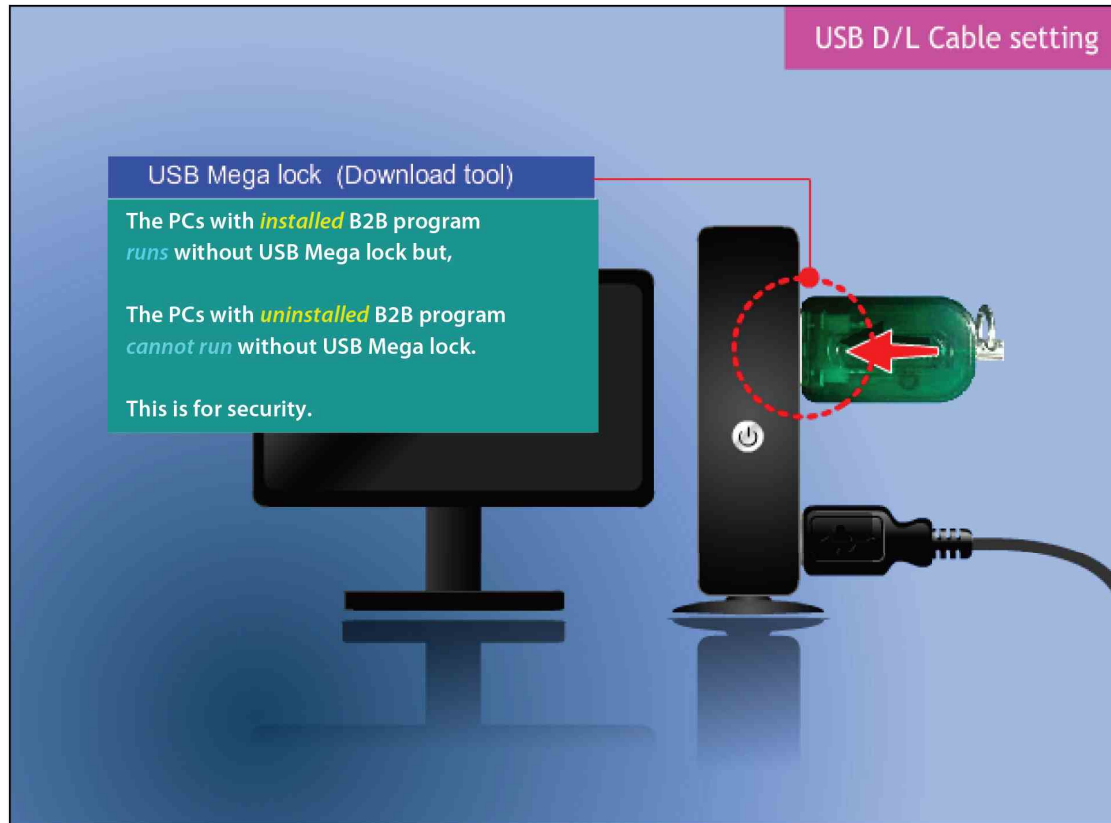
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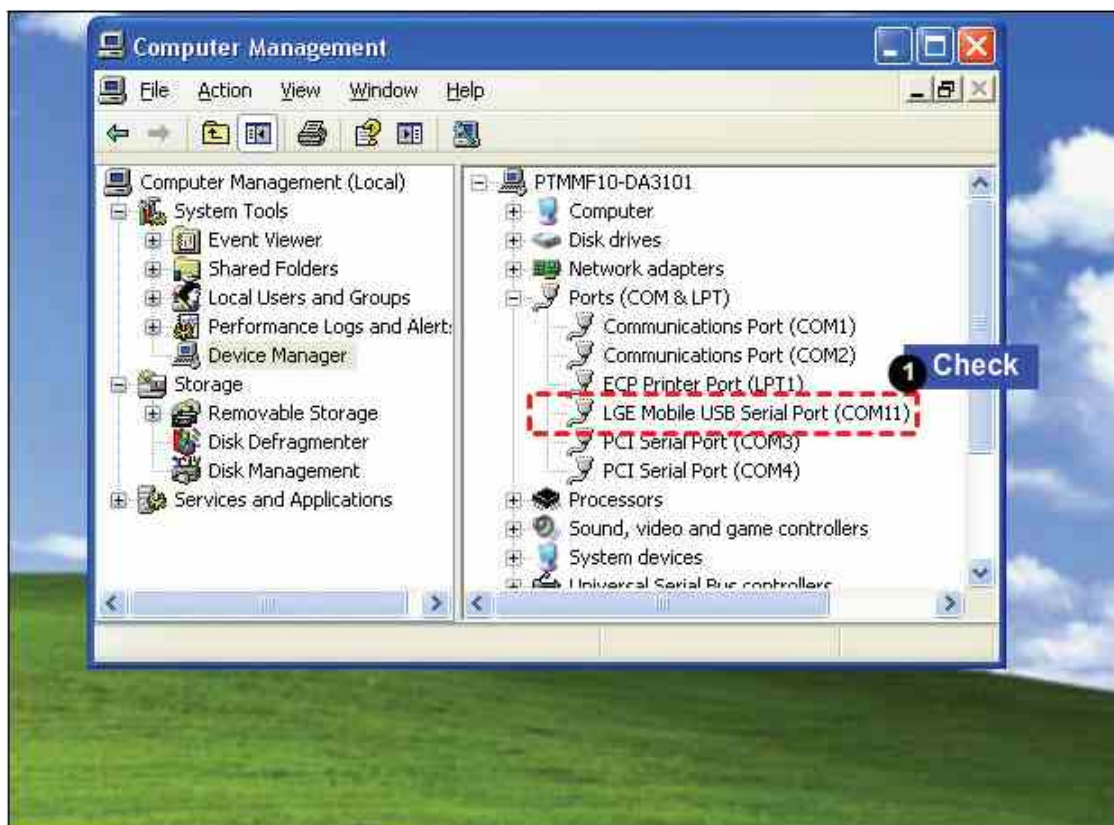
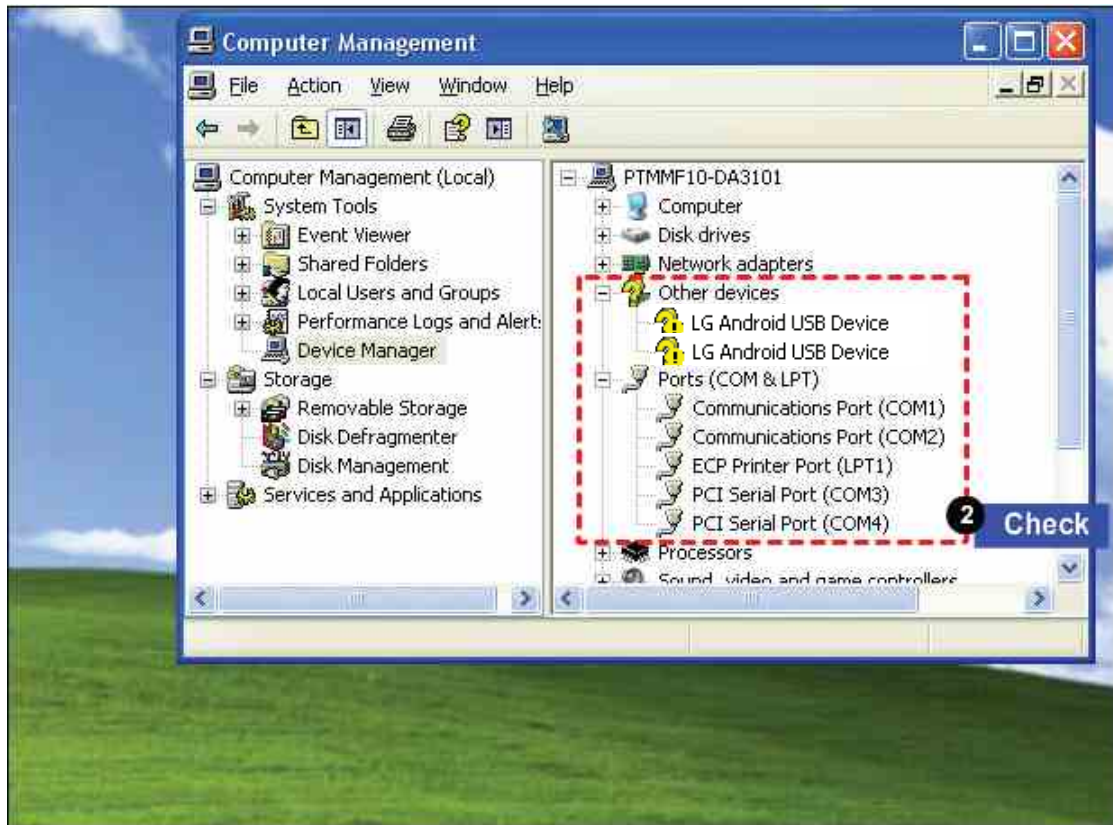


4. DOWNLOAD

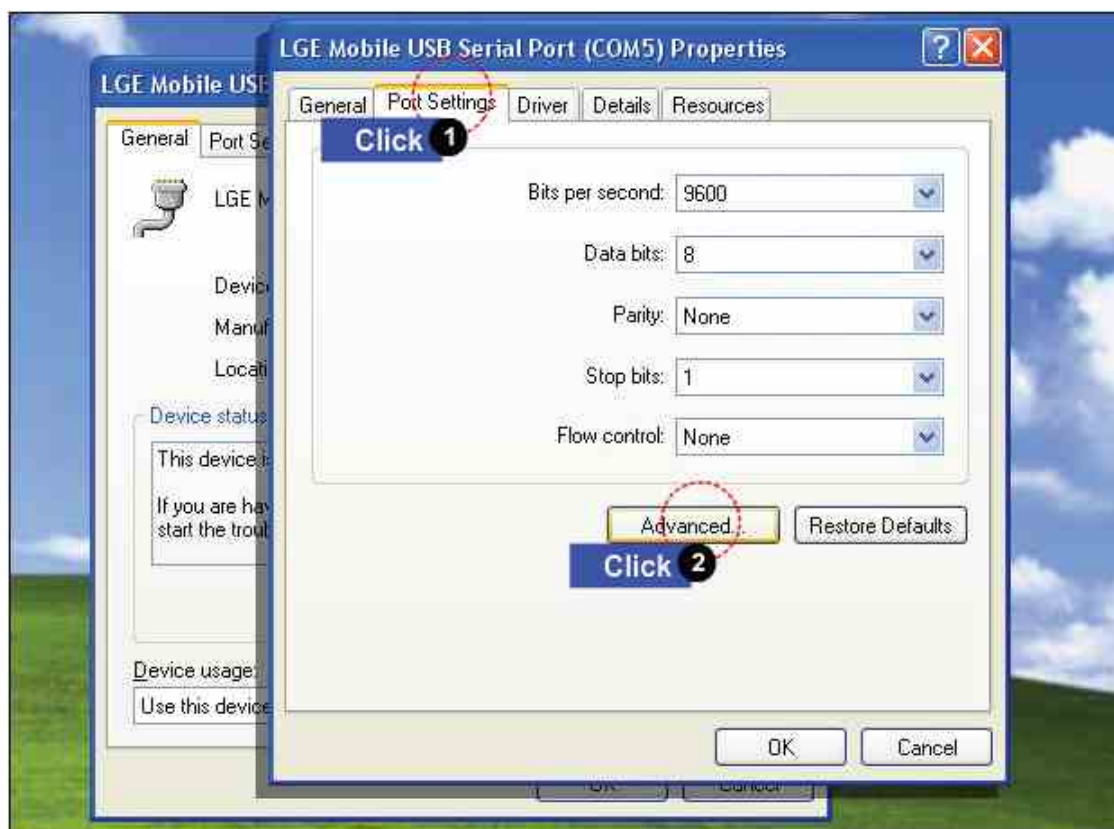
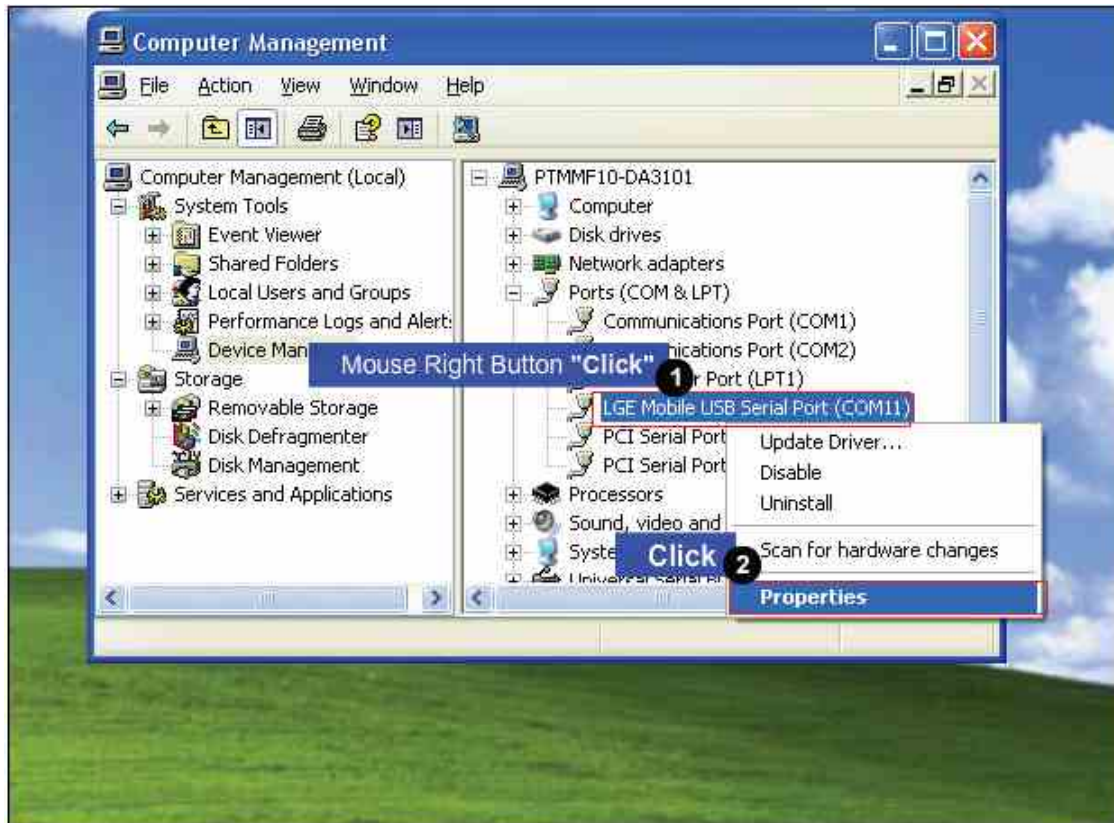


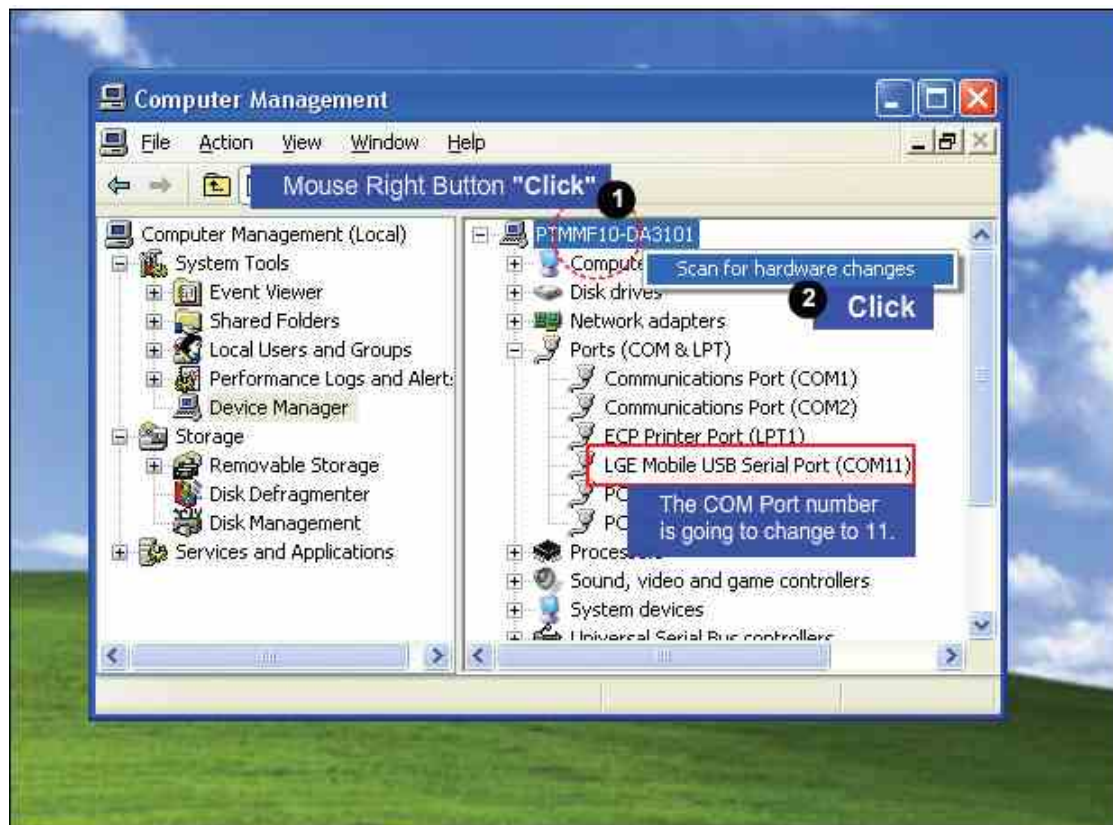
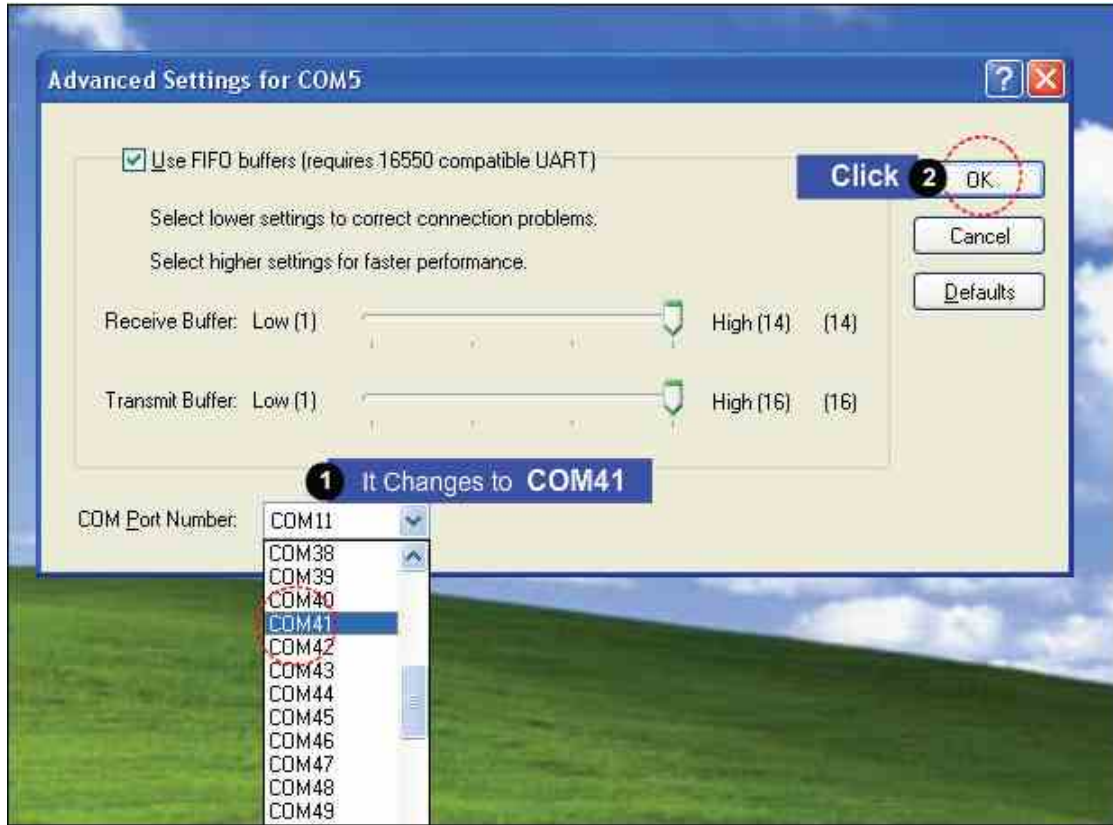
4. DOWNLOAD



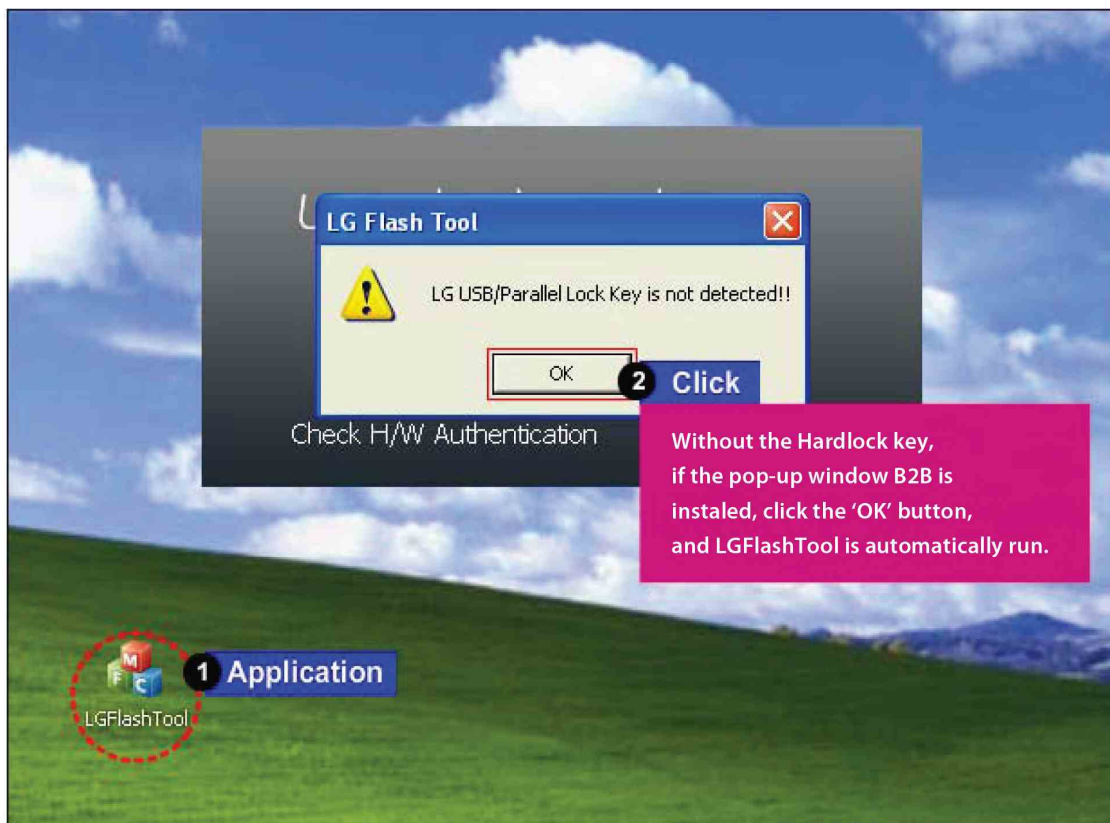
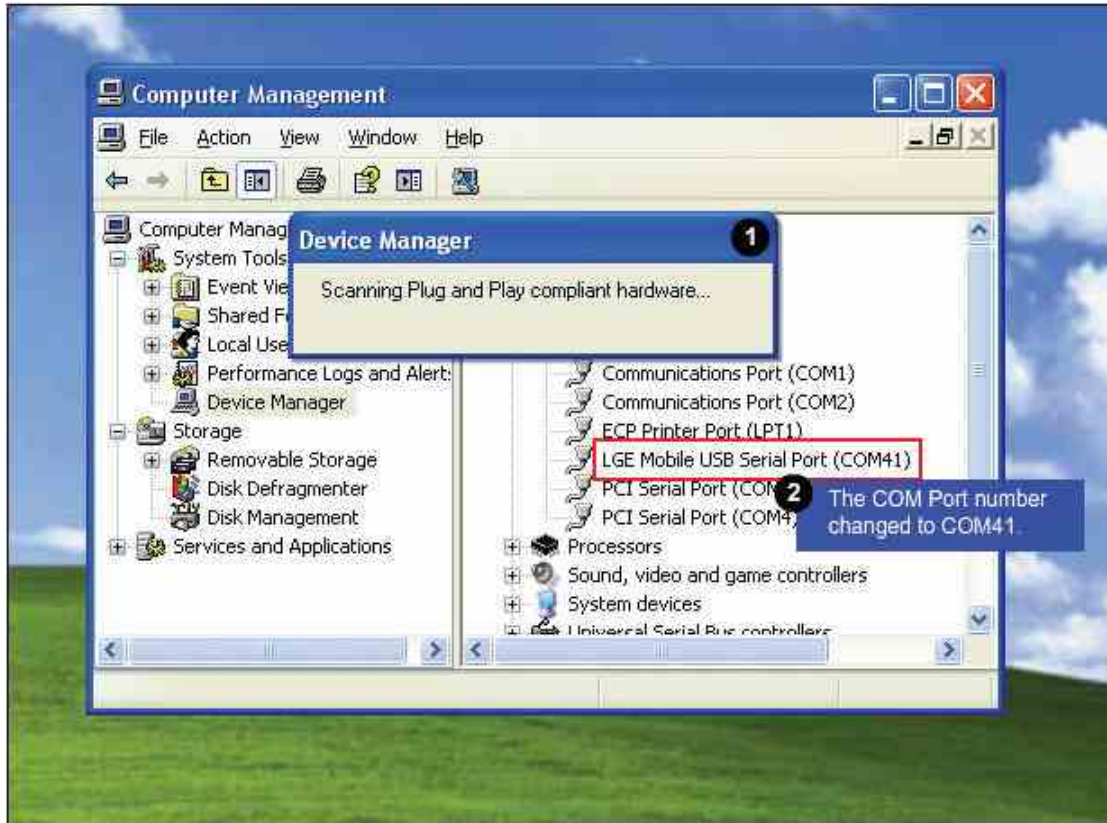


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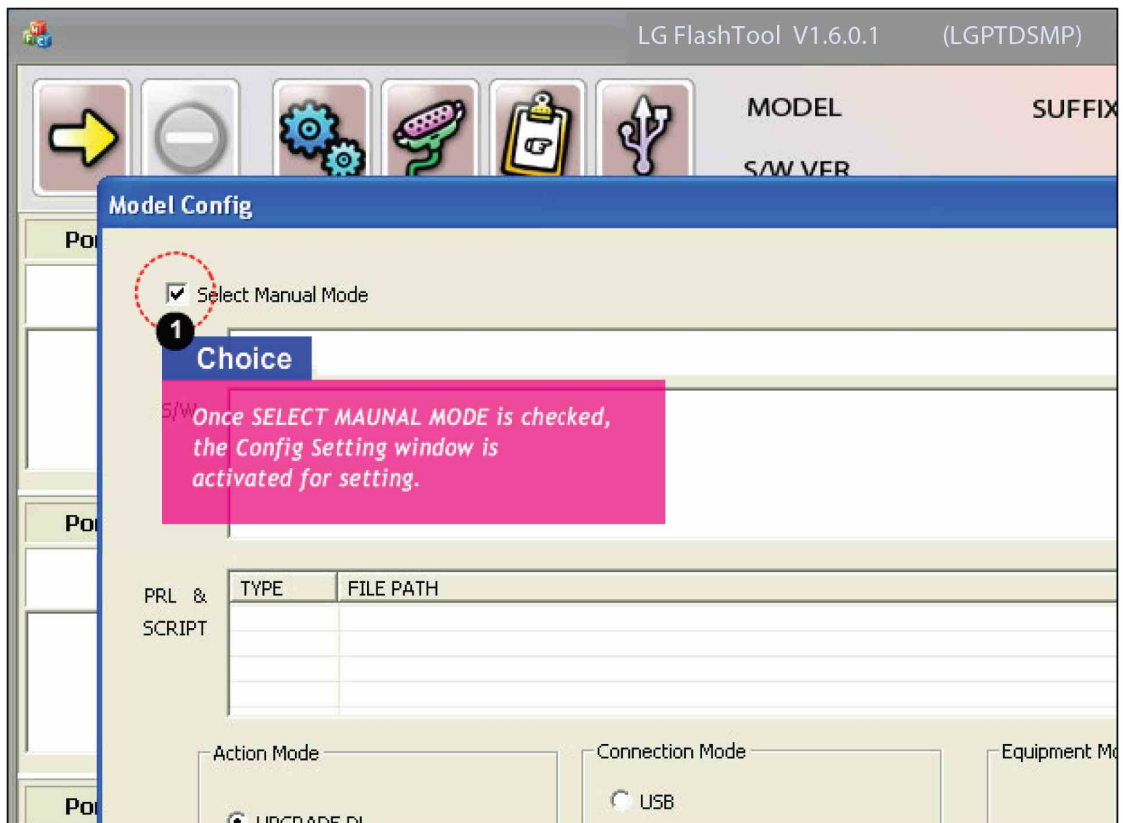
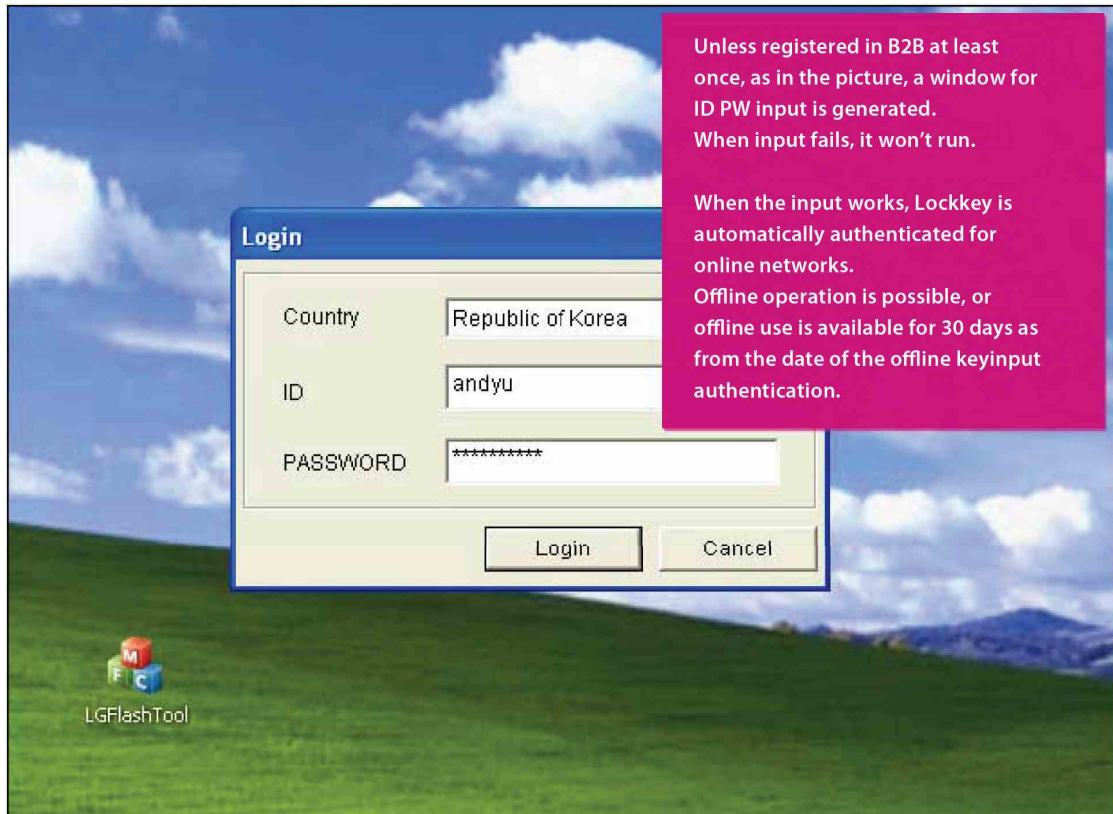




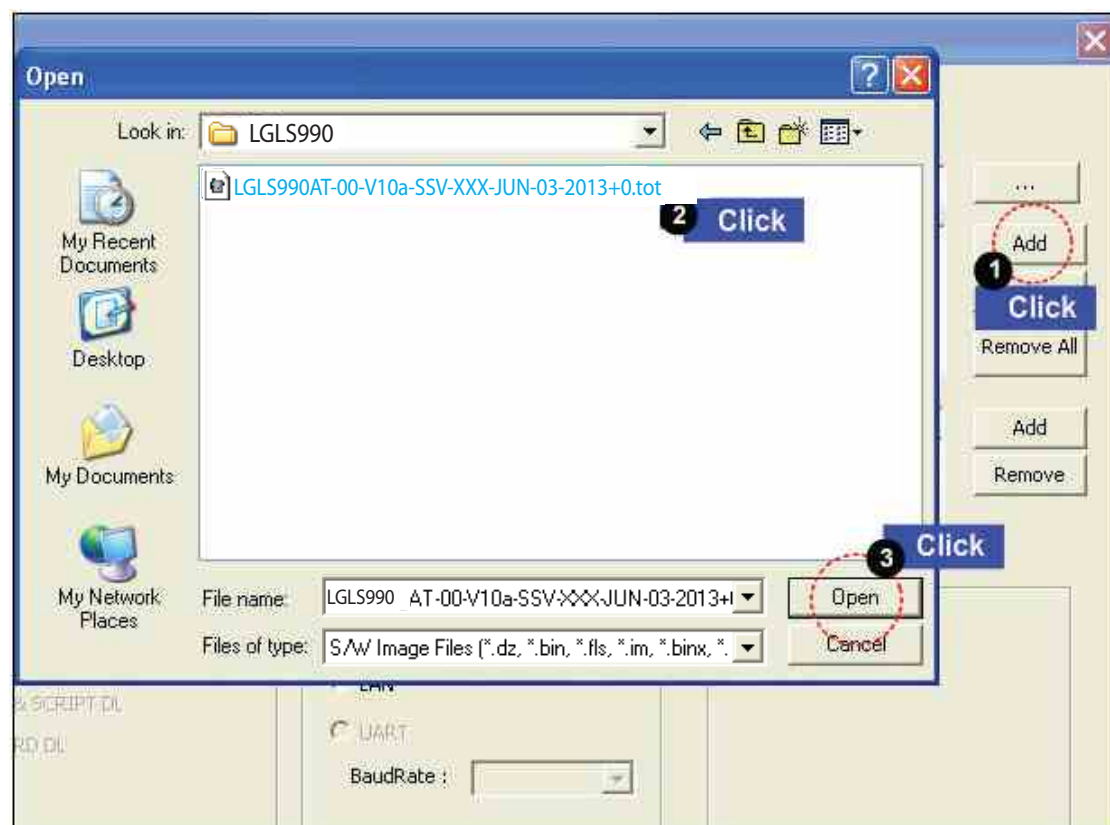
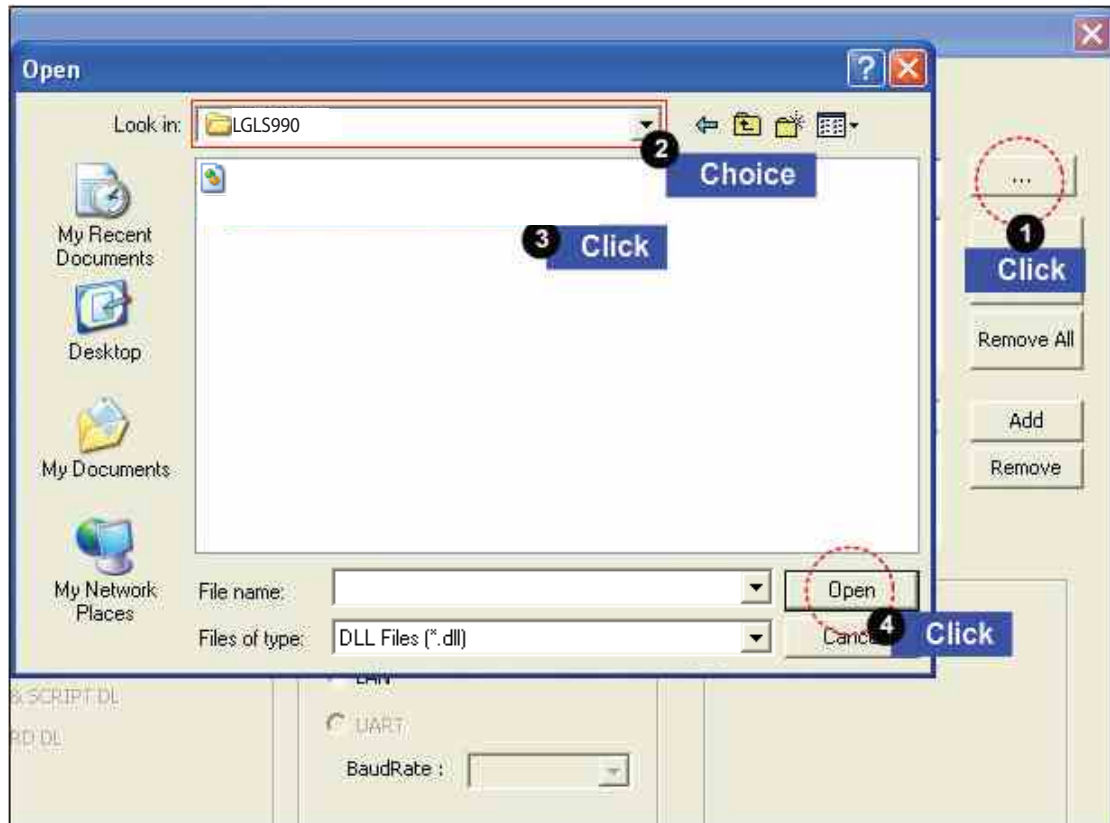
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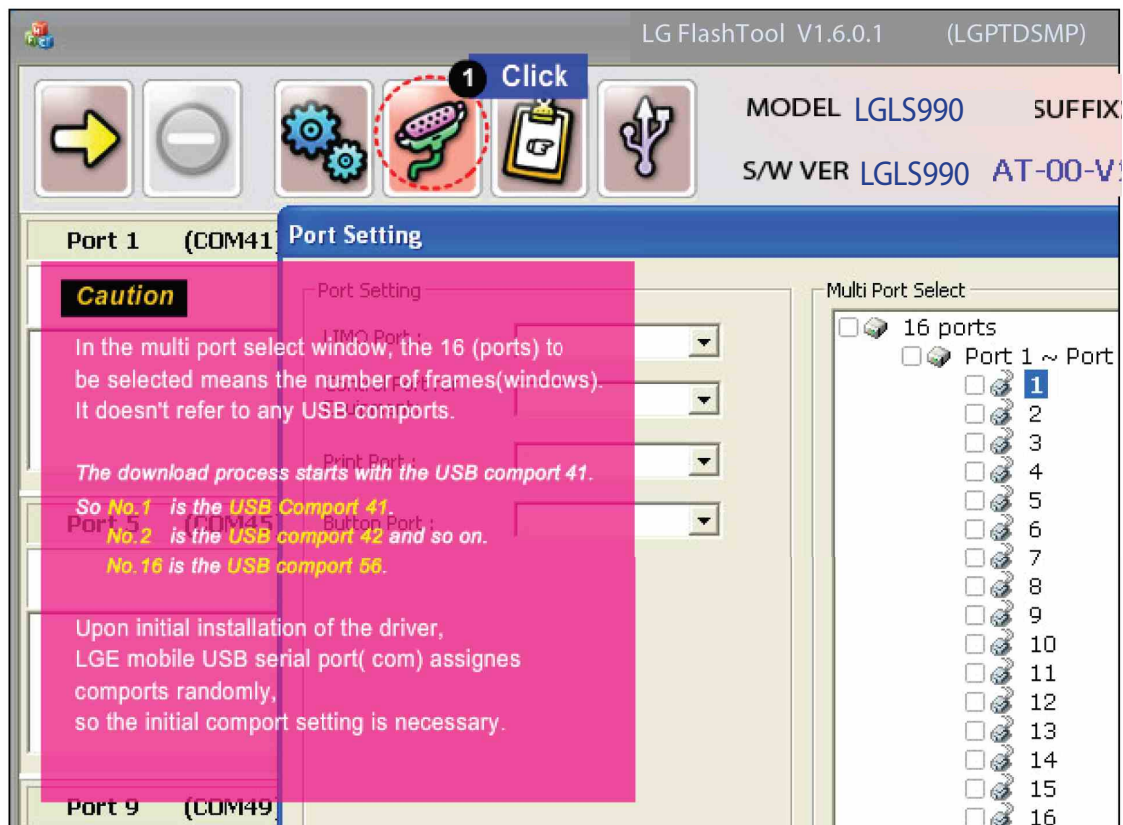
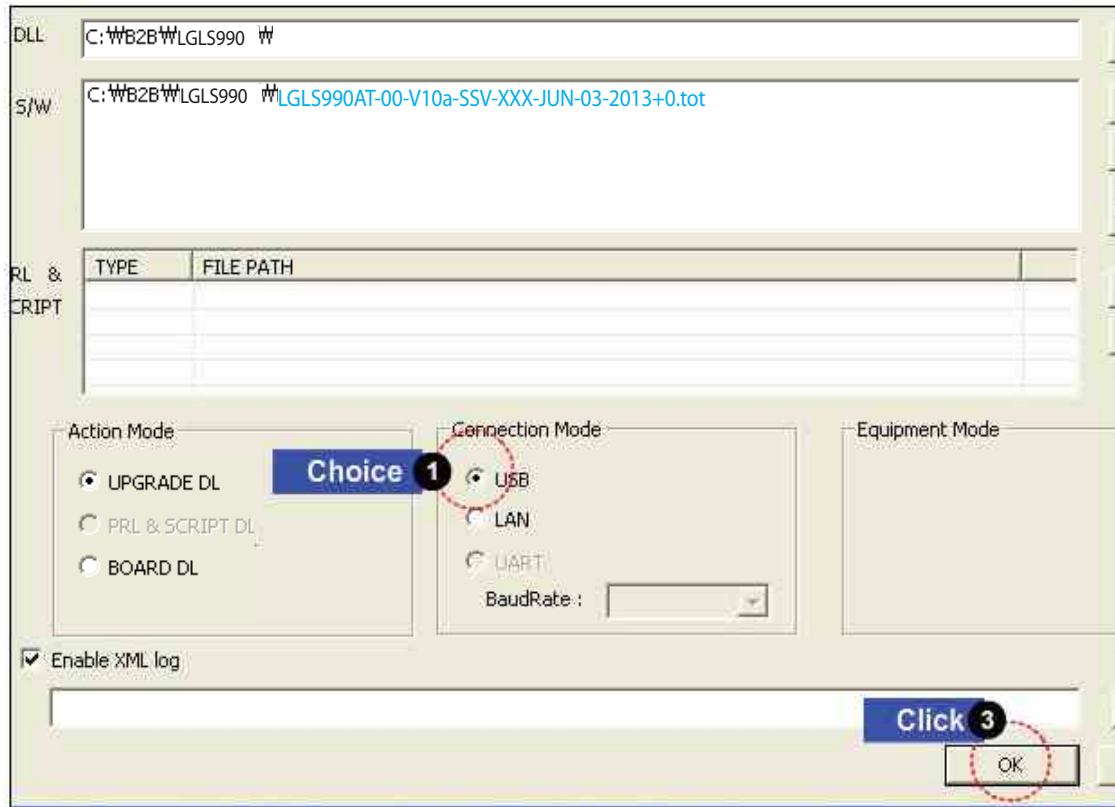
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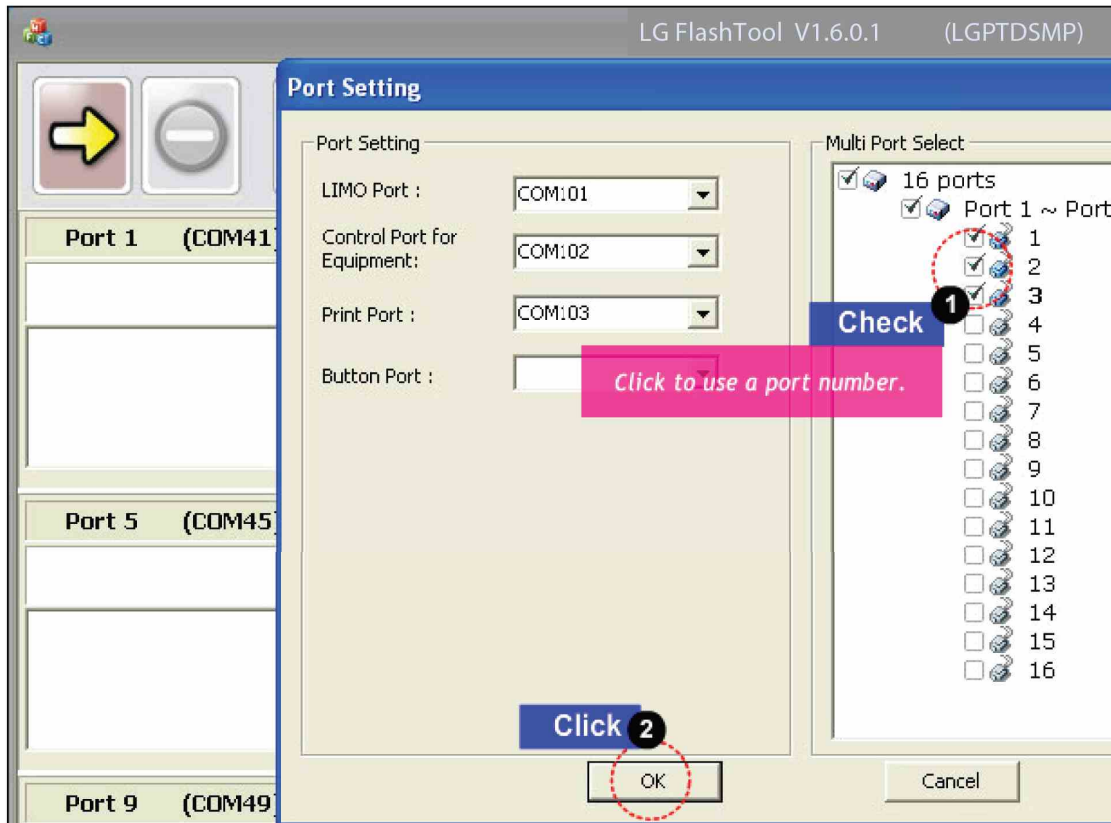
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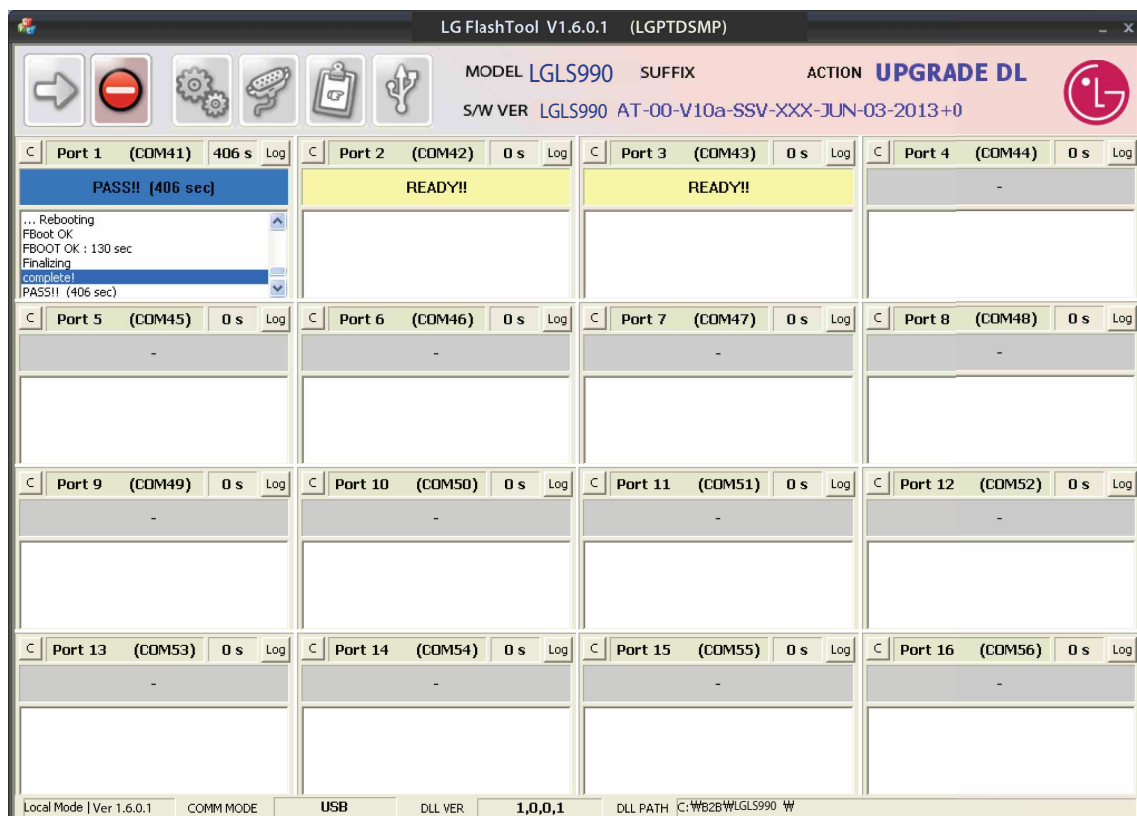
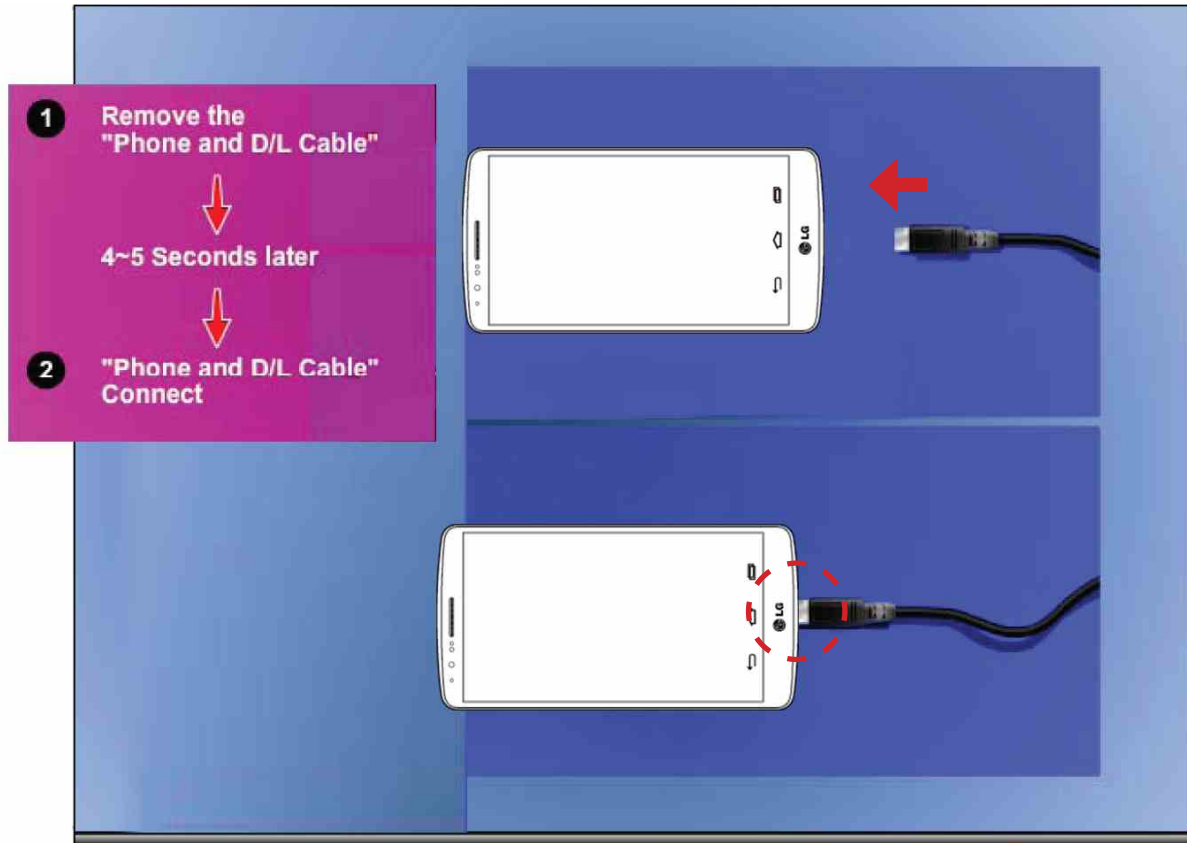
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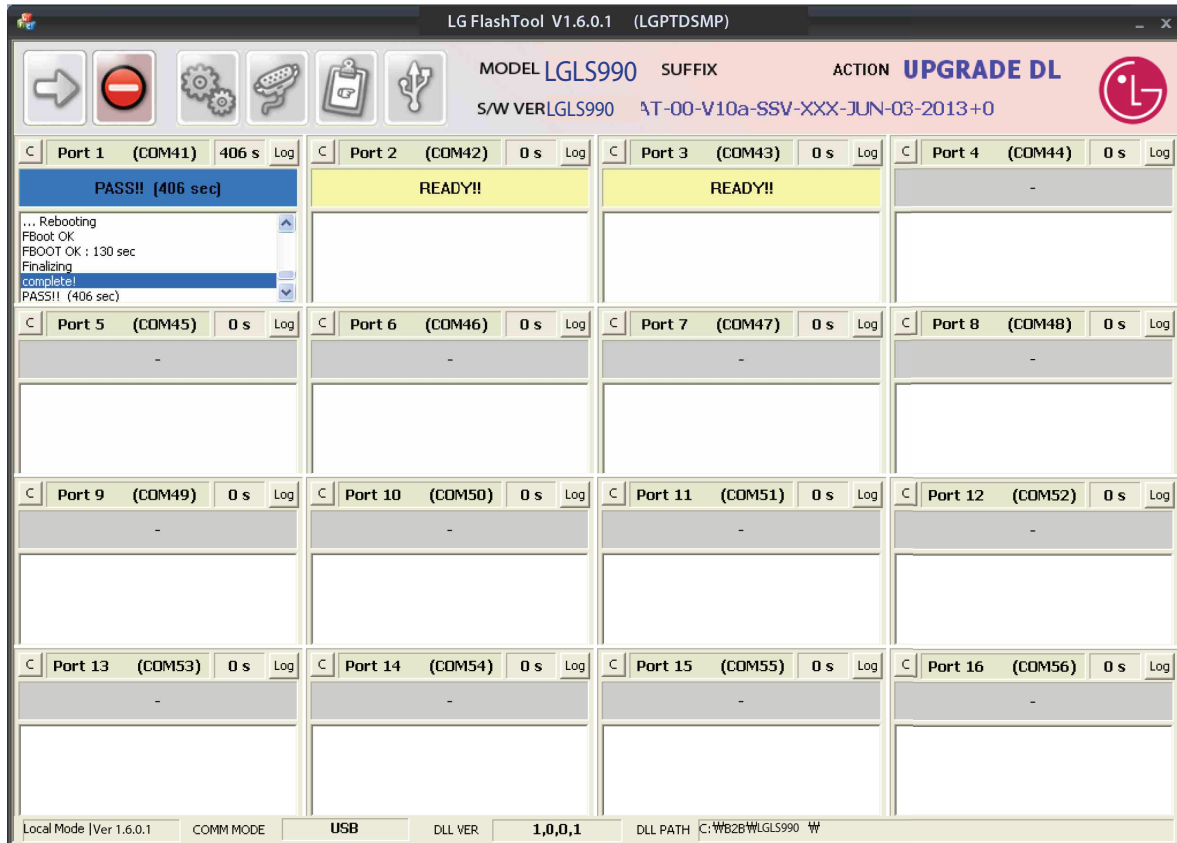
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4. DOWNLOAD

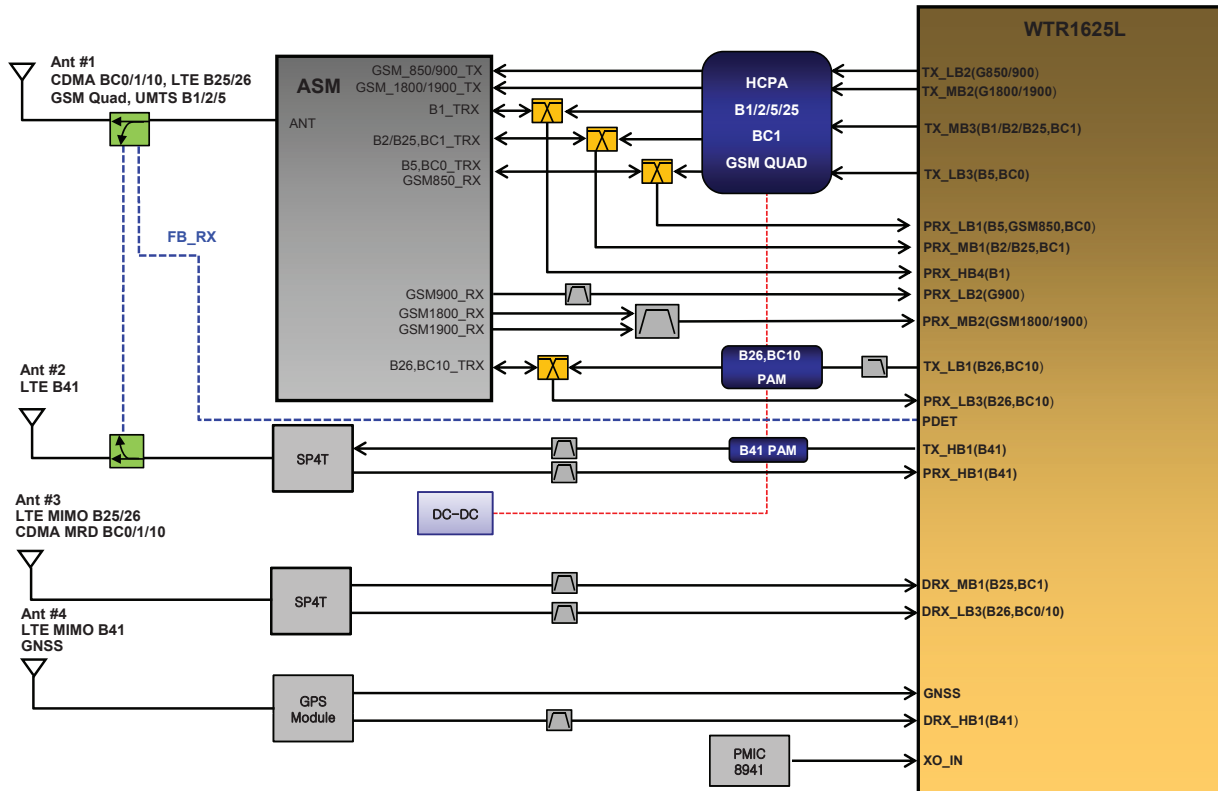


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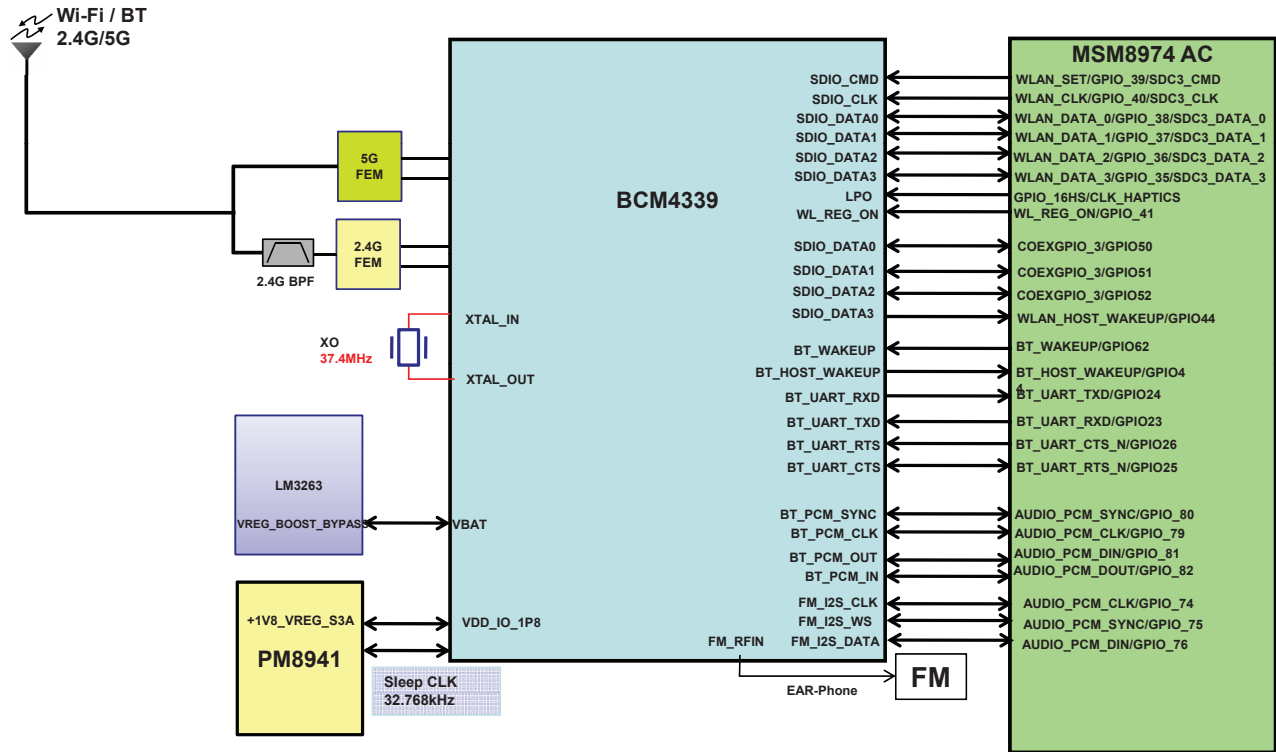
5. BLOCK DIAGRAM

[LS990] RF Block Diagram



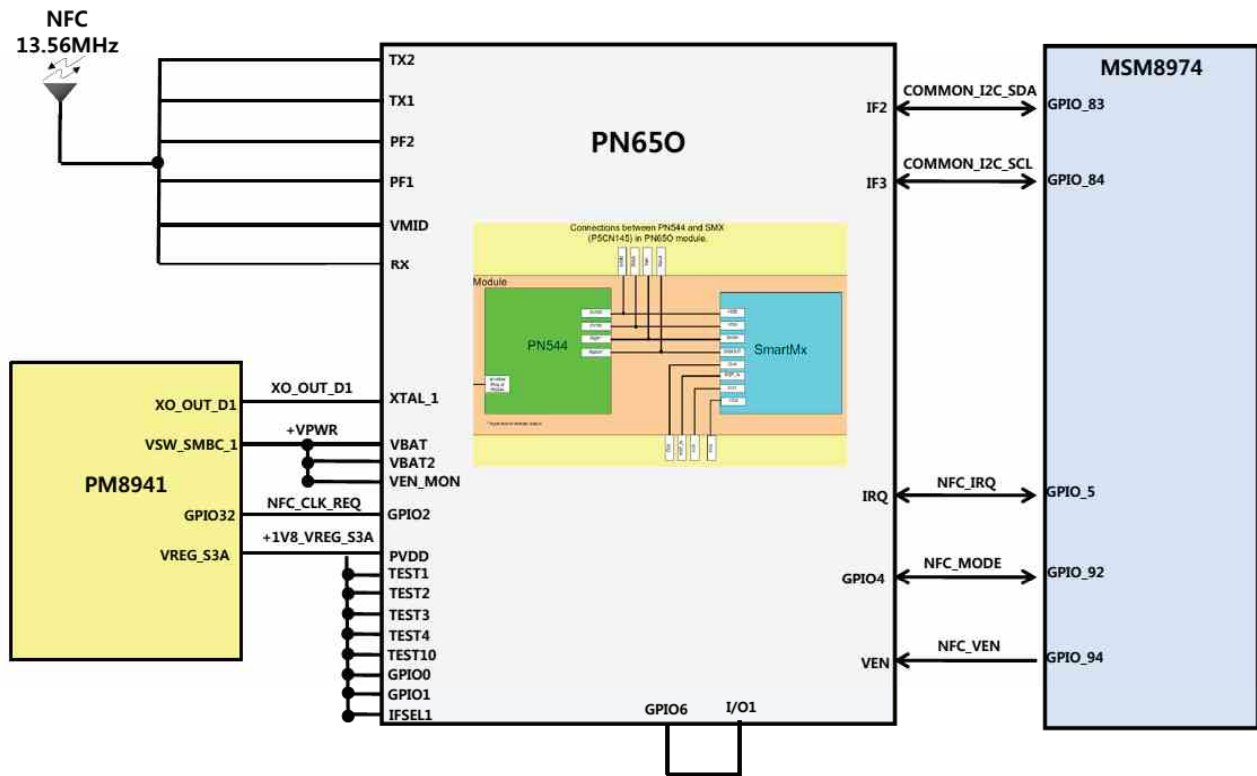
5. BLOCK DIAGRAM

[LS990] BT/Wi-Fi

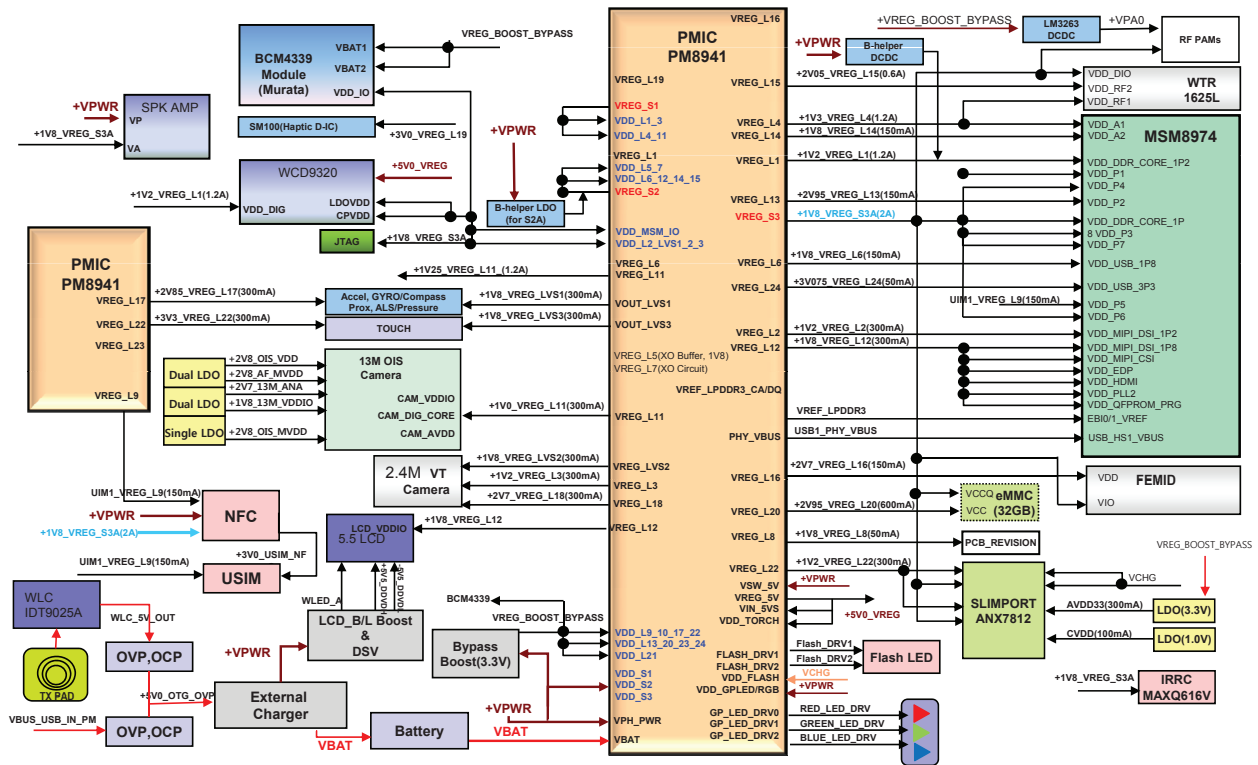


5. BLOCK DIAGRAM

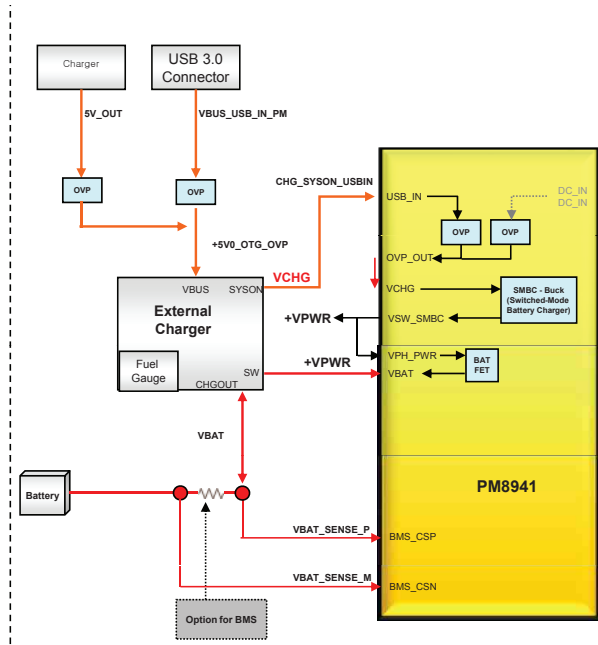
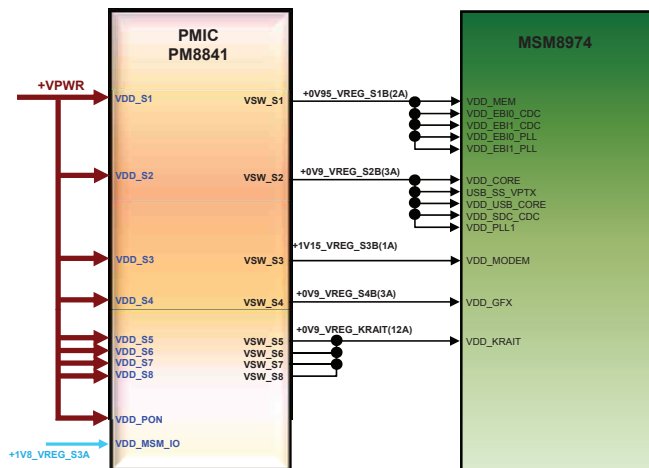
[LS990] NFC



[LS990] PM8941 Output Power Management

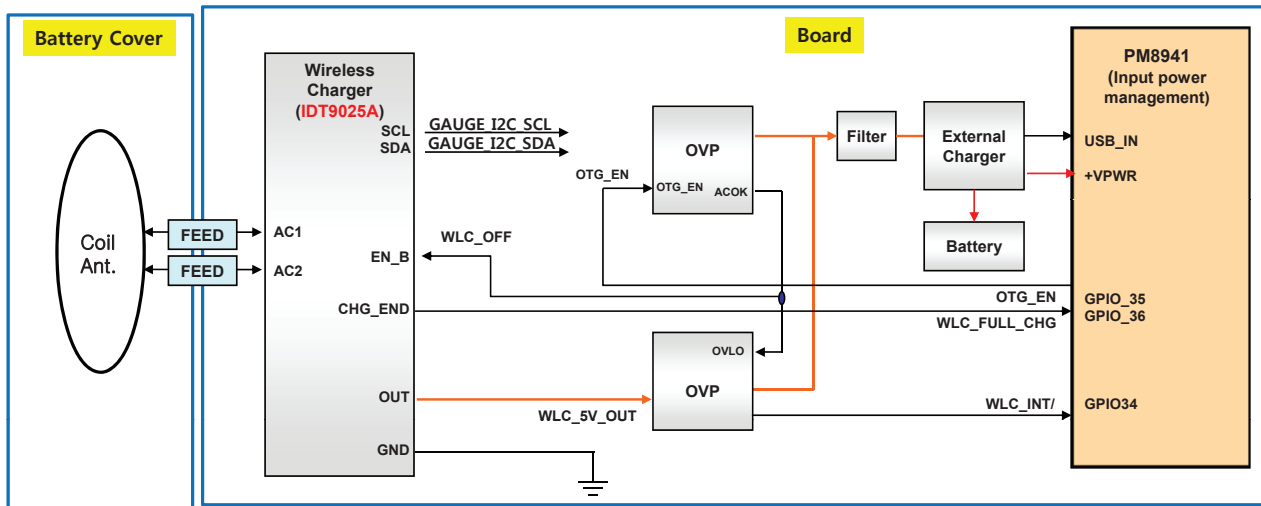


[LS990] PM8841 SMPS/PM8941 Input Power Management



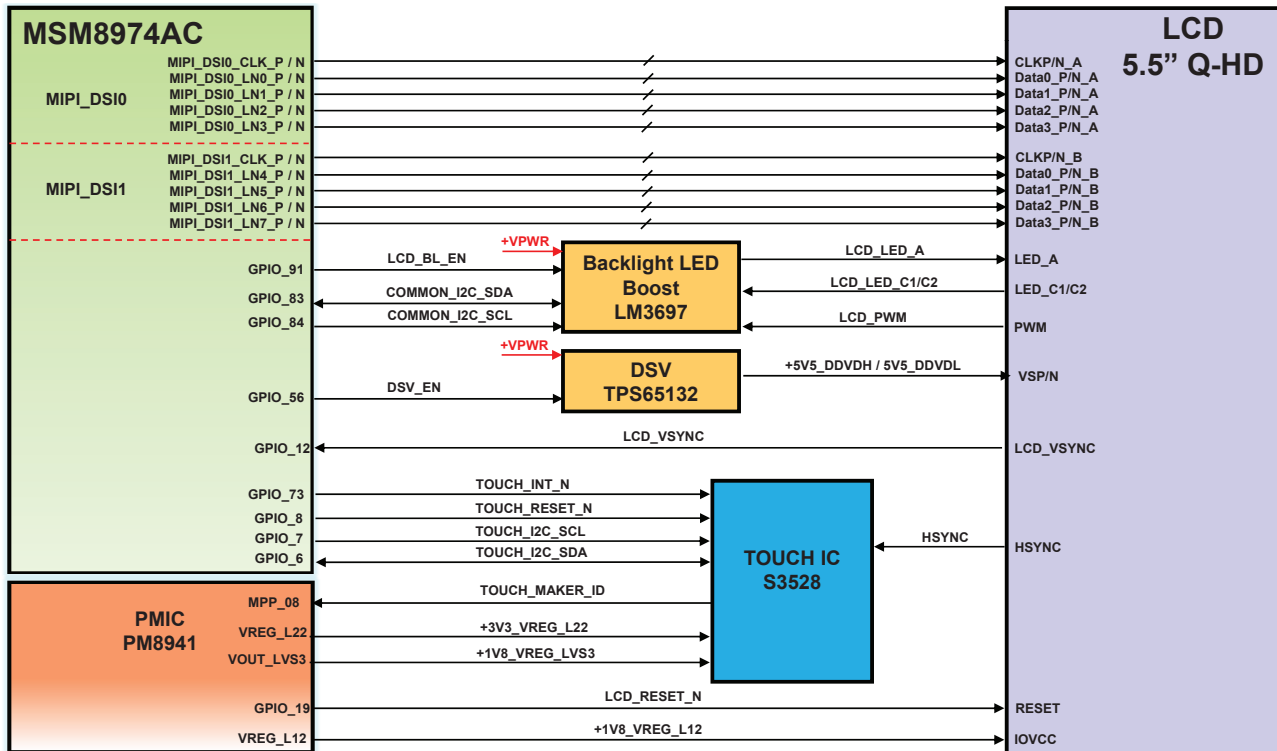
5. BLOCK DIAGRAM

[LS990] Wireless charger



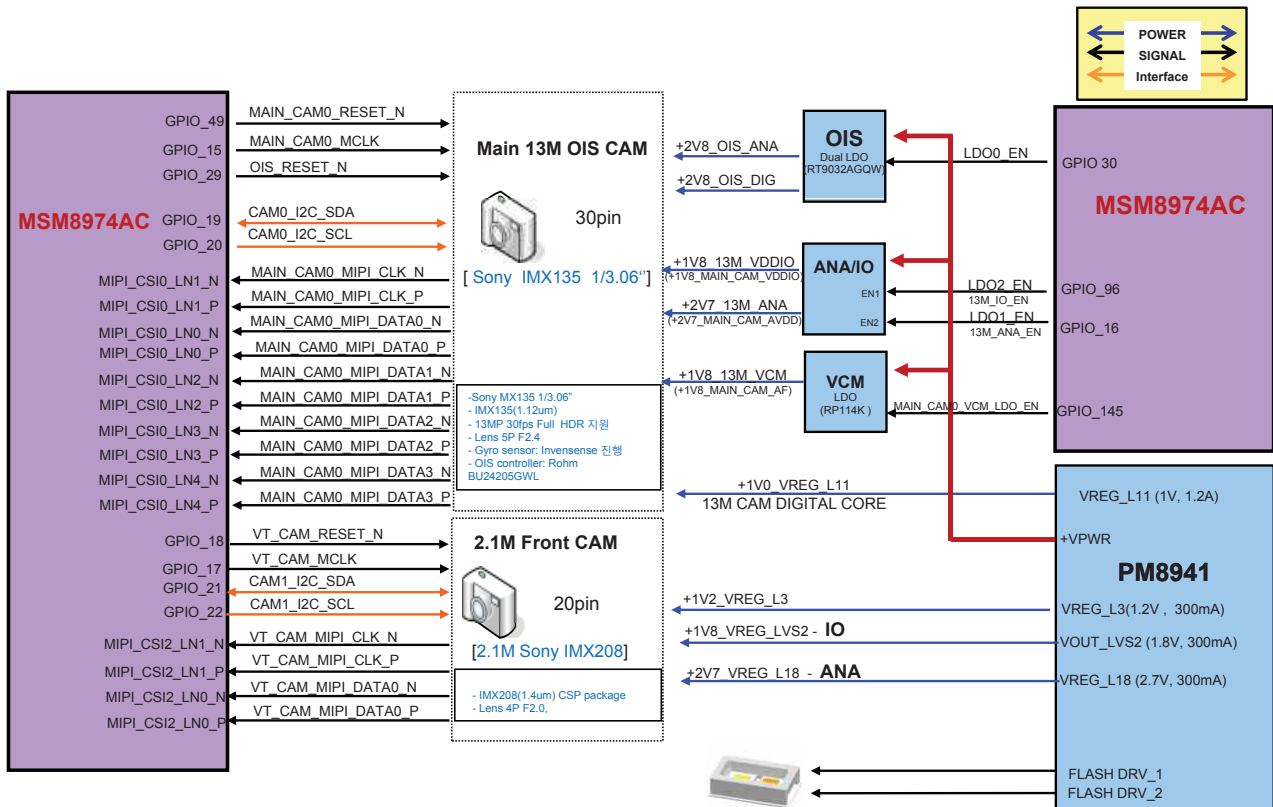
5. BLOCK DIAGRAM

[LS990] Main Display I/F



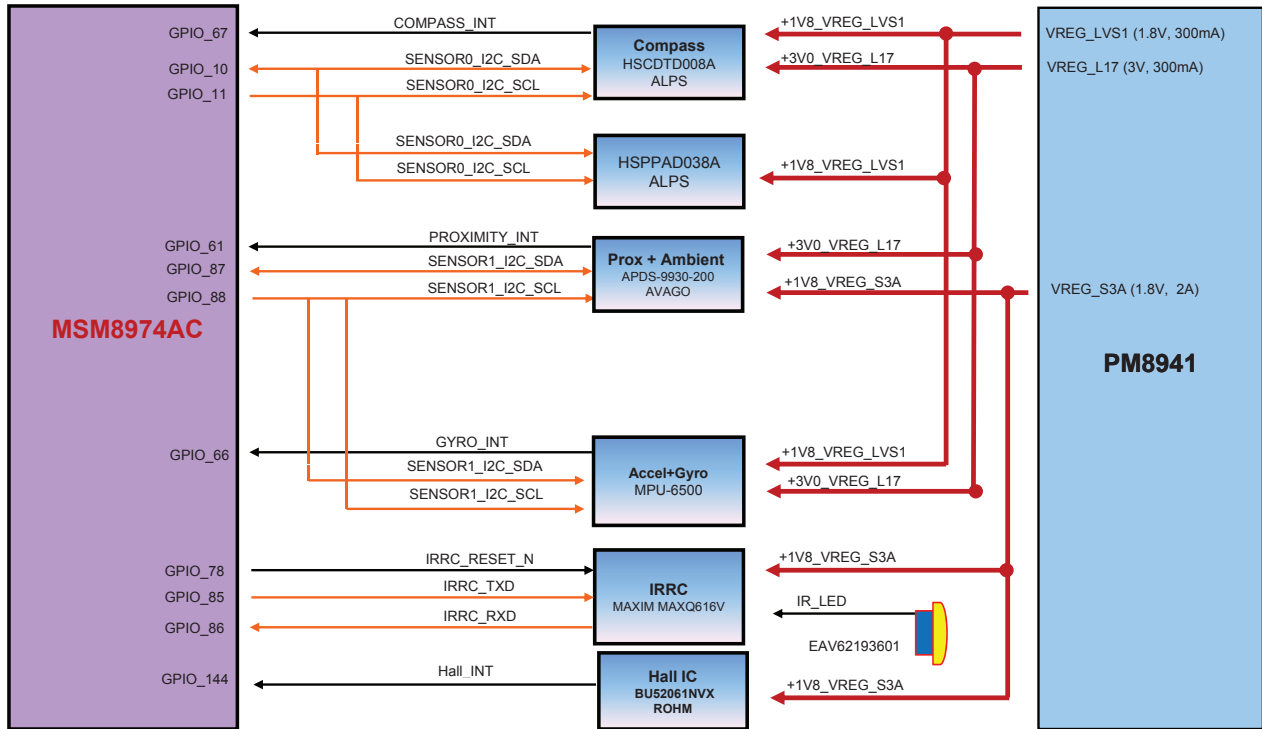
5. BLOCK DIAGRAM

[LS990] Camera I/F



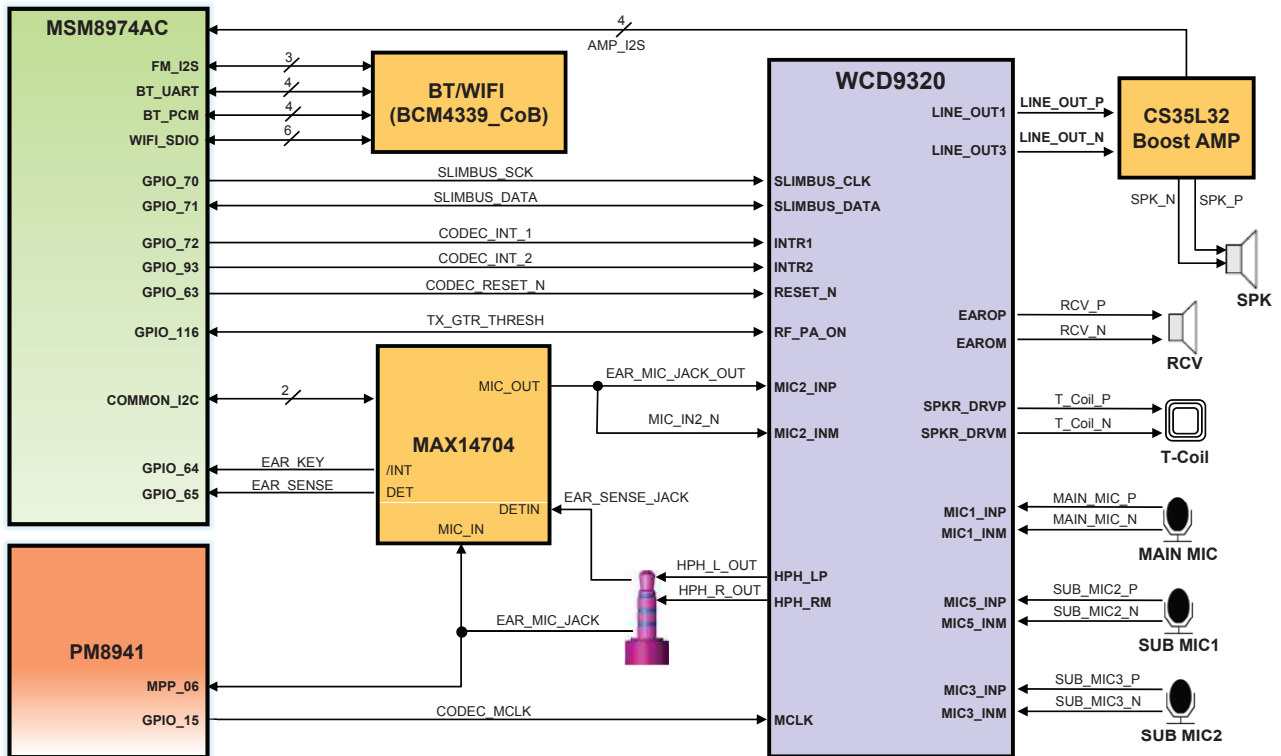
5. BLOCK DIAGRAM

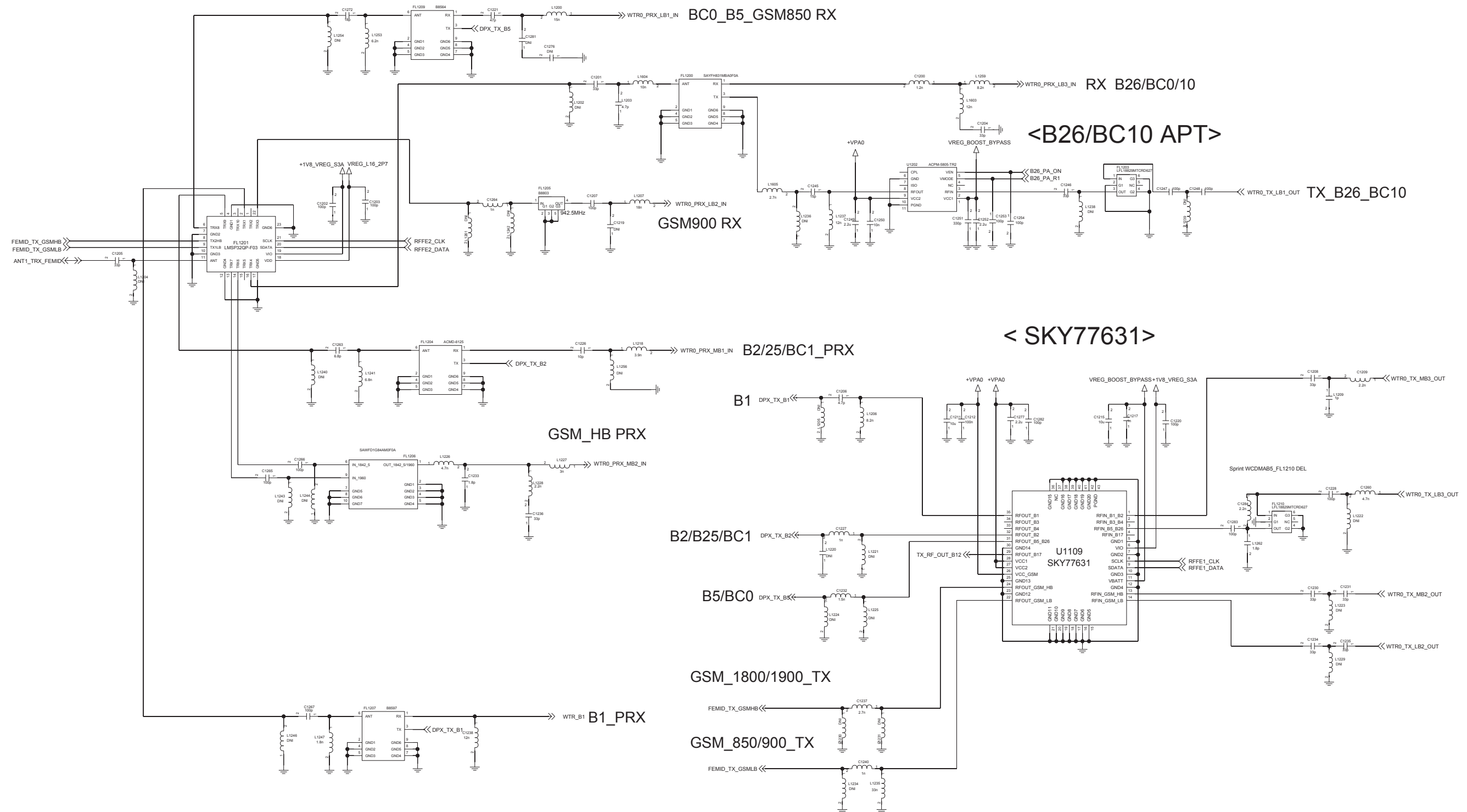
[LS990] Sensor



5. BLOCK DIAGRAM

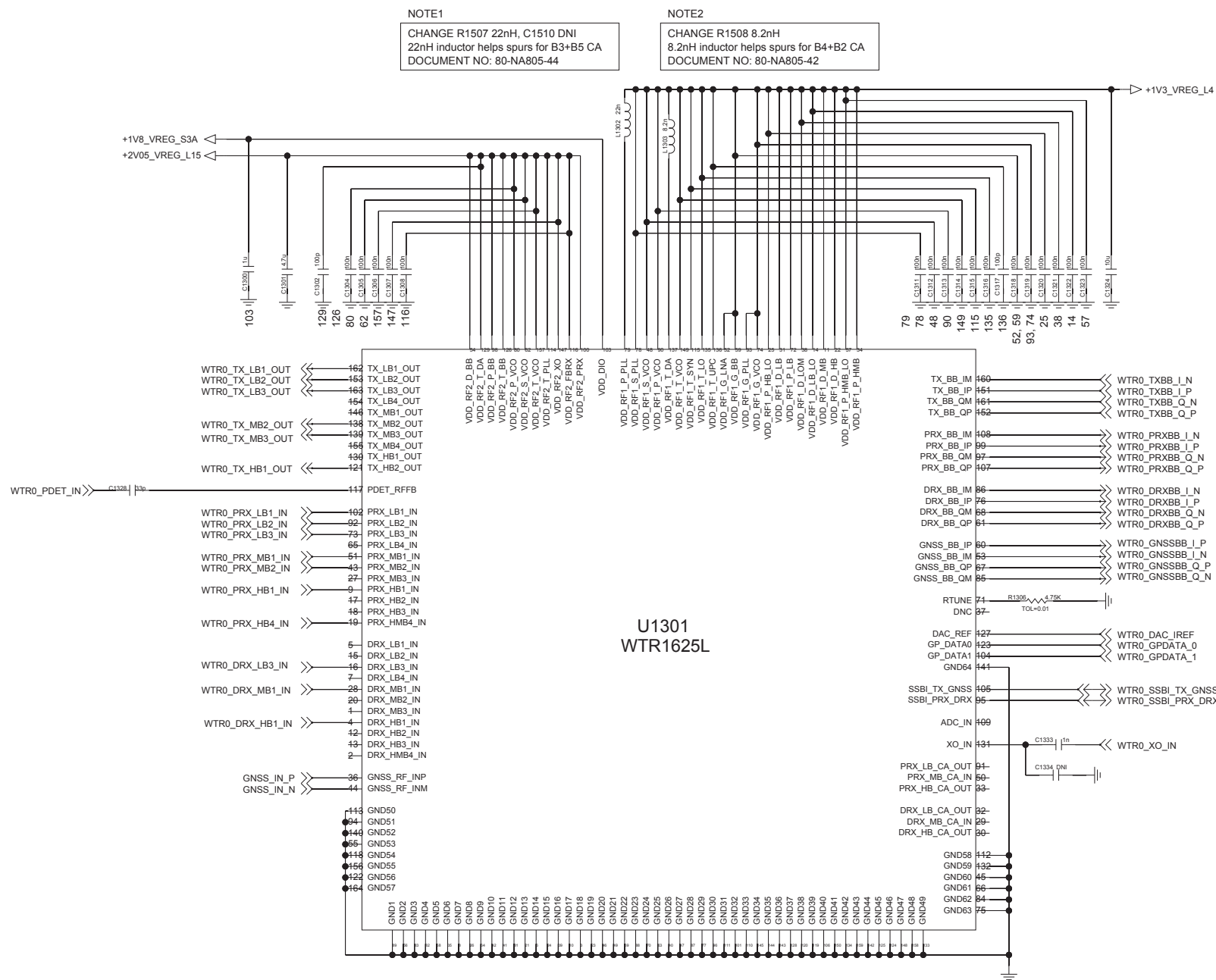
[LS990] Audio Codec I/F



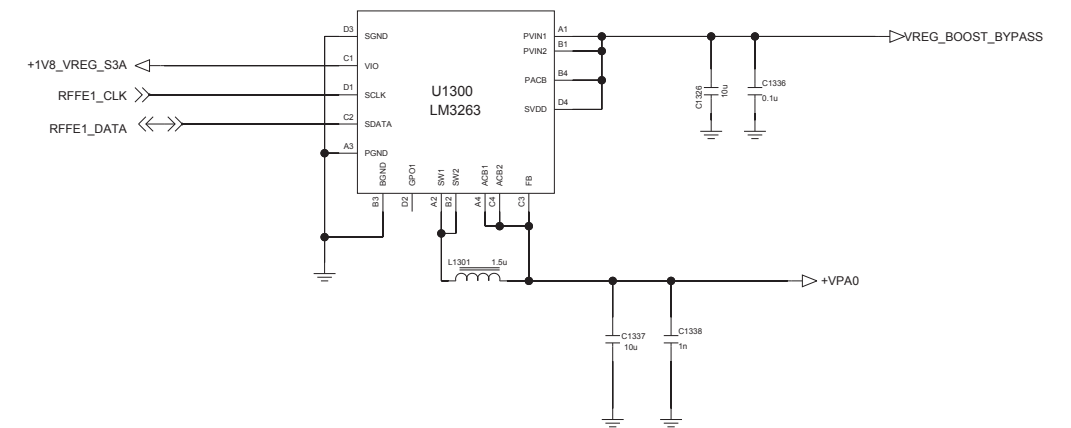


< 1-3-5-1_RFIC_CA_nCA_WTR1625 >

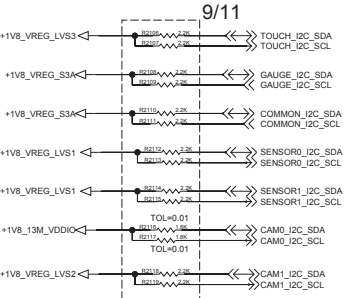
CAUTION: YOU MUST CHECK & MODIFY RF PORT AND GPIO ON YOUR SCHEMATICS.



< 1-3-4-3_APT_LM3263 >



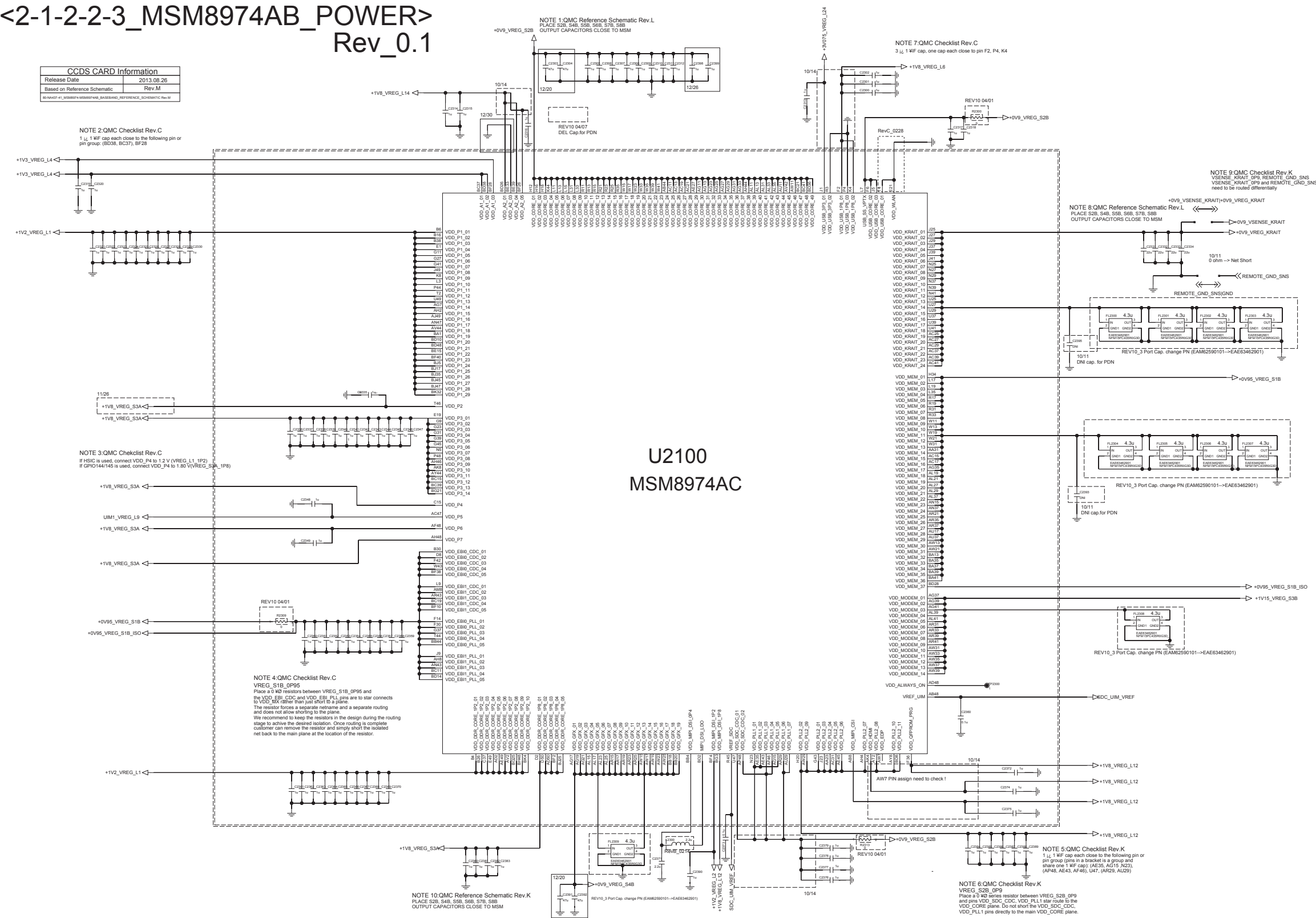
I2C Pull-Up



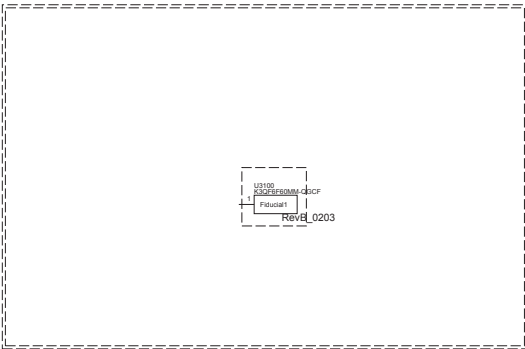
eMMC v4.5 : NC / eMMC v5.0 : RCLK

< 9-3-1_JTAG >
Rev 0.4

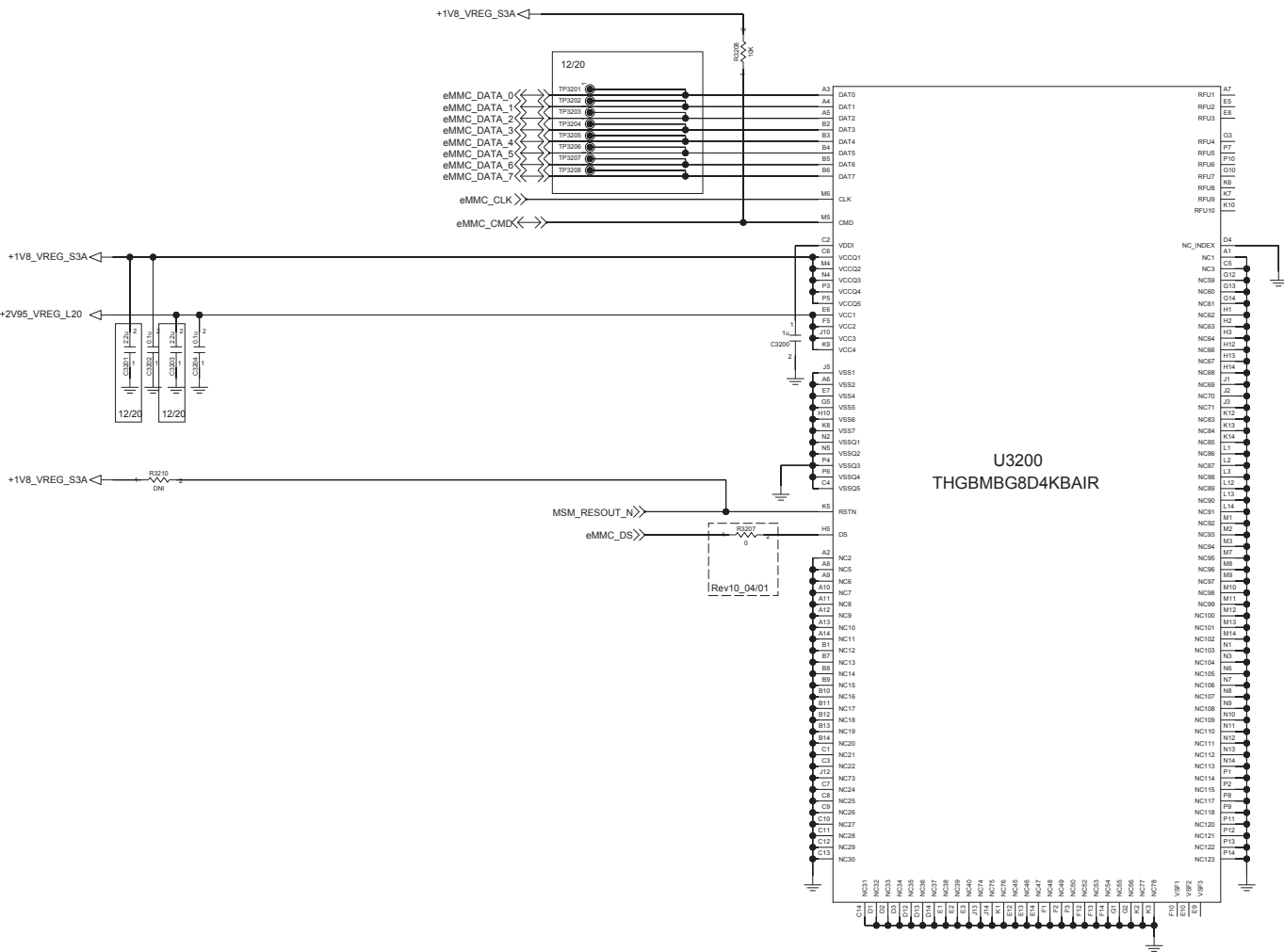
<2-1-2-2-3_MSM8974AB_POWER>
Rev_0.1



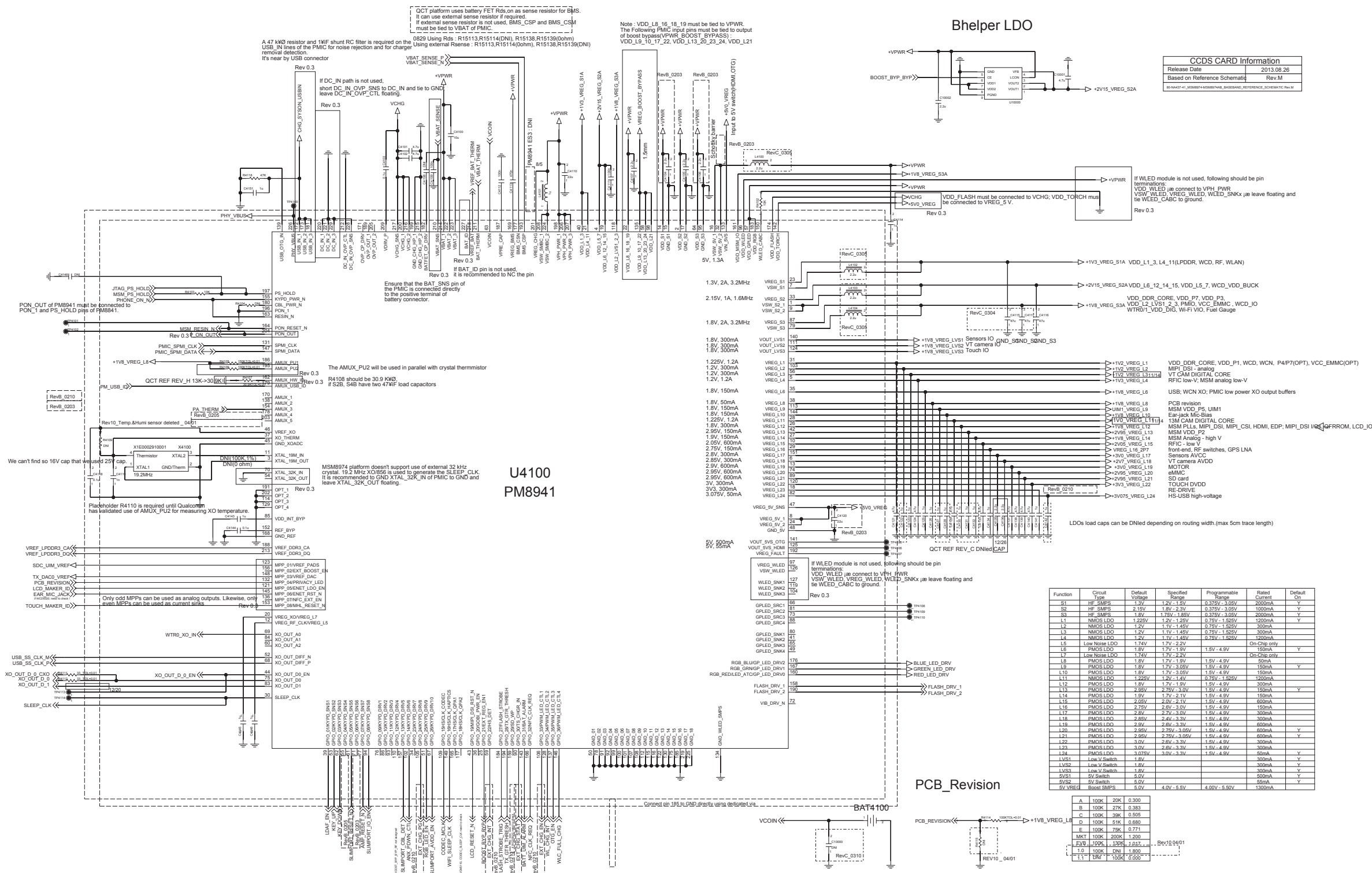
<3-1-4-3-1_LPDDR3_POP_16G-Bit> Rev_0.3



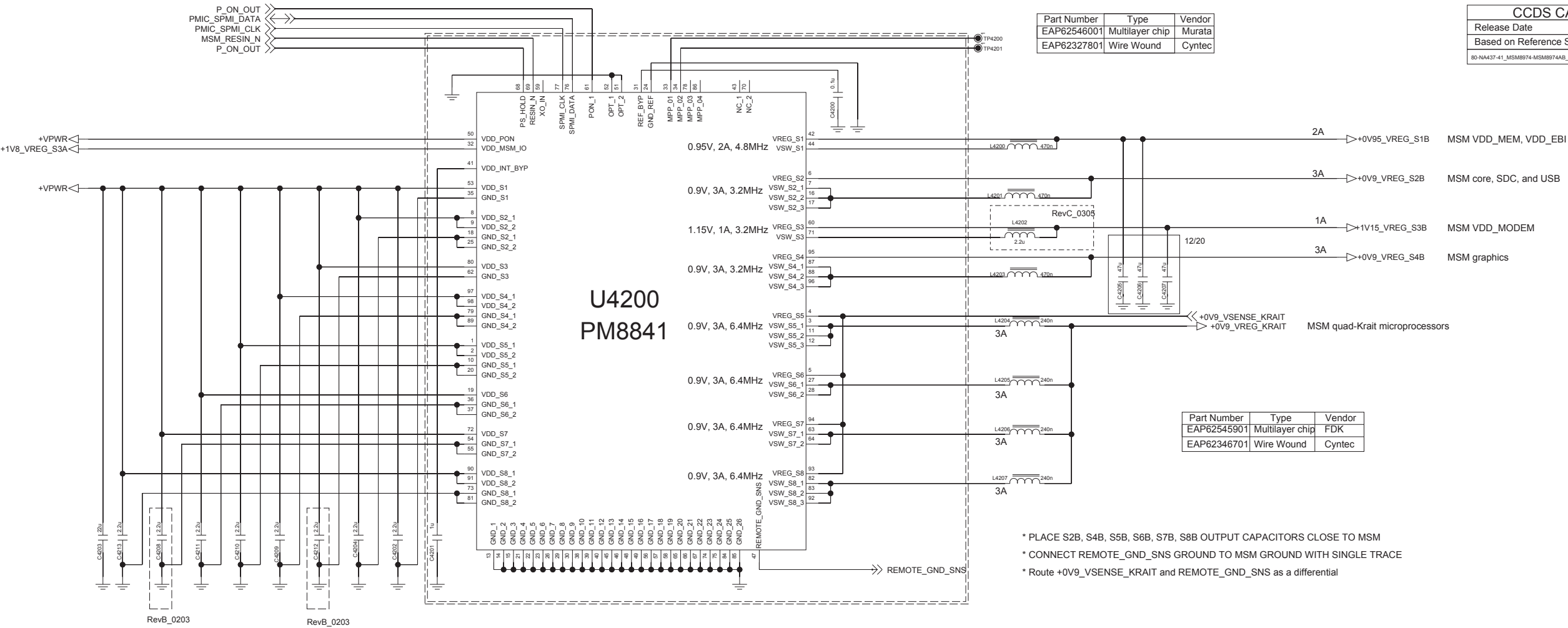
<3-2-2-3-1_eMMC_4_5_Toshiba_32G-Byte> Rev_0.4



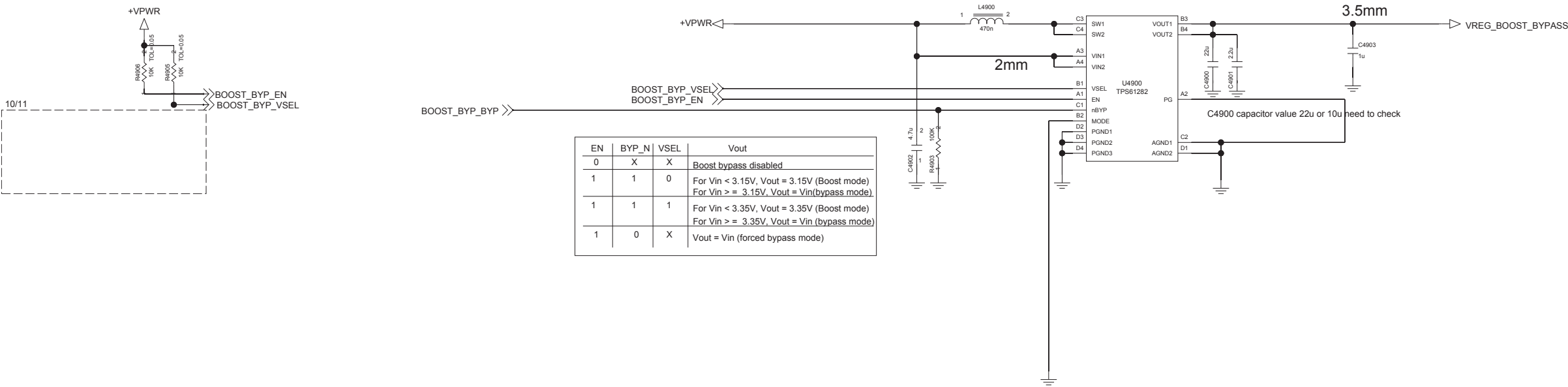
<4-1-2_PMIC_PM8941> Rev_0.3



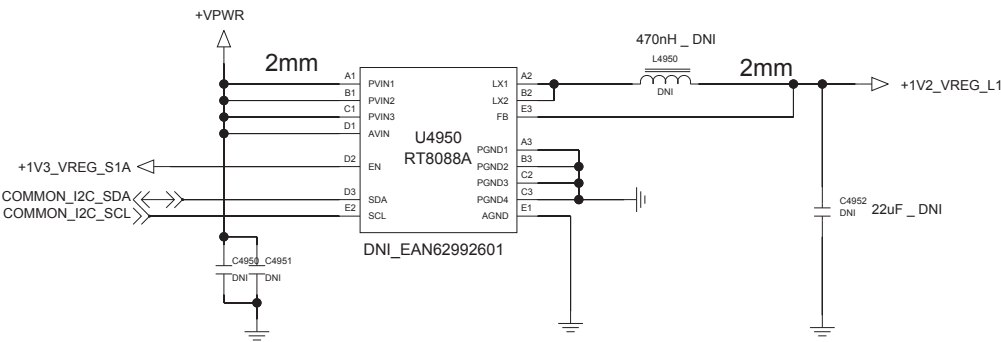
<4-2-1_PMIC_PM8841> Rev_0.3

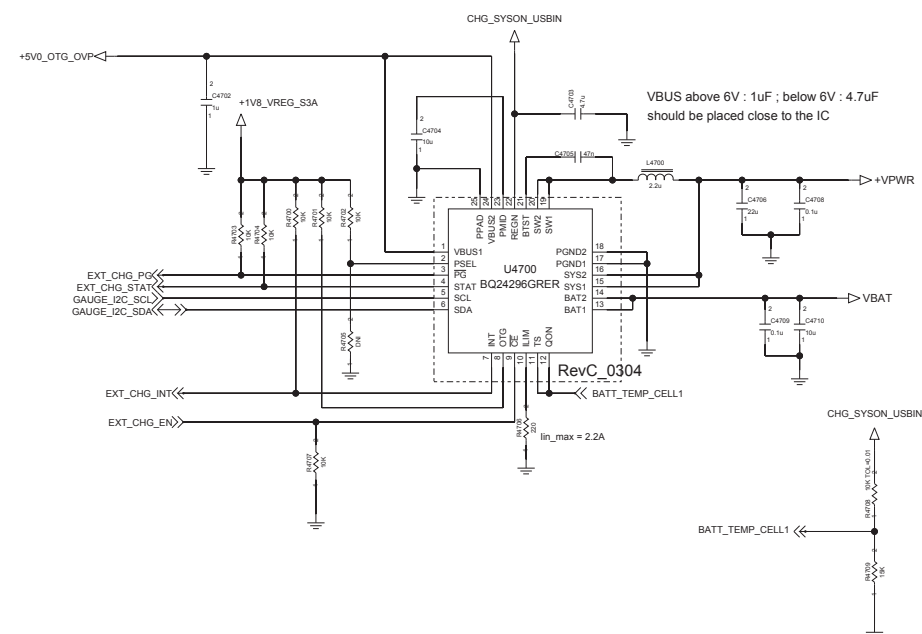
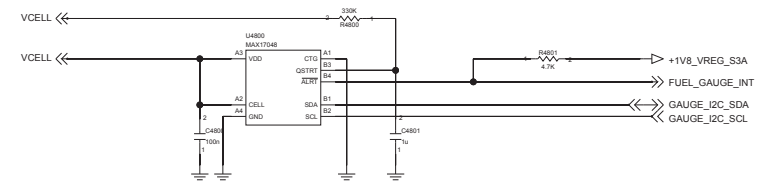
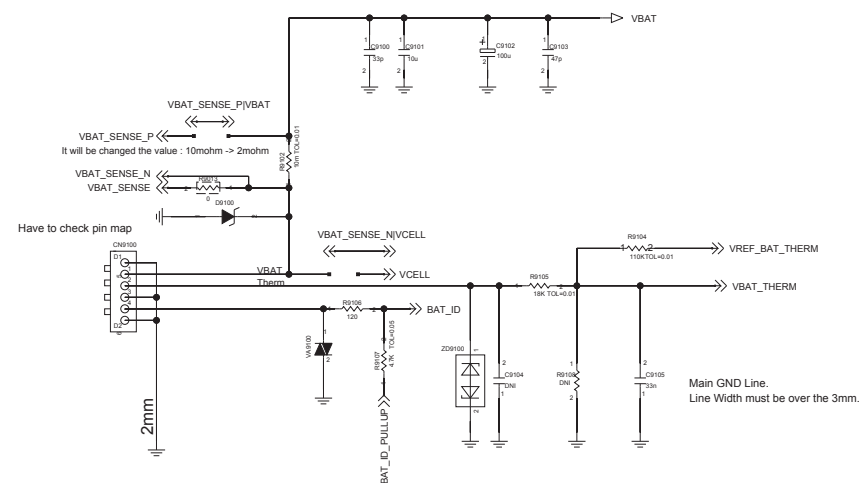


<4-9-1-4-3_Bypass_Booster_Single_TPS61282> Rev_0.3

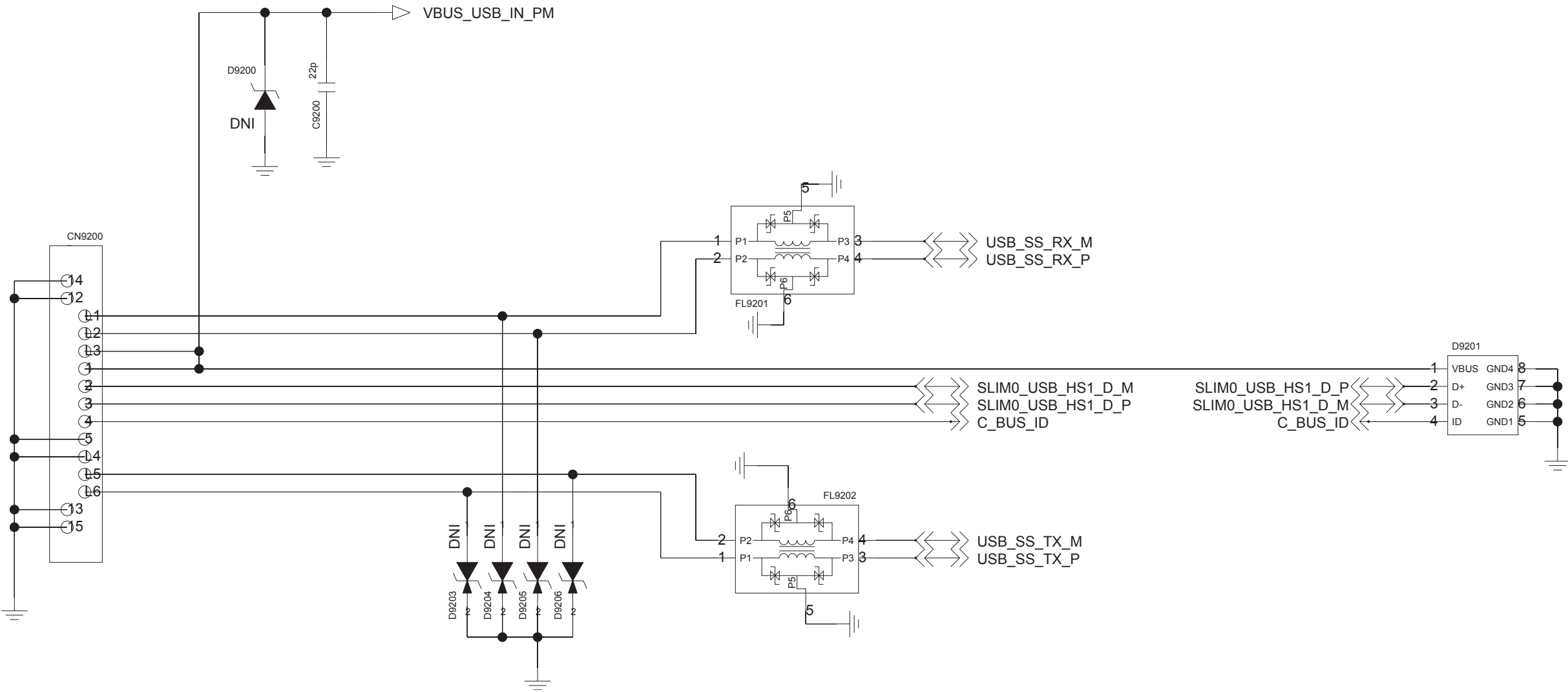


Bhelper LDO for 3GB LPDDR3





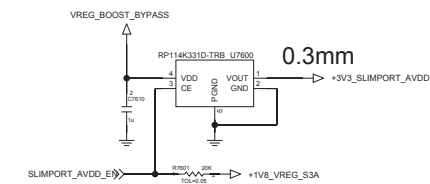
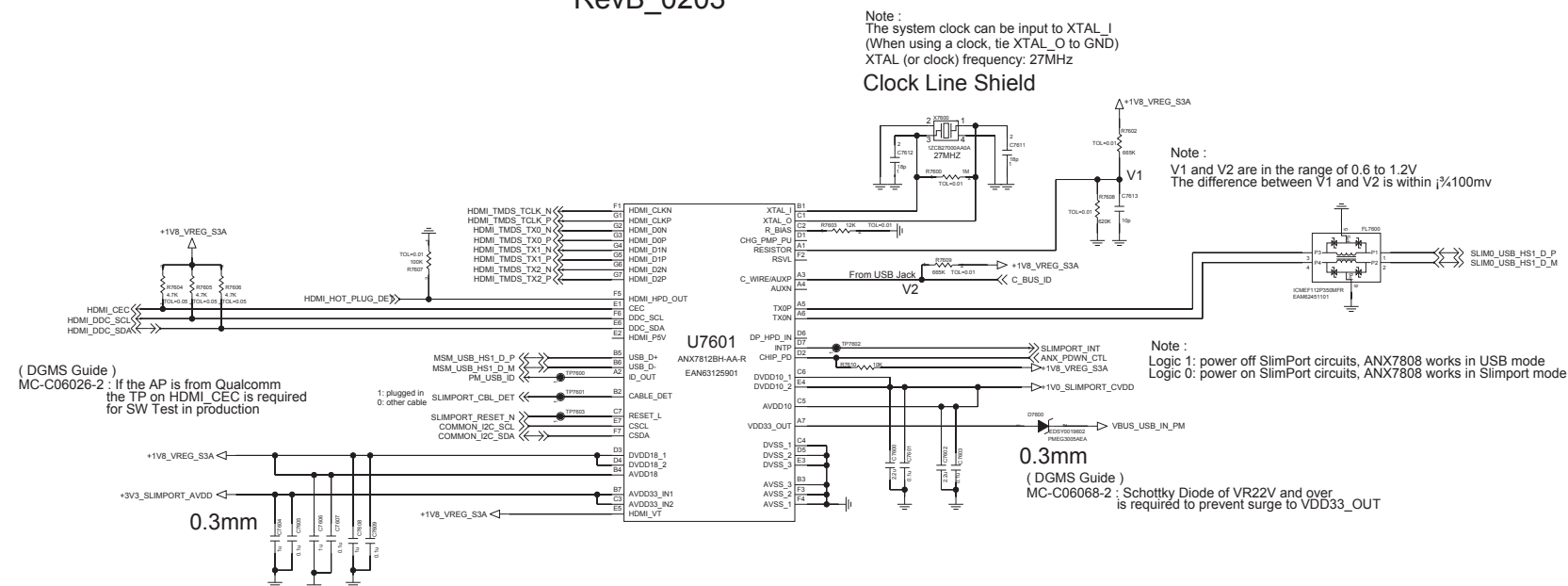
<9-2-2_Multi_IO_USB3_0> Rev_0.3



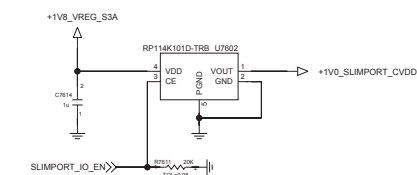
SLIMPORT/USB3.0 Rev_0.3

<7-6-2-1_SlimPort_ANX7812BH-AA-R>

RevB_0203

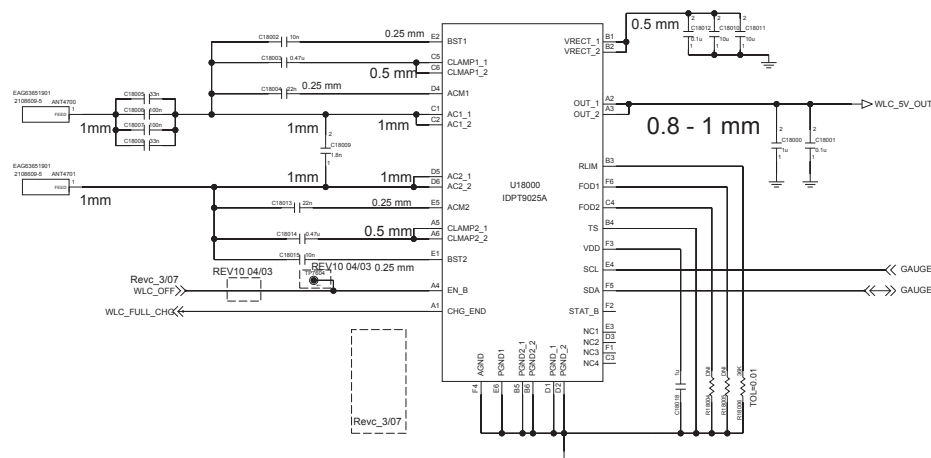


(Slimport)+1V0_SLIMPORT_IOVCC

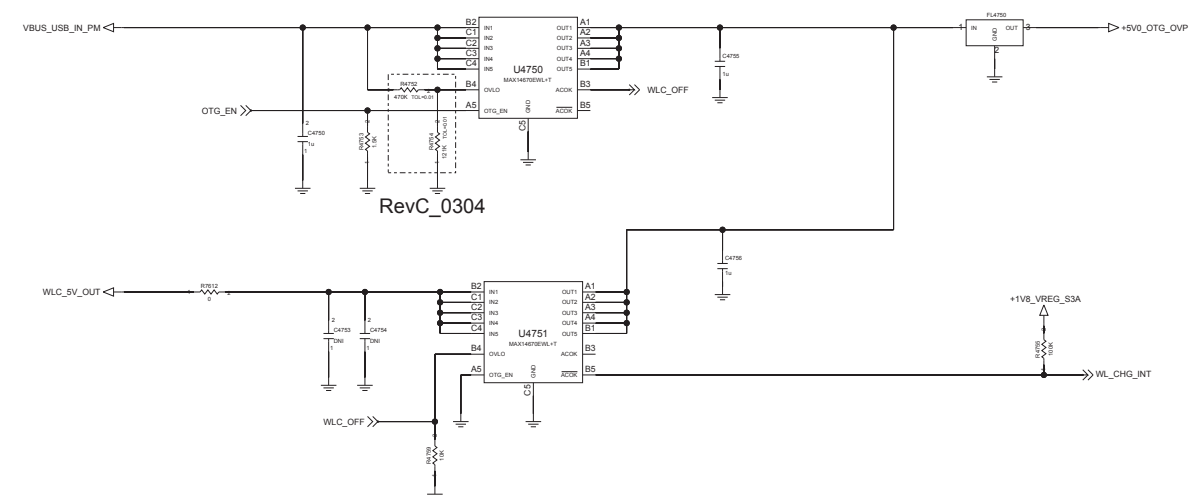


Wireless Charger

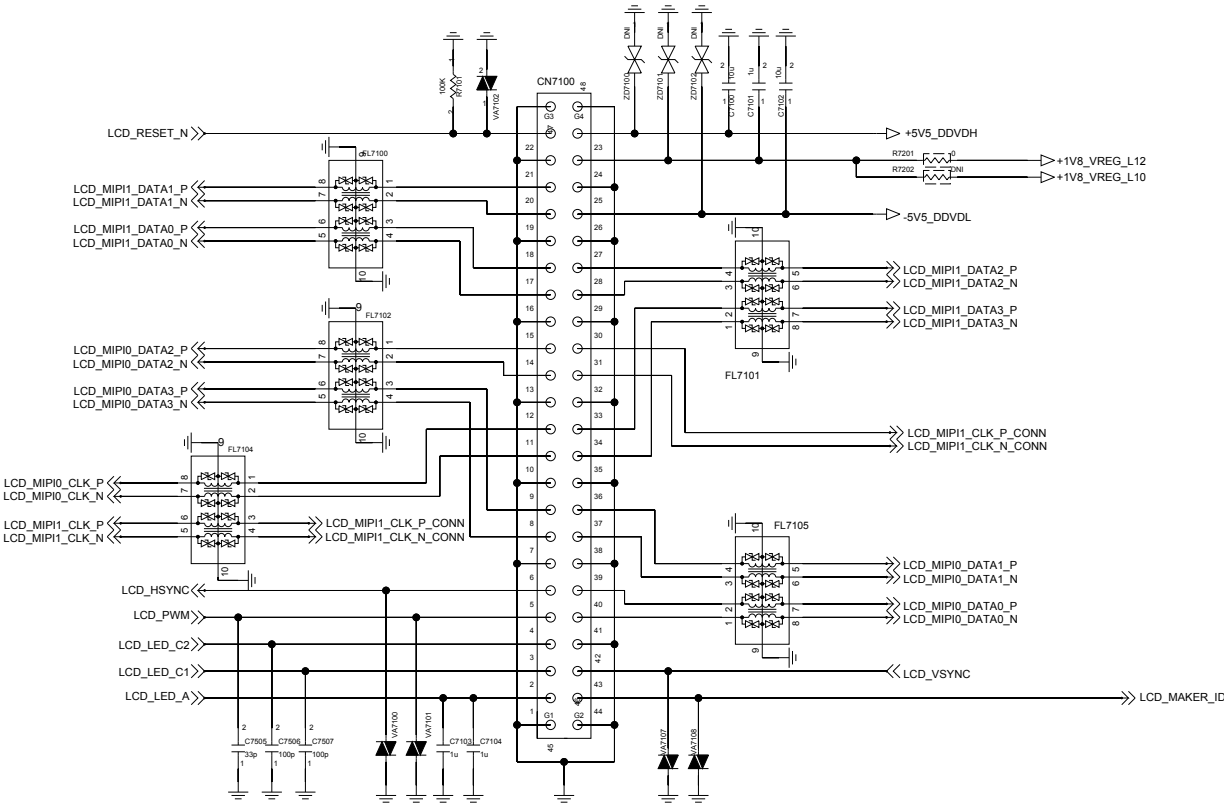
< IDT9025A >



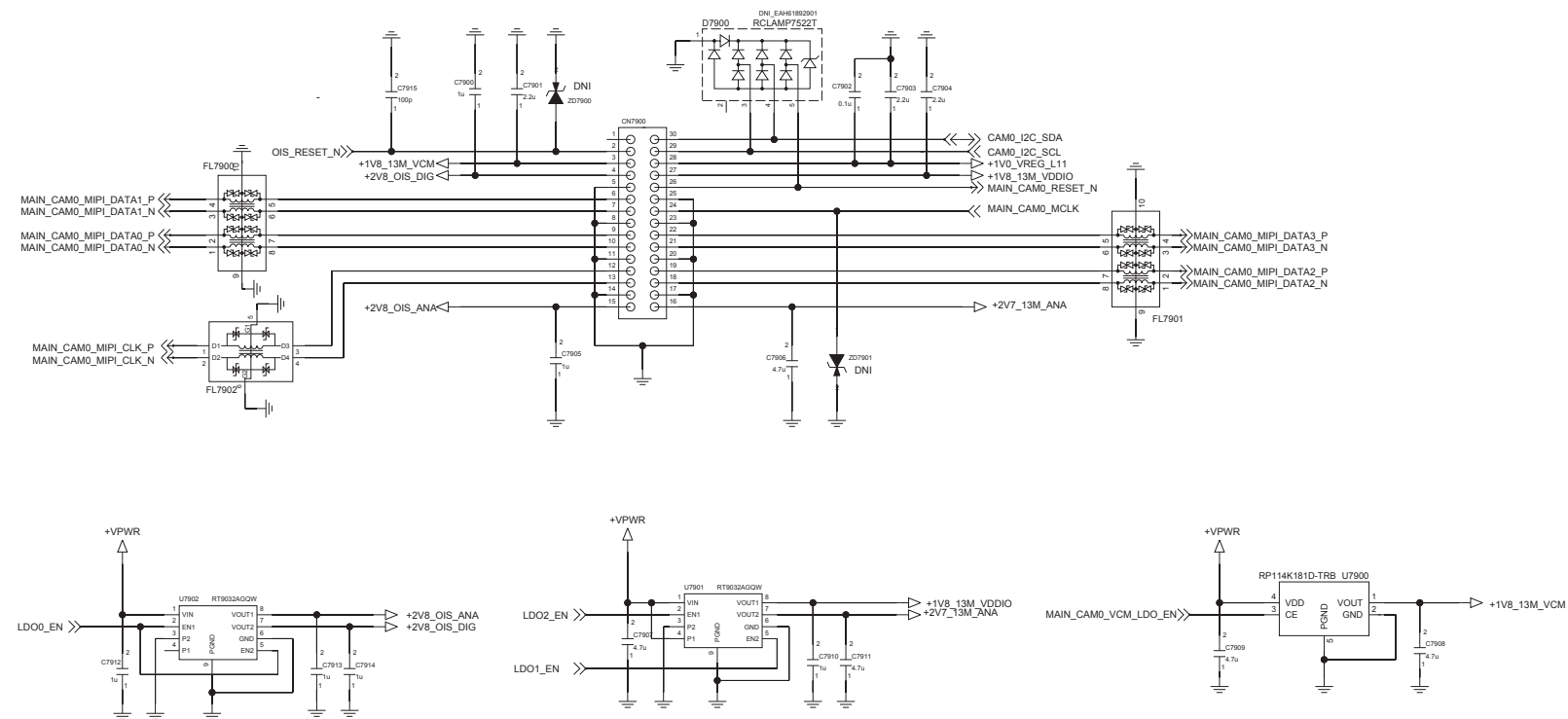
<4-7-3-1_Wireless_Charger_MAX14670EWL_T>
Rev_0.3



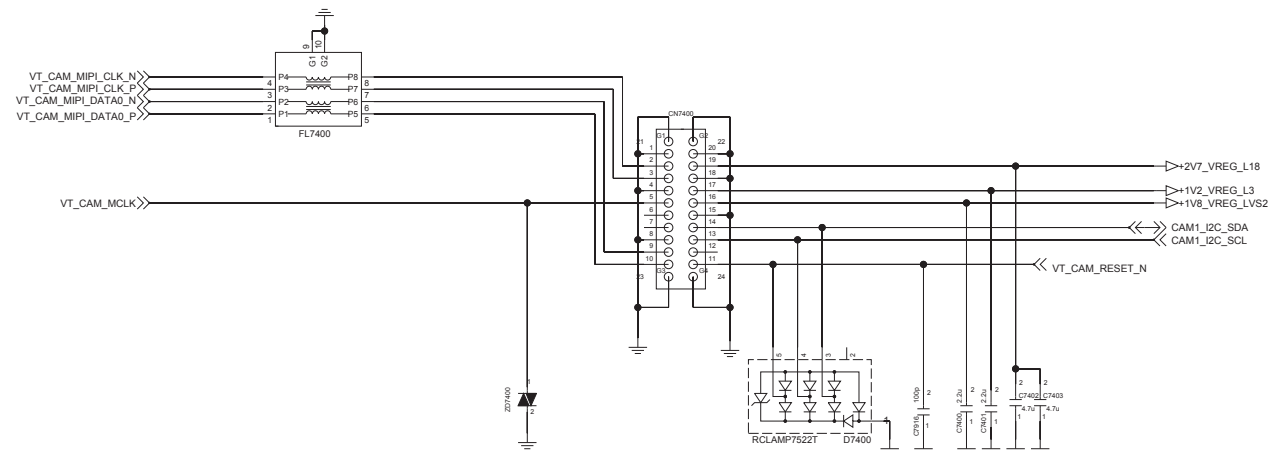
<7-1-8-4-2_Quad_LCD_5.5"> Rev_0.1



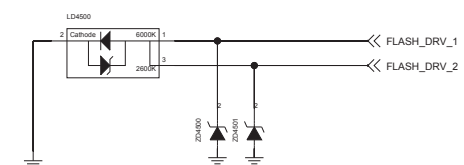
13M_OIS_CAMERA_same as B1(F350L)



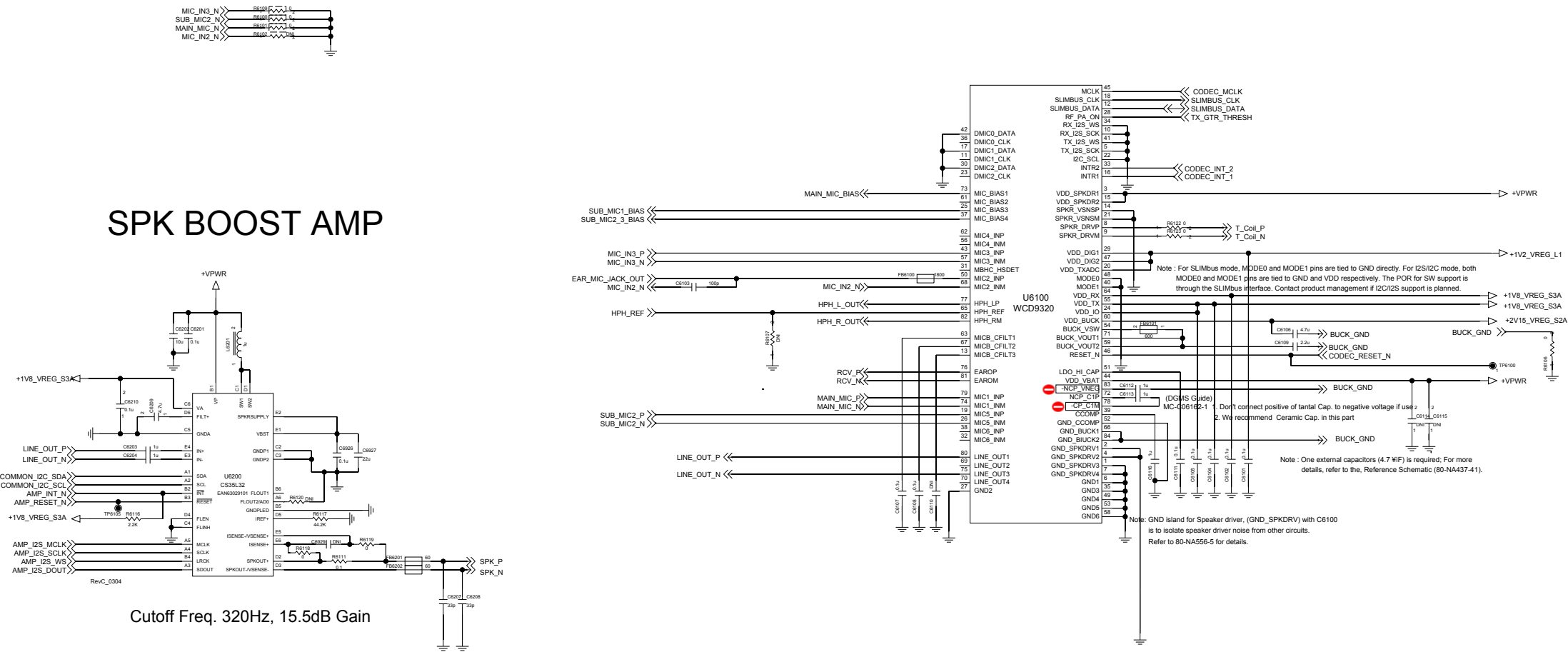
< 7-4-1_VT_Camera_20pin_VGA_1.26M_1.3M_2.4M> Rev_0.3



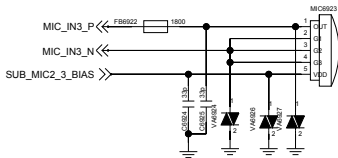
< Flash LED >



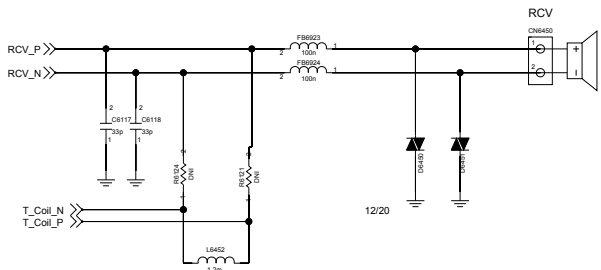
<6-1-1-5_WCD9320>REV_0.3



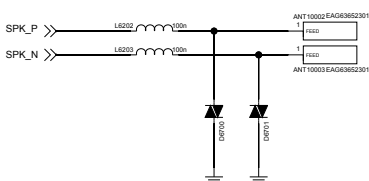
Sub_MIC2



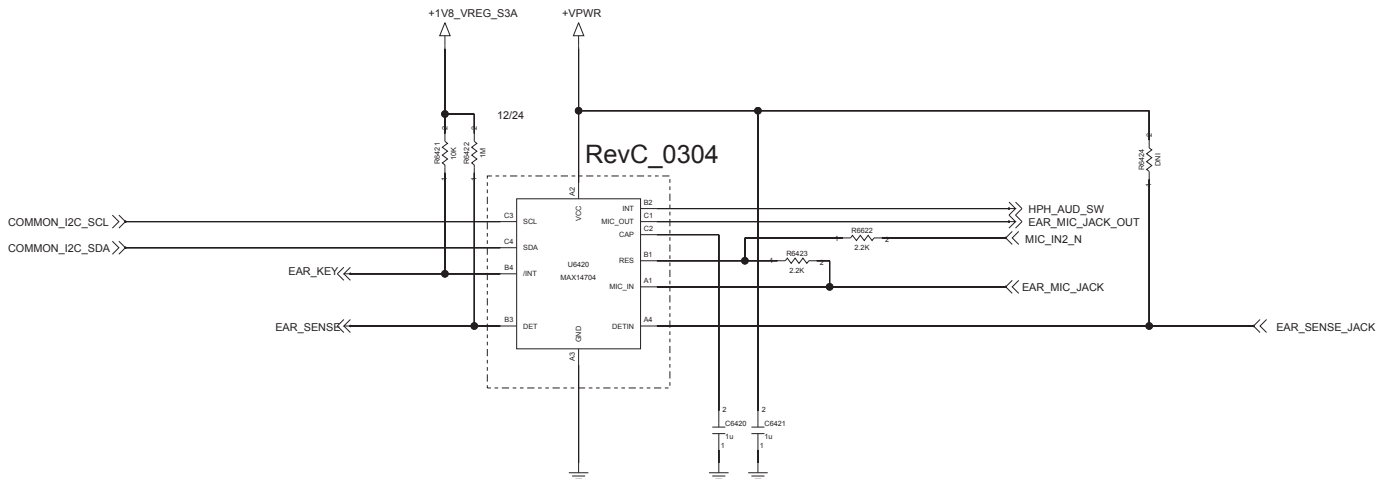
< 6-8-2_Receiver > Rev_0.4



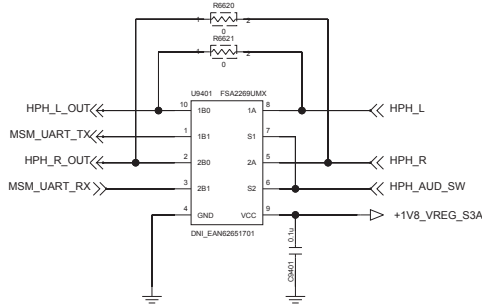
< 6-7-1_Speaker > Rev_0.4



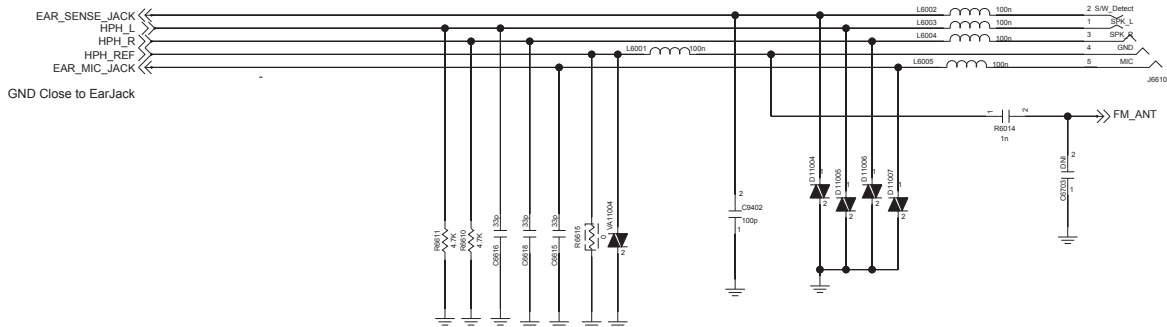
< Earjack detect IC >



<Earjack_Debug_Port >



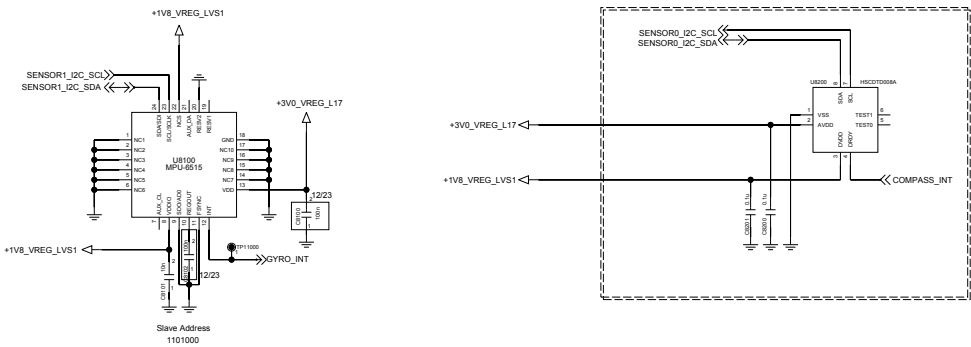
< 6-6-1_Earjack > Rev_0.5



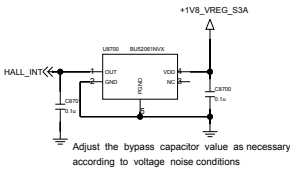
SENSOR/MOTOR/IRRRC/SIM/SD/LED

I2C	Devices
Sensor0	Compass, Temp+Humid+Pressure
Sensor1	Accel+Gyro, Pressure, UV, Prox+RGB+IR Gesture

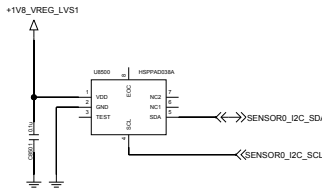
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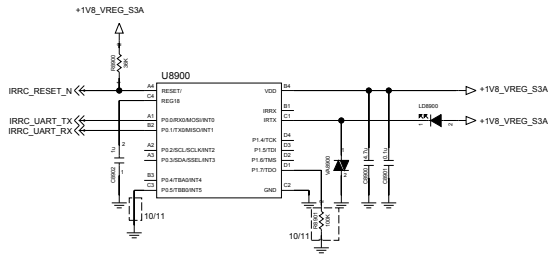
<8-7-1-1_Hall IC_BU52061NVX>



<Pressure_HSPPAD038A >

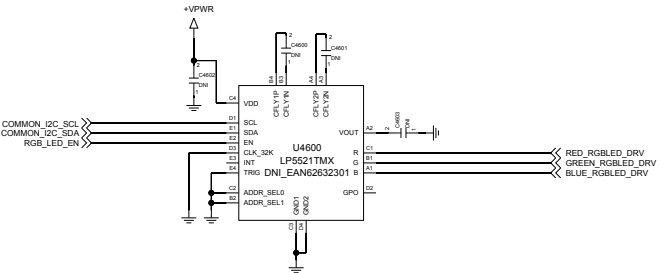


< 8-9-1-1_IRRC_MAXQ616V > Rev_0.4

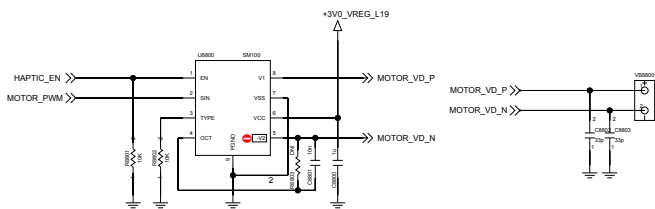


REV.1.0

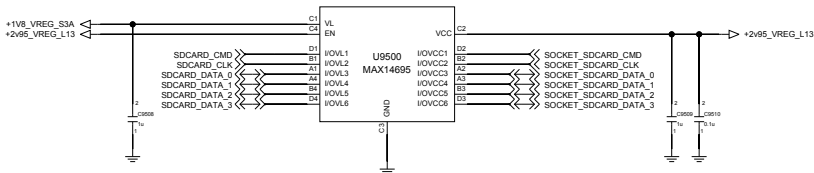
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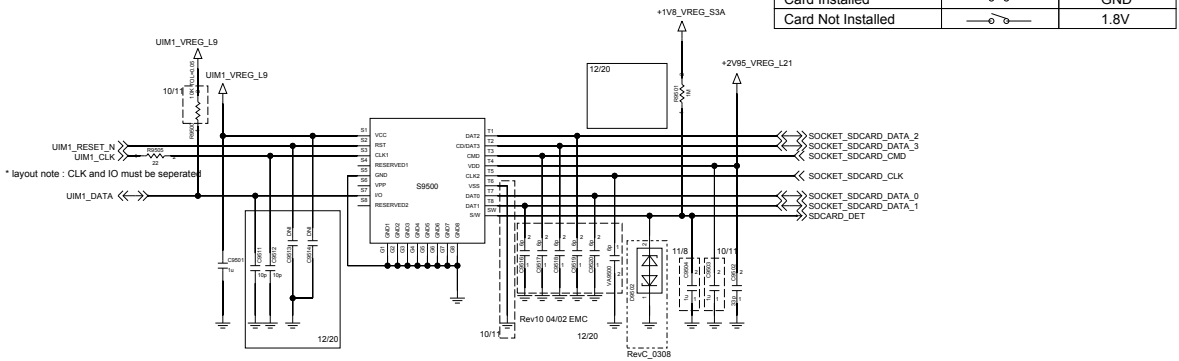
< 8-8-1-1_Haptic_SM100 >



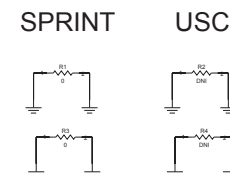
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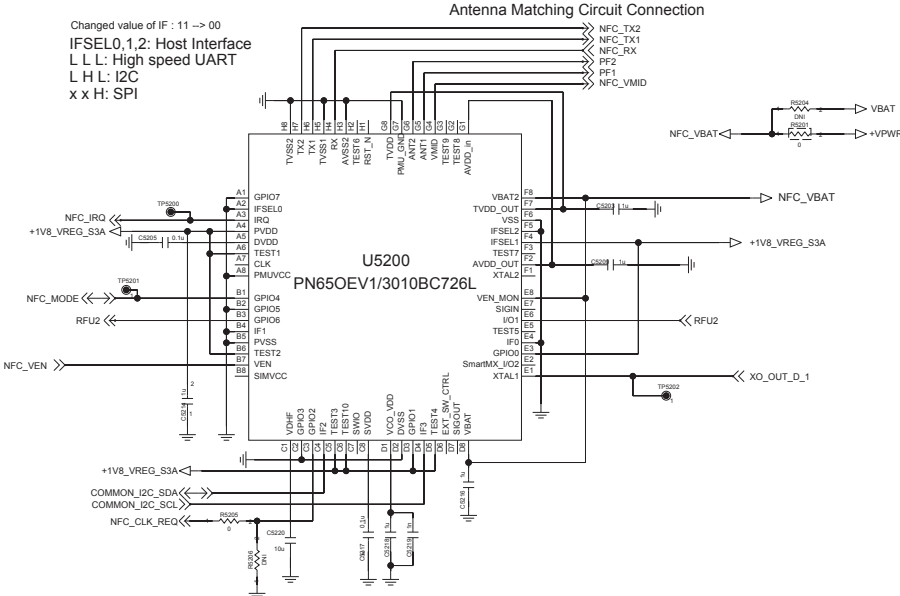
< 9-5-3_Micro_SD_Combo > Rev_0.3



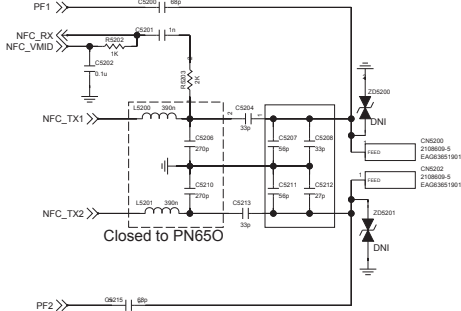
Card Detect	SW	VSS(T6)	SD_CARD_DET
Card Installed	—○—	—○—	GND
Card Not Installed	—○—	—○—	1.8V

[illegible]

NFC - PN650



NFC Antenna

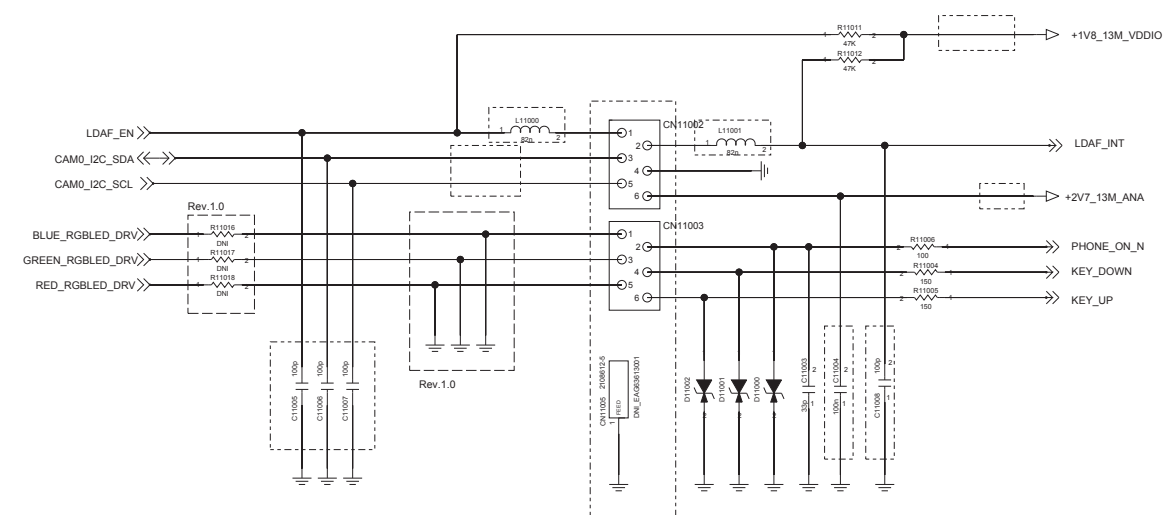


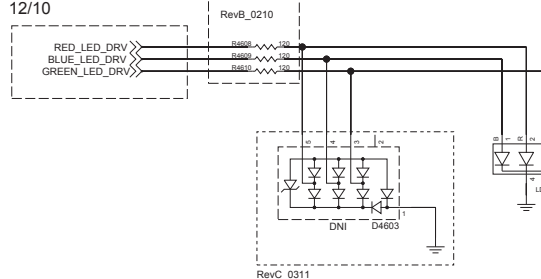
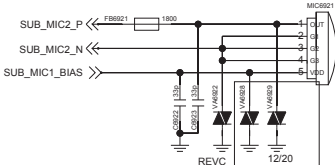
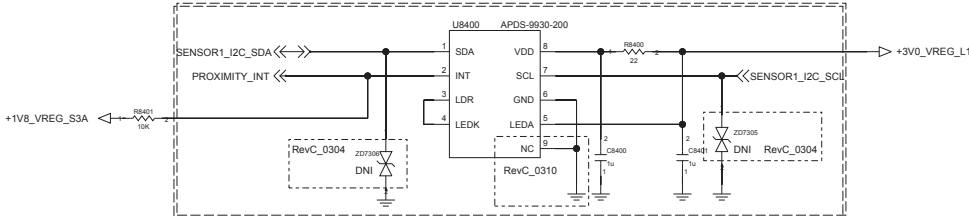
FPCB_CNT

I2C	Devices
Sensor0	Compass, Temp+Humid+Pressure
Sensor1	Accel+Gyro, Pressure, UV, Prox+RGB+IR Gesture

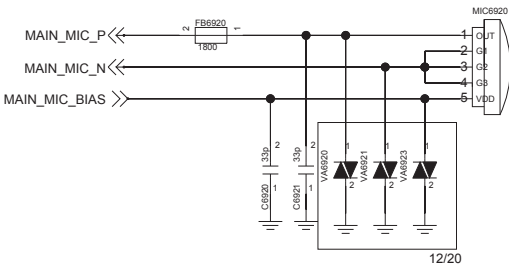
RevC_0302

Back_Key_LDAP_FPCB



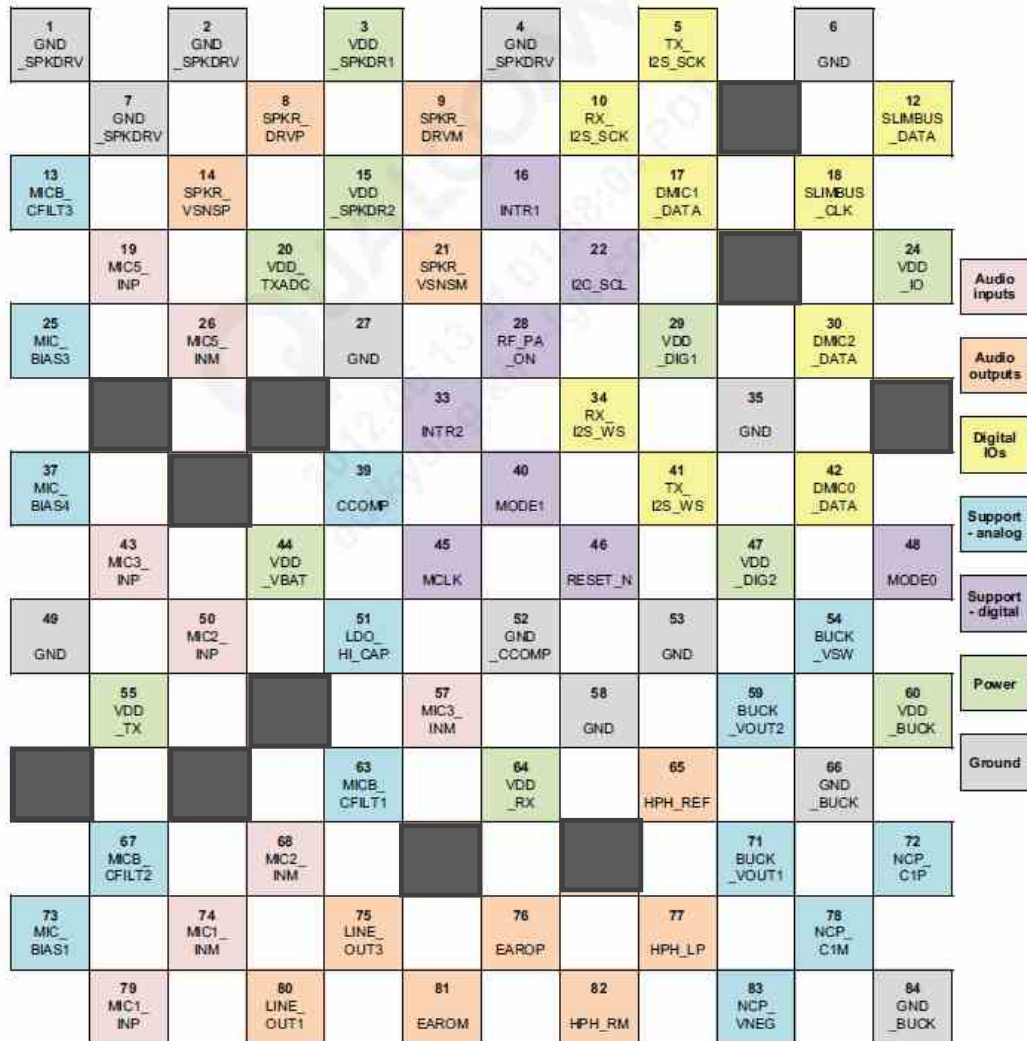


From previously Lower FPCB to Main PCB (11/13 now)



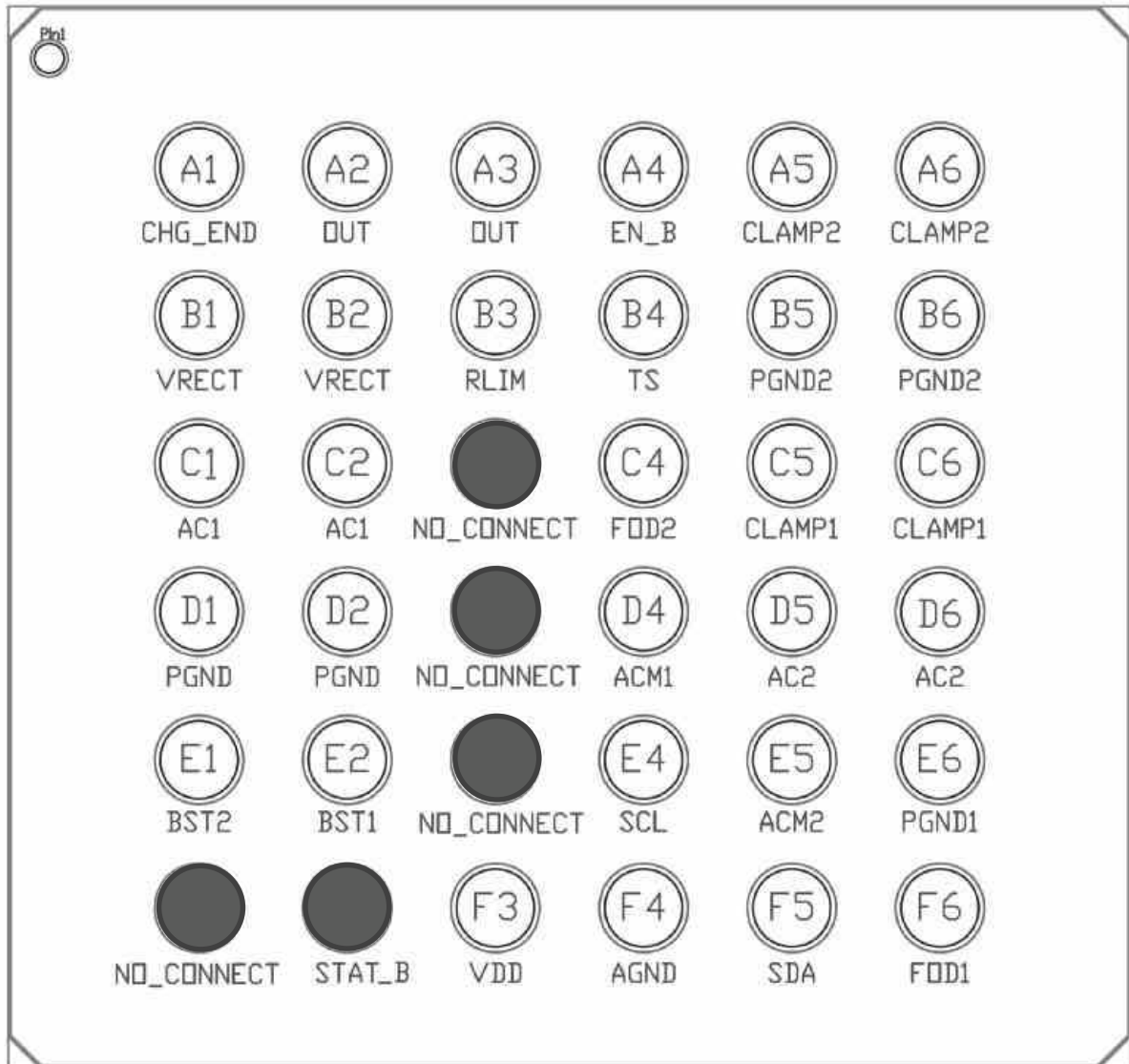
7. BGA PIN MAP

U6100_WCD9320_IC,Audio Codec(Top view)



Not Used

U18000_IC,Charger_IDTP9025A (Top View)



● Not Used

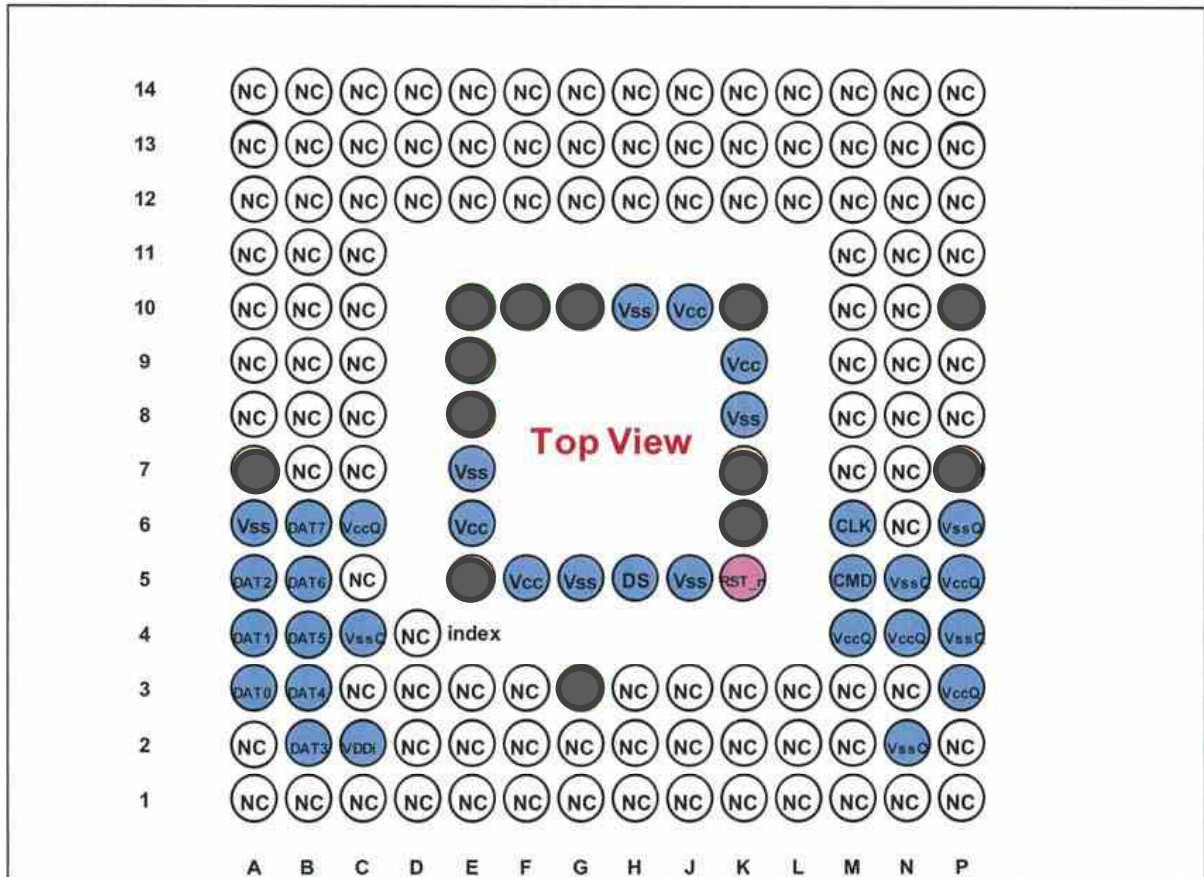
7. BGA PIN MAP

U4200_PM8841_PMIC(TOP view)

1 VDD _S5	2 VDD _S5	3 VSW _S5	4 VREG _S5	5 VREG _S6	6 VREG _S2	7 VSW _S2	8 VDD _S2	9 VDD _S2
10 GND _S5	11 VSW _S5	12 VSW _S5	13 GND	14 GND	15 GND	16 VSW _S2	17 VSW _S2	18 GND _S2
19 VDD _S6	20 GND _S5	21 GND	22 GND	23 GND	24 GND _REF		25 GND _S2	26 GND
27 VSW _S6	28 VSW _S6	29 GND	30 GND	31 REF _BYP	32 VDD _MSM_IO	33 MPP _01	34 MPP _02	35 GND _S1
36 GND _S6	37 GND _S6	38 GND	39 GND	40 GND	41 VDD _INT_BYP	42 VREG _S1		44 VSW _S1
45 GND	46 GND	47 REMOTE _GND_SNS	48 GND	49 DNC	50 VDD _PON	51 OPT_2	52 OPT_1	53 VDD _S1
54 GND _S7	55 GND _S7	56 GND	57 GND	58 DNC		60 VREG _S3	61 PON_1	62 GND _S3
63 VSW _S7	64 VSW _S7	65 GND	66 GND	67 GND	68 PS _HOLD	69 RESN _N		71 VSW _S3
72 VDD _S7	73 GND _S8	74 GND	75 GND	76 SPMI _DATA	77 SPMI _CLK		79 GND _S4	80 VDD _S3
81 GND _S8	82 VSW _S8	83 VSW _S8	84 GND	85 GND		87 VSW _S4	88 VSW _S4	89 GND _S4
90 VDD _S8	91 VDD _S8	92 VSW _S8	93 VREG _S8	94 VREG _S7	95 VREG _S4	96 VSW _S4	97 VDD _S4	98 VDD _S4
OUTPUT PWR MGT	IC1/F	MPPs and GPIOs	No Connection	Power	Ground			

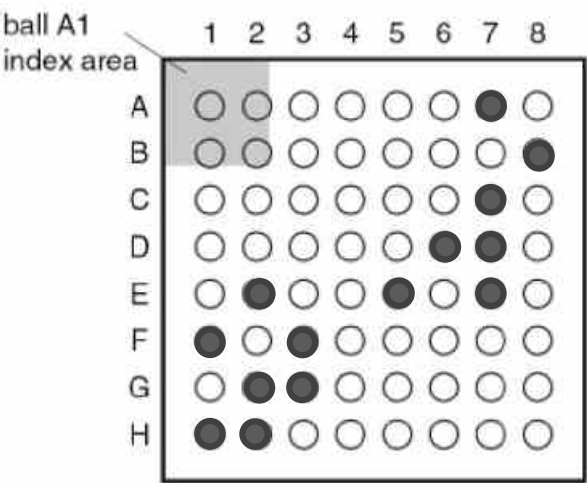
 Not Used

U3200_eMMC(TOP_view)



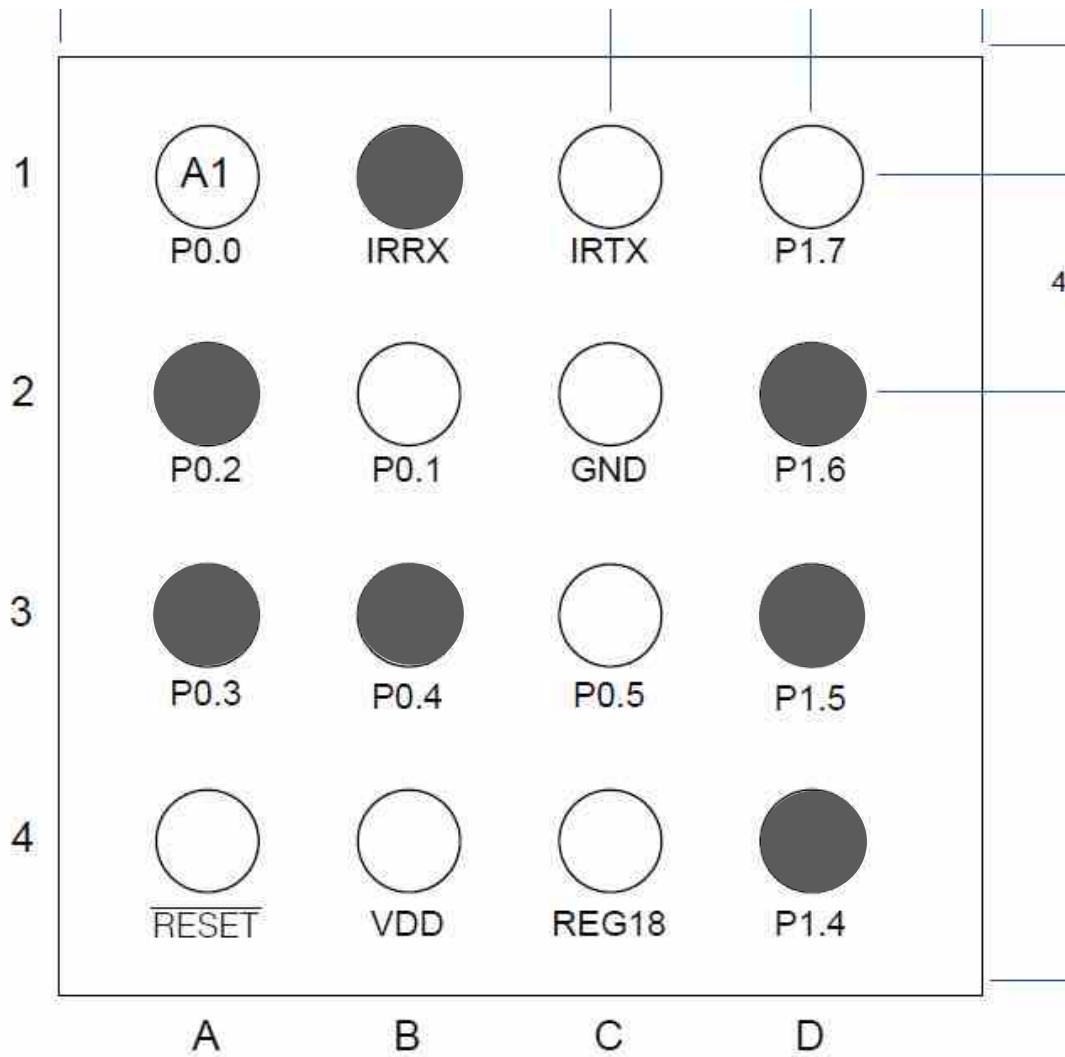
● Not Used

U5200_PN650/NFC Controller



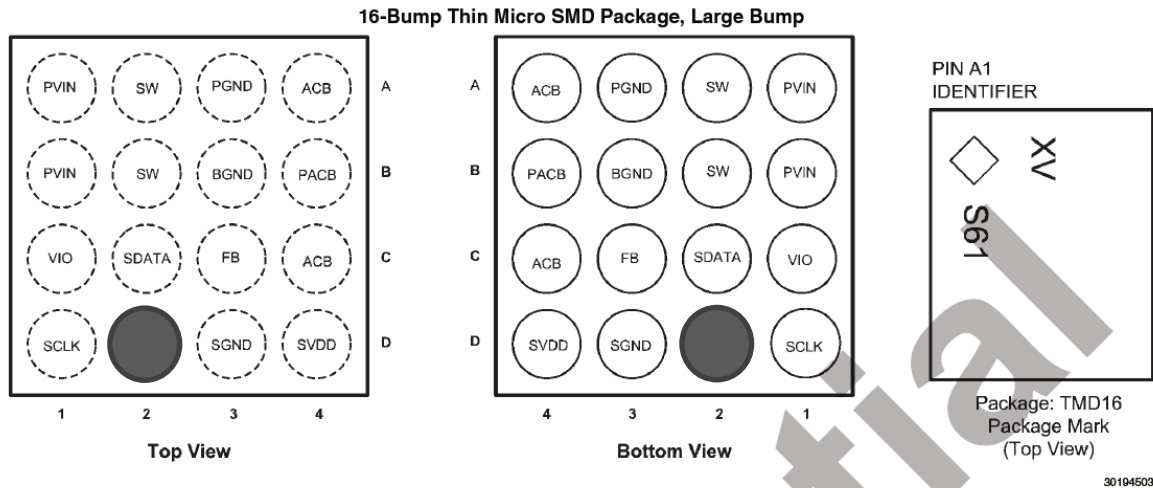
● Not Used

U8900_IC, Microprocessors (Bottom view)



● Not Used

U1300_IC,DC-DC Converter



Pin Descriptions

Pin #	Name	Description
A1	PVIN	Power Supply Voltage Input to the internal PFET switch and ACB.
B1		
C1	VIO	VIO functions as the RFFE interface reference voltage. VIO also functions as reset and enable input to the LM3263. Typically connected to voltage regulator controlled by RF or Baseband IC.
D1	SCLK	Digital control interface RFFE Bus clock input. Typically connected to RFFE master on RF or Baseband IC. SCLK must be held low when VIO is not applied.
A2	SW	Switching Node connection to the internal PFET switch and NFET synchronous rectifier.
B2		
C2	SDATA	Digital control interface RFFE Bus data input/output. Typically connected to RFFE master on RF or Baseband IC. SDATA must be held low when VIO is not applied.
D2	GPO1	General Purpose Output. Also used to reconfigure USID.
A3	PGND	Power Ground to the internal NFET switch.
B3	BGND	Active Current assist and Analog Bypass Ground and Digital Ground.
C3	FB	Feedback Analog Input. Connect to the output at the output filter capacitor.
D3	SGND	Signal Analog Ground (Low Current).
A4	ACB	Active Current assist and analog Bypass output. Connect to the output at the output filter capacitor.
C4		
B4	PACB	ACB Power Supply Input.
D4	SVDD	Analog Power Supply Voltage.



U7601_IC,Signal Bridge

	1	2	3	4	5	6	7	
A	RESISTOR	ID_OUT	C-WIRE /AUXP (AUX_HPD)		TX0P	TX0N	VDD33_OUT	A
B	XTAL_I	CABLE_DET	AVSS	AVDD18	USB_D+	USB_D-	AVDD33_IN	B
C	XTAL_O	R_BIAS	AVDD33_IN	DVSS	AVDD10	DVDD10	RESET_L	C
D		CHIP_PD	DVDD18	DVDD18	DVSS		INTP	D
E	CEC		DVSS	DVDD10	HDMI_VT	DDC_SDA	CSCL	E
F	HDMI_CLKN		AVSS	AVSS	HDMI_HPD_OUT	DDC_SCL	CSDA	F
G	HDMI_CLKP	HDMI_D0N	HDMI_D0P	HDMI_D1N	HDMI_D1P	HDMI_D2N	HDMI_D2P	G
	1	2	3	4	5	6	7	

 Not Used

7. BGA PIN MAP

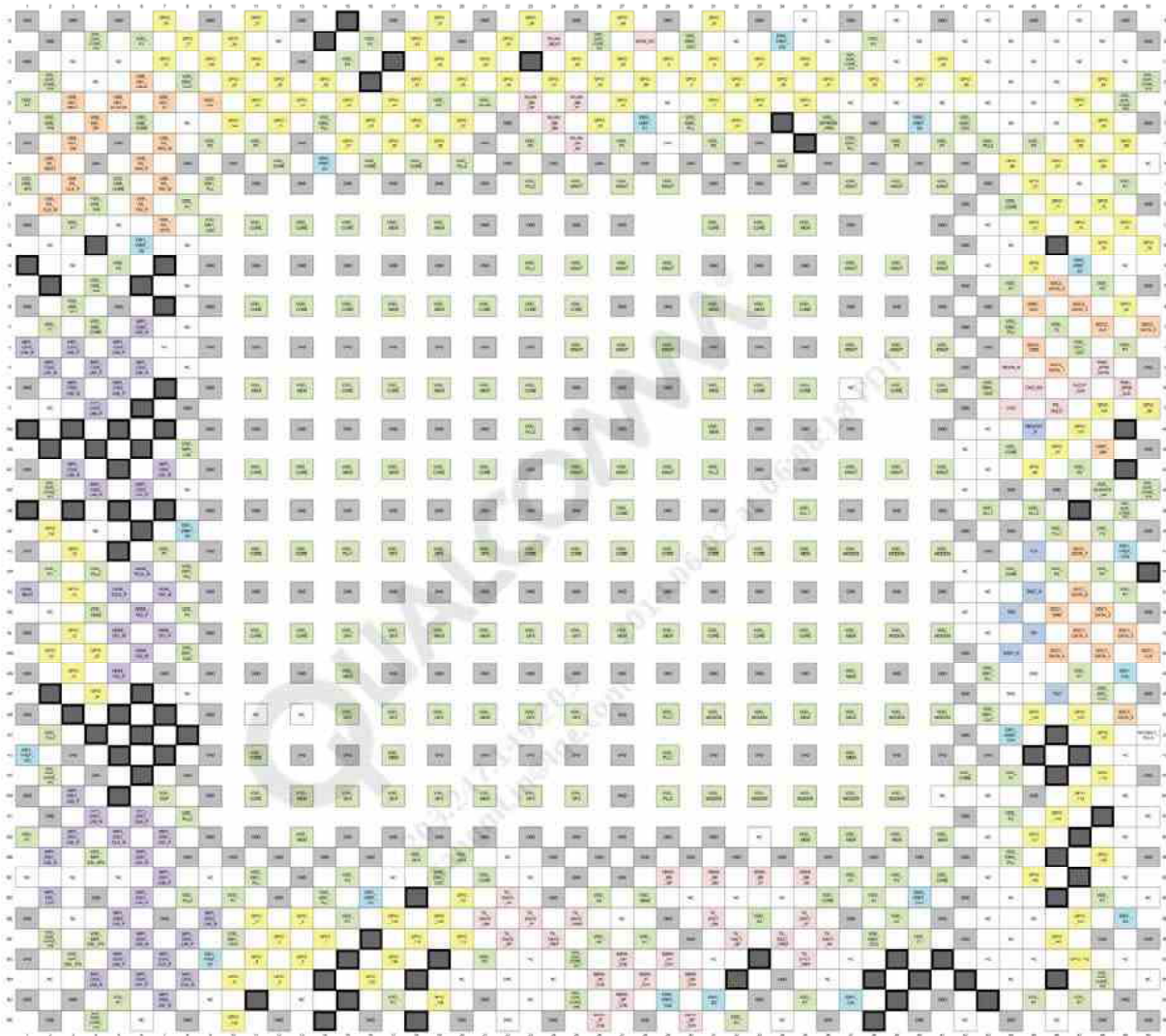
U21000_BCM4339,WiFi (Top View)

	1	2	3	4	5	6	7	8	9	10	11	12	
A		NO CONNECT	NO CONNECT	NO CONNECT		NO CONNECT	NO CONNECT		NO CONNECT	NO CONNECT	NO CONNECT		A
B	SR_PVSS	SR_VLX	WL_REG_ON	LPO_IN	GPIO_3	GPIO_0	HSIC_DATA	HSIC_STROBE		SDIO_DATA_0	SDIO_CLK	SDIO_CMD	B
C	SR_VDDBATPSV	SR_VDDBATASV	PMU_AVSS		GPIO_4		WL_VDDC	HSIC_AVDD12PLL	HSIC_DVDD12	SDIO_DATA_1	SDIO_DATA_3	WL_VDDC	C
D	LDO_VDD1P5	VOUT_CLDO	BT_REG_ON		GPIO_5		VSSC	HSIC_AGN0PLL	VDDIO_5D	SDIO_DATA_2	VSSC		D
E	VOUT_3P3	VOUT_LN1D0	VSSC			BT_UART_CTS	VDDIO_RF	VSSC		RF_SW_CTRL_8	RF_SW_CTRL_3	RF_SW_CTRL_2	E
F	VOUT_BTLD02P5	LDO_VDDBATSV	VDDIO		RF_SW_CTRL_9	BT_UART_RTS	BT_UART_TXD				RF_SW_CTRL_1	RF_SW_CTRL_0	F
G	BT_PCM_IN	BT_PCM_CLK		WL_VDDC	WL_VDDC	BT_UART_RXD	RF_SW_CTRL_7	WL_VDDC		BBPLL_AV5	WRF_XTAL_GND1P2	BBPLL_AVDD1P2	G
H		BT_PCM_SYNC		BT_VDDIO	BT_VDDC	BT_I2S_WS	WRF_GPIO_OUT	WRF_WL_LN1D0M_VDD1P5		WRF_VCO_GND	WRF_XTAL_VDD1P5	WRF_XTAL_IN	H
J	FM_AUDIOVDD1P2	BT_HOST_WAKE	BT_PCM_OUT	BT_VDDC	VSSC	BT_I2S_CLK		WRF_BUCK_GND1P5	WRF_MMIO_GND1P2	WRF_PFD_GND1P2	WRF_CP_GND	WRF_XTAL_OUT	J
K		FM_AUDIOVSS	BT_DEV_WAKE	VSSC	BT_I2S_DI	BT_I2S_DO	WRF_AFE_GND1P2	WRF_IO_GND1P2_2	WRF_SYNTH_VBAT_VDD3P3	WRF_MMIO_VDD1P2	WRF_PFD_VDD1P2	WRF_XTAL_VDD1P2	K
L		FM_PLLVDD1P2	FM_PLLVSS	BT_PVDD1P2	BT_PLLVSS	BT_PVSS	WRF_RX2G_GND1P2	WRF_TX_GND1P2	WRF_PADREV_VBAT_VDD3P3	WRF_PADREV_VBAT_GND3P3	WRF_LO_GND1P2_2	WRF_RXSG_GND1P2	L
M	FM_VCOVSS	FM_LNAVSS	BT_VCOVSS	BT_PLLVDD1P2	BT_PAVSS		WRF_LNA_2G_GND1P2	WRF_PA_VBAT_GND3P3_4	WRF_PA_VBAT_GND3P3_3	WRF_PA_VBAT_GND3P3_2	WRF_PA_VBAT_GND3P3_1	WRF_LNA_5G_GND1P2	M
N	FM_LNAVCOVDD1P2	FM_RFIN	BT_VCOVDD1P2	BT_LNAVDD1P2	BT_RF	BT_PAVDD2P5	WRF_RFIN_2G	WRF_RFOUT_2G	WRF_PA2G_VBAT_VDD3P3	WRF_PA5G_VBAT_VDD3P3	WRF_RFOUT_5G	WRF_RFIN_5G	N
	1	2	3	4	5	6	7	8	9	10	11	12	



Not Used

U2100- MSM8974AC (TOP view)



 Not Used

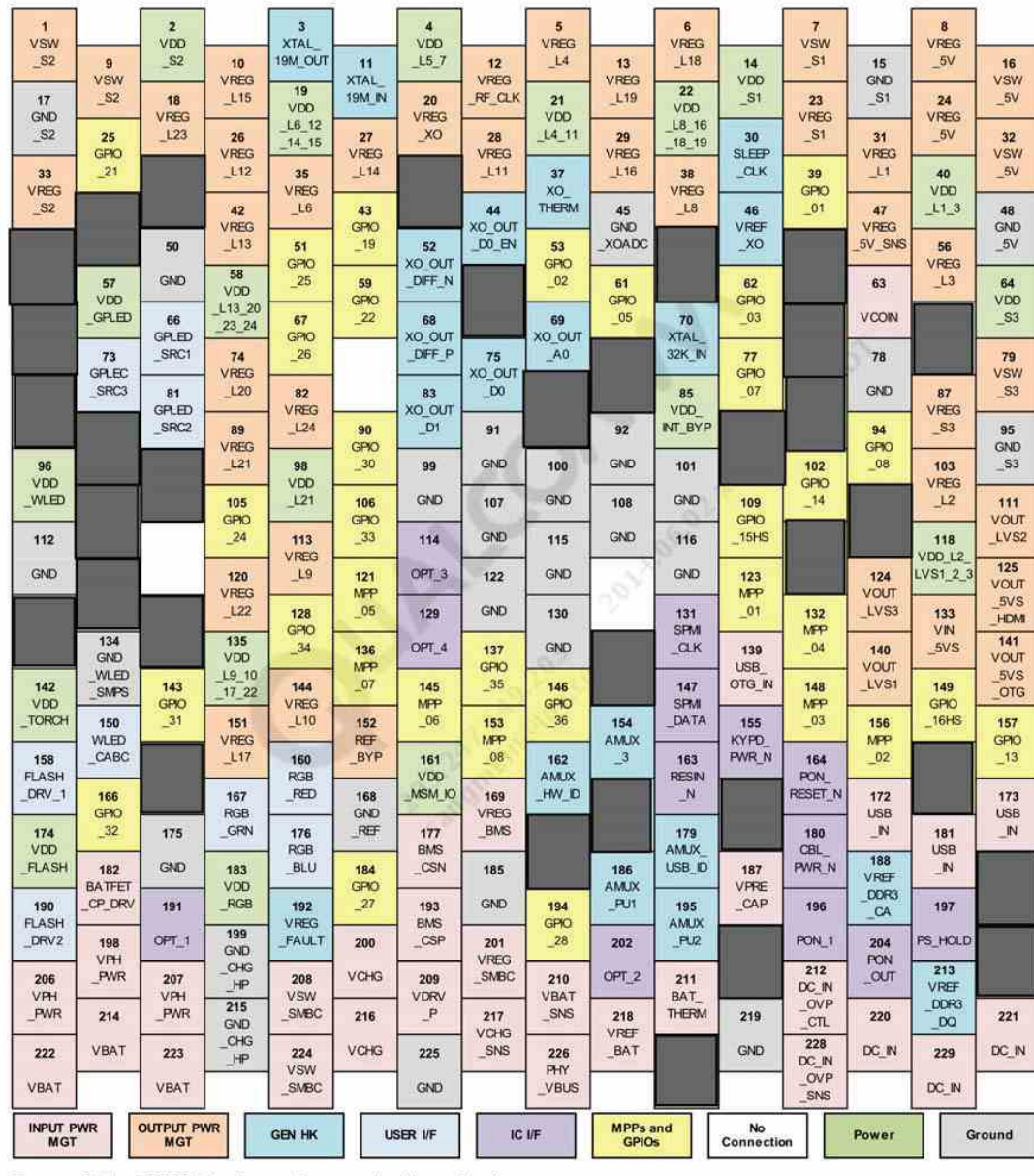
U1500 – WTR1625L (Top View)

			2 DRX_HMB4_IN		3 GND		4 DRX_HB1_IN			6 GND		7 DRX_LB4_IN		8 GND		9 PRX_HB1_IN
10 GND		11 VDD_RF1_D_MB1					14 VDD_RF1_D_LB_LO			16 DRX_LB3_IN						19 PRX_HMB4_IN
			21 GND		22 VDD_RF1_D_HB		23 GND			24 GND		25 VDD_RF1_P_HB_LO		26 GND		
28 DRX_MB1_IN									31 VDD_RF1_D_LB						34 VDD_RF1_P_HMB	35 GND
	36 GNSS_RF_INP				38 VDD_RF1_D_LOM		39 GND		40 GND			41 GND		42 GND		43 PRX_MB2_IN
44 GNSS_RF_IM		45 GND		46 GND		47 GND		48 VDD_RF1_S_VCO	49 GND							51 PRX_MB1_IN
	52 VDD_RF1_G_LNA		53 GNSS_BB_IM		54 VDD_RF2_D_BB				55 GND			56 GND		57 VDD_RF1_P_HMB_LO		58 GND
59 VDD_RF1_G_BB		60 GNSS_BB_IP		61 DRX_BB_OP				62 VDD_RF2_S_VCO	63 GND			64 GND				65 PRX_LB4_IN
	66 GND		67 GNSS_BB_OP		68 DRX_BB_OM				69 GND		70 GND	71 RTUNE		72 VDD_RF1_P_LB		73 PRX_LB3_IN
74 VDD_RF1_G_VCO		75 GND		76 DRX_BB_IP		77 GND		78 VDD_RF1_S_PLL	79 VDD_RF1_P_PLL	80 VDD_RF2_P_VCO		81 GND		82 GND		83 GND
	84 GND		85 GNSS_BB_OM		86 DRX_BB_IM		87 GND		88 GND		89 GND		90 VDD_RF1_P_VCO			92 PRX_LB2_IN
93 VDD_RF1_D_PLL		94 GND		95 SSBI_PRX_DRX		96 GND		97 PRX_BB_OM	98 VDD_RF2_P_BB		99 PRX_BB_IP		100 VDD_RF2_P_RX		101 GND	102 PRX_LB1_IN
	103 VDD_DIO		104 GP_DATAT		105 SSBI_TX_GNSS		106 GND		107 PRX_BB_OP		108 PRX_BB_IM		109 ADC_IN		110 GND	111 GND
112 GND		113 GND		114 VDD_RF2_T_PLL		115 VDD_RF1_T_SYN		116 VDD_RF2_FBRX		117 POET_RFFB		118 GND		119 GND	120 GND	121 TX_HB2_OUT
	122 GND		123 GP_DATA0		124 GND		125 GND		126 VDD_RF2_T_BB		127 DAC_REF		128 GND		129 VDD_RF2_T_DA	130 TX_HB1_OUT
131 XO_IN		132 GND		133 GND				134 GND	135 VDD_RF1_T_LO			136 VDD_RF1_T_UPC		137 VDD_RF1_T_DA	138 TX_MB2_OUT	139 TX_MB3_OUT
		140 GND		141 GND			142 GND		143 GND				144 GND	145 GND		146 TX_MB1_OUT
147 VDD_RF2_XO			148 GND		149 VDD_RF1_T_VCO				150 GND	151 TX_BB_IP		152 TX_BB_OP		153 TX_LB2_OUT	154 TX_LB4_OUT	155 TX_MB4_OUT
	156 GND		157 VDD_RF2_T_VCO		158 GND		159 GND		160 TX_BB_IM		161 TX_BB_OM		162 TX_LB1_OUT		163 TX_LB3_OUT	164 GND

 Not Used

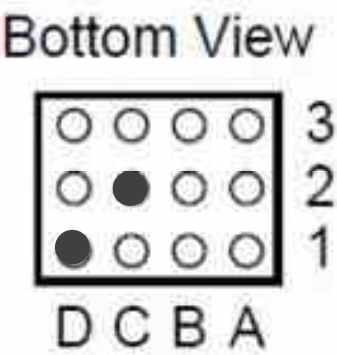
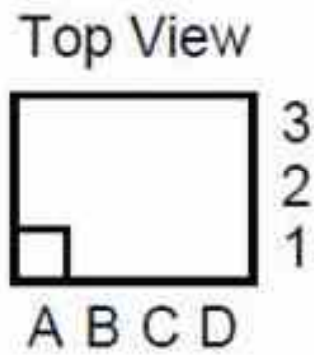
7. BGA PIN MAP

U4100-PM8941 (TOP VIEW)



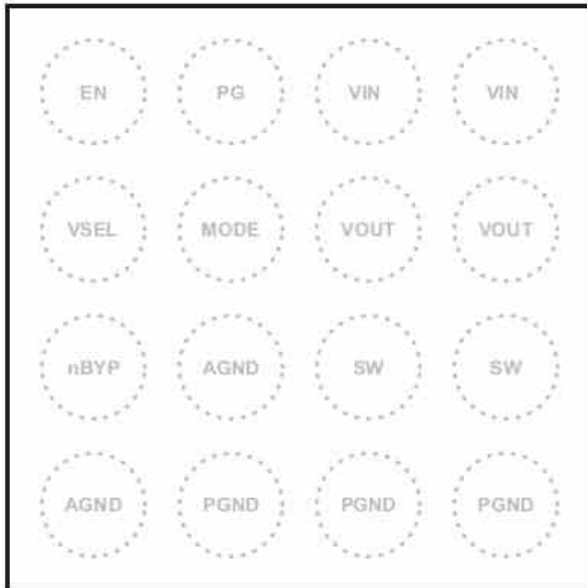
Not Used

U4400-IC,DC-DC Converter

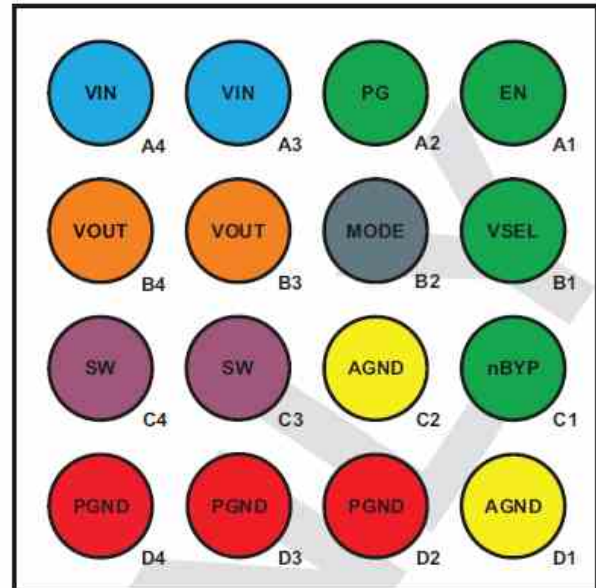


 Not Used

U4900_DC-DC Converter(TOP view)



TOP VIEW



BOTTOM VIEW

All Used

U6200_IC,Speaker Amplifier(Top view)

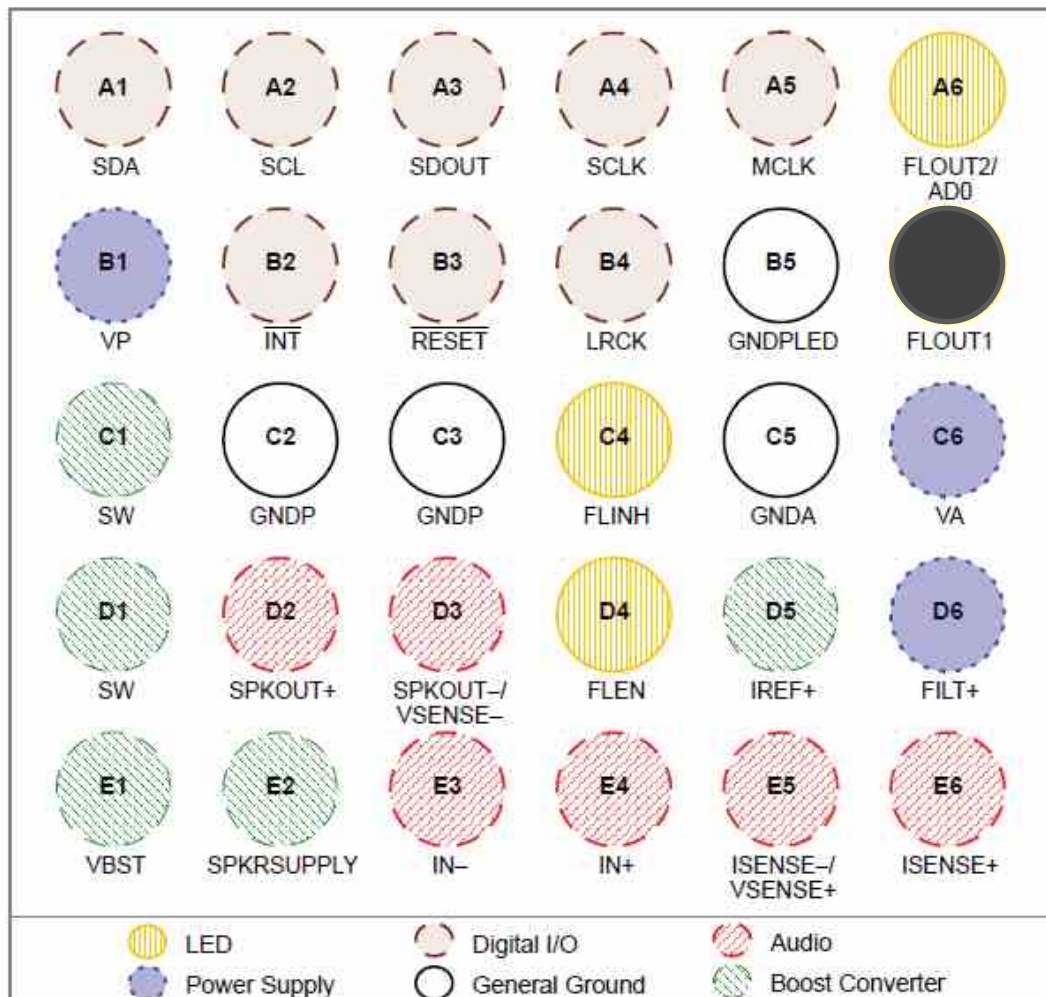
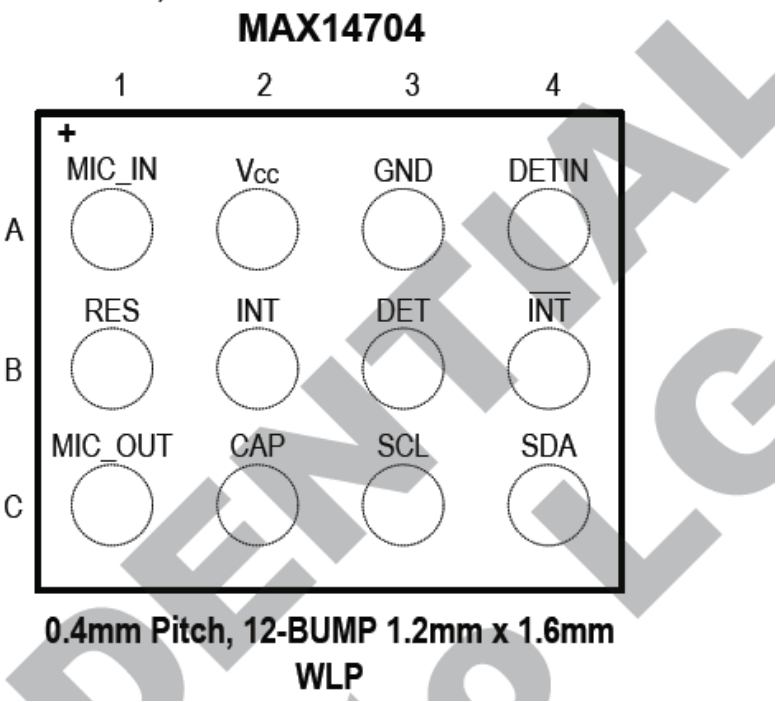


Figure 1-1. Top-Down (Through-Package) View—30-Ball WLCSP Package

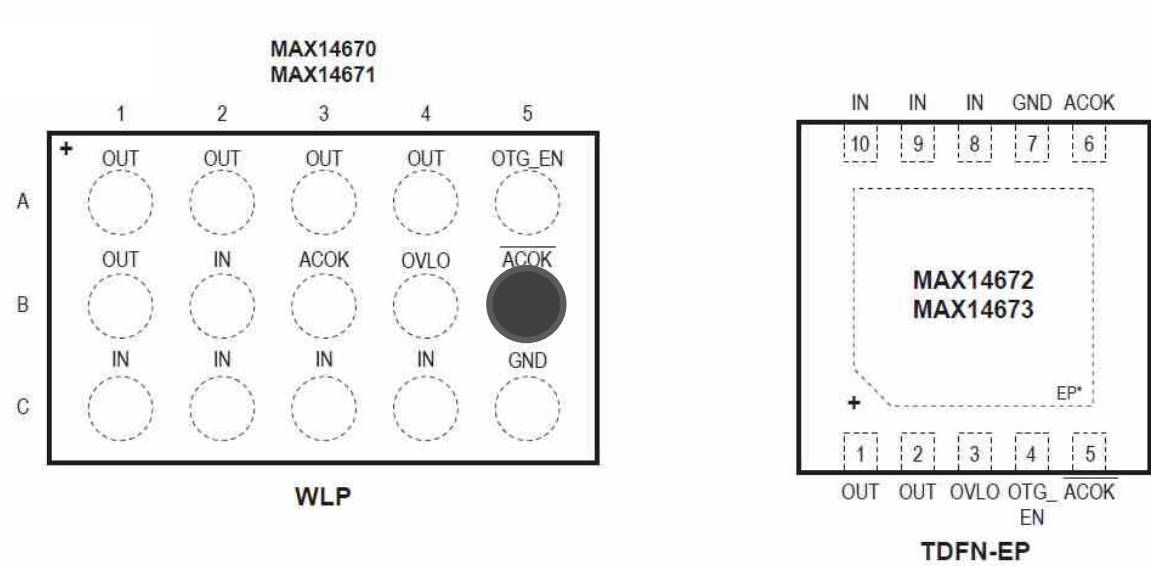
● Not Used

U6420_IC,Comparator(Top view)



ALL USED

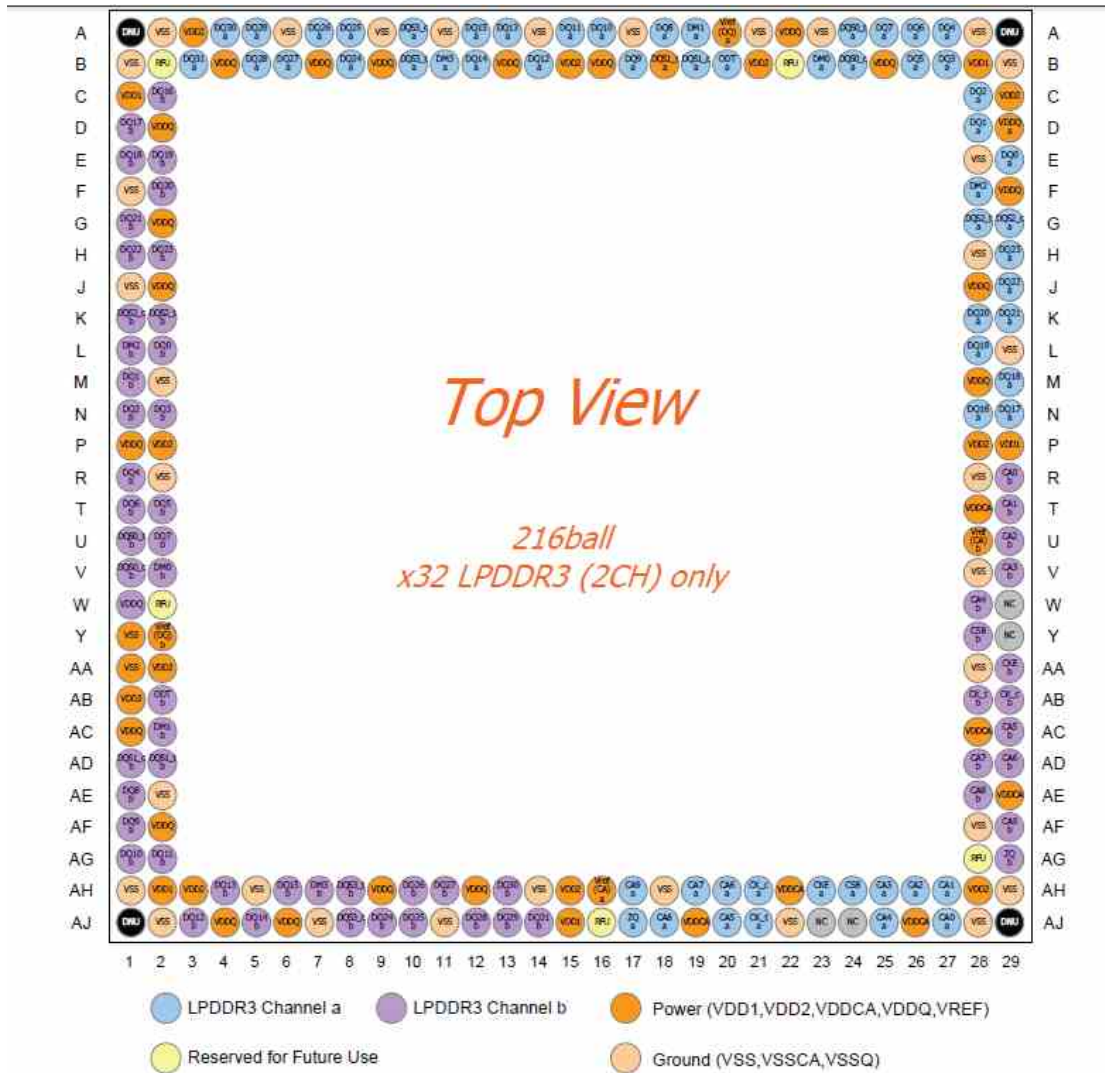
U4750,U4751_IC,over voltage protection(Top view)



Not Used

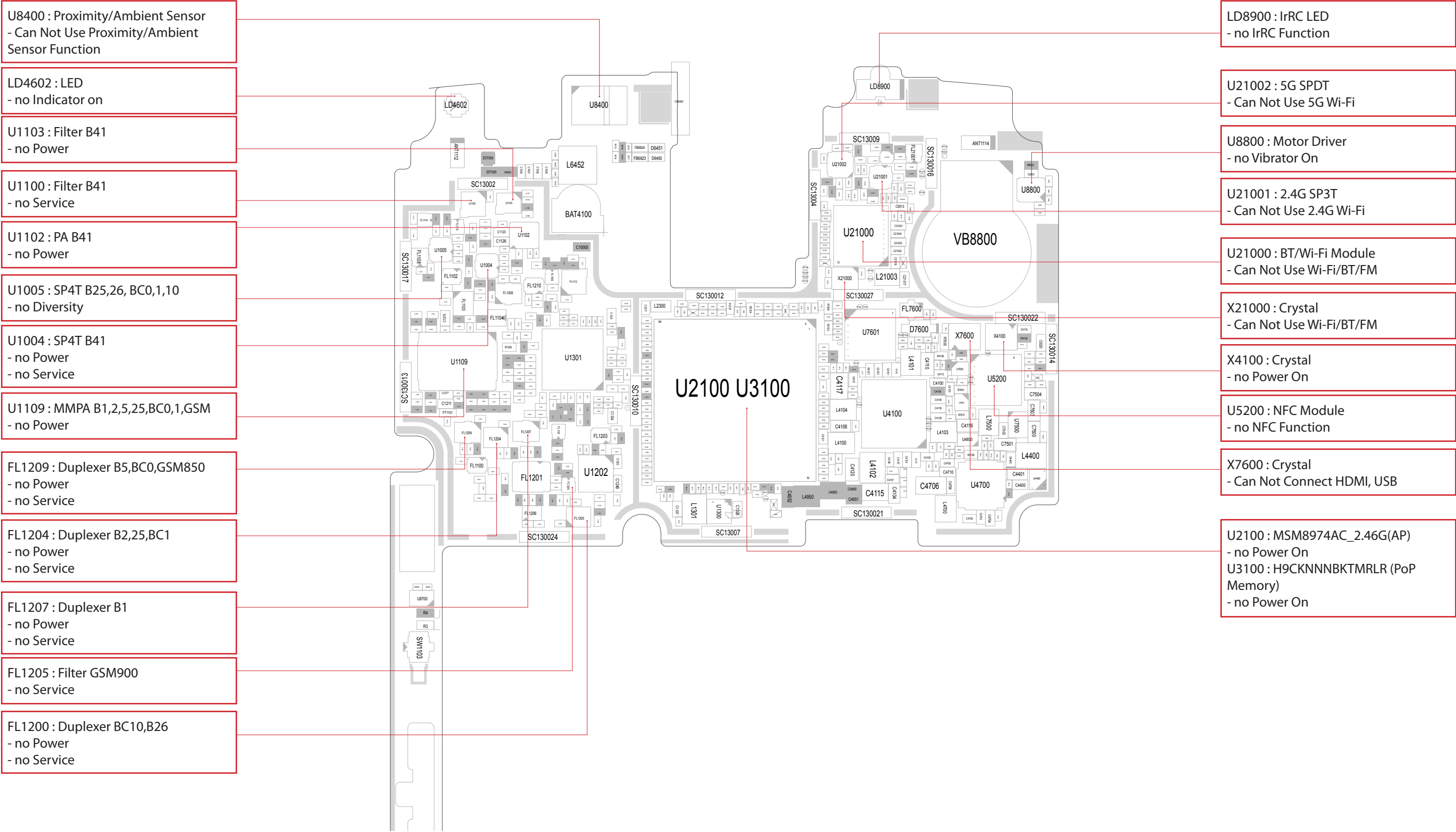
7. BGA PIN MAP

U3100_LPDDR3_SDRAM(Top View)

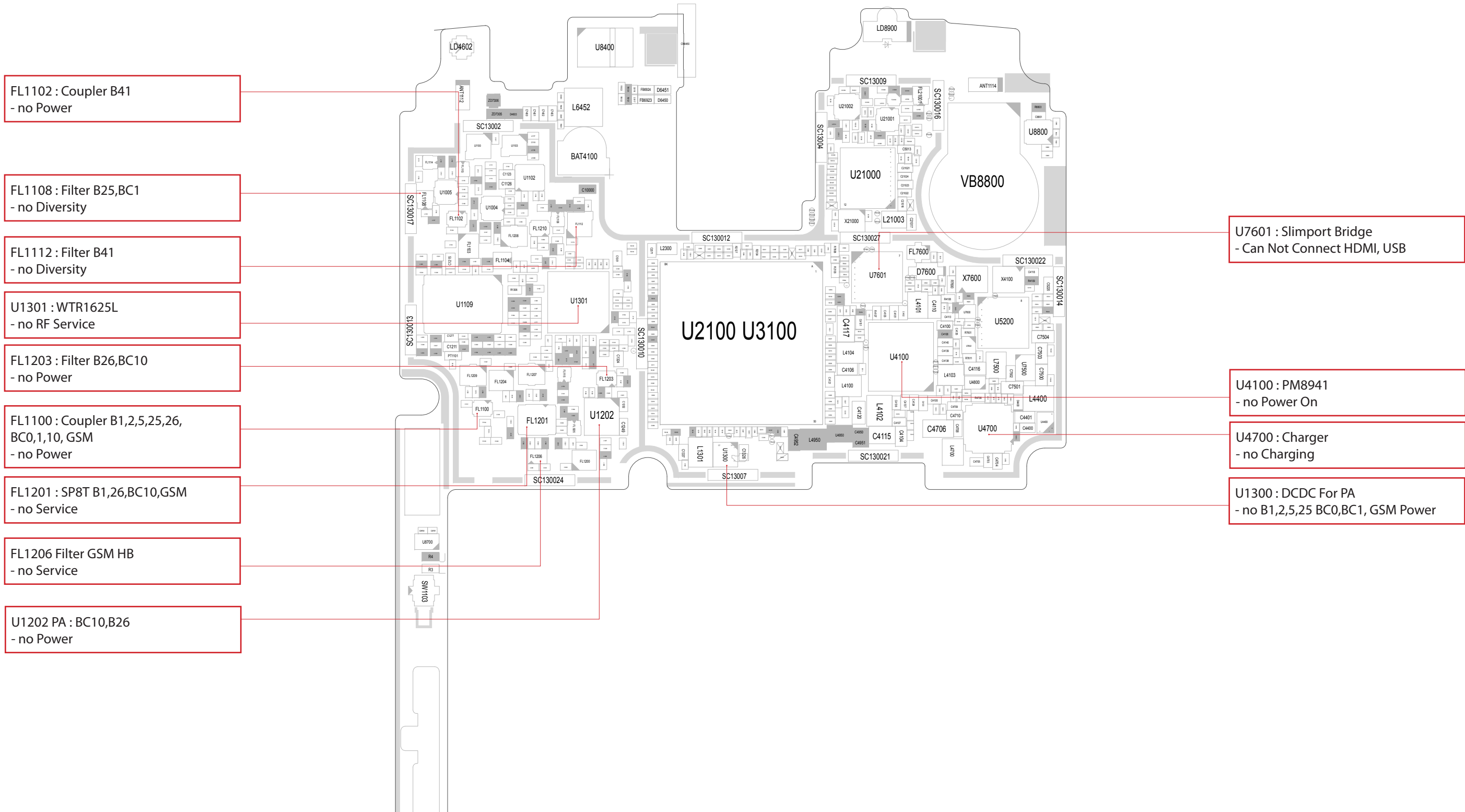


ALL USED

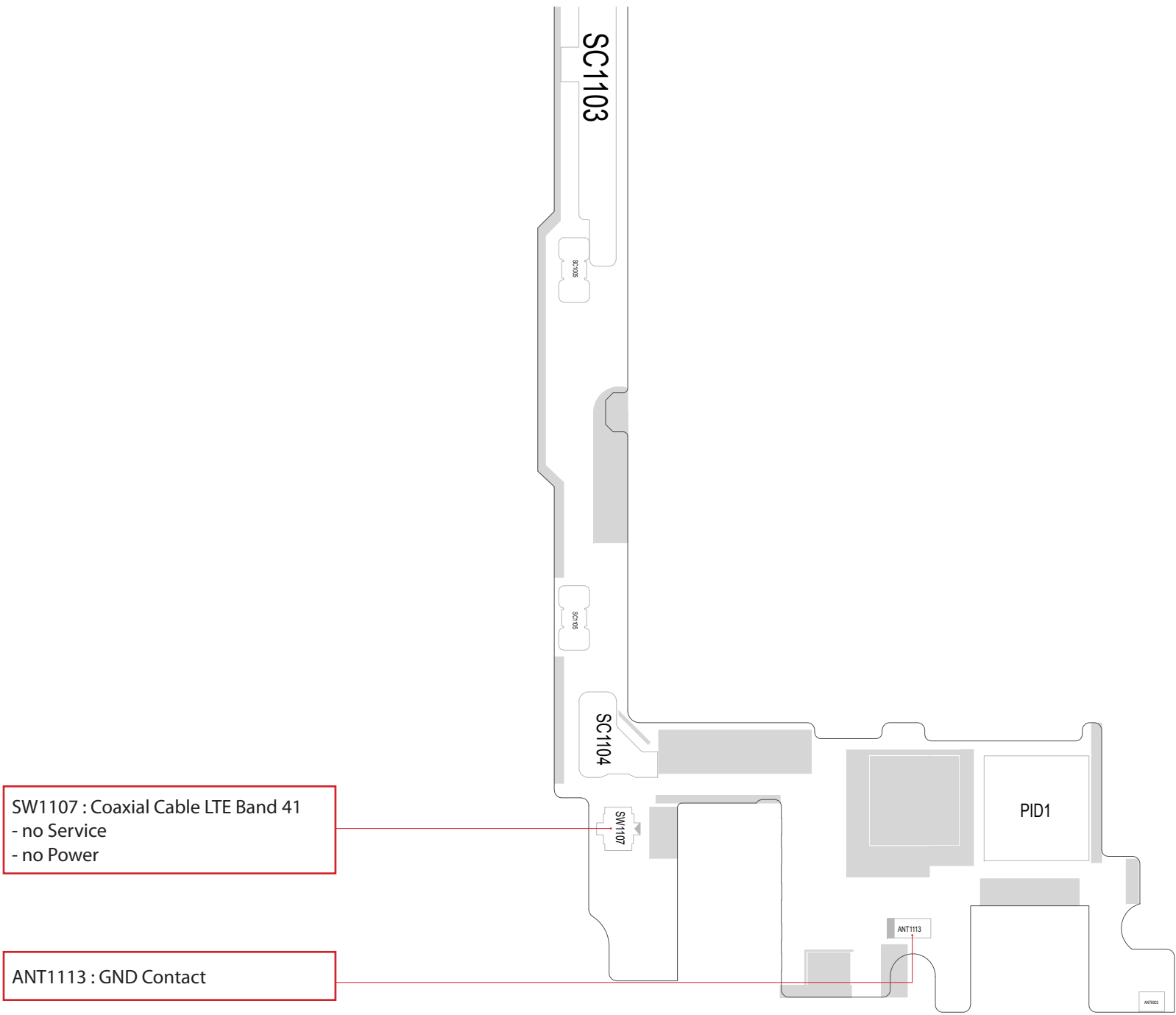
8. PCB LAYOUT



LG-LS990-MAIN_EAX65632401_1.0_TOP-1

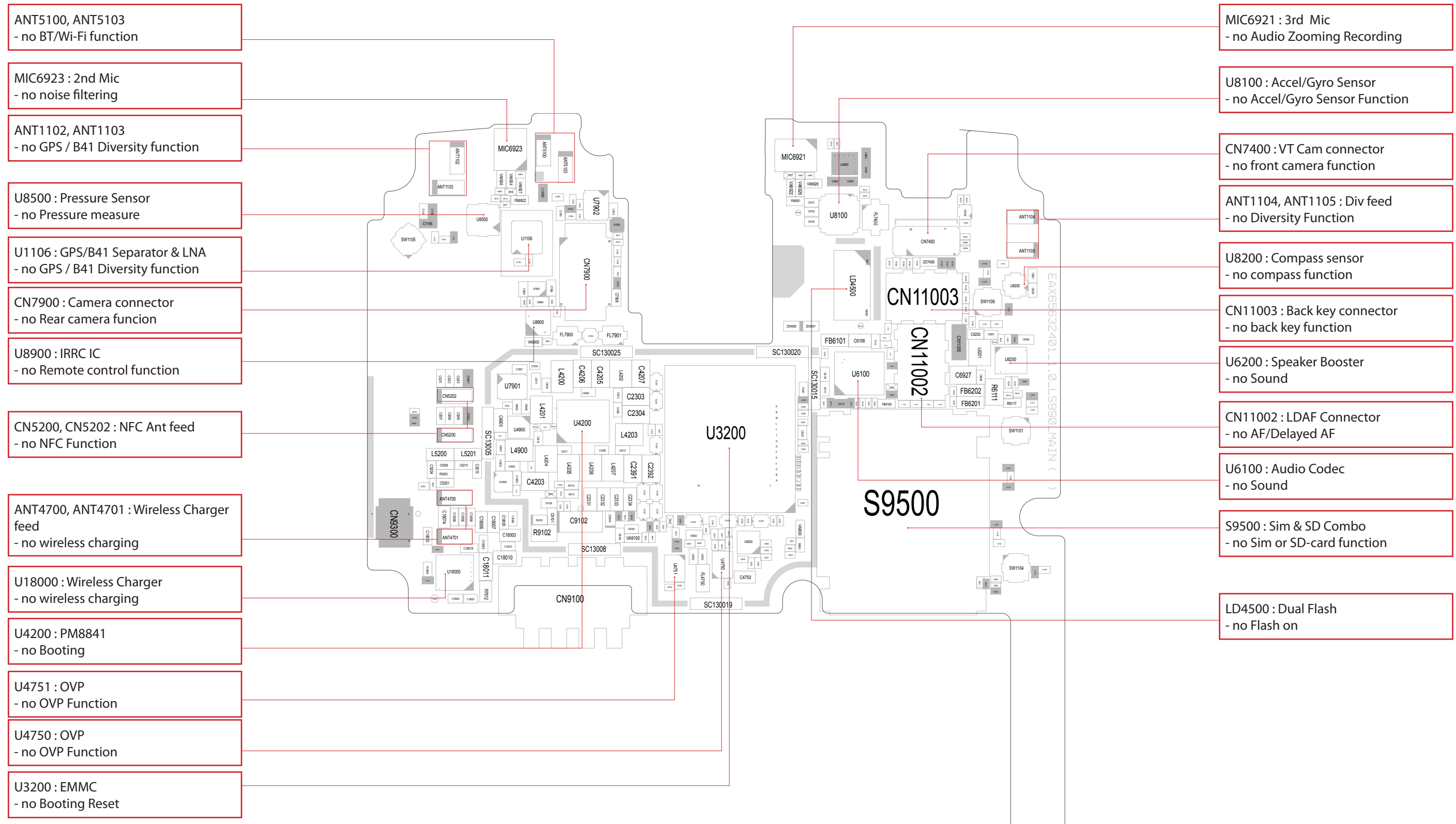


LG-LS990-MAIN_EAX65632401_1.0_TOP-2



LG-LS990-MAIN_EAX65632401_1.0_TOP-3

8. PCB LAYOUT



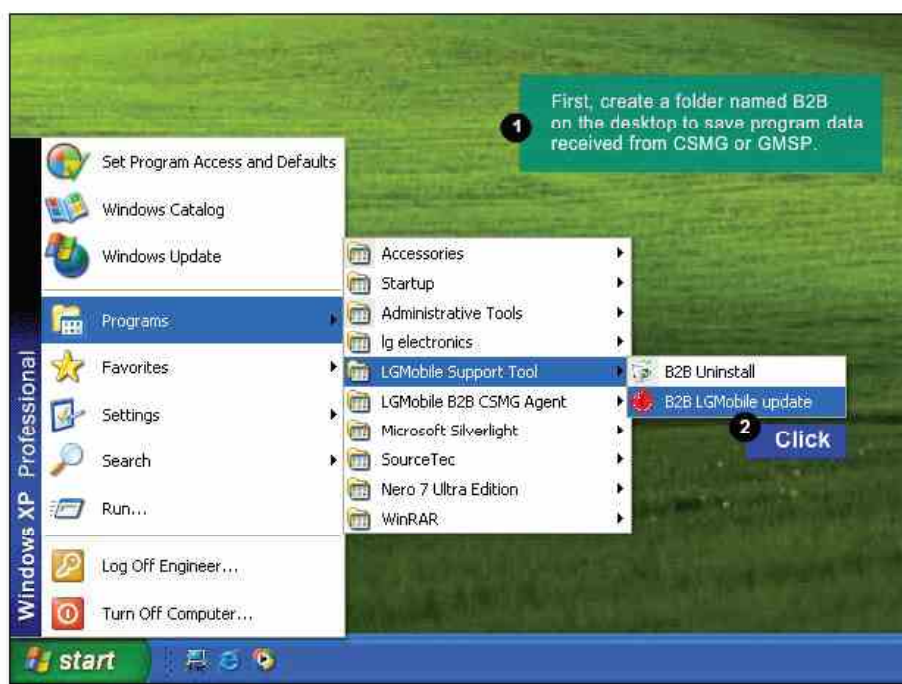
LG-LS990-MAIN_EAX65632401_1.0_BOT-1

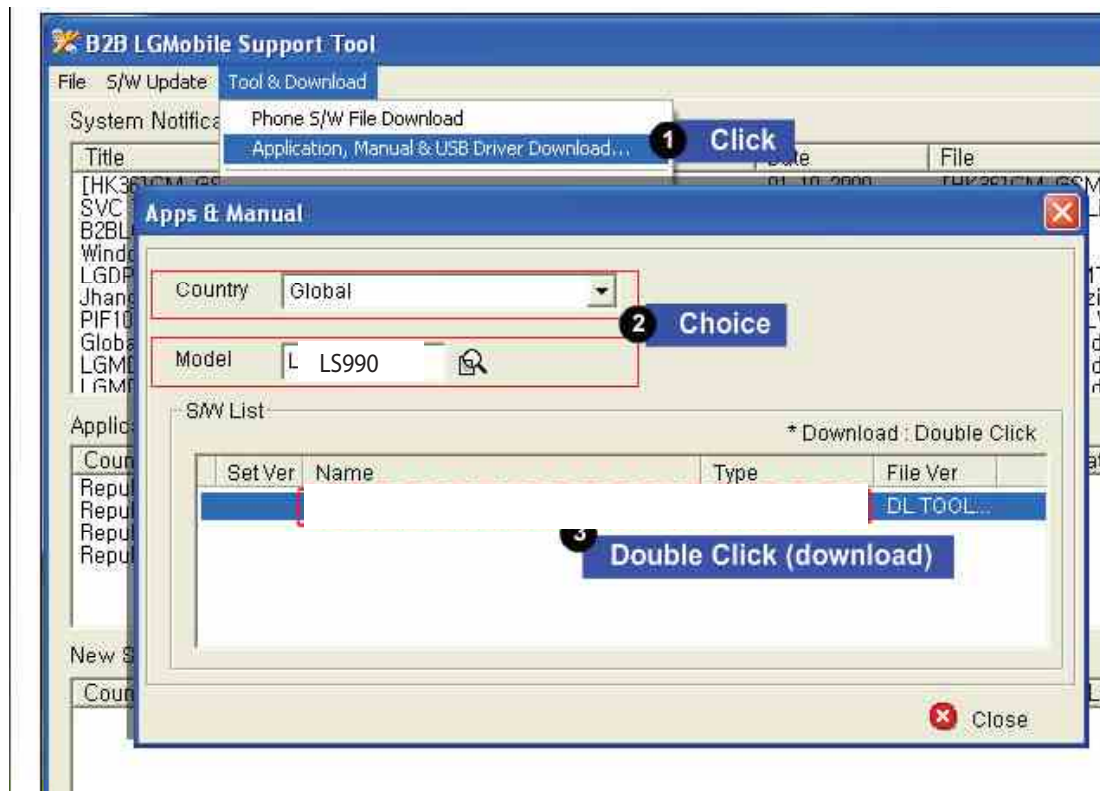


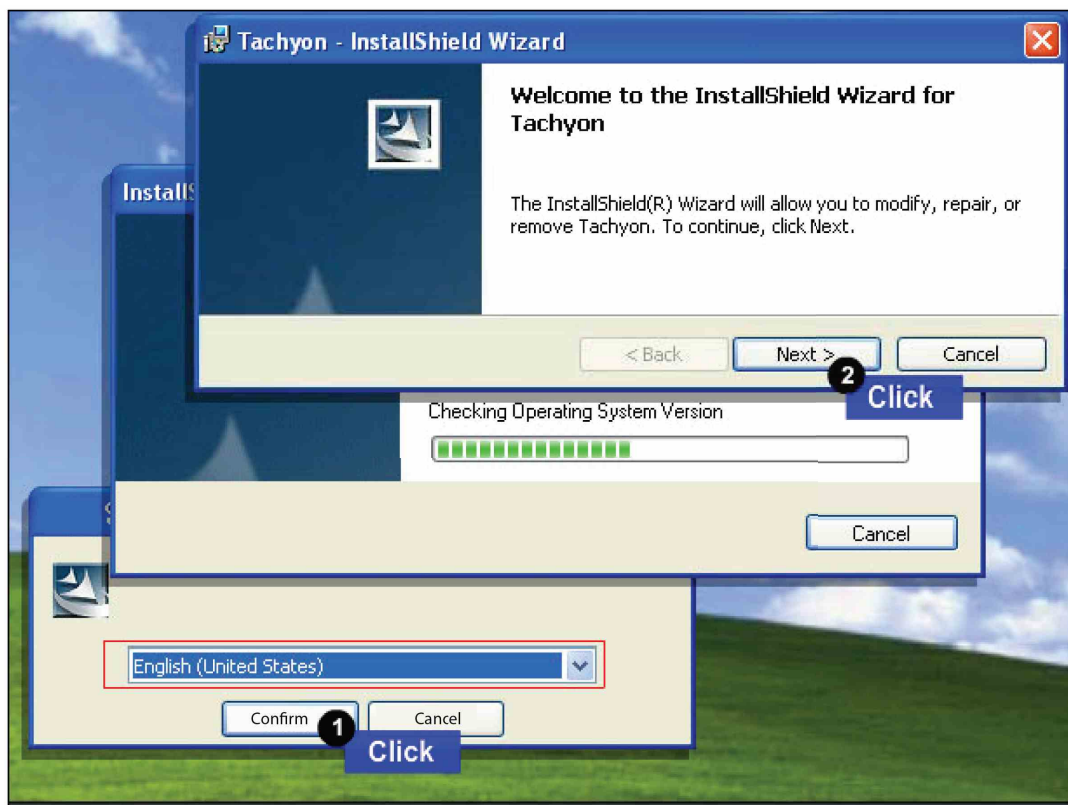
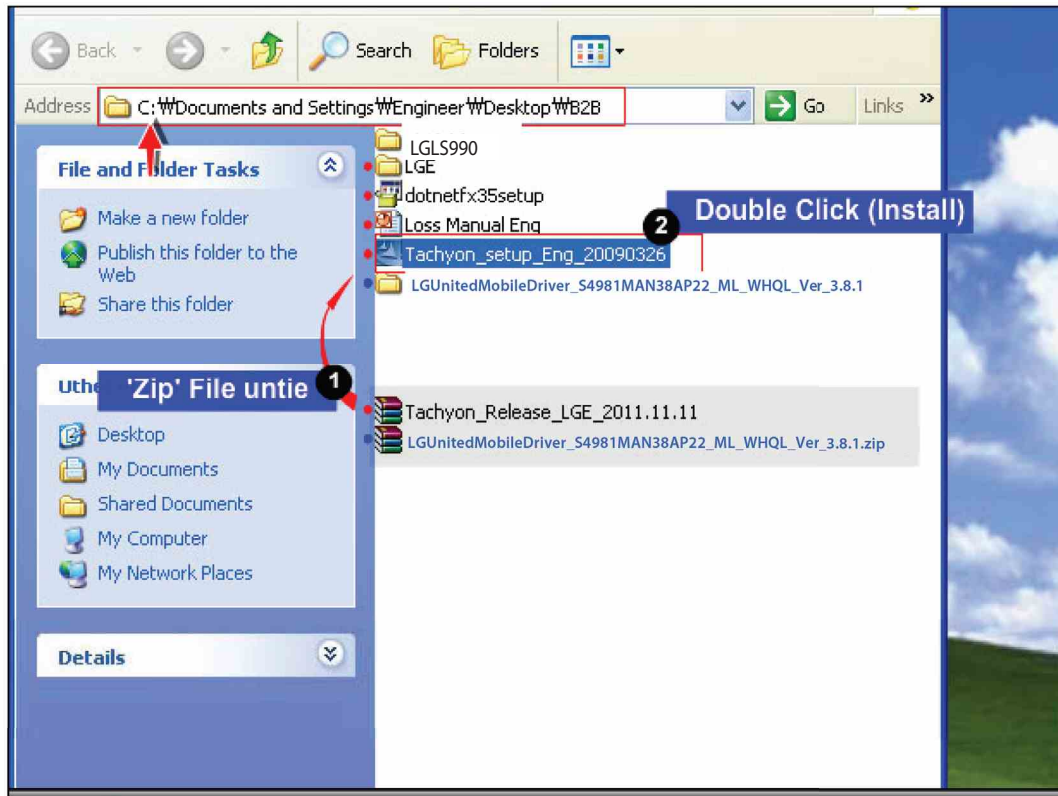
9. CALIBRATION

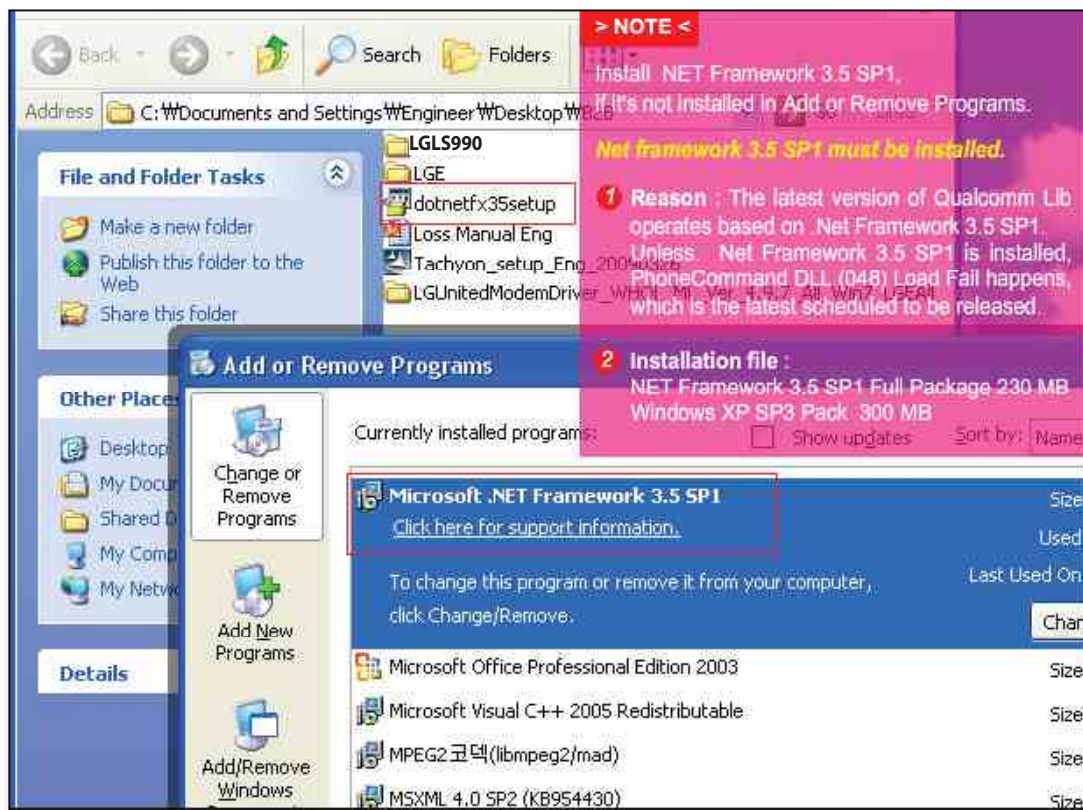
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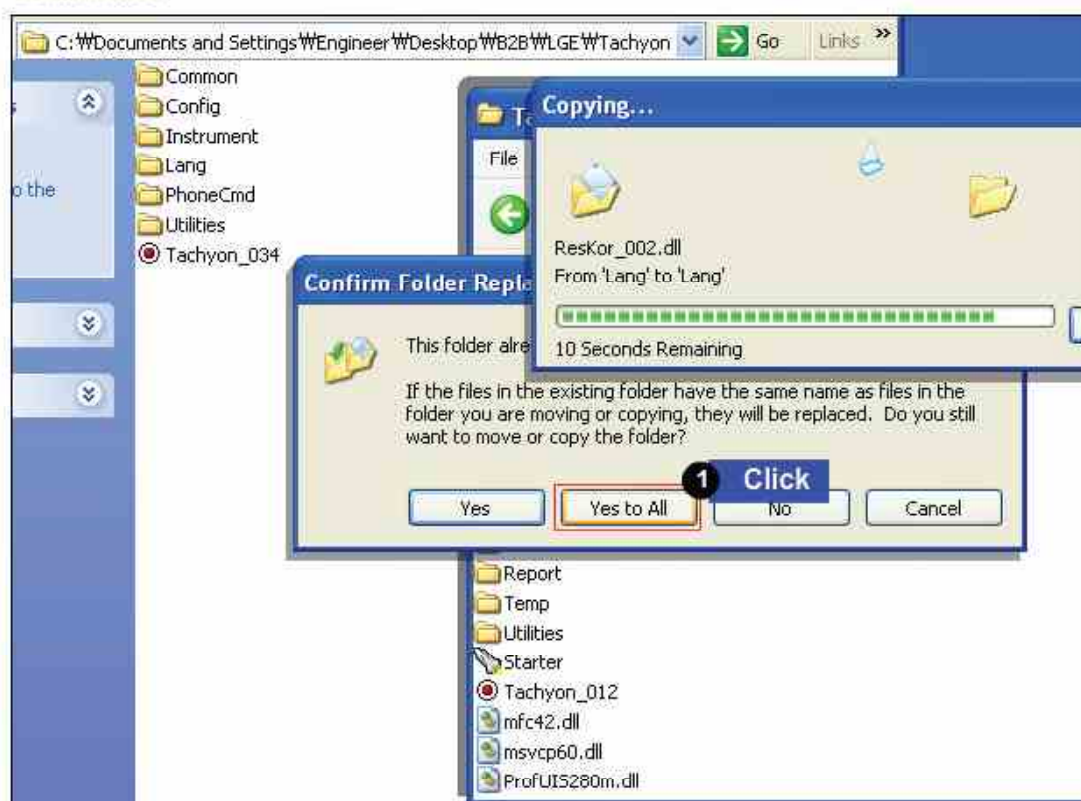
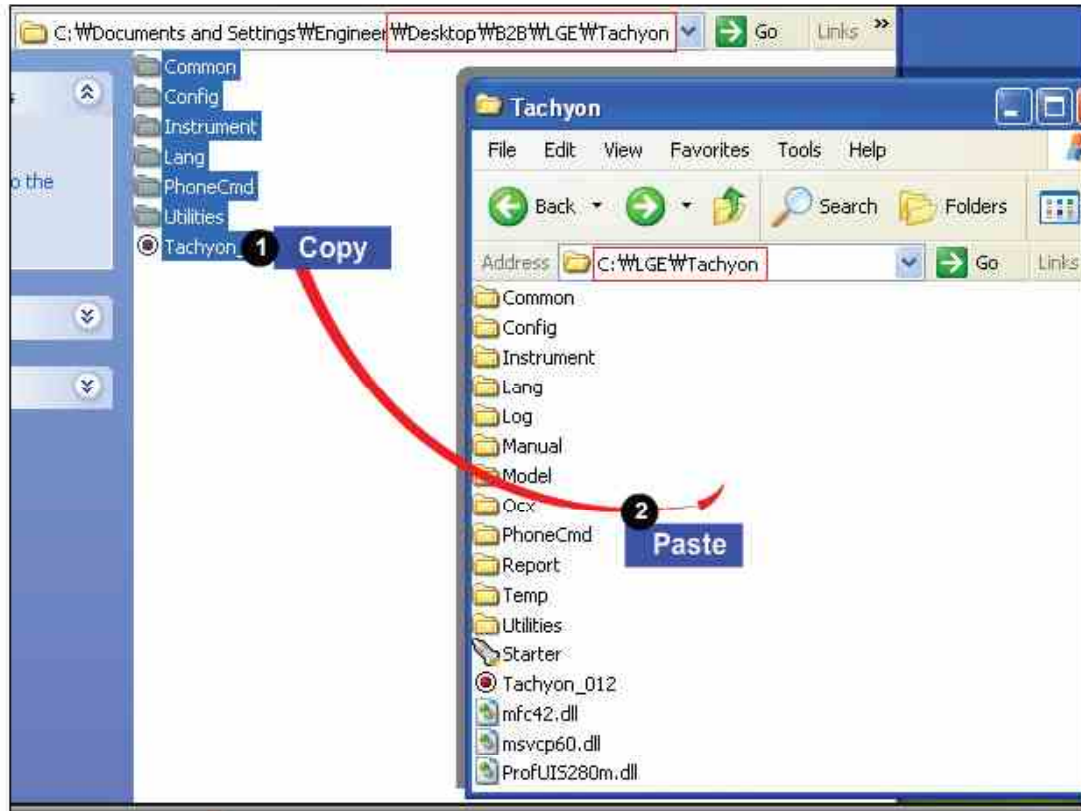
CAL INFORMATION		
S/W VERSION		
Please Check the Version to "B2B"		
H/W		
	Name	Part No.
PIF	PIF200(All Type)	BJAY0024021
USB Cable	USB Cable	RAD32247898
Power Cable	DC Power Cable	RAD32247878
I/O Cable		
RF Cable_Main		
Power Supply_PIF	Power Supply 5.3V	
Power Supply_Phone	Power Supply 5.0V	
RF Test Equipment	CMW500, RF800A	
NOTICE		
CMW500 RF Cable connection		



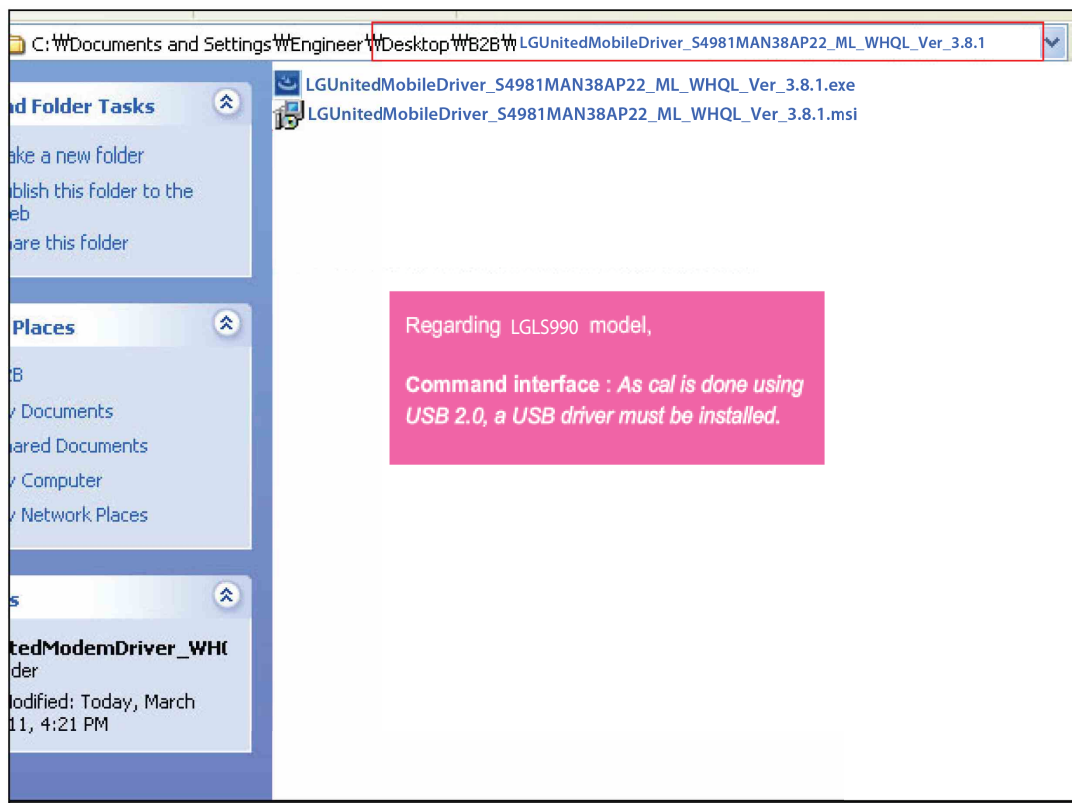
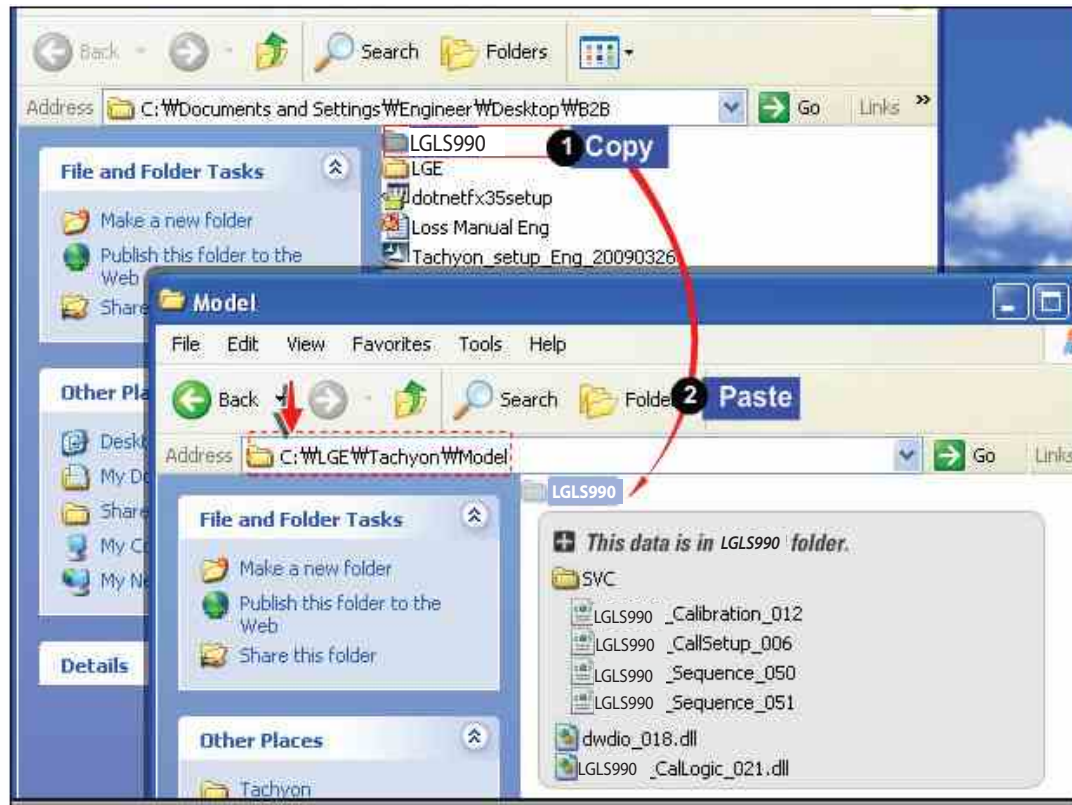


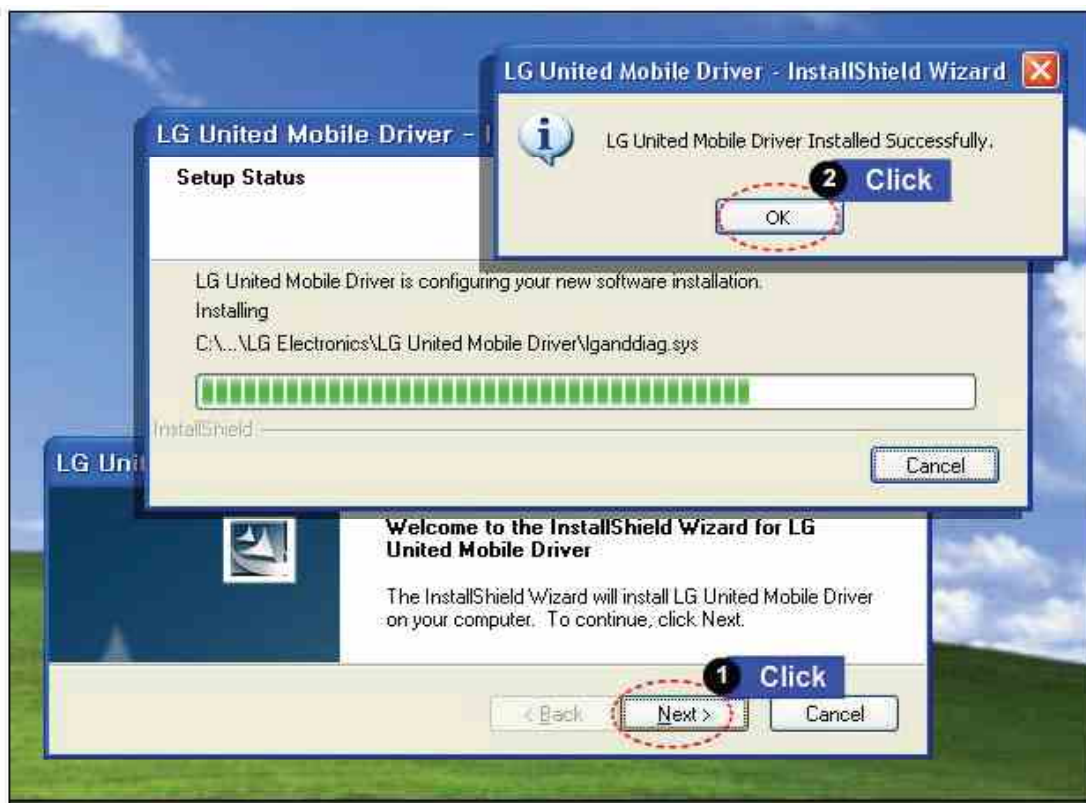
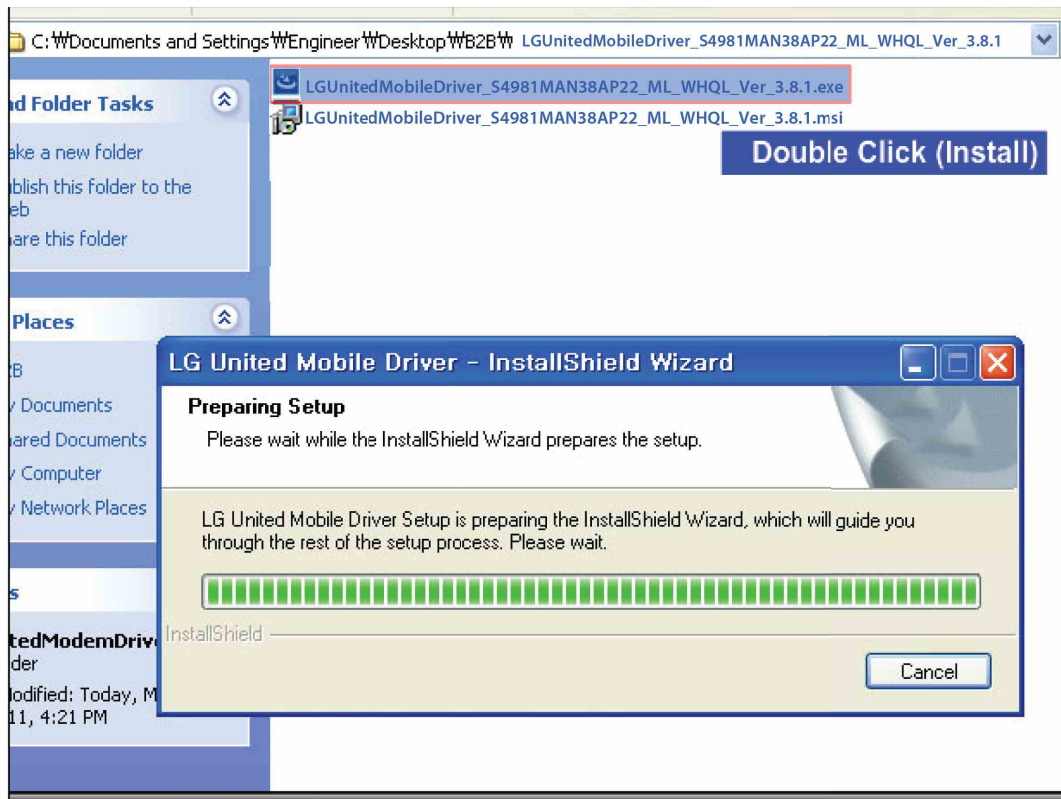




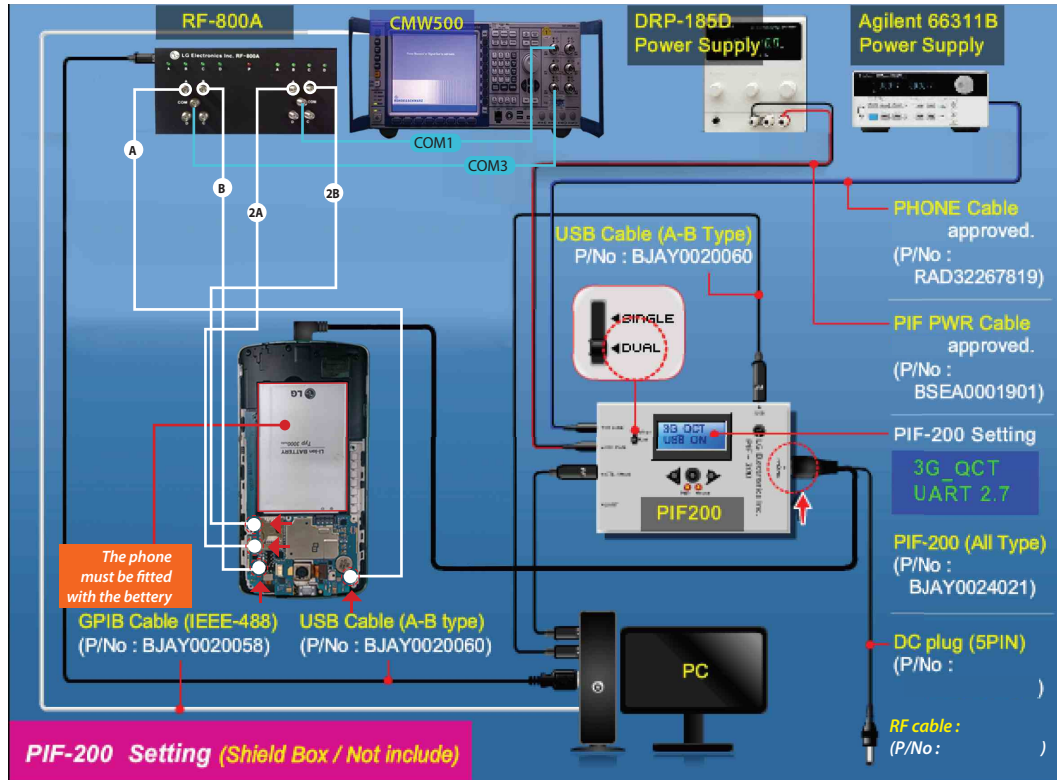


9. CALIBRATION

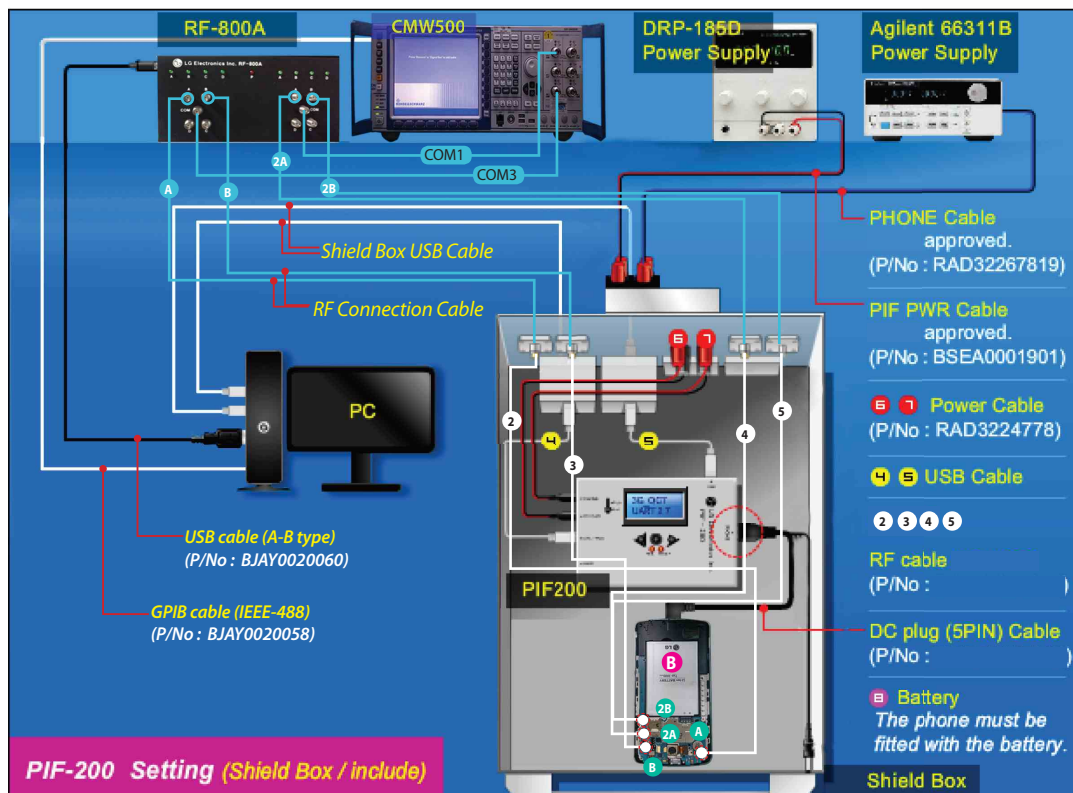


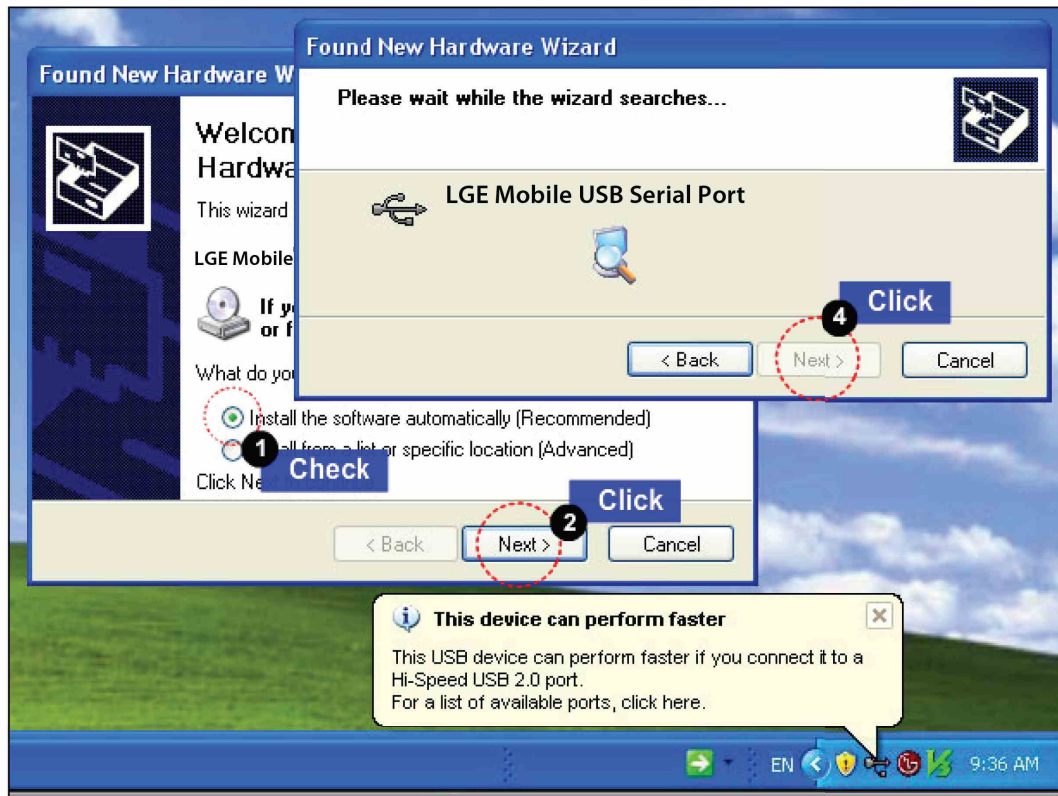


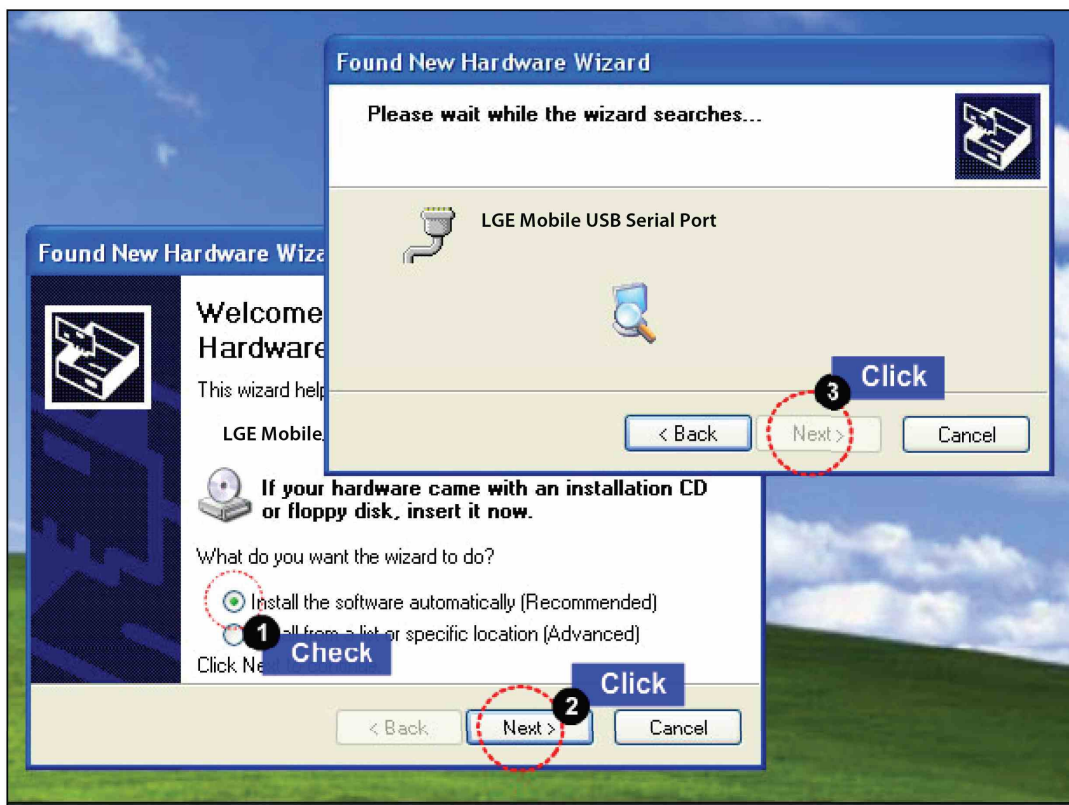
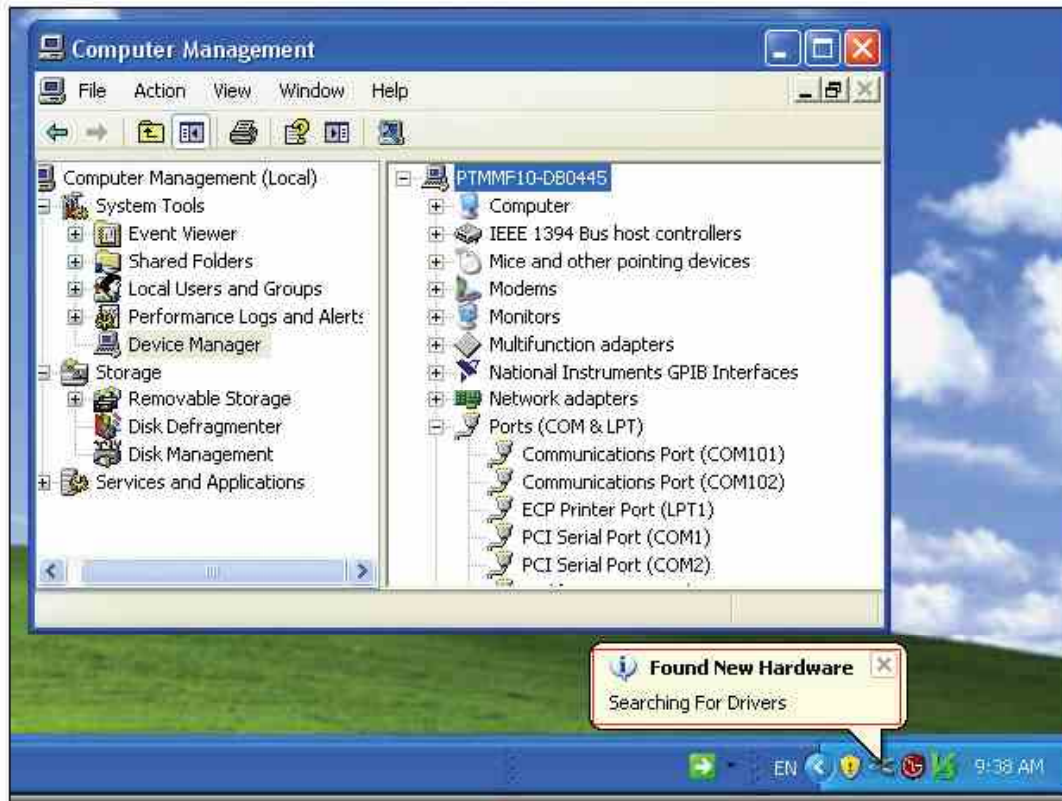
9. CALIBRATION

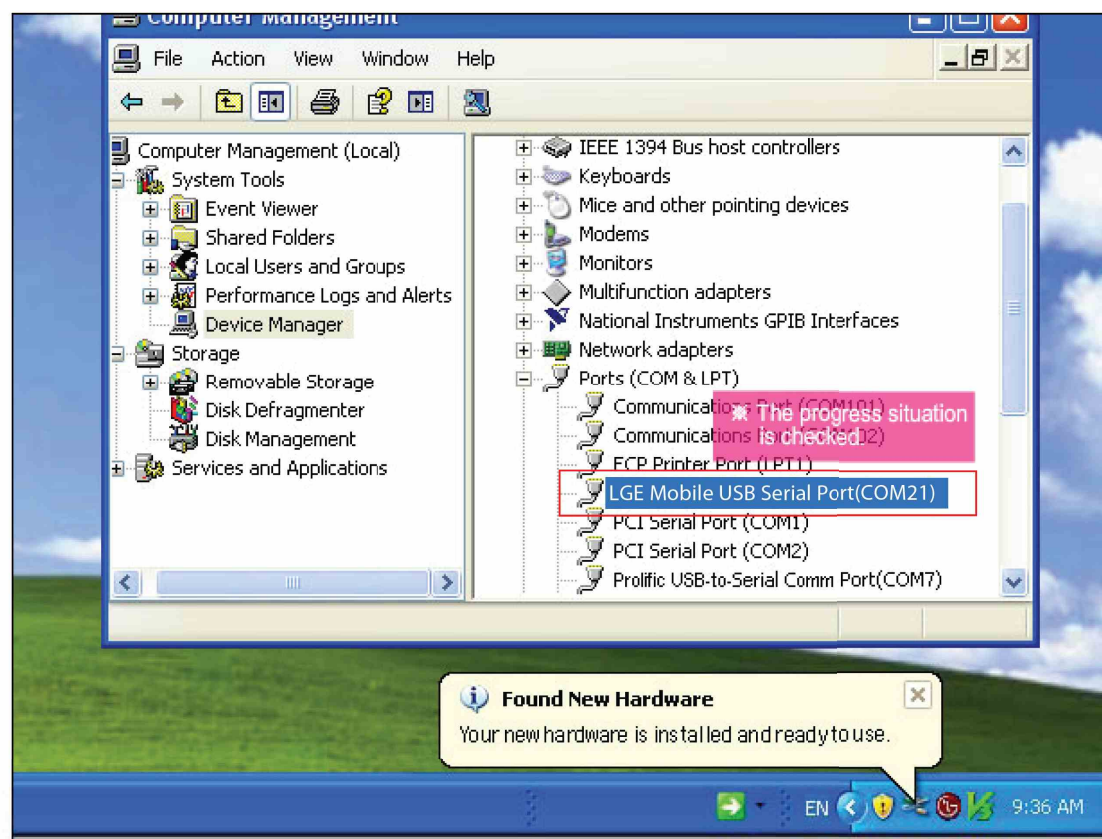


→ This is for example.
You can check "C:\LGE\Tachyon\Model\LS990\Setting Guide" ←

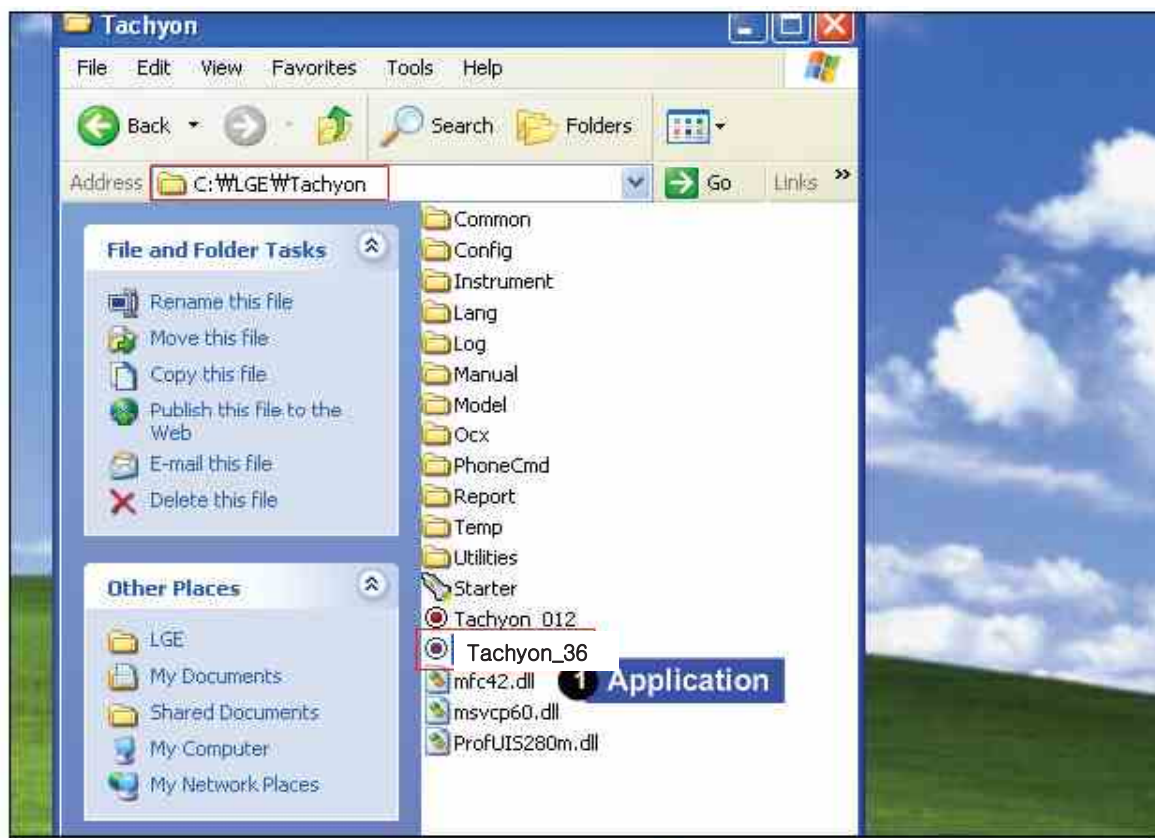
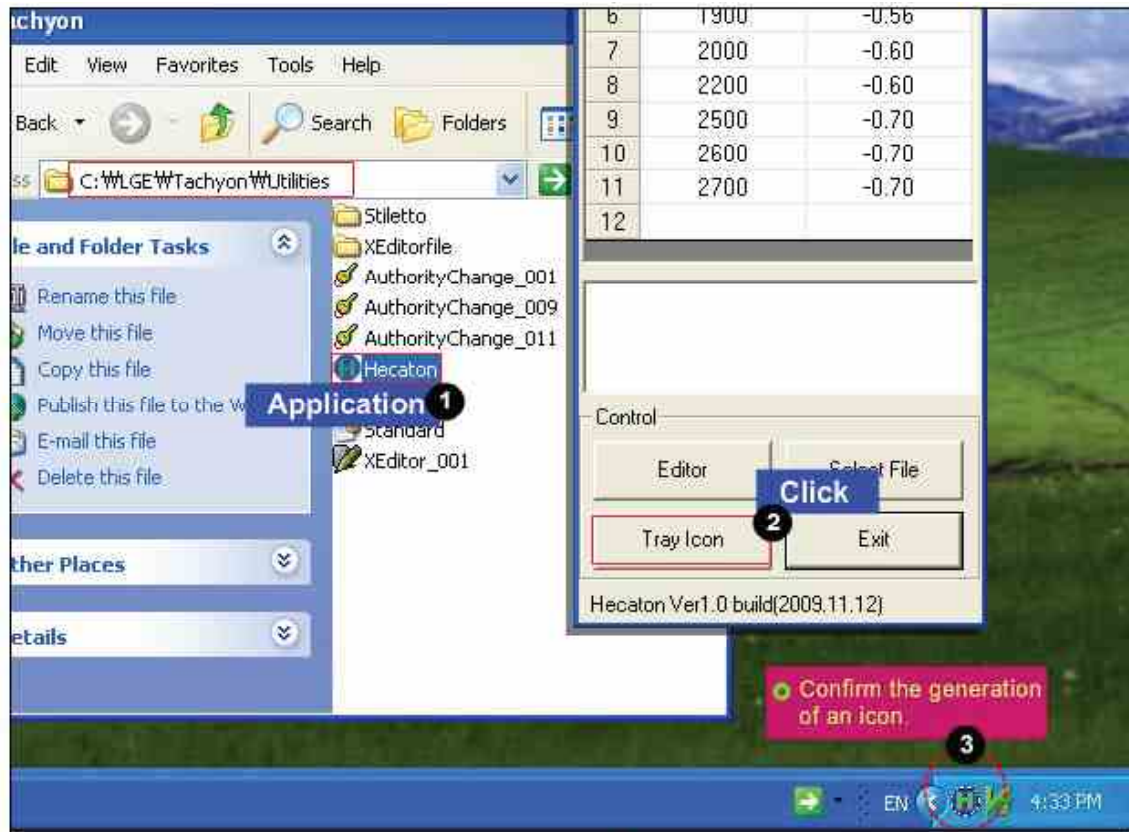




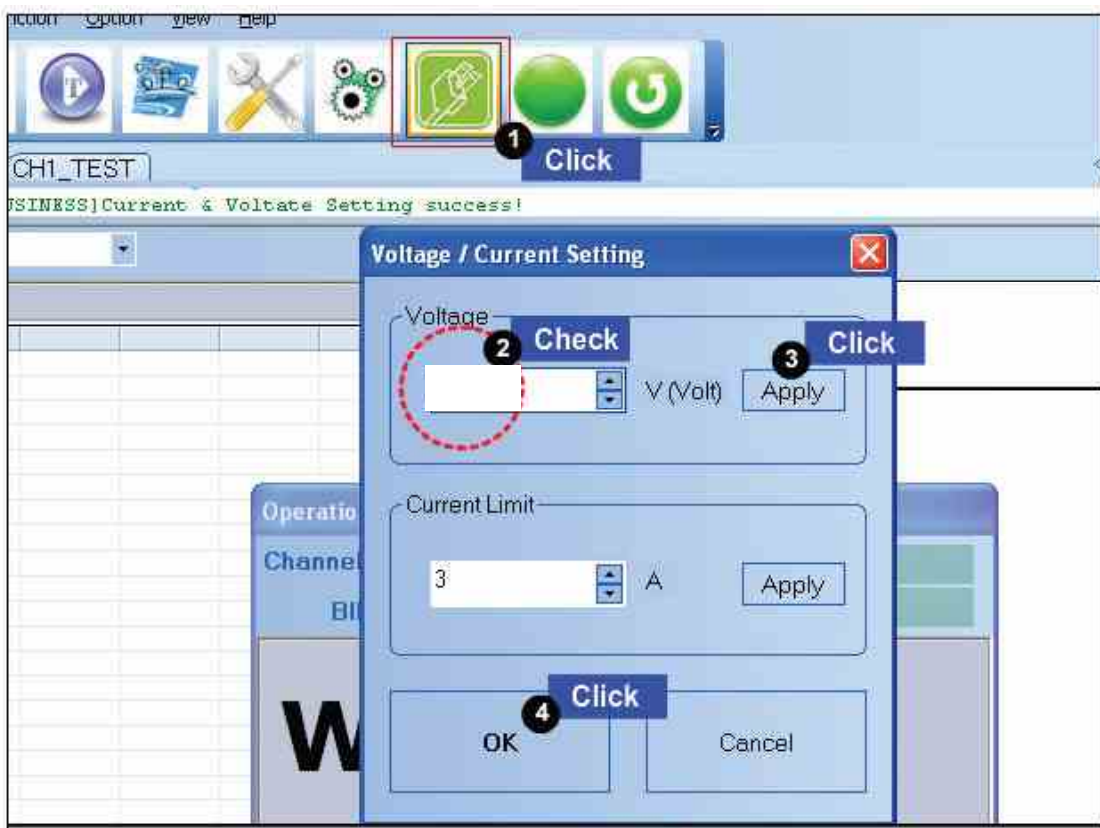
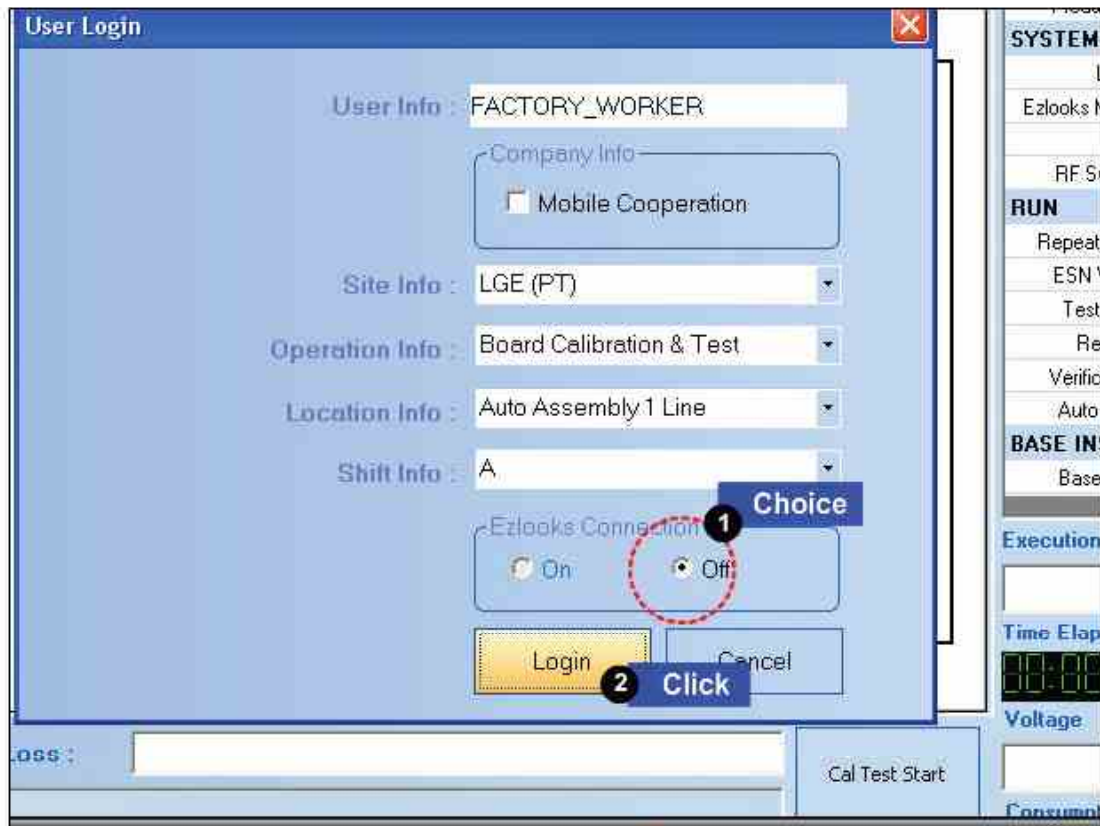


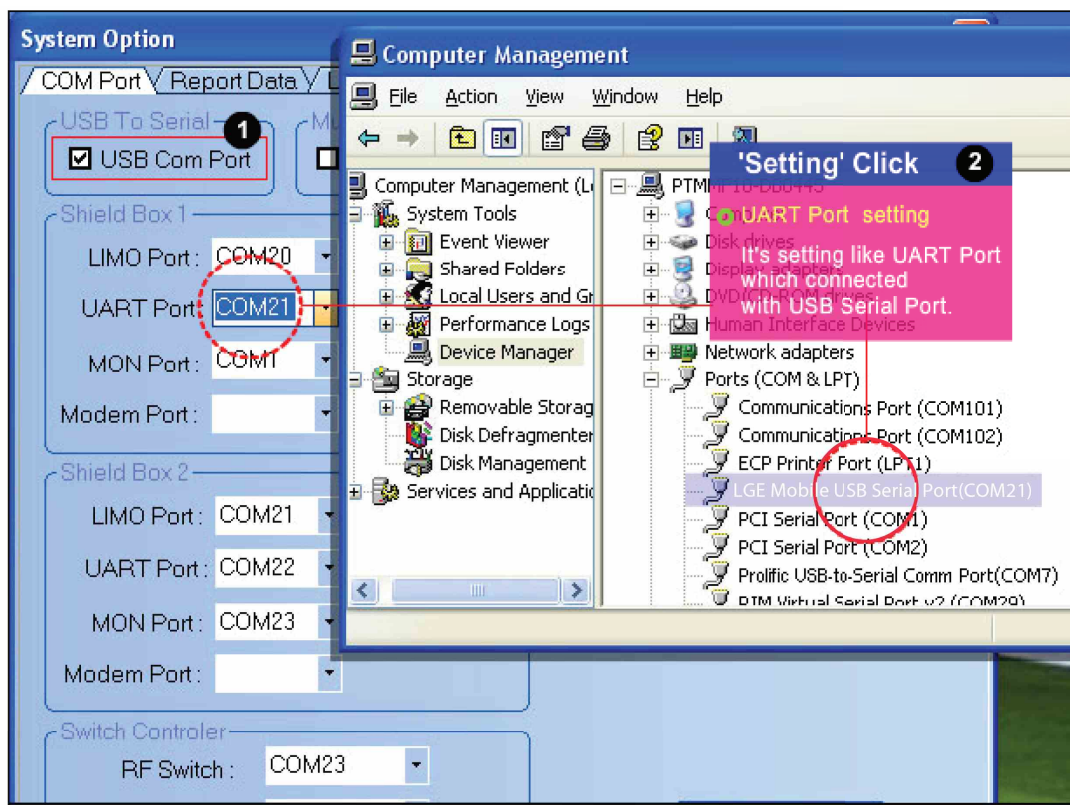
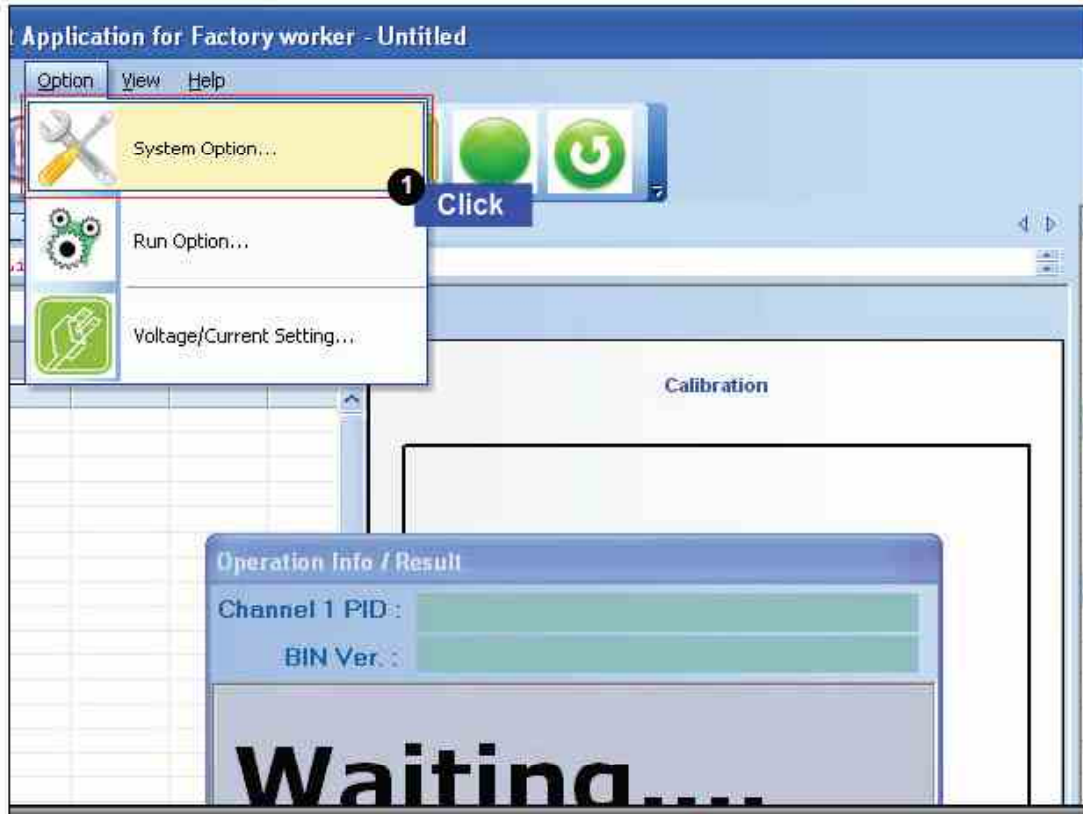


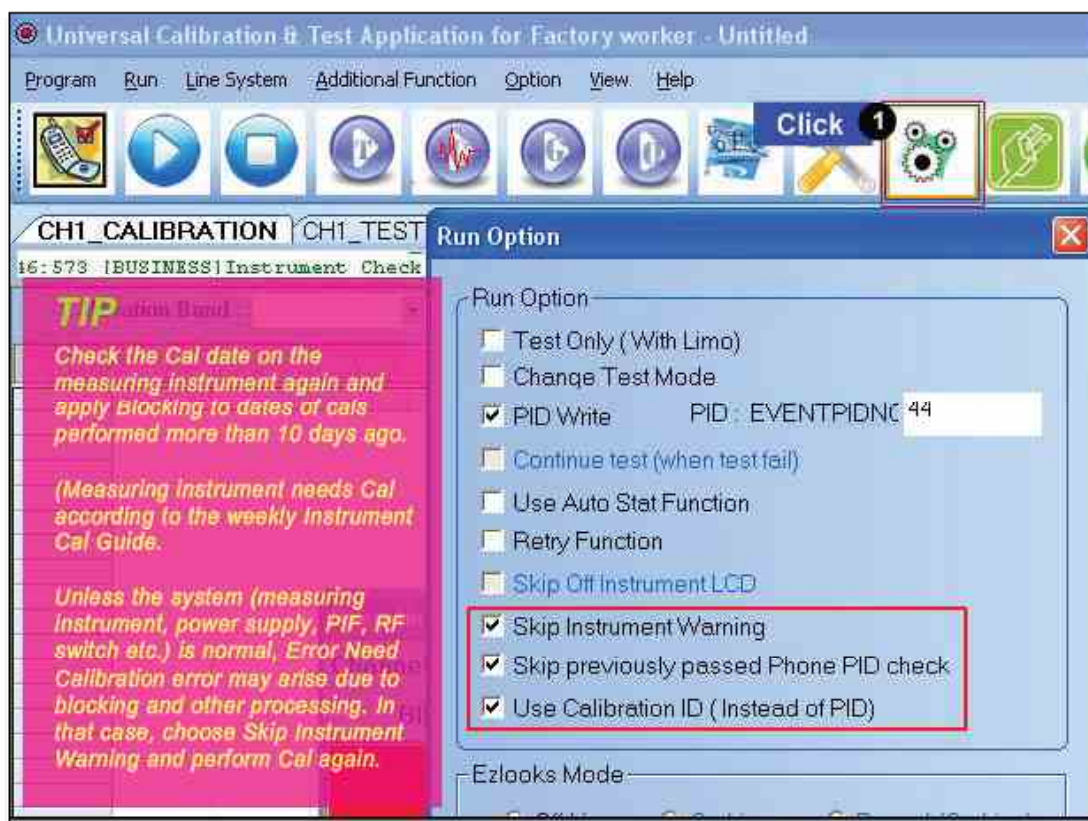
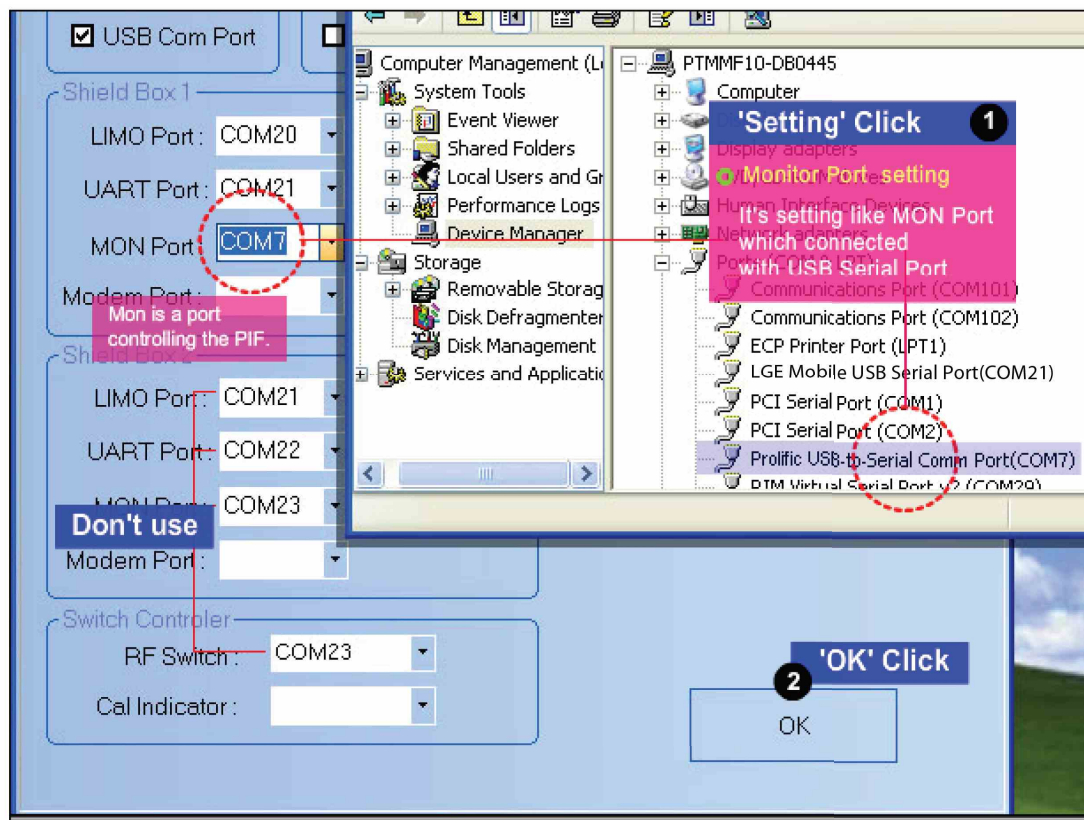
9. CALIBRATION



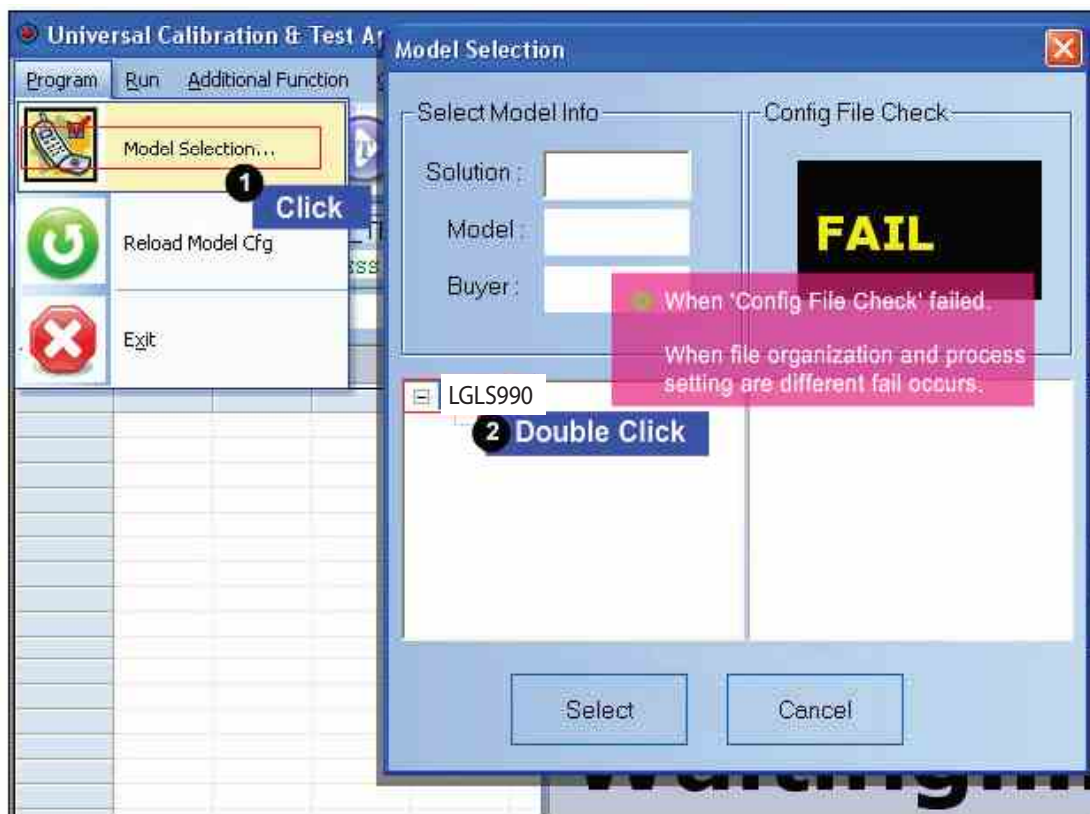
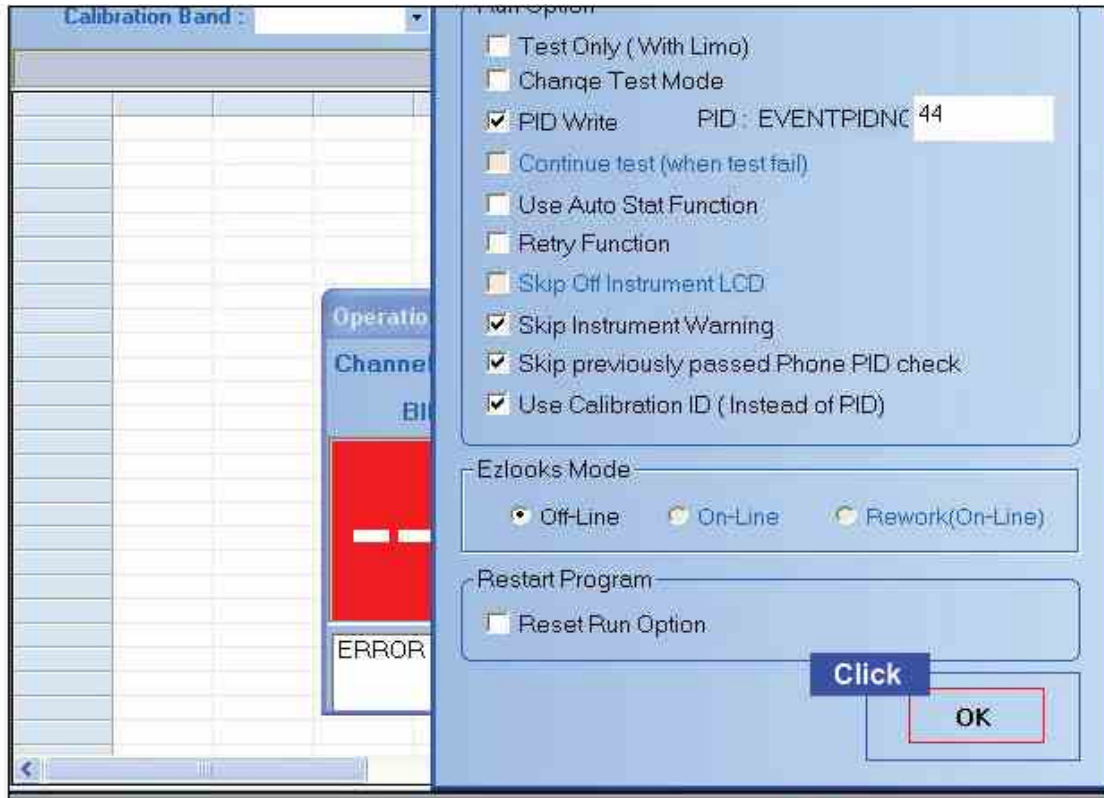
9. CALIBRATION



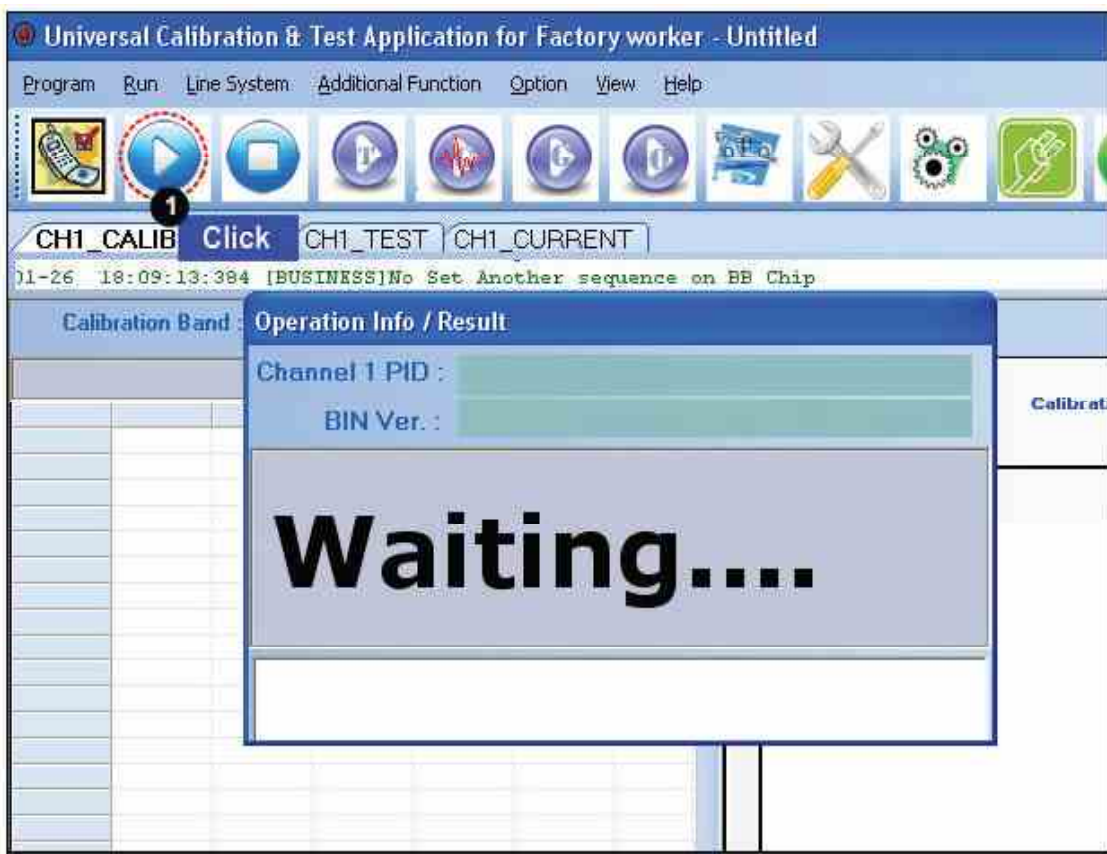
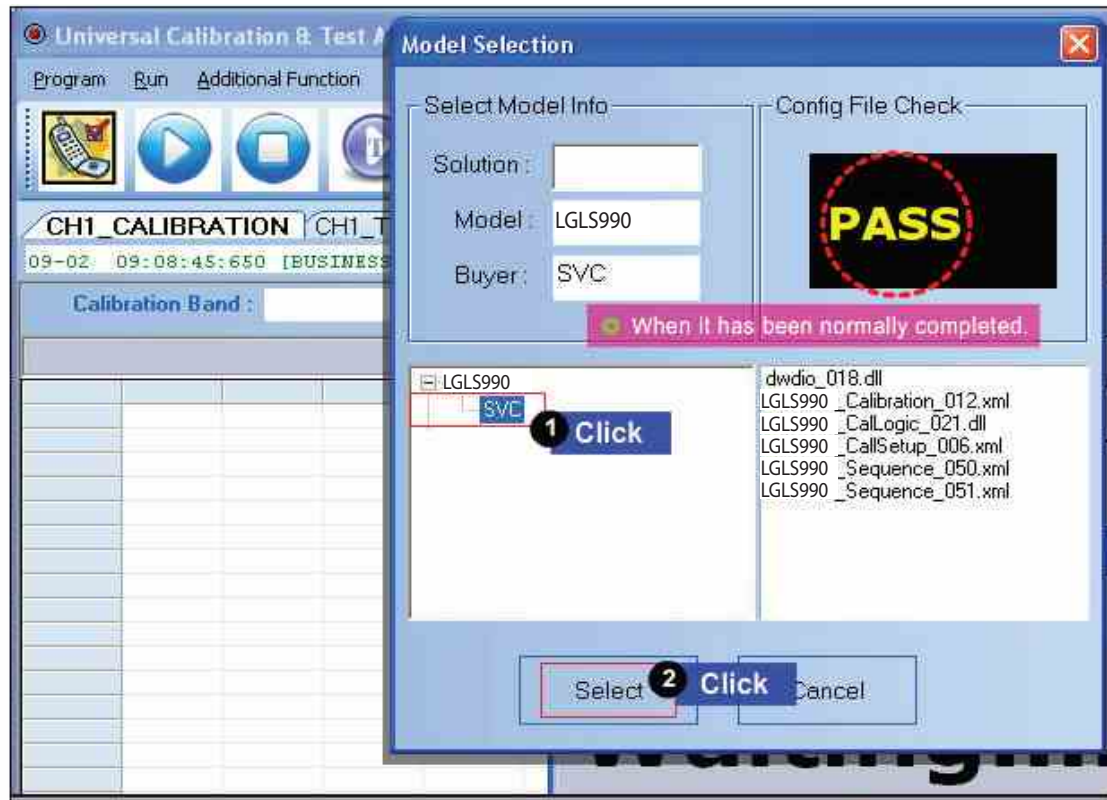


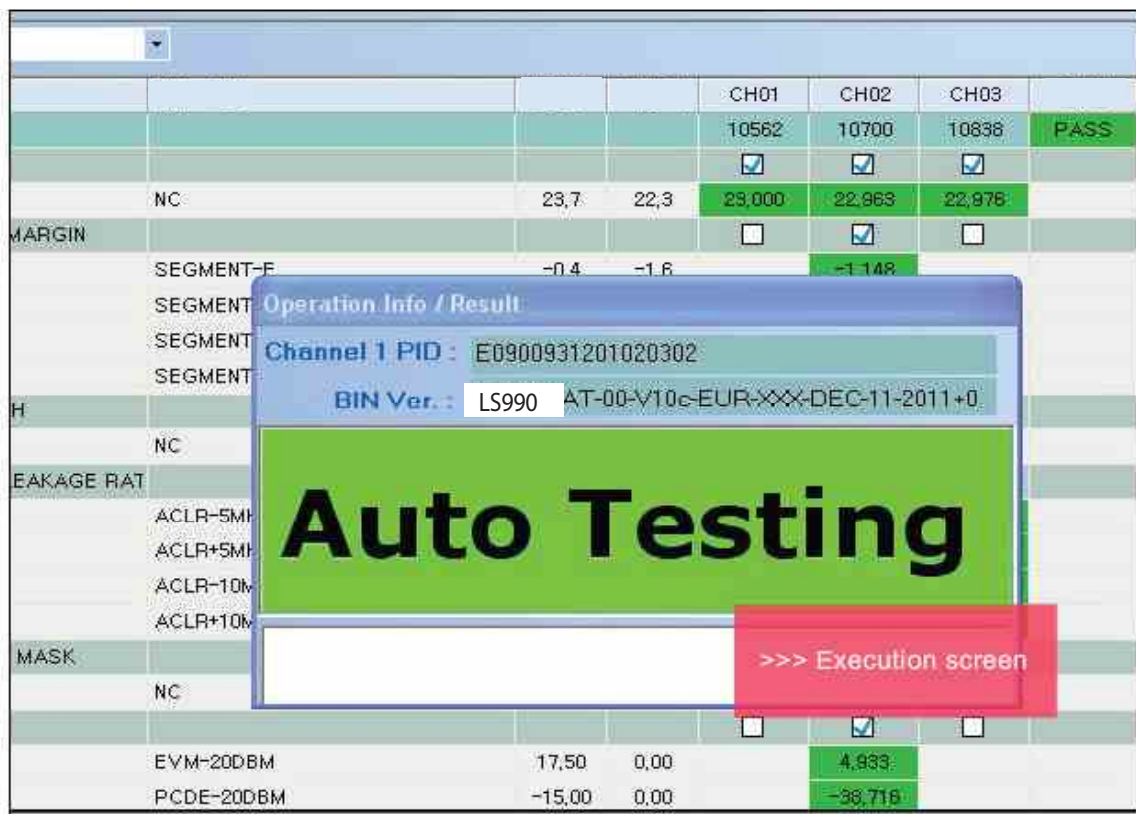
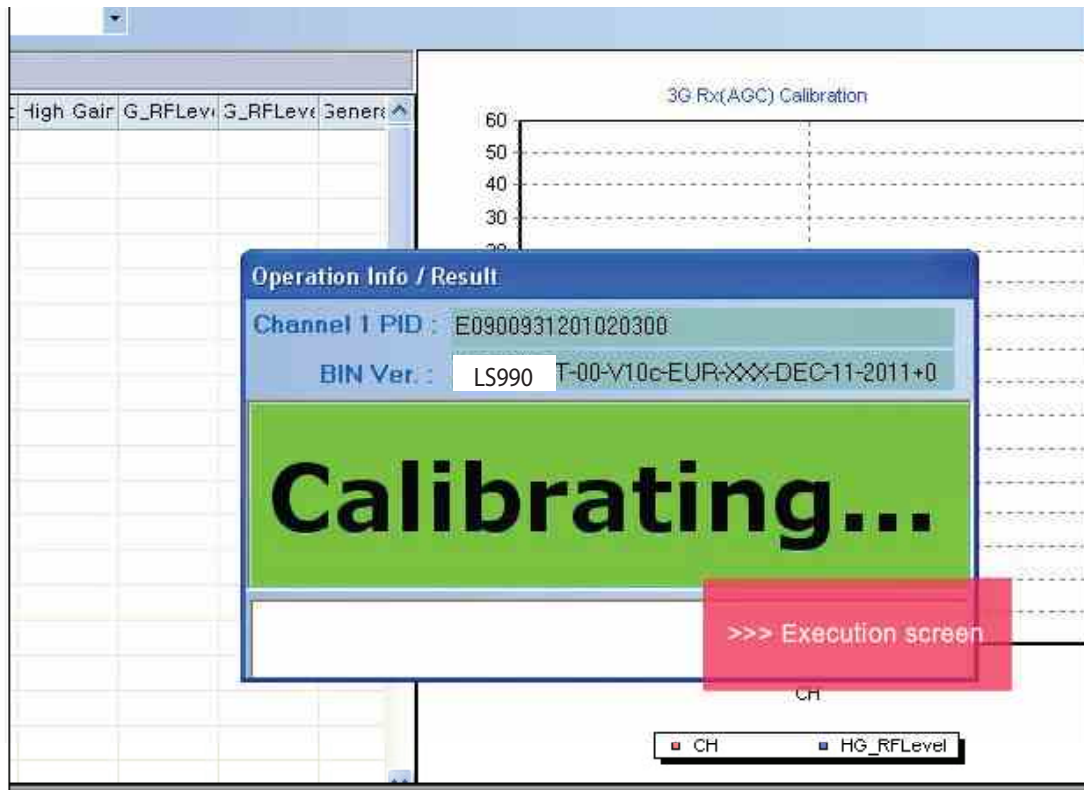


9. CALIBRATION



9. CALIBRATION

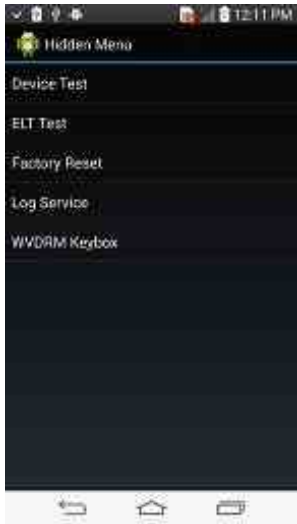
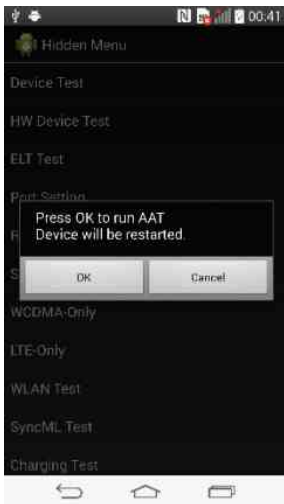
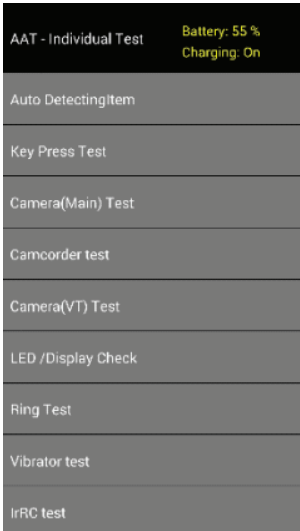




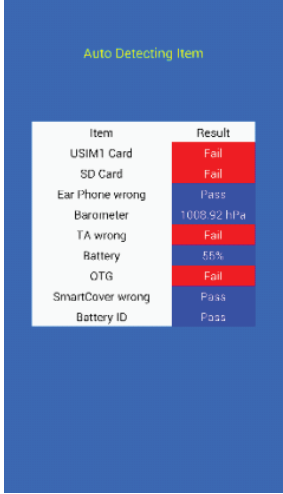
	Condition	USL	LSL	CH01	CH02	CH03	P/F
				10562	10700	10838	PASS
				☒	☒	☒	
	NC	23,7	22,3	23,065	22,963	23,013	
MARGIN				☐	☒	☐	
SEGMENT	Operation Info / Result Channel 1 PID : F0900931201020299 BIN Ver. : LS990 AT-00-V10c-EUR-XXX-DEC-11-2011+0 --- PASS ---						
SEGMENT							
SEGMENT							
SEGMENT							
OTH							
	NC						
LEAKAGE RAT							
ACLR-5M							
ACLR+5M							
ACLR-10M							
ACLR+10M							
IN MASK							
	NC	0,5	-0,5				
Y							☐
	EVM-20DBM	17,50	0,00		4,720		
	PCDE-20DBM	-15,00	0,00		-38,441		
System Loss :		MySystem(MS).gms RF900 6C.grf					

'PASS' The End

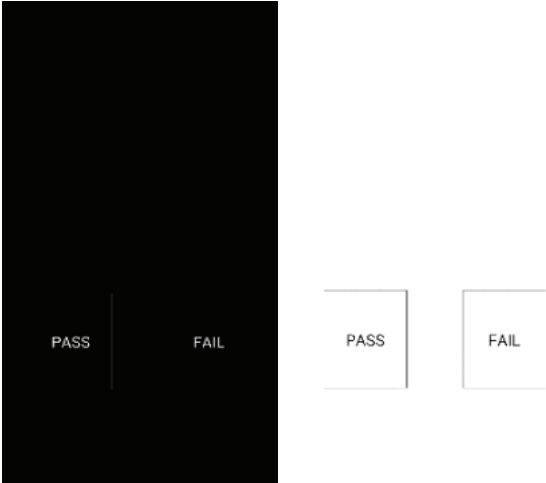
10. HIDDEN MENU

	Hidden Menu Start Start shortcut keys:3845#*990# Hidden Menu List Start the desired menu: Menu, click	
		Device Test List: All Auto Test – Partial : AATSET needed All Auto Test Result – Partial : AATSET needed All Auto all Test: Device functionality testing at the factory to use Normal Boot(+power key) FTM Boot(+power Key)
	Device Test List All Auto Test - Full: -> All Auto Test - Full menu click -> Continuous information on the menu, giving you ability test	

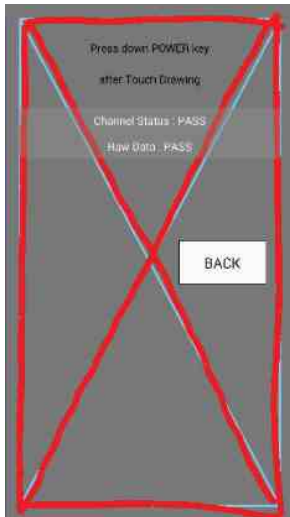
10. HIDDEN MENU

 Auto Detecting Item <table border="1"> <thead> <tr> <th>Item</th> <th>Result</th> </tr> </thead> <tbody> <tr> <td>USIM1 Card</td> <td>Fail</td> </tr> <tr> <td>SD Card</td> <td>Fail</td> </tr> <tr> <td>Ear Phone wrong</td> <td>Pass</td> </tr> <tr> <td>Barometer</td> <td>1008.92 hPa</td> </tr> <tr> <td>TA wrong</td> <td>Fail</td> </tr> <tr> <td>Battery</td> <td>55%</td> </tr> <tr> <td>OTG</td> <td>Fail</td> </tr> <tr> <td>SmartCover wrong</td> <td>Pass</td> </tr> <tr> <td>Battery ID</td> <td>Pass</td> </tr> </tbody> </table>	Item	Result	USIM1 Card	Fail	SD Card	Fail	Ear Phone wrong	Pass	Barometer	1008.92 hPa	TA wrong	Fail	Battery	55%	OTG	Fail	SmartCover wrong	Pass	Battery ID	Pass	USIM, SD Card, Ear Phone wrong, TA, Battery , OTG Test
Item	Result																				
USIM1 Card	Fail																				
SD Card	Fail																				
Ear Phone wrong	Pass																				
Barometer	1008.92 hPa																				
TA wrong	Fail																				
Battery	55%																				
OTG	Fail																				
SmartCover wrong	Pass																				
Battery ID	Pass																				
 Slide the "EXIT" button to exit Proximity: Cal. Pass: 110 Block the sensor with your hand Proximity: FAR Block the light with your hand Light: 384.00	Key Press Test - Check Key Press, Proximity sensor																				
 Touch to take a picture 4180a3120 Verify the picture min: 170, max: 262 value: 222 RETRY FAIL	Camera(Main)Test - Main Camera Test Preview and Result																				

10. HIDDEN MENU

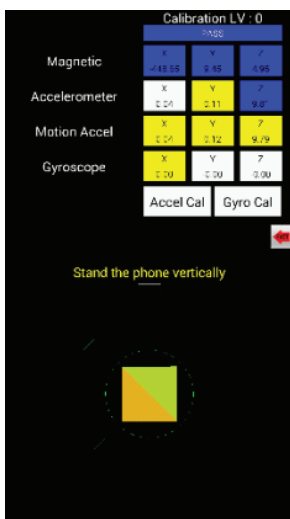
	Camcorder Test
	Camera(VT) Test
	LED/Display Check Test - Check Front-Bottom 3 LED (Left/Right menu & Home Button) - Touch 'Pass' -> Check Change the White color

	Ring test
	Vibrator test
	IrRC Master Phone IrRC Test



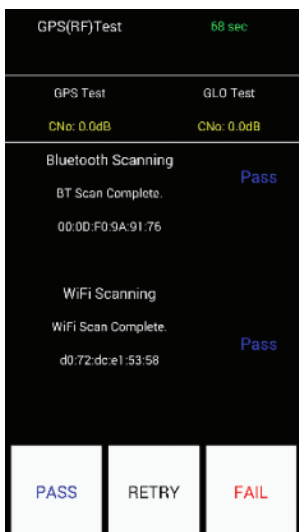
Touch window test

- Write with finger

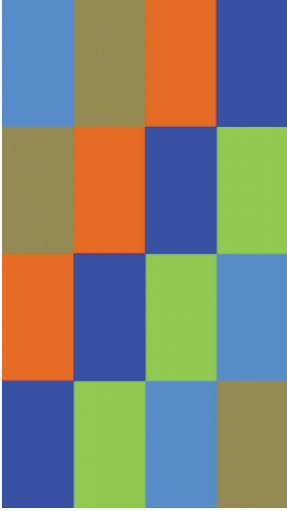
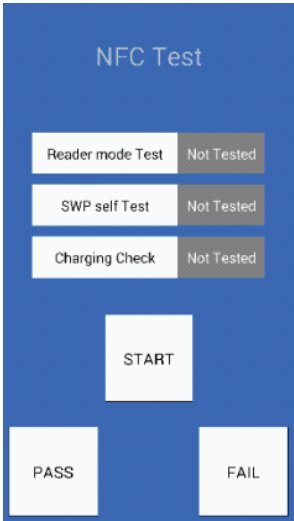
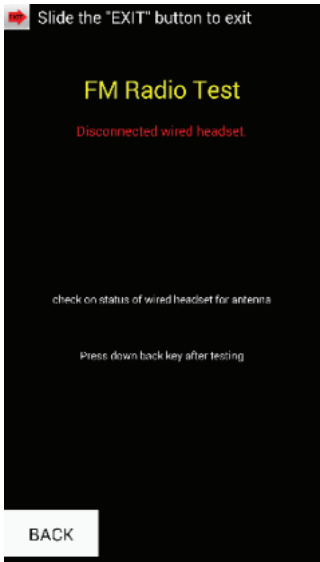


Motion Sensor Test

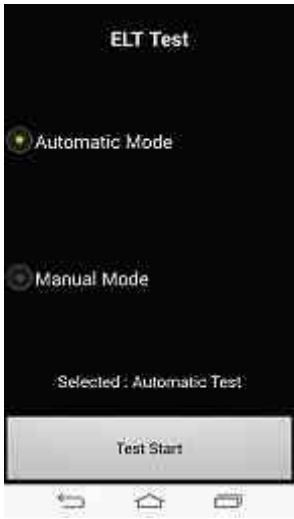
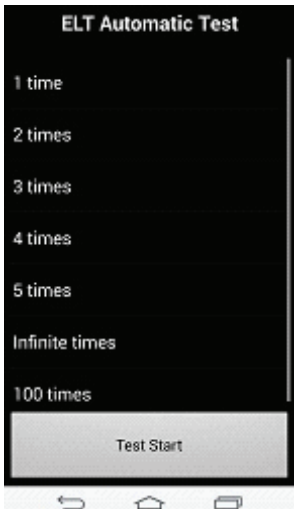
- Check Magnetic, Accelerometer, Gyroscope Sensor

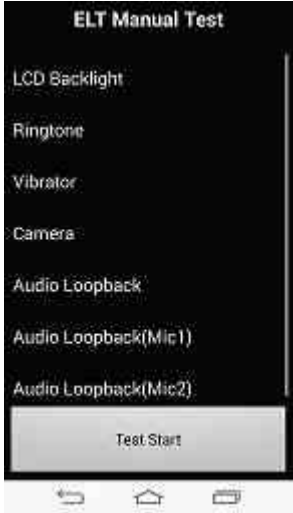
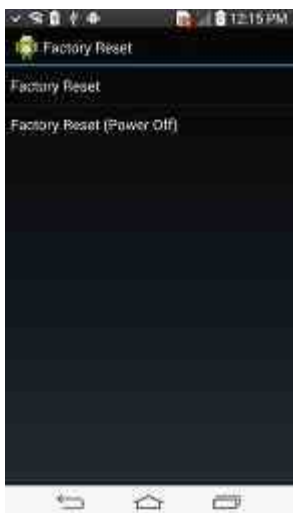


GPS BT WIFI Test

	HDMI Test : Using HDMI output
	NFC Test
	FM Radio

	Loopback Test : Audio
	Sensor Test
	Touch Draw Auto

 The screenshot shows the 'Camera OIS Test' menu. At the top, it says 'Camera OIS Test' and 'Fujifilm_1080M'. Below this are three rows of test results: 'Gyro' with a green background and 'FAIL' in red; 'Roll' with a blue background and 'PASS' in white; and 'Pitch' with a red background and 'FAIL' in white. At the bottom, there is a 'START' button and a 'FAIL' button.	Camera OIS Test
 The screenshot shows the 'ELT Test' menu. It has two options: 'Automatic Mode' and 'Manual Mode', both with a green dot icon. Below these options, it says 'Selected : Automatic Test'. At the bottom, there is a 'Test Start' button and three navigation icons (back, home, recent apps).	ELT Test Automatic Mode : Test Automatically Manual Mode : Test Selectivity
 The screenshot shows the 'ELT Automatic Test' menu. It has a list of test times: '1 time', '2 times', '3 times', '4 times', '5 times', 'Infinite times', and '100 times'. At the bottom, there is a 'Test Start' button and three navigation icons (back, home, recent apps).	ELT Automatic Test 1 time 2 times 3 times 4 times 5 times Infinite times 100 times

	ELT Manual Test LCD Backlight Ringtone Vibrator Camera Audio Loopback Audio Loopback(Mic1) Audio Loopback(Mic2)
	Factory Reset Factory Reset Factory Reset (Power Off)
	Factory Reset Factory Reset and Reboot

	Factory Reset (Power Off) Factory Reset and Power Off
	Log Service Log enable / Main log System log / Radio log Events log / Kernel log Power log / Packet log Crash Handler / Subsystem restart SSR Ramdump App QuickDump enable Copy Log files
	WVDRM Keybox Keybox Check

11. DISASSEMBLE GUIDE

1. Disassemble Rear Cover, Main antenna



Disassemble screw (12ea)



Disassemble Rear Cover

11. DISASSEMBLE GUIDE

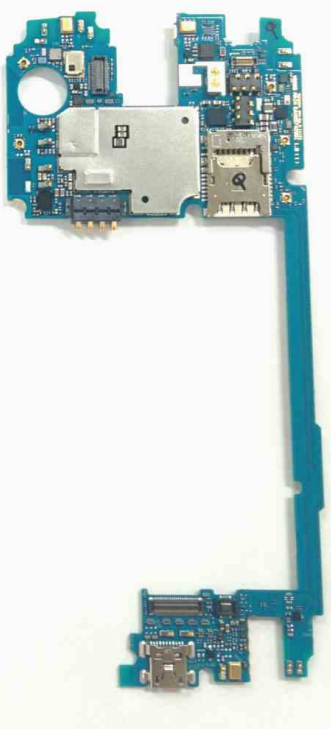
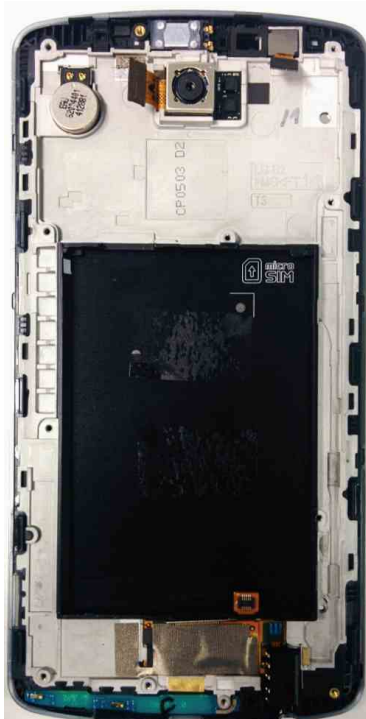


Disassemble Main Antenna

2. Disassemble Main PCB



Disassemble connector

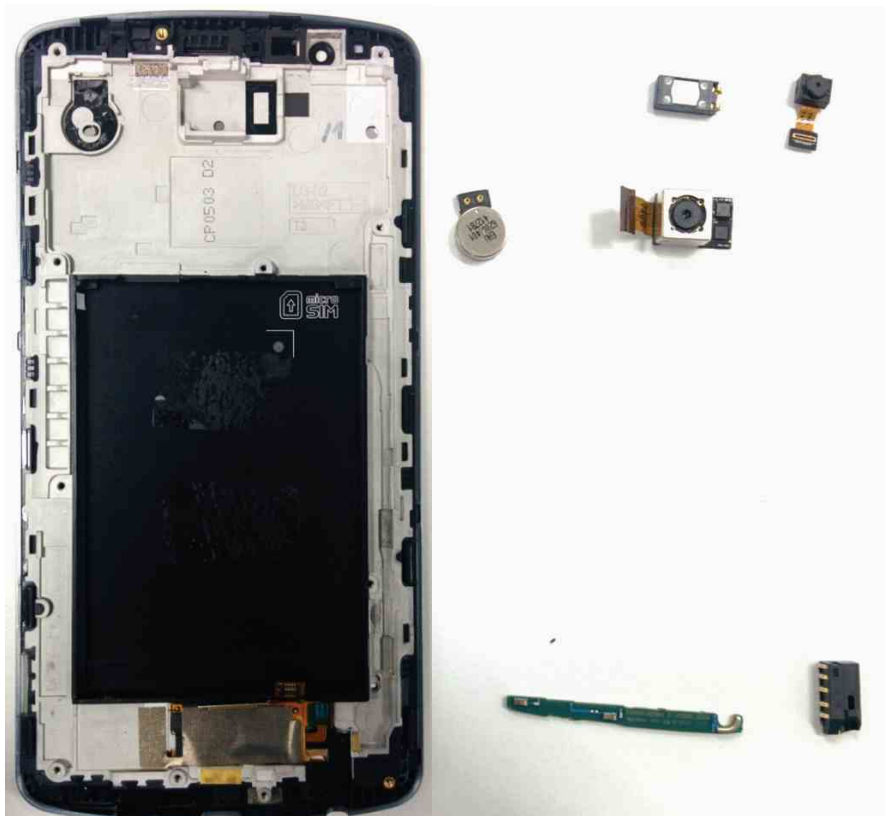


Disassemble Main PCB

3. Disassemble H/W parts



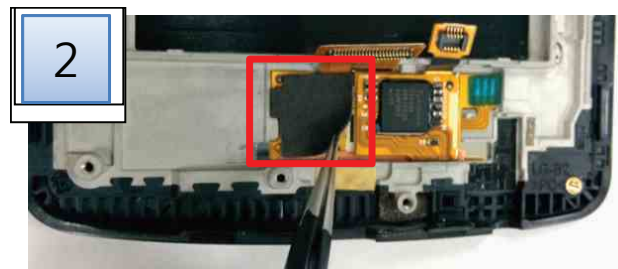
Disassemble H/W parts (6ea)



4. Disassemble Gasket, pad



Disassemble Gasket touch IC



Disassemble Pad LCD FPCB



Disassemble Gasket LCD FPCB



5. Complete disassembling LS990



12. EXPLODED VIEW & REPLACEMENT PART LIST

12.1 EXPLODED VIEW(SBOM)

12. EXPLODED VIEW & REPLACEMENT PART LIST

This document is draft version, so the contents will be updated after registering the final one.